# Conformance Test Report

IEC 61851-1, DIN 70121 and ISO 15118 (SLAC & V2G message) test cases — SECC under test (VTB)

Report generated: 2026-10-28 08:13:11

**Overall: FAIL**

## System Under Test

| | |
|---|---|
| Manufacturer | - |
| Model / type | - |
| Software / firmware | EVerest testing/main @ 0fa14b932e1774f354b283c596d3c9687bc0cc3a |
| Serial / identification | - |
| Notes | Nightly CI, all SECC stacks Evse15118D20. SUT config: config-test-conformance-dc-full-ocpp.yaml (one of the nightly's rig cycles, each with its own report) |
| Transfer mode | DC |

Black-box DUT: the internal software stack is not observable by the test system; the identification above was entered by the test operator.

## Declared capability profile (PICS/PIXIT)

| Capability | PICS/PIXIT | Declared |
|---|---|---|
| Energy transfer mode | `PICS_CMN_CMN_ChargingMode` | **dc** |
| Payment options offered | `PIXIT_SECC_CMN_PaymentOption` / `PICS_CMN_CMN_IdentificationMode` | **ExternalPayment, Contract** |
| Certificate services offered | `PIXIT_SECC_CMN_CertificateHandling` | **Install, Update** |
| Certificate validity checked against current time | `PICS_SECC_CertificateValidationCheck` | **yes** |
| Tester CSMS is the DUT's OCPP backend | `PIXIT_SECC_TesterCSMS` | **yes** |
| Transport security accepted | `PIXIT_SECC_CMN_TLS` | **tcp, tls** |
| Value-added service list (VAS) | `PIXIT_SECC_CMN_VAS == serviceDetail` / `PICS_SECC_VAS` | **no** |
| Internet access service | `PIXIT_SECC_CMN_InternetAccess == internetAccess` | **no** |
| Metering receipts requested | `PIXIT_SECC_CMN_ReceiptRequired` | **yes** |

| Capability | PICS/PIXIT | Declared |
|---|---|---|
| MeterInfo in charge-loop responses | `PIXIT_SECC_CMN_MeterInfo == meterInfo` | **yes** |
| MeteringConfirmation requested (ISO 15118-20) | `PICS_CMN_MeteringConfirmationRequested` | **no** |
| SalesTariff in schedules | `PIXIT_SECC_CMN_SalesTariff == salesTariff` | **no** |
| Control modes offered (ISO 15118-20) | `PICS_CMN_ControlMode` | **Dynamic, Scheduled** |
| Schedule initially unavailable (ISO 15118-20) | `PIXIT_SECC_ScheduleRetrievalSuspended` | **no** |
| DepartureTime considered | `PICS_SECC_CMN_DepartureTime` | **yes** |
| MaxEntriesSAScheduleTuple considered | `PICS_SECC_CMN_MaxEntriesSAScheduleTuple` | **yes** |
| Renegotiation supported | `PICS_CMN_CMN_Renegotiation` | **no** |
| CP State D (ventilation) during charging | `PICS_SECC_CMN_EvStateInChargeProgress == D` | **no** |
| Pause / resume supported | `PICS_SECC_CMN_Pause` | **yes** |
| Welding detection supported | `PICS_SECC_DC_WeldingDetection` | **yes** |
| ISO 15118-2 supported | `PICS_SECC_CMN_ISO15118_2` | **yes** |
| ISO 15118-20 supported | `PICS_SECC_CMN_ISO15118_20` | **yes** |
| DIN SPEC 70121 supported (DC) | `PICS_SECC_DC_DIN70121` | **yes** |
| DC energy transfer mode(s) advertised | `PIXIT_SECC_DC_SupportedEnergyTransferMode` | **DC_extended** |
| CableCheck answers Ongoing | `PIXIT_SECC_DC_CableCheckOngoing == ongoing` | **yes** |
| Authorization answers Ongoing | `PIXIT_SECC_CMN_AuthorizationOngoing == ongoing` | **no** |
| ChargeParameterDiscovery answers Ongoing | `PIXIT_SECC_CMN_ChargeParameterDiscoveryOngoing == ongoing` | **no** |
| CM_VALIDATE (BCB-toggle validation) | `PIXIT_SECC_CMN_CmValidate == cmValidate` | **yes** |
| Architecture makes validation unnecessary | `PIXIT_SECC_CMN_ArchitectureValidationNotRequired` | **no** |
| CM_AMP_MAP supported | `PIXIT_CMN_CMN_CmAmpMap` | **no** |
| SECC initiates CM_AMP_MAP | `PICS_CMN_CMN_InitiateCmAmpMap` | **no** |

| Capability | PICS/PIXIT | Declared |
|---|---|---|
| Self-initiated wake-up from pause | `PIXIT_CMN_CMN_WakeUp < PICS_CMN_CMN_WakeUp` | **no** |
| AC connection-loss handling | `PIXIT_SECC_AC_ConnectionLossHandling` | **n/a (AC-only)** |
| AC CP duty cycle after plug-in | `PIXIT_SECC_AC_InitialDutyCyle` | **n/a (AC-only)** |
| EIM completion handled by the SUT | `PICS_SECC_CMN_EIMDone` | **n/a (AC-only)** |
| IEC 61851-1 basic charging (AC) | `PIXIT_SECC_IEC61851_1_BasicCharging` | **n/a (AC-only)** |
| Identification modes supported | `PICS_CMN_CMN_IdentificationMode` | **ExternalPayment** |
| Certificate services used | `PIXIT_EVCC_CMN_CertificateHandling` | **none** |
| Renegotiation supported | `PICS_CMN_CMN_Renegotiation` | **no** |
| Pause / resume (sleep mode) supported | `PIXIT_EVCC_CMN_Pause` | **no** |
| Optimises the charging profile against sales tariffs | `PIXIT_EVCC_CMN_ChargingProfileOptimization` | **unknown** |
| States MaxNumberOfPhasesInUse (AC) | `PIXIT_EVCC_AC_MaxNumberOfPhasesInUse` | **n/a (AC-only)** |
| Queries value-added services (ServiceDetail) | `PIXIT_EVCC_CMN_VAS` | **no** |
| Transport security requested | `PIXIT_EVCC_CMN_TLS` | **none_** |
| Welding detection performed (DC) | `PIXIT_EVCC_DC_WeldingDetection` | **weldingDetection** |
| Reaction to a Pmax=0 W schedule | `PICS_EVCC_CMN_PmaxSchedulewithZeroPow` | **noSleep** |
| SLAC validation (CM_VALIDATE / BCB toggle) | `PIXIT_EVCC_CMN_CmValidate` | **unknown** |
| AC connection-loss handling | `PIXIT_EVCC_AC_ConnectionLossHandling` | **n/a (AC-only)** |
| CM_AMP_MAP supported | `PIXIT_CMN_CMN_CmAmpMap` | **no** |
| EV initiates CM_AMP_MAP | `PICS_CMN_CMN_InitiateCmAmpMap` | **no** |

Declared by the test operator (profile v1); the test-case selection of this run was derived from this declaration, mirroring the ISO 15118-4 ATS control part ([V2G4-018]/[V2G4-019]). Test cases the declaration marks not applicable were deselected by default; any that were explicitly force-run are annotated “declared N/A (operator force-run)” at their result.

## Test families & variants

| | |
|---|---|
| ISO 15118-3 SLAC (-5) | DUT (dut) |
| ISO 15118-2 message layer (-4) | DUT (dut) |
| DIN SPEC 70121 DC charging (DIN 70122) | DUT (dut) |
| ISO 15118-20 (ISO 15118-21 ATS) | DUT (dut) |
| PIONIX interop testing (vendor-defined) | DUT (dut) |
| PIONIX use cases (vendor-defined end-to-end scenarios) | DUT (dut) |

## Test Environment

| | |
|---|---|
| Test suite version | - |
| Test suite git SHA | - |
| Test suite branch | - |
| Mode | HW (CHARGEBRIDGE, BLACK-BOX DUT) |
| Host | CI runner |
| Kernel | 6.8.0-86-generic |
| PLC taps (EVSE / EV) | - / cb_plc_ev |
| MQTT broker | /tmp/everest-conformance-mqtt.sock:1883 |
| EVerest prefix | - |
| Pause between TCs | 3 s |
| EV BSP module id | ev_bsp |

## Totals

### ISO 15118-3 SLAC (-5)

DUT (dut)
62 pass
0 fail
0 not run
0 inconclusive

### ISO 15118-2 message layer (-4)

DUT (dut)
158 pass
0 fail
1 not run
0 inconclusive

### DIN SPEC 70121 DC charging (DIN 70122)

**DUT** (dut)
**103** pass
**12** fail
**0** not run
**0** inconclusive

### ISO 15118-20 (ISO 15118-21 ATS)

**DUT** (dut)
**131** pass
**1** fail
**0** not run
**0** inconclusive

### PIONIX interop testing (vendor-defined)

**DUT** (dut)
**8** pass
**0** fail
**0** not run
**0** inconclusive

### PIONIX use cases (vendor-defined end-to-end scenarios)

**DUT** (dut)
**4** pass
**4** fail
**0** not run
**0** inconclusive

Verdict semantics: expected failures (author-flagged known deviations / non-deterministic cases) are reported as **FAIL EXPECTED** — genuine reds that are not hidden and must be explicitly accepted by the operator. A FAIL (or FAIL EXPECTED) accepted by the operator is reported as **ACCEPTED FAIL** with the mandatory justification and does not count against the overall verdict. Overall verdict is PASS only if no test case FAILs (or is left FAIL EXPECTED without acceptance) on any variant.

## Summary matrix — ISO 15118-3 SLAC (-5)

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| **CmSlacParm** | | |
| TC_SECC_CMN_VTB_CmSlacParm_001 | PASS | 4.17 s |
| TC_SECC_CMN_VTB_CmSlacParm_002 | PASS | 4.00 s |
| TC_SECC_CMN_VTB_CmSlacParm_003 | PASS | 4.00 s |
| TC_SECC_CMN_VTB_CmSlacParm_004 | PASS | 54.49 s |
| TC_SECC_CMN_VTB_CmSlacParm_005 | PASS | 54.90 s |
| TC_SECC_CMN_VTB_CmSlacParm_006 | PASS | 55.10 s |
| TC_SECC_CMN_VTB_CmSlacParm_007 | PASS | 21.05 s |
| TC_SECC_CMN_VTB_CmSlacParm_008 | PASS | 9.72 s |
| TC_SECC_CMN_VTB_CmSlacParm_009 | PASS | 9.75 s |
| **AttenuationCharacterization** | | |
| TC_SECC_CMN_VTB_AttenuationCharacterization_001 | PASS | 5.69 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_002 | PASS | 5.00 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_003 | PASS | 5.23 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_004 | PASS | 6.16 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_005 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_006 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_007 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_008 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_009 | PASS | 6.40 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_010 | PASS | 6.60 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_011 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_012 | PASS | 5.79 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_013 | PASS | 5.60 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_014 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_015 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_016 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_017 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_018 | PASS | 6.00 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC_SECC_CMN_VTB_AttenuationCharacterization_019 | PASS | 21.02 s |
| TC_SECC_CMN_VTB_AttenuationCharacterization_020 | PASS | 7.31 s |
| **CmValidate** | | |
| TC_SECC_CMN_VTB_CmValidate_001 | PASS | 6.41 s |
| TC_SECC_CMN_VTB_CmValidate_002 | PASS | 5.79 s |
| TC_SECC_CMN_VTB_CmValidate_003 | PASS | 9.20 s |
| TC_SECC_CMN_VTB_CmValidate_004 | PASS | 8.80 s |
| TC_SECC_CMN_VTB_CmValidate_005 | PASS | 5.79 s |
| TC_SECC_CMN_VTB_CmValidate_006 | PASS | 5.00 s |
| TC_SECC_CMN_VTB_CmValidate_007 | PASS | 5.00 s |
| TC_SECC_CMN_VTB_CmValidate_008 | PASS | 5.00 s |
| TC_SECC_CMN_VTB_CmValidate_009 | PASS | 6.81 s |
| TC_SECC_CMN_VTB_CmValidate_013 | PASS | 20.01 s |
| **CmSlacMatch** | | |
| TC_SECC_CMN_VTB_CmSlacMatch_002 | PASS | 6.89 s |
| TC_SECC_CMN_VTB_CmSlacMatch_004 | PASS | 7.20 s |
| TC_SECC_CMN_VTB_CmSlacMatch_006 | PASS | 7.99 s |
| TC_SECC_CMN_VTB_CmSlacMatch_008 | PASS | 8.00 s |
| TC_SECC_CMN_VTB_CmSlacMatch_010 | PASS | 8.00 s |
| TC_SECC_CMN_VTB_CmSlacMatch_012 | PASS | 8.60 s |
| TC_SECC_CMN_VTB_CmSlacMatch_014 | PASS | 8.40 s |
| TC_SECC_CMN_VTB_CmSlacMatch_016 | PASS | 8.00 s |
| TC_SECC_CMN_VTB_CmSlacMatch_018 | PASS | 8.60 s |
| TC_SECC_CMN_VTB_CmSlacMatch_020 | PASS | 8.40 s |
| TC_SECC_CMN_VTB_CmSlacMatch_022 | PASS | 8.00 s |
| TC_SECC_CMN_VTB_CmSlacMatch_024 | PASS | 22.22 s |
| **PLCLinkStatus** | | |
| TC_SECC_CMN_VTB_PLCLinkStatus_001 | PASS | 4.46 s |
| TC_SECC_CMN_VTB_PLCLinkStatus_002 | PASS | 20.12 s |
| TC_SECC_CMN_VTB_PLCLinkStatus_003 | PASS | 5.28 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC_SECC_CMN_VTB_PLCLinkStatus_004 | PASS | 5.41 s |
| TC_SECC_CMN_VTB_PLCLinkStatus_005 | PASS | 10.70 s |
| TC_SECC_DC_VTB_PLCLinkStatus_001 | PASS | 22.36 s |
| TC_SECC_DC_VTB_PLCLinkStatus_003 | PASS | 32.32 s |
| TC_SECC_DC_VTB_PLCLinkStatus_004 | PASS | 10.71 s |
| TC_SECC_DC_VTB_PLCLinkStatus_005 | PASS | 16.60 s |
| TC_SECC_DC_VTB_PLCLinkStatus_006 | PASS | 22.12 s |
| TC_SECC_DC_VTB_PLCLinkStatus_007 | PASS | 20.02 s |

## Summary matrix — ISO 15118-2 message layer (-4)

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| **SupportedAppProtocol** | | |
| TC_SECC_CMN_VTB_SupportedAppProtocol_001 | PASS | 6.08 s |
| TC_SECC_CMN_VTB_SupportedAppProtocol_002 | PASS | 6.39 s |
| TC_SECC_CMN_VTB_SupportedAppProtocol_003 | PASS | 23.83 s |
| TC_SECC_CMN_VTB_SupportedAppProtocol_004 | PASS | 6.77 s |
| TC_SECC_CMN_VTB_SupportedAppProtocol_005 | PASS | 30.64 s |
| TC_SECC_CMN_VTB_SupportedAppProtocol_006 | PASS | 5.75 s |
| **SessionSetup** | | |
| TC_SECC_CMN_VTB_SessionSetup_001 | PASS | 5.56 s |
| TC_SECC_CMN_VTB_SessionSetup_002 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_SessionSetup_003 | PASS | 23.02 s |
| TC_SECC_CMN_VTB_SessionSetup_004 | PASS | 23.01 s |
| TC_SECC_DC_VTB_SessionSetup_001 | PASS | 20.19 s |
| TC_SECC_DC_VTB_SessionSetup_002 | PASS | 20.00 s |
| TC_SECC_DC_VTB_SessionSetup_003 | PASS | 20.00 s |
| **ServiceDiscovery** | | |
| TC_SECC_CMN_VTB_ServiceDiscovery_001 | PASS | 6.09 s |
| TC_SECC_CMN_VTB_ServiceDiscovery_002 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_ServiceDiscovery_003 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_ServiceDiscovery_004 | PASS | 67.90 s |
| TC_SECC_DC_VTB_ServiceDiscovery_001 | PASS | 20.32 s |
| **PaymentServiceSelection** | | |
| TC_SECC_CMN_VTB_PaymentServiceSelection_001 | PASS | 5.80 s |
| TC_SECC_CMN_VTB_PaymentServiceSelection_003 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_PaymentServiceSelection_004 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_PaymentServiceSelection_005 | PASS | 5.99 s |
| TC_SECC_CMN_VTB_PaymentServiceSelection_006 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_PaymentServiceSelection_007 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_PaymentServiceSelection_008 | PASS | 68.10 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC_SECC_CMN_VTB_PaymentServiceSelection_010 | PASS | 5.90 s |
| **PaymentDetails** | | |
| TC_SECC_CMN_VTB_PaymentDetails_001 | PASS | 5.98 s |
| TC_SECC_CMN_VTB_PaymentDetails_002 | PASS | 6.01 s |
| TC_SECC_CMN_VTB_PaymentDetails_003 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_PaymentDetails_004 | PASS | 5.99 s |
| TC_SECC_CMN_VTB_PaymentDetails_005 | PASS | 6.20 s |
| TC_SECC_CMN_VTB_PaymentDetails_006 | PASS | 68.70 s |
| TC_SECC_CMN_VTB_PaymentDetails_007 | PASS | 6.11 s |
| TC_SECC_CMN_VTB_PaymentDetails_008 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_PaymentDetails_009 | PASS | 5.99 s |
| TC_SECC_CMN_VTB_PaymentDetails_010 | PASS | 6.00 s |
| **Authorization** | | |
| TC_SECC_CMN_VTB_Authorization_001 | PASS | 7.17 s |
| TC_SECC_CMN_VTB_Authorization_002 | PASS | 7.00 s |
| TC_SECC_CMN_VTB_Authorization_003 | PASS | 6.40 s |
| TC_SECC_CMN_VTB_Authorization_004 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_Authorization_005 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_Authorization_006 | PASS | 6.20 s |
| TC_SECC_CMN_VTB_Authorization_007 | PASS | 6.80 s |
| TC_SECC_CMN_VTB_Authorization_008 | PASS | 67.88 s |
| TC_SECC_CMN_VTB_Authorization_009 | PASS | 68.00 s |
| TC_SECC_CMN_VTB_Authorization_010 | PASS | 6.11 s |
| TC_SECC_CMN_VTB_Authorization_011 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_Authorization_012 | PASS | 6.81 s |
| TC_SECC_CMN_VTB_Authorization_013 | PASS | 7.00 s |
| **ChargeParameterDiscovery** | | |
| TC_SECC_CMN_VTB_ChargeParameterDiscovery_001 | PASS | 6.60 s |
| TC_SECC_CMN_VTB_ChargeParameterDiscovery_002 | PASS | 69.70 s |
| TC_SECC_DC_VTB_ChargeParameterDiscovery_001 | PASS | 7.30 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC_SECC_DC_VTB_ChargeParameterDiscovery_002 | PASS | 7.00 s |
| TC_SECC_DC_VTB_ChargeParameterDiscovery_003 | PASS | 7.00 s |
| TC_SECC_DC_VTB_ChargeParameterDiscovery_004 | PASS | 7.00 s |
| TC_SECC_DC_VTB_ChargeParameterDiscovery_006 | PASS | 7.00 s |
| TC_SECC_DC_VTB_ChargeParameterDiscovery_007 | PASS | 21.22 s |
| TC_SECC_DC_VTB_ChargeParameterDiscovery_012 | PASS | 6.77 s |
| TC_SECC_DC_VTB_ChargeParameterDiscovery_016 | PASS | 7.00 s |
| TC_SECC_DC_VTB_ChargeParameterDiscovery_017 | PASS | 21.22 s |
| TC_SECC_DC_VTB_ChargeParameterDiscovery_019 | PASS | 21.40 s |
| TC_SECC_DC_VTB_ChargeParameterDiscovery_022 | PASS | 22.15 s |
| **CableCheck** | | |
| TC_SECC_DC_VTB_CableCheck_001 | PASS | 11.40 s |
| TC_SECC_DC_VTB_CableCheck_003 | PASS | 7.99 s |
| TC_SECC_DC_VTB_CableCheck_004 | PASS | 8.00 s |
| TC_SECC_DC_VTB_CableCheck_005 | PASS | 69.49 s |
| TC_SECC_DC_VTB_CableCheck_006 | PASS | 10.71 s |
| TC_SECC_DC_VTB_CableCheck_008 | PASS | 8.80 s |
| **PreCharge** | | |
| TC_SECC_DC_VTB_PreCharge_001 | PASS | 11.80 s |
| TC_SECC_DC_VTB_PreCharge_002 | PASS | 11.59 s |
| TC_SECC_DC_VTB_PreCharge_003 | PASS | 11.00 s |
| TC_SECC_DC_VTB_PreCharge_004 | PASS | 72.90 s |
| **PowerDelivery** | | |
| TC_SECC_DC_VTB_PowerDelivery_001 | PASS | 11.41 s |
| TC_SECC_DC_VTB_PowerDelivery_002 | PASS | 13.20 s |
| TC_SECC_DC_VTB_PowerDelivery_003 | PASS | 12.20 s |
| TC_SECC_DC_VTB_PowerDelivery_004 | PASS | 12.79 s |
| TC_SECC_DC_VTB_PowerDelivery_005 | PASS | 11.80 s |
| TC_SECC_DC_VTB_PowerDelivery_006 | PASS | 13.20 s |
| TC_SECC_DC_VTB_PowerDelivery_007 | PASS | 11.79 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC_SECC_DC_VTB_PowerDelivery_008 | PASS | 12.41 s |
| TC_SECC_DC_VTB_PowerDelivery_009 | PASS | 73.89 s |
| TC_SECC_DC_VTB_PowerDelivery_010 | PASS | 75.01 s |
| TC_SECC_DC_VTB_PowerDelivery_013 | PASS | 12.09 s |
| TC_SECC_DC_VTB_PowerDelivery_014 | PASS | 13.20 s |
| TC_SECC_DC_VTB_PowerDelivery_015 | PASS | 13.00 s |
| TC_SECC_DC_VTB_PowerDelivery_016 | PASS | 13.00 s |
| TC_SECC_DC_VTB_PowerDelivery_017 | PASS | 12.60 s |
| TC_SECC_DC_VTB_PowerDelivery_018 | PASS | 13.00 s |
| TC_SECC_DC_VTB_PowerDelivery_019 | PASS | 13.00 s |
| TC_SECC_DC_VTB_PowerDelivery_020 | PASS | 13.00 s |
| TC_SECC_DC_VTB_PowerDelivery_021 | PASS | 13.00 s |
| **CurrentDemand** | | |
| TC_SECC_DC_VTB_CurrentDemand_001 | PASS | 12.24 s |
| TC_SECC_DC_VTB_CurrentDemand_002 | PASS | 12.00 s |
| TC_SECC_DC_VTB_CurrentDemand_003 | PASS | 12.39 s |
| TC_SECC_DC_VTB_CurrentDemand_004 | PASS | 11.60 s |
| TC_SECC_DC_VTB_CurrentDemand_005 | PASS | 73.90 s |
| TC_SECC_DC_VTB_CurrentDemand_006 | PASS | 12.30 s |
| TC_SECC_DC_VTB_CurrentDemand_007 | PASS | 11.99 s |
| TC_SECC_DC_VTB_CurrentDemand_008 | PASS | 12.40 s |
| TC_SECC_DC_VTB_CurrentDemand_009 | PASS | 12.60 s |
| **MeteringReceipt** | | |
| TC_SECC_DC_VTB_MeteringReceipt_001 | PASS | 12.40 s |
| TC_SECC_DC_VTB_MeteringReceipt_002 | PASS | 12.59 s |
| TC_SECC_DC_VTB_MeteringReceipt_003 | PASS | 12.00 s |
| TC_SECC_DC_VTB_MeteringReceipt_004 | PASS | 12.00 s |
| TC_SECC_DC_VTB_MeteringReceipt_005 | PASS | 74.10 s |
| TC_SECC_DC_VTB_MeteringReceipt_006 | PASS | 12.09 s |
| TC_SECC_DC_VTB_MeteringReceipt_007 | PASS | 12.00 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| **WeldingDetection** | | |
| TC_SECC_DC_VTB_WeldingDetection_001 | PASS | 13.40 s |
| TC_SECC_DC_VTB_WeldingDetection_002 | PASS | 12.99 s |
| TC_SECC_DC_VTB_WeldingDetection_003 | PASS | 13.00 s |
| TC_SECC_DC_VTB_WeldingDetection_004 | PASS | 75.51 s |
| TC_SECC_DC_VTB_WeldingDetection_005 | PASS | 14.70 s |
| **SessionStop** | | |
| TC_SECC_DC_VTB_SessionStop_002 | PASS | 13.21 s |
| TC_SECC_DC_VTB_SessionStop_006 | PASS | 13.98 s |
| TC_SECC_DC_VTB_SessionStop_007 | PASS | 13.29 s |
| TC_SECC_DC_VTB_SessionStop_008 | PASS | 13.90 s |
| TC_SECC_DC_VTB_SessionStop_009 | PASS | 75.90 s |
| **V2GTPSessionSetup** | | |
| TC_SECC_CMN_VTB_V2GTPSessionSetup_001 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_V2GTPSessionSetup_002 | PASS | 9.99 s |
| TC_SECC_CMN_VTB_V2GTPSessionSetup_003 | PASS | 10.01 s |
| TC_SECC_CMN_VTB_V2GTPSessionSetup_004 | PASS | 10.00 s |
| TC_SECC_CMN_VTB_V2GTPSessionSetup_005 | PASS | 10.60 s |
| **V2GTPSDP** | | |
| TC_SECC_CMN_VTB_V2GTPSDP_001 | PASS | 6.99 s |
| TC_SECC_CMN_VTB_V2GTPSDP_002 | PASS | 8.20 s |
| TC_SECC_CMN_VTB_V2GTPSDP_003 | PASS | 8.80 s |
| TC_SECC_CMN_VTB_V2GTPSDP_004 | PASS | 8.21 s |
| TC_SECC_CMN_VTB_V2GTPSDP_005 | PASS | 8.00 s |
| **SDP** | | |
| TC_SECC_CMN_VTB_SDP_001 | PASS | 5.80 s |
| TC_SECC_CMN_VTB_SDP_002 | PASS | 5.99 s |
| TC_SECC_CMN_VTB_SDP_003 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_SDP_004 | PASS | 40.11 s |
| TC_SECC_CMN_VTB_SDP_005 | PASS | 40.03 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC_SECC_CMN_VTB_SDP_006 | PASS | 40.04 s |
| **DinChargeParameterDiscovery** | | |
| TC_SECC_DC_VTB_DinChargeParameterDiscovery_008 | NOT RUN<br>not applicable (ATS selection) | 0.00 s |
| **CertificateInstallation** | | |
| TC_SECC_CMN_VTB_CertificateInstallation_001 | PASS | 6.80 s |
| TC_SECC_CMN_VTB_CertificateInstallation_002 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_CertificateInstallation_003 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_CertificateInstallation_004 | PASS | 67.89 s |
| TC_SECC_CMN_VTB_CertificateInstallation_005 | PASS | 6.11 s |
| TC_SECC_CMN_VTB_CertificateInstallation_006 | PASS | 5.99 s |
| TC_SECC_CMN_VTB_CertificateInstallation_007 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_CertificateInstallation_008 | PASS | 5.99 s |
| **CertificateUpdate** | | |
| TC_SECC_CMN_VTB_CertificateUpdate_001 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_CertificateUpdate_002 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_CertificateUpdate_003 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_CertificateUpdate_004 | PASS | 68.30 s |
| TC_SECC_CMN_VTB_CertificateUpdate_005 | PASS | 6.70 s |
| TC_SECC_CMN_VTB_CertificateUpdate_006 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_CertificateUpdate_007 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_CertificateUpdate_008 | PASS | 6.03 s |
| TC_SECC_CMN_VTB_CertificateUpdate_009 | PASS | 5.97 s |
| TC_SECC_CMN_VTB_CertificateUpdate_010 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_CertificateUpdate_011 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_CertificateUpdate_012 | PASS | 5.99 s |
| TC_SECC_CMN_VTB_CertificateUpdate_013 | PASS | 6.20 s |
| TC_SECC_CMN_VTB_CertificateUpdate_014 | PASS | 6.80 s |
| TC_SECC_CMN_VTB_CertificateUpdate_015 | PASS | 6.00 s |
| TC_SECC_CMN_VTB_CertificateUpdate_016 | PASS | 6.20 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC_SECC_CMN_VTB_CertificateUpdate_017 | PASS | 6.79 s |

## Summary matrix — DIN SPEC 70121 DC charging (DIN 70122)

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| **DinV2gtpSessionSetup** | | |
| TC_SECC_DC_VTB_DinV2gtpSessionSetup_001 | PASS | 6.20 s |
| TC_SECC_DC_VTB_DinV2gtpSessionSetup_002 | PASS | 6.00 s |
| TC_SECC_DC_VTB_DinV2gtpSessionSetup_003 | PASS | 5.99 s |
| TC_SECC_DC_VTB_DinV2gtpSessionSetup_004 | PASS | 6.00 s |
| **DinV2gtpSdp** | | |
| TC_SECC_DC_VTB_DinV2gtpSdp_001 | PASS | 5.80 s |
| **DinSdp** | | |
| TC_SECC_DC_VTB_DinSdp_001 | PASS | 7.00 s |
| TC_SECC_DC_VTB_DinSdp_002 | PASS | 7.00 s |
| **DinSupportedAppProtocol** | | |
| TC_SECC_DC_VTB_DinSupportedAppProtocol_001 | PASS | 6.42 s |
| TC_SECC_DC_VTB_DinSupportedAppProtocol_002 | PASS | 6.00 s |
| TC_SECC_DC_VTB_DinSupportedAppProtocol_003 | PASS | 23.82 s |
| TC_SECC_DC_VTB_DinSupportedAppProtocol_004 | PASS | 6.17 s |
| TC_SECC_DC_VTB_DinSupportedAppProtocol_005 | PASS | 6.00 s |
| **DinSessionSetup** | | |
| TC_SECC_DC_VTB_DinSessionSetup_001 | PASS | 6.37 s |
| TC_SECC_DC_VTB_DinSessionSetup_002 | PASS | 6.40 s |
| TC_SECC_DC_VTB_DinSessionSetup_003 | PASS | 23.02 s |
| TC_SECC_DC_VTB_DinSessionSetup_004 | PASS | 21.00 s |
| **DinServiceDiscovery** | | |
| TC_SECC_DC_VTB_DinServiceDiscovery_001 | PASS | 6.60 s |
| TC_SECC_DC_VTB_DinServiceDiscovery_002 | PASS | 6.00 s |
| TC_SECC_DC_VTB_DinServiceDiscovery_003 | PASS | 6.00 s |
| TC_SECC_DC_VTB_DinServiceDiscovery_004 | FAIL | 68.29 s |
| TC_SECC_DC_VTB_DinServiceDiscovery_005 | PASS | 8.11 s |
| TC_SECC_DC_VTB_DinServiceDiscovery_006 | PASS | 22.02 s |
| TC_SECC_DC_VTB_DinServiceDiscovery_008 | PASS | 21.50 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| **DinServicePaymentSelection** | | |
| TC_SECC_DC_VTB_DinServicePaymentSelection_001 | PASS | 6.07 s |
| TC_SECC_DC_VTB_DinServicePaymentSelection_002 | PASS | 6.00 s |
| TC_SECC_DC_VTB_DinServicePaymentSelection_003 | PASS | 6.00 s |
| TC_SECC_DC_VTB_DinServicePaymentSelection_004 | PASS | 6.00 s |
| TC_SECC_DC_VTB_DinServicePaymentSelection_005 | FAIL | 68.48 s |
| TC_SECC_DC_VTB_DinServicePaymentSelection_006 | PASS | 21.34 s |
| TC_SECC_DC_VTB_DinServicePaymentSelection_007 | PASS | 6.17 s |
| TC_SECC_DC_VTB_DinServicePaymentSelection_008 | PASS | 21.03 s |
| **DinContractAuthentication** | | |
| TC_SECC_DC_VTB_DinContractAuthentication_001 | PASS | 6.96 s |
| TC_SECC_DC_VTB_DinContractAuthentication_002 | PASS | 6.39 s |
| TC_SECC_DC_VTB_DinContractAuthentication_003 | PASS | 6.00 s |
| TC_SECC_DC_VTB_DinContractAuthentication_004 | FAIL | 69.29 s |
| TC_SECC_DC_VTB_DinContractAuthentication_005 | PASS | 21.93 s |
| TC_SECC_DC_VTB_DinContractAuthentication_006 | PASS | 6.97 s |
| TC_SECC_DC_VTB_DinContractAuthentication_007 | PASS | 21.73 s |
| **DinChargeParameterDiscovery** | | |
| TC_SECC_DC_VTB_DinChargeParameterDiscovery_001 | PASS | 6.86 s |
| TC_SECC_DC_VTB_DinChargeParameterDiscovery_002 | PASS | 7.00 s |
| TC_SECC_DC_VTB_DinChargeParameterDiscovery_003 | PASS | 7.60 s |
| TC_SECC_DC_VTB_DinChargeParameterDiscovery_004 | PASS | 7.40 s |
| TC_SECC_DC_VTB_DinChargeParameterDiscovery_005 | PASS | 7.00 s |
| TC_SECC_DC_VTB_DinChargeParameterDiscovery_006 | FAIL | 69.69 s |
| TC_SECC_DC_VTB_DinChargeParameterDiscovery_007 | PASS | 22.93 s |
| TC_SECC_DC_VTB_DinChargeParameterDiscovery_009 | PASS | 22.41 s |
| **DinCableCheck** | | |
| TC_SECC_DC_VTB_DinCableCheck_001 | PASS | 10.99 s |
| TC_SECC_DC_VTB_DinCableCheck_003 | PASS | 7.39 s |
| TC_SECC_DC_VTB_DinCableCheck_004 | PASS | 8.00 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC_SECC_DC_VTB_DinCableCheck_005 | FAIL | 69.88 s |
| TC_SECC_DC_VTB_DinCableCheck_006 | PASS | 8.11 s |
| TC_SECC_DC_VTB_DinCableCheck_007 | PASS | 12.00 s |
| TC_SECC_DC_VTB_DinCableCheck_008 | PASS | 12.00 s |
| TC_SECC_DC_VTB_DinCableCheck_009 | PASS | 11.80 s |
| TC_SECC_DC_VTB_DinCableCheck_010 | PASS | 11.00 s |
| TC_SECC_DC_VTB_DinCableCheck_011 | PASS | 11.00 s |
| TC_SECC_DC_VTB_DinCableCheck_012 | PASS | 11.00 s |
| TC_SECC_DC_VTB_DinCableCheck_013 | PASS | 11.00 s |
| TC_SECC_DC_VTB_DinCableCheck_014 | PASS | 11.00 s |
| TC_SECC_DC_VTB_DinCableCheck_015 | PASS | 21.61 s |
| TC_SECC_DC_VTB_DinCableCheck_016 | PASS | 11.38 s |
| TC_SECC_DC_VTB_DinCableCheck_017 | PASS | 11.00 s |
| TC_SECC_DC_VTB_DinCableCheck_018 | PASS | 11.00 s |
| TC_SECC_DC_VTB_DinCableCheck_019 | PASS | 11.20 s |
| TC_SECC_DC_VTB_DinCableCheck_020 | PASS | 11.80 s |
| TC_SECC_DC_VTB_DinCableCheck_021 | PASS | 7.39 s |
| TC_SECC_DC_VTB_DinCableCheck_023 | PASS | 22.34 s |
| **DinPreCharge** | | |
| TC_SECC_DC_VTB_DinPreCharge_001 | PASS | 11.74 s |
| TC_SECC_DC_VTB_DinPreCharge_002 | PASS | 11.60 s |
| TC_SECC_DC_VTB_DinPreCharge_003 | PASS | 11.40 s |
| TC_SECC_DC_VTB_DinPreCharge_004 | FAIL | 74.09 s |
| TC_SECC_DC_VTB_DinPreCharge_005 | PASS | 11.51 s |
| TC_SECC_DC_VTB_DinPreCharge_006 | PASS | 12.60 s |
| TC_SECC_DC_VTB_DinPreCharge_007 | PASS | 26.22 s |
| TC_SECC_DC_VTB_DinPreCharge_008 | PASS | 26.57 s |
| **DinPowerDelivery** | | |
| TC_SECC_DC_VTB_DinPowerDelivery_001 | PASS | 11.85 s |
| TC_SECC_DC_VTB_DinPowerDelivery_002 | PASS | 12.00 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC_SECC_DC_VTB_DinPowerDelivery_003 | PASS | 12.00 s |
| TC_SECC_DC_VTB_DinPowerDelivery_004 | PASS | 12.00 s |
| TC_SECC_DC_VTB_DinPowerDelivery_005 | PASS | 12.00 s |
| TC_SECC_DC_VTB_DinPowerDelivery_006 | PASS | 12.00 s |
| TC_SECC_DC_VTB_DinPowerDelivery_007 | PASS | 12.00 s |
| TC_SECC_DC_VTB_DinPowerDelivery_008 | PASS | 12.00 s |
| TC_SECC_DC_VTB_DinPowerDelivery_009 | FAIL | 34.03 s |
| TC_SECC_DC_VTB_DinPowerDelivery_010 | FAIL | 22.58 s |
| TC_SECC_DC_VTB_DinPowerDelivery_011 | PASS | 26.21 s |
| TC_SECC_DC_VTB_DinPowerDelivery_012 | PASS | 26.20 s |
| TC_SECC_DC_VTB_DinPowerDelivery_013 | PASS | 27.03 s |
| TC_SECC_DC_VTB_DinPowerDelivery_014 | PASS | 27.20 s |
| **DinCurrentDemand** | | |
| TC_SECC_DC_VTB_DinCurrentDemand_001 | PASS | 11.47 s |
| TC_SECC_DC_VTB_DinCurrentDemand_002 | PASS | 12.40 s |
| TC_SECC_DC_VTB_DinCurrentDemand_003 | PASS | 11.60 s |
| TC_SECC_DC_VTB_DinCurrentDemand_004 | FAIL | 74.28 s |
| TC_SECC_DC_VTB_DinCurrentDemand_005 | PASS | 26.53 s |
| TC_SECC_DC_VTB_DinCurrentDemand_006 | PASS | 27.13 s |
| **DinWeldingDetection** | | |
| TC_SECC_DC_VTB_DinWeldingDetection_001 | PASS | 13.41 s |
| TC_SECC_DC_VTB_DinWeldingDetection_002 | PASS | 13.30 s |
| TC_SECC_DC_VTB_DinWeldingDetection_003 | PASS | 13.70 s |
| TC_SECC_DC_VTB_DinWeldingDetection_004 | FAIL | 75.48 s |
| TC_SECC_DC_VTB_DinWeldingDetection_005 | PASS | 15.31 s |
| TC_SECC_DC_VTB_DinWeldingDetection_006 | PASS | 28.42 s |
| TC_SECC_DC_VTB_DinWeldingDetection_007 | PASS | 29.15 s |
| **DinSessionStop** | | |
| TC_SECC_DC_VTB_DinSessionStop_001 | PASS | 14.99 s |
| TC_SECC_DC_VTB_DinSessionStop_002 | PASS | 15.19 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC_SECC_DC_VTB_DinSessionStop_003 | PASS | 13.20 s |
| TC_SECC_DC_VTB_DinSessionStop_004 | PASS | 13.00 s |
| TC_SECC_DC_VTB_DinSessionStop_005 | FAIL | 75.49 s |
| TC_SECC_DC_VTB_DinSessionStop_006 | PASS | 13.71 s |
| TC_SECC_DC_VTB_DinSessionStop_007 | PASS | 14.00 s |
| TC_SECC_DC_VTB_DinSessionStop_008 | FAIL | 36.23 s |
| TC_SECC_DC_VTB_DinSessionStop_009 | PASS | 14.97 s |
| TC_SECC_DC_VTB_DinSessionStop_010 | PASS | 28.42 s |
| TC_SECC_DC_VTB_DinSessionStop_011 | PASS | 28.20 s |
| TC_SECC_DC_VTB_DinSessionStop_012 | PASS | 28.05 s |
| TC_SECC_DC_VTB_DinSessionStop_013 | PASS | 28.10 s |

## Summary matrix — ISO 15118-20 (ISO 15118-21 ATS)

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| **SDP** | | |
| TC21_SECC_SDP_001 | PASS | 5.60 s |
| TC21_SECC_SDP_002 | PASS | 5.99 s |
| TC21_SECC_SDP_003 | PASS | 18.82 s |
| TC21_SECC_SDP_004 | PASS | 18.80 s |
| **V2GTPSDP** | | |
| TC21_SECC_V2GTPSDP_001 | PASS | 5.97 s |
| TC21_SECC_V2GTPSDP_002 | PASS | 18.62 s |
| TC21_SECC_V2GTPSDP_003 | PASS | 17.99 s |
| TC21_SECC_V2GTPSDP_004 | PASS | 18.00 s |
| TC21_SECC_V2GTPSDP_005 | PASS | 18.00 s |
| TC21_SECC_V2GTPSDP_006 | PASS | 18.00 s |
| TC21_SECC_V2GTPSDP_007 | PASS | 18.00 s |
| **V2GTPSupportedAppProtocol** | | |
| TC21_SECC_V2GTPSupportedAppProtocol_001 | PASS | 6.18 s |
| TC21_SECC_V2GTPSupportedAppProtocol_002 | PASS | 6.40 s |
| TC21_SECC_V2GTPSupportedAppProtocol_003 | PASS | 23.62 s |
| TC21_SECC_V2GTPSupportedAppProtocol_004 | PASS | 23.20 s |
| TC21_SECC_V2GTPSupportedAppProtocol_005 | PASS | 23.00 s |
| TC21_SECC_V2GTPSupportedAppProtocol_006 | PASS | 23.00 s |
| TC21_SECC_V2GTPSupportedAppProtocol_007 | PASS | 23.00 s |
| TC21_SECC_V2GTPSupportedAppProtocol_008 | PASS | 23.00 s |
| **V2GTPCommonMessages** | | |
| TC21_SECC_V2GTPCommonMessages_001 | PASS | 6.00 s |
| TC21_SECC_V2GTPCommonMessages_002 | PASS | 6.00 s |
| TC21_SECC_V2GTPCommonMessages_003 | PASS | 22.82 s |
| TC21_SECC_V2GTPCommonMessages_004 | PASS | 23.00 s |
| TC21_SECC_V2GTPCommonMessages_005 | PASS | 23.00 s |
| TC21_SECC_V2GTPCommonMessages_006 | PASS | 23.80 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC21_SECC_V2GTPCommonMessages_007 | PASS | 23.19 s |
| TC21_SECC_V2GTPCommonMessages_008 | PASS | 23.00 s |
| **SupportedAppProtocol** | | |
| TC21_SECC_SupportedAppProtocol_001 | PASS | 6.19 s |
| TC21_SECC_SupportedAppProtocol_002 | PASS | 6.40 s |
| TC21_SECC_SupportedAppProtocol_003 | PASS | 6.00 s |
| TC21_SECC_SupportedAppProtocol_004 | PASS | 6.00 s |
| **SessionSetup** | | |
| TC21_SECC_SessionSetup_002 | PASS | 6.00 s |
| TC21_SECC_SessionSetup_003 | PASS | 23.83 s |
| TC21_SECC_SessionSetup_004 | PASS | 6.17 s |
| TC21_SECC_SessionSetup_005 | PASS | 6.00 s |
| **AuthorizationSetup** | | |
| TC21_SECC_AuthorizationSetup_001 | PASS | 6.00 s |
| TC21_SECC_AuthorizationSetup_002 | PASS | 6.00 s |
| TC21_SECC_AuthorizationSetup_003 | PASS | 6.00 s |
| TC21_SECC_AuthorizationSetup_004 | PASS | 67.88 s |
| TC21_SECC_AuthorizationSetup_005 | PASS | 6.11 s |
| **CertificateInstallation** | | |
| TC21_SECC_CertificateInstallation_001 | PASS | 6.61 s |
| TC21_SECC_CertificateInstallation_002 | PASS | 6.99 s |
| TC21_SECC_CertificateInstallation_003 | PASS | 7.00 s |
| TC21_SECC_CertificateInstallation_004 | PASS | 6.99 s |
| TC21_SECC_CertificateInstallation_005 | PASS | 7.00 s |
| TC21_SECC_CertificateInstallation_006 | PASS | 6.40 s |
| TC21_SECC_CertificateInstallation_007 | PASS | 6.00 s |
| TC21_SECC_CertificateInstallation_008 | PASS | 6.00 s |
| TC21_SECC_CertificateInstallation_009 | PASS | 6.00 s |
| TC21_SECC_CertificateInstallation_010 | PASS | 6.00 s |
| TC21_SECC_CertificateInstallation_011 | PASS | 6.00 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC21_SECC_CertificateInstallation_012 | PASS | 5.99 s |
| TC21_SECC_CertificateInstallation_013 | PASS | 6.00 s |
| TC21_SECC_CertificateInstallation_014 | PASS | 6.00 s |
| TC21_SECC_CertificateInstallation_015 | PASS | 6.00 s |
| TC21_SECC_CertificateInstallation_016 | PASS | 6.00 s |
| TC21_SECC_CertificateInstallation_017 | PASS | 6.00 s |
| TC21_SECC_CertificateInstallation_018 | PASS | 6.00 s |
| TC21_SECC_CertificateInstallation_019 | PASS | 6.00 s |
| TC21_SECC_CertificateInstallation_020 | PASS | 5.99 s |
| TC21_SECC_CertificateInstallation_021 | PASS | 6.40 s |
| **Authorization** | | |
| TC21_SECC_Authorization_001 | PASS | 7.02 s |
| TC21_SECC_Authorization_003 | PASS | 7.00 s |
| TC21_SECC_Authorization_004 | PASS | 6.40 s |
| TC21_SECC_Authorization_005 | PASS | 5.99 s |
| TC21_SECC_Authorization_006 | PASS | 6.00 s |
| TC21_SECC_Authorization_007 | PASS | 6.00 s |
| TC21_SECC_Authorization_008 | PASS | 6.00 s |
| TC21_SECC_Authorization_009 | PASS | 6.00 s |
| TC21_SECC_Authorization_010 | PASS | 6.40 s |
| TC21_SECC_Authorization_011 | PASS | 6.60 s |
| TC21_SECC_Authorization_012 | PASS | 6.00 s |
| TC21_SECC_Authorization_013 | PASS | 6.00 s |
| TC21_SECC_Authorization_014 | PASS | 6.00 s |
| TC21_SECC_Authorization_015 | PASS | 5.99 s |
| TC21_SECC_Authorization_016 | PASS | 6.00 s |
| TC21_SECC_Authorization_017 | PASS | 6.00 s |
| TC21_SECC_Authorization_018 | PASS | 6.00 s |
| TC21_SECC_Authorization_021 | PASS | 6.00 s |
| TC21_SECC_Authorization_023 | PASS | 7.40 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC21_SECC_Authorization_024 | PASS | 6.60 s |
| **CertificateInstallationOrAuthorization** | | |
| TC21_SECC_CertificateInstallationOrAuthorization_001 | PASS | 6.59 s |
| TC21_SECC_CertificateInstallationOrAuthorization_002 | PASS | 67.89 s |
| **ServiceDiscovery** | | |
| TC21_SECC_ServiceDiscovery_001 | PASS | 6.79 s |
| TC21_SECC_ServiceDiscovery_002 | PASS | 7.00 s |
| TC21_SECC_ServiceDiscovery_003 | PASS | 7.00 s |
| TC21_SECC_ServiceDiscovery_004 | PASS | 68.89 s |
| TC21_SECC_ServiceDiscovery_005 | PASS | 7.11 s |
| **ServiceDetail** | | |
| TC21_SECC_ServiceDetail_001 | PASS | 6.58 s |
| TC21_SECC_ServiceDetail_002 | PASS | 7.60 s |
| TC21_SECC_ServiceDetail_003 | PASS | 7.40 s |
| TC21_SECC_ServiceDetail_004 | PASS | 7.00 s |
| TC21_SECC_ServiceDetail_005 | PASS | 68.89 s |
| TC21_SECC_ServiceDetail_006 | PASS | 7.11 s |
| **ServiceSelection** | | |
| TC21_SECC_ServiceSelection_001 | PASS | 7.20 s |
| TC21_SECC_ServiceSelection_002 | PASS | 7.60 s |
| TC21_SECC_ServiceSelection_003 | PASS | 7.40 s |
| TC21_SECC_ServiceSelection_004 | PASS | 7.00 s |
| TC21_SECC_ServiceSelection_006 | PASS | 68.89 s |
| TC21_SECC_ServiceSelection_007 | PASS | 7.11 s |
| **ChargeParameterDiscovery** | | |
| TC21_SECC_ChargeParameterDiscovery_001 | PASS | 7.30 s |
| TC21_SECC_ChargeParameterDiscovery_002 | PASS | 7.00 s |
| TC21_SECC_ChargeParameterDiscovery_003 | PASS | 69.49 s |
| TC21_SECC_ChargeParameterDiscovery_004 | PASS | 7.51 s |
| **ScheduleExchange** | | |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC21_SECC_ScheduleExchange_002 | PASS | 7.39 s |
| TC21_SECC_ScheduleExchange_003 | PASS | 7.00 s |
| TC21_SECC_ScheduleExchange_004 | PASS | 7.00 s |
| TC21_SECC_ScheduleExchange_005 | PASS | 68.89 s |
| TC21_SECC_ScheduleExchange_006 | PASS | 7.11 s |
| TC21_SECC_ScheduleExchange_008 | PASS | 7.00 s |
| **PowerDelivery** | | |
| TC21_SECC_PowerDelivery_001 | PASS | 12.41 s |
| TC21_SECC_PowerDelivery_002 | PASS | 13.40 s |
| TC21_SECC_PowerDelivery_003 | PASS | 12.00 s |
| TC21_SECC_PowerDelivery_004 | PASS | 13.60 s |
| TC21_SECC_PowerDelivery_005 | PASS | 12.60 s |
| TC21_SECC_PowerDelivery_006 | PASS | 13.40 s |
| TC21_SECC_PowerDelivery_007 | FAIL | 12.60 s |
| TC21_SECC_PowerDelivery_008 | PASS | 12.00 s |
| TC21_SECC_PowerDelivery_009 | PASS | 73.49 s |
| TC21_SECC_PowerDelivery_010 | PASS | 74.60 s |
| TC21_SECC_PowerDelivery_011 | PASS | 11.51 s |
| TC21_SECC_PowerDelivery_012 | PASS | 12.00 s |
| TC21_SECC_PowerDelivery_013 | PASS | 12.00 s |
| **ChargeLoop** | | |
| TC21_SECC_ChargeLoop_001 | PASS | 11.92 s |
| TC21_SECC_ChargeLoop_002 | PASS | 73.89 s |
| **SessionStop** | | |
| TC21_SECC_SessionStop_001 | PASS | 13.21 s |
| TC21_SECC_SessionStop_002 | PASS | 13.00 s |
| TC21_SECC_SessionStop_003 | PASS | 13.00 s |
| TC21_SECC_SessionStop_004 | PASS | 75.09 s |
| TC21_SECC_SessionStop_005 | PASS | 13.11 s |
| TC21_SECC_SessionStop_006 | PASS | 13.00 s |

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| TC21_SECC_SessionStop_007 | PASS | 12.99 s |

## Summary matrix — PIONIX interop testing (vendor-defined)

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| **PionixInterop** | | |
| TC_SECC_CMN_VTB_PionixInterop_010 | PASS | 4.38 s |
| TC_SECC_DC_VTB_PionixInterop_001 | PASS | 12.41 s |
| TC_SECC_DC_VTB_PionixInterop_003 | PASS | 52.57 s |
| TC_SECC_DC_VTB_PionixInterop_004 | PASS | 4.62 s |
| TC_SECC_DC_VTB_PionixInterop_007 | PASS | 11.60 s |
| TC_SECC_DC_VTB_PionixInterop_014 | PASS | 28.43 s |
| TC_SECC_DC_VTB_PionixInterop_017 | PASS | 42.80 s |
| TC_SECC_DC_VTB_PionixInterop_019 | PASS | 15.03 s |

## Summary matrix — PIONIX use cases (vendor-defined end-to-end scenarios)

| Test case | DUT (dut) | |
|---|---|---|
| | Verdict | Duration |
| **PionixUseCase** | | |
| TC_SECC_DC_VTB_PionixUseCase_001 | PASS | 28.32 s |
| TC_SECC_DC_VTB_PionixUseCase_003 | PASS | 29.47 s |
| TC_SECC_DC_VTB_PionixUseCase_005 | PASS | 26.17 s |
| TC_SECC_DC_VTB_PionixUseCase_007 | PASS | 27.50 s |
| TC_SECC_DC_VTB_PionixUseCase_009 | FAIL | 28.10 s |
| TC_SECC_DC_VTB_PionixUseCase_010 | FAIL | 27.99 s |
| TC_SECC_DC_VTB_PionixUseCase_011 | FAIL | 28.60 s |
| TC_SECC_DC_VTB_PionixUseCase_012 | FAIL | 26.90 s |

# Test case details — ISO 15118-3 SLAC (-5)

## TC_SECC_CMN_VTB_CmSlacParm_001

TC_SECC_CMN_VTB_CmSlacParm_001.

Test objective: Test System executes GoodCase procedure, indicates the initial CP State B transition independent of duty cycle. Test System then sends a CM_SLAC_PARM.REQ message with a valid runID and all additional valid parameters and waits for the CM_SLAC_PARM.CNF message. Test System then checks that a CM_SLAC_PARM.CNF message with the current runID, EV MAC and all additional valid parameters is sent by the SUT.

**DUT** (dut) **PASS** 4.17 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_SLAC_PARM.CNF runID = REQ runID | cnf=1122334455667788<br>req=1122334455667788 | ✓ |
| CM_SLAC_PARM.CNF applicationType = COMMON | got 0 | ✓ |
| CM_SLAC_PARM.CNF securityType = COMMON | got 0 | ✓ |
| CM_SLAC_PARM.CNF M-SOUND target = broadcast | got ffffffffffff | ✓ |
| CM_SLAC_PARM.CNF num_sounds = expected | got 10 | ✓ |
| CM_SLAC_PARM.CNF timeout = expected | got 6 | ✓ |
| CM_SLAC_PARM.CNF resp_type = expected | got 1 | ✓ |
| CM_SLAC_PARM.CNF forwarding_sta = EV MAC | got 02005e00cafe | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF num_sounds | 10 | = 10 | ✓ |
| CM_SLAC_PARM.CNF timeout | 6 | = 6 | ✓ |
| CM_SLAC_PARM.CNF resp_type | 1 | = 1 | ✓ |

## TC_SECC_CMN_VTB_CmSlacParm_002

TC_SECC_CMN_VTB_CmSlacParm_002.

Test objective: as _001 but Test System indicates the initial CP State C transition independent of duty cycle before sending CM_SLAC_PARM.REQ, and checks the CM_SLAC_PARM.CNF with the current runID, EV MAC and all additional valid parameters is sent by the SUT. Annex C sets vc_state := C before the StateB precondition: the plug-in transition is A→C directly.

**DUT (dut)** **PASS** 4.00 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_SLAC_PARM.CNF runID = REQ runID | cnf=1122334455667788 req=1122334455667788 | ✓ |
| CM_SLAC_PARM.CNF applicationType = COMMON | got 0 | ✓ |
| CM_SLAC_PARM.CNF securityType = COMMON | got 0 | ✓ |
| CM_SLAC_PARM.CNF M-SOUND target = broadcast | got ffffffffffff | ✓ |
| CM_SLAC_PARM.CNF num_sounds = expected | got 10 | ✓ |
| CM_SLAC_PARM.CNF timeout = expected | got 6 | ✓ |
| CM_SLAC_PARM.CNF resp_type = expected | got 1 | ✓ |
| CM_SLAC_PARM.CNF forwarding_sta = EV MAC | got 02005e00cafe | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF num_sounds | 10 | = 10 | ✓ |
| CM_SLAC_PARM.CNF timeout | 6 | = 6 | ✓ |
| CM_SLAC_PARM.CNF resp_type | 1 | = 1 | ✓ |

## TC_SECC_CMN_VTB_CmSlacParm_003

TC_SECC_CMN_VTB_CmSlacParm_003.

Test objective: as _001 but Test System indicates the initial CP State D transition independent of duty cycle before sending CM_SLAC_PARM.REQ, and checks the CM_SLAC_PARM.CNF with the current runID, EV MAC and all additional valid parameters is sent by the SUT. Annex C sets vc_state := D before the StateB precondition: the plug-in transition is A→D directly.

**DUT (dut)** **PASS** 4.00 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_SLAC_PARM.CNF runID = REQ runID | cnf=1122334455667788 req=1122334455667788 | ✓ |
| CM_SLAC_PARM.CNF applicationType = COMMON | got 0 | ✓ |
| CM_SLAC_PARM.CNF securityType = COMMON | got 0 | ✓ |
| CM_SLAC_PARM.CNF M-SOUND target = broadcast | got ffffffffffff | ✓ |
| CM_SLAC_PARM.CNF num_sounds = expected | got 10 | ✓ |
| CM_SLAC_PARM.CNF timeout = expected | got 6 | ✓ |
| CM_SLAC_PARM.CNF resp_type = expected | got 1 | ✓ |
| CM_SLAC_PARM.CNF forwarding_sta = EV MAC | got 02005e00cafe | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF num_sounds | 10 | = 10 | ✓ |
| CM_SLAC_PARM.CNF timeout | 6 | = 6 | ✓ |
| CM_SLAC_PARM.CNF resp_type | 1 | = 1 | ✓ |

## TC_SECC_CMN_VTB_CmSlacParm_004

TC_SECC_CMN_VTB_CmSlacParm_004.

Test objective: Test System executes GoodCase procedure and indicates the initial CP State B transition. Test System then waits until the TT_EVSE_SLAC_init timer (maximum value) has expired before sending the CM_SLAC_PARM.REQ message with a valid runID and all additional valid parameters. Test System then checks that no CM_SLAC_PARM.CNF message is sent by the SUT until the TT_match_response timer has expired.

**DUT (dut)** **PASS** 54.49 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 50s | silent for 50s | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

## TC_SECC_CMN_VTB_CmSlacParm_005

TC_SECC_CMN_VTB_CmSlacParm_005.

Test objective: Test System executes GoodCase procedure and indicates the initial CP State B transition. Test System then waits until the TT_EVSE_SLAC_init timer (maximum value) has expired before sending the CM_SLAC_PARM.REQ message with a valid runID and all additional valid parameters. Furthermore an additional CM_SLAC_PARM.REQ message with an invalid 'applicationType' equals to 'FF'H was sent before timeout was triggered. Test System then checks that no CM_SLAC_PARM.CNF message is sent by the SUT until the TT_match_response timer has expired.

**DUT (dut)** **PASS** 54.90 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 50s | silent for 50s | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

## TC_SECC_CMN_VTB_CmSlacParm_006

TC_SECC_CMN_VTB_CmSlacParm_006.

Test objective: Test System executes GoodCase procedure and indicates the initial CP State B transition. Test System then waits until the TT_EVSE_SLAC_init timer (maximum value) has expired before sending the CM_SLAC_PARM.REQ message with a valid runID and all additional valid parameters. Furthermore an additional CM_SLAC_PARM.REQ message with an invalid 'securityType' equals to 'FF'H was sent before timeout was triggered. Test System then checks that no CM_SLAC_PARM.CNF message is sent by the SUT until the TT_match_response timer has expired.

**DUT** (dut) **PASS** 55.10 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 50s | silent for 50s | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

## TC_SECC_CMN_VTB_CmSlacParm_007

TC_SECC_CMN_VTB_CmSlacParm_007.

Test objective: Test System executes GoodCase procedure, indicates the initial CP State B transition, CP State A again and sends a CM_SLAC_PARM.REQ message with a valid runID and all additional valid parameters. Test System then checks that no CM_SLAC_PARM.CNF message is sent by the SUT until the TT_match_response timer has expired.

**DUT** (dut) **PASS** 21.05 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

## TC_SECC_CMN_VTB_CmSlacParm_008

TC_SECC_CMN_VTB_CmSlacParm_008.

Test objective: Test System executes GoodCase procedure and stays for 'par_SECC_waitForPlugin' in CP State A (SUT goes to sleep after time period in CP State A). After 'par_SECC_waitForPlugin' Test System indicates the initial CP State B transition, sends CM_SLAC_PARM.REQ messages with a valid runID and all additional valid parameters and waits for the CM_SLAC_PARM.CNF message as long as 'par_T_conn_max_comm' is running. Test System then checks that a CM_SLAC_PARM.CNF message with the current runID, EV MAC and all additional valid parameters is sent by the SUT after wakeup by plug-in.

**DUT** (dut) **PASS** 9.72 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_SLAC_PARM.CNF runID = REQ runID | cnf=1122334455667788<br>req=1122334455667788 | ✓ |
| CM_SLAC_PARM.CNF applicationType = COMMON | got 0 | ✓ |
| CM_SLAC_PARM.CNF securityType = COMMON | got 0 | ✓ |
| CM_SLAC_PARM.CNF M-SOUND target = broadcast | got ffffffffffff | ✓ |
| CM_SLAC_PARM.CNF num_sounds = expected | got 10 | ✓ |
| CM_SLAC_PARM.CNF timeout = expected | got 6 | ✓ |
| CM_SLAC_PARM.CNF resp_type = expected | got 1 | ✓ |
| CM_SLAC_PARM.CNF forwarding_sta = EV MAC | got 02005e00cafe | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| readiness after wakeup by plug-in | 452.7 ms | ≤ 8000 ms (T_conn_max_comm) | ✓ |
| CM_SLAC_PARM.CNF num_sounds | 10 | = 10 | ✓ |
| CM_SLAC_PARM.CNF timeout | 6 | = 6 | ✓ |
| CM_SLAC_PARM.CNF resp_type | 1 | = 1 | ✓ |

## TC_SECC_CMN_VTB_CmSlacParm_009

TC_SECC_CMN_VTB_CmSlacParm_009.

Test objective: Test System executes GoodCase procedure and stays for 'par_SECC_waitForPlugin' in CP State A (SUT goes to sleep after time period in CP State A). After 'par_SECC_waitForPlugin' Test System indicates the initial CP State B transition and waits for a 5 % duty cycle as long as 'par_T_conn_max_comm' is running. After duty cycle detection Test System sends one CM_SLAC_PARM.REQ message with a valid runID and all additional valid parameters and waits for the CM_SLAC_PARM.CNF message. Test System then checks that a CM_SLAC_PARM.CNF message with the current runID, EV MAC and all additional valid parameters is sent by the SUT after wakeup by plug-in.

### DUT (dut) PASS 9.75 s

| Conformance check | Observed | Result |
|---|---|---|
| 5% duty cycle detected within par_T_conn_max_comm (8.0s) after wakeup by plug-in | | ✓ |
| CM_SLAC_PARM.CNF runID = REQ runID | cnf=1122334455667788<br>req=1122334455667788 | ✓ |
| CM_SLAC_PARM.CNF applicationType = COMMON | got 0 | ✓ |
| CM_SLAC_PARM.CNF securityType = COMMON | got 0 | ✓ |
| CM_SLAC_PARM.CNF M-SOUND target = broadcast | got ffffffffffff | ✓ |
| CM_SLAC_PARM.CNF num_sounds = expected | got 10 | ✓ |
| CM_SLAC_PARM.CNF timeout = expected | got 6 | ✓ |
| CM_SLAC_PARM.CNF resp_type = expected | got 1 | ✓ |
| CM_SLAC_PARM.CNF forwarding_sta = EV MAC | got 02005e00cafe | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| 5% duty after wakeup by plug-in | 601.6 ms | ≤ 8000 ms (T_conn_max_comm) | ✓ |
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF num_sounds | 10 | = 10 | ✓ |
| CM_SLAC_PARM.CNF timeout | 6 | = 6 | ✓ |
| CM_SLAC_PARM.CNF resp_type | 1 | = 1 | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_001

TC_SECC_CMN_VTB_AttenuationCharacterization_001.

Test objective: Test System executes GoodCase procedure, sends 3 CM_START_ATTEN_CHAR.IND messages with the current runID and all additional valid parameters and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters, so that the SUT can measure the individual attenuation values. Test System then waits for the CM_ATTEN_CHAR.IND message. Test System then checks that a CM_ATTEN_CHAR.IND message with the current runID, EV MAC, 58 attenuation entries and all additional valid parameters is sent by the SUT.

**DUT** (dut) **PASS** 5.69 s

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_ATTEN_CHAR.IND num_sounds (template wildcard) | 10 | - | info |
| CM_ATTEN_CHAR.IND num_groups | 58 | = 58 | ✓ |
| CM_ATTEN_CHAR.IND mean attenuation | 25.0 dB | - | info |

## TC_SECC_CMN_VTB_AttenuationCharacterization_002

TC_SECC_CMN_VTB_AttenuationCharacterization_002.

Test objective: Test System executes GoodCase procedure and sends a new CM_SLAC_PARM.REQ message with a new runID and all additional valid parameters after receiving a valid CM_ATTEN_CHAR.IND message with the current runID, 58 attenuation entries, EV MAC and all additional valid parameters. Test System then waits for the confirmation of a new SLAC process, receiving a CM_SLAC_PARM.CNF message because it shall be considered as a new retry by the EV. Test System then checks that a CM_SLAC_PARM.CNF message with current runID, EV MAC and all additional valid parameters is sent by the SUT.

f_..._002 sends NO CM_ATTEN_CHAR.RSP: on IND receipt it immediately regenerates the runID and sends the new CM_SLAC_PARM.REQ -- the restart from the awaiting-RSP state ([V2G3-A09-16]) is what is under test.

### DUT (dut) PASS 5.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends CM_SLAC_PARM.CNF for the new (retry) runID | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF (new runID) response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_003

> TC_SECC_CMN_VTB_AttenuationCharacterization_003.
>
> Test objective: Test System executes GoodCase procedure, sends 3 CM_START_ATTEN_CHAR.IND messages with the current runID and all additional valid parameters and 9 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters, so that the SUT can measure the individual attenuation values. Test System then waits for the CM_ATTEN_CHAR.IND message. Test System then checks that a CM_ATTEN_CHAR.IND message with the current runID, EV MAC, 58 attenuation entries and all additional valid parameters is sent by the SUT if the anticipated number of CM_ATTEN_PROFILE.IND messages is not achieved (n := 9 soundings).

**DUT** (dut) **PASS** 5.23 s

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding (9 of 10) | 645.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_ATTEN_CHAR.IND num_sounds (template wildcard) | 9 | - | info |

## TC_SECC_CMN_VTB_AttenuationCharacterization_004

TC_SECC_CMN_VTB_AttenuationCharacterization_004.

Test objective: Test System executes GoodCase procedure, sends 3 CM_START_ATTEN_CHAR.IND and 10 CM_MNBC_SOUND.IND messages (current runID, decrementing counter) so the SUT measures the attenuation values, then counts the number of CM_ATTEN_CHAR.IND repetitions (current runID, EV MAC, 58 attenuation entries) WITHOUT sending a CM_ATTEN_CHAR.RSP until the TT_match_response timer has expired. Test System checks that the CM_ATTEN_CHAR.IND retransmission is limited to 2 retries (C_EV_match_retry) by the SUT. After the last retry window has expired, a valid CM_ATTEN_CHAR.RSP and a valid CM_SLAC_MATCH.REQ are sent: the matching process is FAILED, so a CM_SLAC_MATCH.CNF must NOT be sent.

### DUT (dut) PASS 6.16 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_ATTEN_CHAR.IND retransmission limited to C_EV_match_retry (2) | got 2 of 2 retransmissions | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_ATTEN_CHAR.IND retransmission count | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_005

> TC_SECC_CMN_VTB_AttenuationCharacterization_005.
>
> Test objective: Test System executes GoodCase procedure, sends 3 CM_START_ATTEN_CHAR.IND messages with the current runID and all additional valid parameters and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters, so that the SUT can measure the individual attenuation values. Test System then counts the number of CM_ATTEN_CHAR.IND repetitions including the current runID, EV MAC, 58 attenuation entries and all additional valid parameters after sending an invalid 'applicationType' equals to 'FF'H in CM_ATTEN_CHAR.RSP after each CM_ATTEN_CHAR.IND message. Test System then checks the repetition of CM_ATTEN_CHAR.IND messages and whether it is limited to 2 retries by the SUT.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_ATTEN_CHAR.IND retransmission limited to C_EV_match_retry (2) | got 2 of 2 retransmissions | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_ATTEN_CHAR.IND retransmission count | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_006

> TC_SECC_CMN_VTB_AttenuationCharacterization_006.
>
> Test objective: Test System executes GoodCase procedure, sends 3 CM_START_ATTEN_CHAR.IND messages with the current runID and all additional valid parameters and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters, so that the SUT can measure the individual attenuation values. Test System then counts the number of CM_ATTEN_CHAR.IND repetitions including the current runID, EV MAC, 58 attenuation entries and all additional valid parameters after sending an invalid 'securityType' equals to 'FF'H in CM_ATTEN_CHAR.RSP after each CM_ATTEN_CHAR.IND message. Test System then checks the repetition of CM_ATTEN_CHAR.IND messages and whether it is limited to 2 retries by the SUT.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_ATTEN_CHAR.IND retransmission limited to C_EV_match_retry (2) | got 2 of 2 retransmissions | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 2.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_ATTEN_CHAR.IND retransmission count | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_007

> TC_SECC_CMN_VTB_AttenuationCharacterization_007.
>
> Test objective: Test System executes GoodCase procedure, sends 3 CM_START_ATTEN_CHAR.IND messages with the current runID and all additional valid parameters and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters, so that the SUT can measure the individual attenuation values. Test System then counts the number of CM_ATTEN_CHAR.IND repetitions including the current runID, EV MAC, 58 attenuation entries and all additional valid parameters after sending an invalid 'sourceAddress' equals to '000000000000'H in CM_ATTEN_CHAR.RSP after each CM_ATTEN_CHAR.IND message. Test System then checks the repetition of CM_ATTEN_CHAR.IND messages and whether it is limited to 2 retries by the SUT.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_ATTEN_CHAR.IND retransmission limited to C_EV_match_retry (2) | got 2 of 2 retransmissions | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_ATTEN_CHAR.IND retransmission count | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_008

TC_SECC_CMN_VTB_AttenuationCharacterization_008.

Test objective: Test System executes GoodCase procedure, sends 3 CM_START_ATTEN_CHAR.IND messages with the current runID and all additional valid parameters and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters, so that the SUT can measure the individual attenuation values. Test System then counts the number of CM_ATTEN_CHAR.IND repetitions including the current runID, EV MAC, 58 attenuation entries and all additional valid parameters after sending an invalid 'runID' in CM_ATTEN_CHAR.RSP after each CM_ATTEN_CHAR.IND message. Test System then checks the repetition of CM_ATTEN_CHAR.IND messages and whether it is limited to 2 retries by the SUT.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_ATTEN_CHAR.IND retransmission limited to C_EV_match_retry (2) | got 2 of 2 retransmissions | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_ATTEN_CHAR.IND retransmission count | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_009

TC_SECC_CMN_VTB_AttenuationCharacterization_009.

Test objective: Test System executes GoodCase procedure, sends 3 CM_START_ATTEN_CHAR.IND messages with the current runID and all additional valid parameters and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters, so that the SUT can measure the individual attenuation values. Test System then counts the number of CM_ATTEN_CHAR.IND repetitions including the current runID, EV MAC, 58 attenuation entries and all additional valid parameters after sending an invalid 'sourceID' equals to '000000000000000000000000000000001'H in CM_ATTEN_CHAR.RSP after each CM_ATTEN_CHAR.IND message. Test System then checks the repetition of CM_ATTEN_CHAR.IND messages and whether it is limited to 2 retries by the SUT.

### DUT (dut) PASS 6.40 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_ATTEN_CHAR.IND retransmission limited to C_EV_match_retry (2) | got 2 of 2 retransmissions | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_ATTEN_CHAR.IND retransmission count | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_010

> TC_SECC_CMN_VTB_AttenuationCharacterization_010.
>
> Test objective: Test System executes GoodCase procedure, sends 3 CM_START_ATTEN_CHAR.IND messages with the current runID and all additional valid parameters and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters, so that the SUT can measure the individual attenuation values. Test System then counts the number of CM_ATTEN_CHAR.IND repetitions including the current runID, EV MAC, 58 attenuation entries and all additional valid parameters after sending an invalid 'respID' equals to '00000000000000000000000000000001'H in CM_ATTEN_CHAR.RSP after each CM_ATTEN_CHAR.IND message. Test System then checks the repetition of CM_ATTEN_CHAR.IND messages and whether it is limited to 2 retries by the SUT.

### DUT (dut) PASS 6.60 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_ATTEN_CHAR.IND retransmission limited to C_EV_match_retry (2) | got 2 of 2 retransmissions | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_ATTEN_CHAR.IND retransmission count | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_011

TC_SECC_CMN_VTB_AttenuationCharacterization_011.

Test objective: Test System executes GoodCase procedure, sends 3 CM_START_ATTEN_CHAR.IND messages with the current runID and all additional valid parameters and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters, so that the SUT can measure the individual attenuation values. Test System then counts the number of CM_ATTEN_CHAR.IND repetitions including the current runID, EV MAC, 58 attenuation entries and all additional valid parameters after sending an invalid 'result' equals to 'FF'H in CM_ATTEN_CHAR.RSP after each CM_ATTEN_CHAR.IND message. Test System then checks the repetition of CM_ATTEN_CHAR.IND messages and whether it is limited to 2 retries by the SUT.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_ATTEN_CHAR.IND retransmission limited to C_EV_match_retry (2) | got 2 of 2 retransmissions | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_ATTEN_CHAR.IND retransmission count | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_012

TC_SECC_CMN_VTB_AttenuationCharacterization_012.

Test objective: Test System executes GoodCase procedure, sends 3 CM_START_ATTEN_CHAR.IND messages with the current runID and all additional valid parameters and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters after the TT_match_sequence timer has expired. Test System then checks that no CM_ATTEN_CHAR.IND message is sent by the SUT until the TT_EV_atten_results timer has expired.

f_..._006 waits exactly tc_TT_match_sequence (par_TT_match_sequence = 400 ms) before the first CM_START_ATTEN_CHAR.IND -- no extra margin.

**DUT (dut)** PASS 5.79 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.805151s | silent for 0.805151s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_013

TC_SECC_CMN_VTB_AttenuationCharacterization_013.

Test objective: Test System executes GoodCase procedure, sends an invalid 'applicationType' equals to 'FF'H in the 3 CM_START_ATTEN_CHAR.IND messages and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameter. Test System then checks that no CM_ATTEN_CHAR.IND message is sent by the SUT until the TT_EV_atten_results timer has expired.

**DUT (dut)** PASS 5.60 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.80542s | silent for 0.80542s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_014

TC_SECC_CMN_VTB_AttenuationCharacterization_014.

Test objective: Test System executes GoodCase procedure, sends an invalid 'securityType' equals to 'FF'H in the 3 CM_START_ATTEN_CHAR.IND messages and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameter. Test System then checks that no CM_ATTEN_CHAR.IND message is sent by the SUT until the TT_EV_atten_results timer has expired.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.806073s | silent for 0.806073s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_015

TC_SECC_CMN_VTB_AttenuationCharacterization_015.

Test objective: Test System executes GoodCase procedure, sends an invalid 'numSounds' equals to '00'H in the 3 CM_START_ATTEN_CHAR.IND messages and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameter. Test System then checks that no CM_ATTEN_CHAR.IND message is sent by the SUT until the TT_EV_atten_results timer has expired.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.805448s | silent for 0.805448s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_016

TC_SECC_CMN_VTB_AttenuationCharacterization_016.

Test objective: Test System executes GoodCase procedure, sends an invalid 'timeOut' equals to '00'H in the 3 CM_START_ATTEN_CHAR.IND messages and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameter. Test System then checks that no CM_ATTEN_CHAR.IND message is sent by the SUT until the TT_EV_atten_results timer has expired.

**DUT (dut)** **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.805661s | silent for 0.805661 s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_017

TC_SECC_CMN_VTB_AttenuationCharacterization_017.

Test objective: Test System executes GoodCase procedure, sends an invalid 'respType' equals to '00'H in the 3 CM_START_ATTEN_CHAR.IND messages and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameter. Test System then checks that no CM_ATTEN_CHAR.IND message is sent by the SUT until the TT_EV_atten_results timer has expired.

**DUT (dut)** **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.805787s | silent for 0.805787 s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_018

TC_SECC_CMN_VTB_AttenuationCharacterization_018.

Test objective: Test System executes GoodCase procedure, sends an invalid 'runID' in the 3 CM_START_ATTEN_CHAR.IND messages and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters. Test System then checks that no CM_ATTEN_CHAR.IND message is sent by the SUT until the TT_EV_atten_results timer has expired.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.806208s | silent for 0.806208s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_019

TC_SECC_CMN_VTB_AttenuationCharacterization_019.

Test objective: Test System executes GoodCase procedure, indicates CP State A and sends 3 CM_START_ATTEN_CHAR.IND messages with the current runID and all additional valid parameters and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters. Test System then checks that no CM_ATTEN_CHAR.IND message is sent by the SUT until the TT_EV_atten_results timer has expired.

**DUT** (dut) **PASS** 21.02 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.80642s | silent for 0.80642s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |

## TC_SECC_CMN_VTB_AttenuationCharacterization_020

TC_SECC_CMN_VTB_AttenuationCharacterization_020.

Test objective: Test System executes GoodCase procedure, sends 3 CM_START_ATTEN_CHAR.IND messages with the current runID and all additional valid parameters and 10 CM_MNBC_SOUND.IND messages with the current runID, a decrementing counter and all additional valid parameters, so that the SUT can measure the individual attenuation values. Test System then waits for the CM_ATTEN_CHAR.IND message. Test System then checks that a CM_ATTEN_CHAR.IND message with the current runID, EV MAC, 58 attenuation entries and all additional valid parameters is sent by the SUT. Furthermore Test System executes a plug-out, injects an 10 dB attenuator on the control pilot and repeats the SLAC attenuation process by initiating a plug-in again. Test System then checks that SUT has sent the measured attenuation values by comparing both processes. (The mean of the attenuation values of test run 1 should be 'par_SECC_attenuationDeviation' smaller than the mean of the attenuation values of test run 2).

Verdict per f_SECC_CMN_compareAttenuationValues_001: FAIL iff (mean2 - mean1) < par_SECC_attenuationDeviation, with the means computed as INTEGER means (f_SECC_CMN_calculateMeanOfAttenuationValues_001: sum / 58, integer division). ONE-SIDED -- the standard has no upper bound. par_SECC_attenuationDeviation is a declared PIXIT (pics.SECC_ATTENUATION_DEVIATION_DB = 5).

The "10 dB attenuator on the control pilot" is emulated by raising the cb_tandem_sim virtual-modem attenuation base from 25 dB (run 1) to 35 dB (run 2) via set_sim_attenuation() -- equivalent in kind to the ATS's own TX-power emulation (f_SECC_CMN_setTXPower(10) in f_SECC_CMN_Reset_001). set_sim_attenuation() waits for the sim's acknowledgement; on a rig whose EV side is real PLC hardware nothing acknowledges and the TC is INCONCLUSIVE (a real RF attenuator would have to be inserted between the two runs, which such a bench cannot do unattended).

**DUT** (dut) **PASS** 7.31 s

| Conformance check | Observed | Result |
|---|---|---|
| mean(run 2) - mean(run 1) >= par_SECC_attenuationDeviation | deviation 10 dB (run1=25, run2=35); required >= 5 dB | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| Run 1 mean attenuation (integer mean, 25 dB base) | 25 dB | - | info |
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| Run 2 mean attenuation (integer mean, 35 dB base) | 35 dB | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| Attenuation increase (run 2 - run 1) | 10 dB | ≥ 5 dB | ✓ |

## TC_SECC_CMN_VTB_CmValidate_001

TC_SECC_CMN_VTB_CmValidate_001.

Test objective: Test System executes GoodCase procedure, sends the CM_VALIDATE.REQ message (step 1) with 'pilotTimer' equals to '00'H and all additional valid parameters and waits for the CM_VALIDATE.CNF message (step 1). Test System then sends a CM_VALIDATE.REQ message (step 2) with a valid 'pilotTimer' unequals to '00'H and all additional valid parameters, indicates the BCB toggle sequence and waits for the CM_VALIDATE.CNF message (step 2). Test System then checks that the CM_VALIDATE.CNF message (step 1) with 'result' equals to '01'H and all additional valid parameters is sent by the SUT. Furthermore the Test System checks that the CM_VALIDATE.CNF message (step 2) with a valid number of toggles, 'result' equals to '02'H and all additional valid parameters is sent by the SUT.

### DUT (dut) PASS 6.41 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_VALIDATE.CNF step 1 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| SUT sends CM_VALIDATE.CNF (step 2) | | ✓ |
| CM_VALIDATE.CNF step-2 result = 0×02 (success) | got 0×02 | ✓ |
| CM_VALIDATE.CNF step-2 toggleNum = C_vald_nb_toggles | got 2, expected 2 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF (step 1) response time | 0.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_VALIDATE.CNF step 1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step-2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step-2 toggle_num | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_CmValidate_002

TC_SECC_CMN_VTB_CmValidate_002.

Test objective: Test System executes GoodCase procedure, sends the CM_VALIDATE.REQ message (step 1) with 'pilotTimer' equals to '00'H and all additional valid parameters and waits for the CM_VALIDATE.CNF message (step 1). Test System then sends another CM_VALIDATE.REQ message (step 1) with a valid 'pilotTimer' equals to '00'H and all additional valid parameters and waits for the repetition of CM_VALIDATE.CNF message (step 1). Test System then checks that a CM_VALIDATE.CNF message (step 1) with 'result' equals to '01'H and all additional valid parameters was repeated by the SUT.

**DUT** (dut) **PASS** 5.79 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_VALIDATE.CNF step 1 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| CM_VALIDATE.CNF step-1 repetition signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step-1 repetition toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step-1 repetition result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF (step 1) response time | 0.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_VALIDATE.CNF step 1 result | 1 | = 1 | ✓ |
| repeated CM_VALIDATE.CNF (step 1) response time | 0.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_VALIDATE.CNF step-1 repetition result | 1 | = 1 | ✓ |

## TC_SECC_CMN_VTB_CmValidate_003

TC_SECC_CMN_VTB_CmValidate_003.

Test objective: Test System executes GoodCase procedure, sends the CM_VALIDATE.REQ message (step 1) with 'pilotTimer' equals to '00'H and all additional valid parameters and waits for the CM_VALIDATE.CNF message (step 1). Test System then counts the number of CM_VALIDATE.CNF (step 1) repetitions including 'result' equals to '01'H and all additional valid parameters without sending a CM_VALIDATE.REQ message (step 2) message until the TT_match_sequence timer has expired. Test System then checks the repetition of CM_VALIDATE.CNF messages (step 1) and whether it is limited to 2 retries by the SUT.

### DUT (dut) PASS 9.20 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_VALIDATE.CNF step 1 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #1 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #1 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #1 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #2 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #2 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #2 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| SUT sends no 0×6079 for 0.65s | none within 0.65s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF (step 1) response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_VALIDATE.CNF step 1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #2 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step-1 repetitions | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_CmValidate_004

TC_SECC_CMN_VTB_CmValidate_004.

Test objective: Test System executes GoodCase procedure, sends the CM_VALIDATE.REQ message (step 1) with 'pilotTimer' equals to '00'H and all additional valid parameters and waits for the CM_VALIDATE.CNF message (step 1). Test System then counts the number of CM_VALIDATE.CNF (step 1) repetitions including 'result' equals to '01'H and all additional valid parameters after sending an invalid 'signalType' equals to 'FF'H in CM_VALIDATE.REQ (step 2) after each CM_VALIDATE.CNF message (step 1). Test System then checks the repetition of CM_VALIDATE.CNF messages (step 1) and whether it is limited to 2 retries by the SUT.

### DUT (dut) PASS 8.80 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_VALIDATE.CNF step 1 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #1 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #1 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #1 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #2 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #2 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #2 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| SUT sends no 0×6079 for 0.65s | none within 0.65s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF (step 1) response time | 0.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_VALIDATE.CNF step 1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step-1 repetition #2 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step-1 repetitions | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_CmValidate_005

TC_SECC_CMN_VTB_CmValidate_005.

Test objective: Test System executes GoodCase procedure, sends the CM_VALIDATE.REQ message (step 1) with 'pilotTimer' equals to '00'H and all additional valid parameters and waits for the CM_VALIDATE.CNF message (step 1). After receipt of a CM_VALIDATE.CNF message (step 1) with 'result' equals to '01'H and all additional valid parameters, Test System sends a CM_VALIDATE.REQ message (step 2) with result equals to '00'H and waits for termination of the SLAC matching process. Test System then checks that no CM_VALIDATE.CNF message is sent by the SUT until the TT_match_sequence timer has expired.

### DUT (dut) PASS 5.79 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_VALIDATE.CNF step 1 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF (step 1) response time | 0.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_VALIDATE.CNF step 1 result | 1 | = 1 | ✓ |

## TC_SECC_CMN_VTB_CmValidate_006

TC_SECC_CMN_VTB_CmValidate_006.

Test objective: Test System executes GoodCase procedure, sends the CM_VALIDATE.REQ message (step 1) with 'pilotTimer' equals to '00'H and all additional valid parameters and waits for the CM_VALIDATE.CNF message (step 1). After receipt of a CM_VALIDATE.CNF message (step 1) with 'result' equals to '01'H and all additional valid parameters, Test System sends a CM_VALIDATE.REQ message (step 2) with result equals to '02'H and waits for termination of the SLAC matching process. Test System then checks that no CM_VALIDATE.CNF message is sent by the SUT until the TT_match_sequence timer has expired.

**DUT** (dut) **PASS** 5.00 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_VALIDATE.CNF step 1 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF (step 1) response time | 0.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_VALIDATE.CNF step 1 result | 1 | = 1 | ✓ |

## TC_SECC_CMN_VTB_CmValidate_007

> TC_SECC_CMN_VTB_CmValidate_007.
>
> Test objective: Test System executes GoodCase procedure, sends the CM_VALIDATE.REQ message (step 1) with 'pilotTimer' equals to '00'H and all additional valid parameters and waits for the CM_VALIDATE.CNF message (step 1). After receipt of a CM_VALIDATE.CNF message (step 1) with 'result' equals to '01'H and all additional valid parameters, Test System sends a CM_VALIDATE.REQ message (step 2) with result equals to '03'H and waits for termination of the SLAC matching process. Test System then checks that no CM_VALIDATE.CNF message is sent by the SUT until the TT_match_sequence timer has expired.

**DUT** (dut) **PASS** 5.00 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_VALIDATE.CNF step 1 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF (step 1) response time | 0.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_VALIDATE.CNF step 1 result | 1 | = 1 | ✓ |

## TC_SECC_CMN_VTB_CmValidate_008

TC_SECC_CMN_VTB_CmValidate_008.

Test objective: Test System executes GoodCase procedure, sends the CM_VALIDATE.REQ message (step 1) with 'pilotTimer' equals to '00'H and all additional valid parameters and waits for the CM_VALIDATE.CNF message (step 1). After receipt of a CM_VALIDATE.CNF message (step 1) with 'result' equals to '01'H and all additional valid parameters, Test System sends a CM_VALIDATE.REQ message (step 2) with result equals to '04'H and waits for termination of the SLAC matching process. Test System then checks that no CM_VALIDATE.CNF message is sent by the SUT until the TT_match_sequence timer has expired.

**DUT** (dut) **PASS** 5.00 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_VALIDATE.CNF step 1 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF (step 1) response time | 0.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_VALIDATE.CNF step 1 result | 1 | = 1 | ✓ |

## TC_SECC_CMN_VTB_CmValidate_009

TC_SECC_CMN_VTB_CmValidate_009.

Test objective: Test System executes GoodCase procedure on a first instance and then executes a GoodCase procedure in parallel on a second instance while the first instance is waiting for CM_VALIDATE.REQ message (step 1). Then Test System executes the CM_VALIDATE.REQ message (step 1) sequence on the first instance, afterwards it sends the CM_VALIDATE.REQ message (step 1) on the second instance in parallel while the Test System is proceeding the validation process on the first instance. Test System then checks that the CM_VALIDATE.CNF message (step 1) with 'result' equals to '00'H and all additional valid parameters is sent to the second instance by the SUT (Processing blocked by first instance).

The "multi-tester rig" (v_SLAC_Tester2) is realised on the single-TAP SIL by a second, independent SLAC identity (own EV MAC + runID) on the same virtual PLC wire. Expected behavior per Table 74: f_..._006 (instance-1 step-1 → result '01'H), then f_..._009 (instance-2 step-1 → result '00'H notReady, blocked by instance 1) concurrently with f_..._007 (instance 1 proceeds step 2 + BCB toggles → result '02'H success with toggleNum == C_vald_nb_toggles). Both outcomes are asserted hard.

### DUT (dut) PASS 6.81 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_VALIDATE.CNF step 1 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF step 1 result = 0×01 (ISO 15118-5) | got 0×01 | ✓ |
| SUT sends CM_VALIDATE.CNF (step 2) | | ✓ |
| CM_VALIDATE.CNF step-2 result = 0×02 (success) | got 0×02 | ✓ |
| CM_VALIDATE.CNF step-2 toggleNum = C_vald_nb_toggles | got 2, expected 2 | ✓ |
| CM_VALIDATE.CNF instance-2 step 1 signalType = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF instance-2 step 1 toggleNum = 0×00 (md_CMN_CMN_CmValidateCnf_001) | got 0×00 | ✓ |
| CM_VALIDATE.CNF instance-2 step 1 result = 0×00 (ISO 15118-5) | got 0×00 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_PARM.CNF response time | 0.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_VALIDATE.CNF (step 1) response time | 0.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_VALIDATE.CNF step 1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step-2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step-2 toggle_num | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF instance-2 step 1 result | 0 | = 0 | ✓ |

## TC_SECC_CMN_VTB_CmValidate_013

TC_SECC_CMN_VTB_CmValidate_013.

Test objective: Test System executes GoodCase procedure, indicates CP State A and sends a CM_VALIDATE.REQ message (step 1) with 'pilotTimer' equals to '00'H and all additional valid parameters. Test System then checks that no CM_VALIDATE.CNF message (step 1) is sent by the SUT until the TT_match_response timer has expired.

Negative check ([V2G3-A09-126]): with the control pilot dropped to State A the SUT aborts SLAC and must not answer CM_VALIDATE.REQ.

### DUT (dut) PASS 20.01 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

## TC_SECC_CMN_VTB_CmSlacMatch_002

TC_SECC_CMN_VTB_CmSlacMatch_002.

Test objective: Test System executes GoodCase procedure with SLAC validation process, sends a CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters and waits for CM_SLAC_MATCH.CNF message. Test System then checks that a CM_SLAC_MATCH.CNF message with the current runID, valid NID and NMK, EV MAC, EVSE MAC, and all additional valid parameter is sent by the SUT.

PIXIT := cmValidate twin of _001: precondition f_SECC_CMN_PR_CmValidate_001 (real BCB-toggle validation, failures INCONCLUSIVE).

### DUT (dut) PASS 6.89 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_SLAC_MATCH.CNF matches the md_CMN_CMN_CmSlacMatchCnf_001 template (runID, EV MAC, EVSE MAC) | runID=0f1e2d3c4b5a6978 evMAC=02005e00cafe evseMAC=4e2035d4ed22 | ✓ |
| CM_SLAC_MATCH.CNF NMK length = expected | got 16 | ✓ |
| CM_SLAC_MATCH.CNF NMK is non-zero | | ✓ |
| CM_SLAC_MATCH.CNF NID length = expected | got 7 | ✓ |
| CM_SLAC_MATCH.CNF NID consistent with NMK (SHA-256 ×5) | nid=93c929b683a305 nmk=4e4543594f495554475a503943544848 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF step1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step2 toggleNum | 2 | = 2 | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_MATCH.CNF NMK length | 16 | = 16 | ✓ |
| CM_SLAC_MATCH.CNF NID length | 7 | = 7 | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_CMN_VTB_CmSlacMatch_004

TC_SECC_CMN_VTB_CmSlacMatch_004.

Test objective: Test System executes GoodCase procedure with SLAC validation process, sends a CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters and ignores the received CM_SLAC_MATCH.CNF message. After 'TT_match_response' timeout, Test System sends another CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters and waits for the repetition of a valid CM_SLAC_MATCH.CNF message. Test System then checks that a CM_SLAC_MATCH.CNF message with the current runID, valid NID and NMK, EV MAC, EVSE MAC, and all additional valid parameters is retransmitted by the SUT.

PIXIT := cmValidate twin of _003 (precondition f_SECC_CMN_PR_CmValidate_001).

### DUT (dut) PASS 7.20 s

| Conformance check | Observed | Result |
|---|---|---|
| first CM_SLAC_MATCH.CNF matches the md_CMN_CMN_CmSlacMatchCnf_001 template | runID=0f1e2d3c4b5a6978 evMAC=02005e00cafe evseMAC=4e2035d4ed22 | ✓ |
| CM_SLAC_MATCH.CNF matches the md_CMN_CMN_CmSlacMatchCnf_001 template (runID, EV MAC, EVSE MAC) | runID=0f1e2d3c4b5a6978 evMAC=02005e00cafe evseMAC=4e2035d4ed22 | ✓ |
| CM_SLAC_MATCH.CNF NMK length = expected | got 16 | ✓ |
| CM_SLAC_MATCH.CNF NMK is non-zero | | ✓ |
| CM_SLAC_MATCH.CNF NID length = expected | got 7 | ✓ |
| CM_SLAC_MATCH.CNF NID consistent with NMK (SHA-256 ×5) | nid=99334a6962100e nmk=314a4d514c4546495442593452515746 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF step1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step2 toggleNum | 2 | = 2 | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_MATCH.CNF NMK length | 16 | = 16 | ✓ |
| CM_SLAC_MATCH.CNF NID length | 7 | = 7 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_CMN_VTB_CmSlacMatch_006

TC_SECC_CMN_VTB_CmSlacMatch_006.

Test objective: Test System executes GoodCase procedure with SLAC validation process and sends a CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters after TT_EVSE_match_session timer has expired. Test System then checks that no CM_SLAC_MATCH.CNF message is sent by the SUT until the TT_match_response timer has expired.

PIXIT := cmValidate twin of _005; per f_SECC_CMN_TB_VTB_CmSlacMatch_003 the session timer is TT_match_sequence (400 ms) in this variant.

### DUT (dut) PASS 7.99 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF step1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step2 toggleNum | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_CmSlacMatch_008

TC_SECC_CMN_VTB_CmSlacMatch_008.

Test objective: Test System executes GoodCase procedure with SLAC validation process and sends a CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters after TT_EVSE_match_session timer has expired. Furthermore an additional CM_SLAC_MATCH.REQ message with an invalid 'applicationType' equals to 'FF'H was sent before timeout was triggered. Test System then checks that no CM_SLAC_MATCH.CNF message is sent by the SUT until the TT_match_response timer has expired.

PIXIT := cmValidate twin of _007 (session timer TT_match_sequence).

### DUT (dut) PASS 8.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF step1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step2 toggleNum | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_CmSlacMatch_010

TC_SECC_CMN_VTB_CmSlacMatch_010.

Test objective: Test System executes GoodCase procedure with SLAC validation process and sends a CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters after TT_EVSE_match_session timer has expired. Furthermore an additional CM_SLAC_MATCH.REQ message with an invalid 'securityType' equals to 'FF'H was sent before timeout was triggered. Test System then checks that no CM_SLAC_MATCH.CNF message is sent by the SUT until the TT_match_response timer has expired.

PIXIT := cmValidate twin of _009 (session timer TT_match_sequence).

**DUT** (dut) **PASS** 8.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF step1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step2 toggleNum | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_CmSlacMatch_012

TC_SECC_CMN_VTB_CmSlacMatch_012.

Test objective: Test System executes GoodCase procedure with SLAC validation process and sends a CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters after TT_EVSE_match_session timer has expired. Furthermore an additional CM_SLAC_MATCH.REQ message with an invalid 'mfvLength' equals to '0000'H was sent before timeout was triggered. Test System then checks that no CM_SLAC_MATCH.CNF message is sent by the SUT until the TT_match_response timer has expired.

PIXIT := cmValidate twin of _011 (session timer TT_match_sequence).

### DUT (dut) PASS 8.60 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF step1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step2 toggleNum | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_CmSlacMatch_014

TC_SECC_CMN_VTB_CmSlacMatch_014.

Test objective: Test System executes GoodCase procedure with SLAC validation process and sends a CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters after TT_EVSE_match_session timer has expired. Furthermore an additional CM_SLAC_MATCH.REQ message with an invalid 'evID' equals to '00000000000000000000000000000001'H was sent before timeout was triggered. Test System then checks that no CM_SLAC_MATCH.CNF message is sent by the SUT until the TT_match_response timer has expired.

PIXIT := cmValidate twin of _013 (session timer TT_match_sequence).

### DUT (dut) PASS 8.40 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF step1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step2 toggleNum | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_CmSlacMatch_016

TC_SECC_CMN_VTB_CmSlacMatch_016.

Test objective: Test System executes GoodCase procedure with SLAC validation process and sends a CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters after TT_EVSE_match_session timer has expired. Furthermore an additional CM_SLAC_MATCH.REQ message with an invalid 'evMac' equals to '000000000000'H was sent before timeout was triggered. Test System then checks that no CM_SLAC_MATCH.CNF message is sent by the SUT until the TT_match_response timer has expired.

PIXIT := cmValidate twin of _015 (session timer TT_match_sequence).

**DUT** (dut) **PASS** 8.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF step1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step2 toggleNum | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_CmSlacMatch_018

TC_SECC_CMN_VTB_CmSlacMatch_018.

Test objective: Test System executes GoodCase procedure with SLAC validation process and sends a CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters after TT_EVSE_match_session timer has expired. Furthermore an additional CM_SLAC_MATCH.REQ message with an invalid 'evseID' equals to '00000000000000000000000000000001'H was sent before timeout was triggered. Test System then checks that no CM_SLAC_MATCH.CNF message is sent by the SUT until the TT_match_response timer has expired.

PIXIT := cmValidate twin of _017 (session timer TT_match_sequence).

**DUT** (dut) **PASS** 8.60 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF step1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step2 toggleNum | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_CmSlacMatch_020

TC_SECC_CMN_VTB_CmSlacMatch_020.

Test objective: Test System executes GoodCase procedure with SLAC validation process and sends a CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters after TT_EVSE_match_session timer has expired. Furthermore an additional CM_SLAC_MATCH.REQ message with an invalid 'evseMac' equals to '000000000000'H was sent before timeout was triggered. Test System then checks that no CM_SLAC_MATCH.CNF message is sent by the SUT until the TT_match_response timer has expired.

PIXIT := cmValidate twin of _019 (session timer TT_match_sequence).

### DUT (dut) PASS 8.40 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF step1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step2 toggleNum | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_CmSlacMatch_022

TC_SECC_CMN_VTB_CmSlacMatch_022.

Test objective: Test System executes GoodCase procedure with SLAC validation process and sends a CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters after TT_EVSE_match_session timer has expired. Furthermore an additional CM_SLAC_MATCH.REQ message with an invalid 'runID' was sent before timeout was triggered. Test System then checks that no CM_SLAC_MATCH.CNF message is sent by the SUT until the TT_match_response timer has expired.

PIXIT := cmValidate twin of _021 (session timer TT_match_sequence).

### DUT (dut) PASS 8.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.65s | silent for 0.65s | ✓ |
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF step1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step2 toggleNum | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_CmSlacMatch_024

TC_SECC_CMN_VTB_CmSlacMatch_024.

Test objective: Test System executes GoodCase procedure with SLAC validation process, indicates CP State A and sends a CM_SLAC_MATCH.REQ message with the current runID, EV MAC, EVSE MAC and all additional valid parameters. Test system then checks that no CM_SLAC_MATCH.CNF message is sent by the SUT until the TT_match_response timer has expired.

PIXIT := cmValidate twin of _023 (precondition f_SECC_CMN_PR_CmValidate_001).

### DUT (dut) PASS 22.22 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_VALIDATE.CNF step1 result | 1 | = 1 | ✓ |
| CM_VALIDATE.CNF step2 result | 2 | = 2 | ✓ |
| CM_VALIDATE.CNF step2 toggleNum | 2 | = 2 | ✓ |

## TC_SECC_CMN_VTB_PLCLinkStatus_001

TC_SECC_CMN_VTB_PLCLinkStatus_001.

Test objective: Test System executes GoodCase procedure, indicates the key at the local PLC node after receipt of a valid CM_SLAC_MATCH.CNF message and then waits for AVLN establishment by exchange of communication with the internal PLC Node. Test System then checks that the data link connection is established within 'TT_match_join'.

**DUT** (dut) **PASS** 4.46 s

| Conformance check | Observed | Result |
|---|---|---|
| data link established within TT_match_join (12.0s) | D-LINK ready 0.001s after match | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| D-LINK ready after match | 1.0 ms | ≤ 12000 ms (TT_match_join) | ✓ |

## TC_SECC_CMN_VTB_PLCLinkStatus_002

TC_SECC_CMN_VTB_PLCLinkStatus_002.

Test objective: Test System executes GoodCase procedure and establishes a new AVLN. Test System then checks that SUT leaves the logical network within 'TP_match_leave' if CP State A was detected before.

**DUT** (dut) **PASS** 20.12 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.102s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| left logical network after CP State A | 0.102 s | ≤ 1.25 s | ✓ |

## TC_SECC_CMN_VTB_PLCLinkStatus_003

> TC_SECC_CMN_VTB_PLCLinkStatus_003.
>
> Test objective: Test System executes GoodCase procedure, establishes a new AVLN and sends a CM_SLAC_PARM.REQ message with a valid runID and all additional valid parameters. Test System then checks that no CM_SLAC_PARM.CNF message is sent by the SUT until the TT_match_response timer has expired if the SUT is in state 'Matched'.

**DUT** (dut) **PASS** 5.28 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends no unexpected SLAC frame for 0.45s | silent for 0.45s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_CMN_VTB_PLCLinkStatus_004

TC_SECC_CMN_VTB_PLCLinkStatus_004.

Test objective: Test System starts GoodCase procedure five times in parallel and starts five independent SLAC processes. Test System then checks that the SUT will respond to each EVCC instance with a CM_ATTEN_CHAR.IND message with the current runID, EV MAC and all additional valid parameters. Futhermore a successful link establishment with the first instance will be checked.

**DUT** (dut) **PASS** 5.41 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT answers each parallel SLAC process with CM_SLAC_PARM.CNF | session run_id=a001020304050607: CNF received | ✓ |
| SUT answers each parallel SLAC process with CM_SLAC_PARM.CNF | session run_id=a101020304050607: CNF received | ✓ |
| SUT answers each parallel SLAC process with CM_SLAC_PARM.CNF | session run_id=a201020304050607: CNF received | ✓ |
| SUT answers each parallel SLAC process with CM_SLAC_PARM.CNF | session run_id=a301020304050607: CNF received | ✓ |
| SUT answers each parallel SLAC process with CM_SLAC_PARM.CNF | session run_id=a401020304050607: CNF received | ✓ |
| SUT responds to each EVCC instance with CM_ATTEN_CHAR.IND within TT_EV_atten_results | session run_id=a001020304050607 IND 0.419s after first START | ✓ |
| SUT responds to each EVCC instance with CM_ATTEN_CHAR.IND within TT_EV_atten_results | session run_id=a101020304050607 IND 0.419s after first START | ✓ |
| SUT responds to each EVCC instance with CM_ATTEN_CHAR.IND within TT_EV_atten_results | session run_id=a201020304050607 IND 0.419s after first START | ✓ |
| SUT responds to each EVCC instance with CM_ATTEN_CHAR.IND within TT_EV_atten_results | session run_id=a301020304050607 IND 0.419s after first START | ✓ |
| SUT responds to each EVCC instance with CM_ATTEN_CHAR.IND within TT_EV_atten_results | session run_id=a401020304050607 IND 0.419s after first START | ✓ |
| all five parallel SLAC processes answered with CM_ATTEN_CHAR.IND | 5/5 answered | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF response time | 0.4 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF response time | 0.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| parallel SLAC sessions answered (CM_ATTEN_CHAR.IND) | 5 | = 5 | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_CMN_VTB_PLCLinkStatus_005

TC_SECC_CMN_VTB_PLCLinkStatus_005.

Test objective: Test System executes GoodCase procedure and establishes a new AVLN. Afterwards Test System initiates a connection loss by setting a new key at the local node. Test System then checks that the SUT leaves the logical network (setting previous key again on Test System side).

**DUT** (dut) **PASS** 10.70 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT leaves the logical network within TP_match_leave (1.25s) on connection loss | left after 0.001s | ✓ |
| SUT sends no SLAC frame during par_TP_match_leave (leaving the network) | silent for 1s | ✓ |
| SUT has left the current logical network (CM_NW_STATS empty) | no stations | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| CM_SET_KEY.CNF response time | 0.5 ms | - | info |
| left logical network on connection loss | 0.001 s | ≤ 1.25 s | ✓ |
| CM_SET_KEY.CNF response time | 0.5 ms | - | info |

## TC_SECC_DC_VTB_PLCLinkStatus_001

TC_SECC_DC_VTB_PLCLinkStatus_001.

Test objective: Test System executes GoodCase procedure and initiates a paused V2G communication session by sending a SessionStopReq message with the current SessionID, ChargingSession 'Pause' and all additional mandatory parameters. After 'PICS_CMN_CMN_WakeUp' Test System resumes the previously paused session by initiating a BCB toggle and waits for successful data link detection triggered by the SUT. Test System checks that the SUT shall not turn off the +12 V supply during the sleeping phase. Furthermore Test System checks the wake-up process (BCB toggle detection) and the successful data link detection within 'par_T_conn_resume'.

**DUT** (dut) **PASS** 22.36 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| data link re-detected within par_T_conn_resume (6.0s) after BCB toggle | detected 0.001s after toggle | ✓ |
| BCB-toggle wake rejoins the paused session (OK_OldSessionJoined) | got 'SessionSetupRes' / 'OK_OldSessionJoined' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.8 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1999.0 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| data link detected after BCB toggle | 0.001 s | ≤ 6.0 s | ✓ |
| PWM re-arm after BCB toggle (informational; the f_..._004 T_conn_resume bound applies to link detection, vacuous on this rig) | 3.113 s | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_DC_VTB_PLCLinkStatus_003

TC_SECC_DC_VTB_PLCLinkStatus_003.

Test objective: Test System executes GoodCase procedure and initiates a paused V2G communication session. After 'PICS_CMN_CMN_WakeUp' Test System resumes the previously paused session by initiating a BCB toggle and waits for failed link detection (new logical network parameter was set from the Test System before). Test System checks that the SUT shall not turn off the +12 V supply during the sleeping phase. Furthermore Test System checks that the SUT applys CP State E or F for T_step_EF and will process the start of a new matching process.

**DUT** (dut) **PASS** 32.32 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| data link establishment fails after wake with mismatched network params | no link within par_T_conn_resume | ✓ |
| CP State E/F applied within par_T_conn_max_comm (8.0s) of the wake toggle | E/F 7.423s after toggle | ✓ |
| SUT sends no unexpected SLAC frame for 3.9879s | silent for 3.9879s | ✓ |
| CP E/F step ends within T_step_EF_max (valid duty follows) | | ✓ |
| CP E/F step held for T_step_EF [4.0, 8.0]s after failed-link wake | E/F step 4.014s | ✓ |
| post-E/F duty is a valid resume duty (failed-link wake) within the rest of T_step_EF_max + par_T_conn_max_comm | 100.0% 0.000s after the E/F end | ✓ |
| SUT processes the start of a new matching process after the E/F cycle | CM_SLAC_PARM.CNF run_id=0dc003abcdef0123 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.6 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 103.2 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1998.6 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| CM_SET_KEY.CNF response time | 0.8 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CP E/F after failed-link wake | 7.423 s | ≤ 8.0 s | ✓ |
| T_step_EF after failed-link wake | 4.014 s | [4.0, 8.0] s | ✓ |
| CP duty after E/F (failed-link wake) | 100.0 % | - | info |
| valid duty after the E/F step (failed-link wake) | 0.0 s | ≤ 11.986 s | ✓ |
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |

## TC_SECC_DC_VTB_PLCLinkStatus_004

TC_SECC_DC_VTB_PLCLinkStatus_004.

Test objective: Test System executes GoodCase procedure, indicates the key at the local PLC node and then checks that the data link connection is established within 'TT_match_join'. Afterwards Test System initiates a connection loss by setting a new key at the local node. Test System then checks that the SUT performs a CP State X2 to X1 to E/F transition. Furthermore Test System checks that the SUT applys CP State E or F for T_step_EF and will switch to CP State X1 or X2 afterwards.

Pure connection-loss CP E/F cycle (f_SECC_CMN_TB_VTB_PLCLinkStatus_007, DC duty ranges): no V2G layer or +12 V/BCB instrumentation needed.

### DUT (dut) PASS 10.71 s

| Conformance check | Observed | Result |
|---|---|---|
| PLC link terminated within TP_match_leave (1.25s) on connection loss | left after 0.001s | ✓ |
| X1 (100%) step within par_T_conn_max_comm (8.0s) | X1 step 0.290s after connection loss | ✓ |
| X1 intermediate step precedes CP State E/F | X1 before E/F | ✓ |
| SUT sends no unexpected SLAC frame for 3.98598s | silent for 3.98598s | ✓ |
| CP E/F step ends within T_step_EF_max (returns to X1/X2) | | ✓ |
| CP E/F step held for T_step_EF [4.0, 8.0]s | E/F step 4.014s | ✓ |
| post-E/F duty is a valid DC duty (3..7% or exactly 100%) | got 100.0% | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| CM_SET_KEY.CNF response time | 0.6 ms | - | info |
| left logical network on connection loss | 0.001 s | ≤ 1.25 s | ✓ |
| X1 (100%) step after connection loss | 0.29 s | ≤ 8.0 s | ✓ |
| T_step_EF on connection loss (DC) | 4.014 s | [4.0, 8.0] s | ✓ |
| CP duty after E/F step (DC: 3..7% or 100%) | 100.0 % | - | info |
| CM_SET_KEY.CNF response time | 0.5 ms | - | info |

## TC_SECC_DC_VTB_PLCLinkStatus_005

TC_SECC_DC_VTB_PLCLinkStatus_005.

Test objective: Test System executes GoodCase procedure and initiates a terminated V2G communication session by sending a SessionStopReq message with the current SessionID, ChargingSession 'Terminate' and all additional mandatory parameters and waits for a SessionStopRes message with the current SessionID and all additional mandatory parameters. Furthermore Test System checks if the SUT terminates the TCP connection after successful charging process. Test System then checks that the SUT leaves the logical network after 'TP_match_leave' by using the CM_NW_STATS message sequence.

**DUT** (dut) **PASS** 16.60 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends SessionStopRes(OK) for the terminating SessionStop | got 'SessionStopRes' / 'OK' | ✓ |
| SessionStopRes carries the current SessionID | got '36ad65de55c42429', expected '36ad65de55c42429' | ✓ |
| SUT sends no SLAC frame during par_TP_match_leave (leaving the network) | silent for 1s | ✓ |
| SUT has left the current logical network (CM_NW_STATS empty) | no stations | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 101.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 101.0 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| WeldingDetectionRes response time | 101.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionStopRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1999.7 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| SUT node absent from tester-node AVLN after Terminate (CM_NW_STATS after TP_match_leave) | 0 stations | = 0 stations | ✓ |

## TC_SECC_DC_VTB_PLCLinkStatus_006

TC_SECC_DC_VTB_PLCLinkStatus_006.

Test objective: Test System executes GoodCase procedure and initiates a paused V2G communication session by sending a SessionStopReq message with the current SessionID, ChargingSession 'Pause' and all additional mandatory parameters. After 'PICS_CMN_CMN_WakeUp' Test System resumes the previously paused session by initiating a BCB toggle. Test System checks that the SUT shall not turn off the +12 V supply during the sleeping phase. Furthermore Test System checks the wake-up process (BCB toggle detection), the 5 % duty cycle detection and the processing of the SDP procedure (SUT is ready for Binding process).

**DUT** (dut) **PASS** 22.12 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| resume duty is the 5% HLC signal (DC valid range 3..7%) | got 5.0% | ✓ |
| 5% duty confirmed within par_T_conn_max_comm (8.0s) | confirmed after 3.112s | ✓ |
| BCB-toggle wake SDP-rejoins the paused session (OK_OldSessionJoined) | got 'SessionSetupRes' / 'OK_OldSessionJoined' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.8 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 101.8 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 2001.0 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| resume CP duty after BCB toggle (DC) | 5.0 % | [3.0, 7.0] % | ✓ |
| 5% duty after BCB toggle | 3.112 s | ≤ 8.0 s | ✓ |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_DC_VTB_PLCLinkStatus_007

TC_SECC_DC_VTB_PLCLinkStatus_007.

Test objective: Test System executes GoodCase procedure and initiates a paused V2G communication session by sending a SessionStopReq message with the current SessionID, ChargingSession 'Pause' and all additional mandatory parameters. After 'PICS_CMN_CMN_WakeUp' < 'par_SECC_T_step_X1' Test System resumes the previously paused session by initiating a BCB toggle. Test System checks that the SUT shall not turn off the +12 V supply during the sleeping phase. Furthermore Test System checks the wake-up process (BCB toggle detection) and the 5 % duty cycle detection by the oscillator. As a result of 'PICS_CMN_CMN_WakeUp' < 'par_SECC_T_step_X1', SUT shall signal 5 % duty cycle not earlier than 'par_SECC_T_step_X1' within B1.

**DUT** (dut) **PASS** 20.02 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| 5% duty signaled within par_T_conn_max_comm (8.0s) | signaled after 3.112s | ✓ |
| resume duty is the 5% HLC signal (DC valid range 3..7%) | got 5.0% | ✓ |
| early BCB-toggle wake rejoins the paused session (OK_OldSessionJoined) | got 'SessionSetupRes' / 'OK_OldSessionJoined' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.7 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.0 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1999.0 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| 5% duty delay after early BCB toggle (informational; the -5 T_step_X1 check is dead code as published) | 3.112 s | - | info |
| resume CP duty after early BCB toggle (DC) | 5.0 % | [3.0, 7.0] % | ✓ |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

# Test case details — ISO 15118-2 message layer (-4)

## TC_SECC_CMN_VTB_SupportedAppProtocol_001

TC_SECC_CMN_VTB_SupportedAppProtocol_001.

Test objective: GoodCase; Test System sends SupportedAppProtocolReq with a valid AppProtocol list including the ISO namespace over TLS and checks the SUT replies SupportedAppProtocolRes with ResponseCode OK_SuccessfulNegotiation or OK_SuccessfulNegotiationWithMinorDeviation. This variant negotiates over TLS (PIXIT_SECC_CMN_TLS := true); the TLS-off twin is _004.

Runs against the PnC SUT config (TLS on): SDP-TLS + TLS 1.2 connect, then the SupportedAppProtocol handshake over the TLS channel.

### DUT (dut) PASS 6.08 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |
| ResponseCode is an OK_* negotiation code (over TLS) | got 'OK_SuccessfulNegotiation' | ✓ |
| supportedAppProtocolRes carries the negotiated SchemaID (TLS) | got 1 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| Negotiated SchemaID (TLS) | 1 | - | info |

## TC_SECC_CMN_VTB_SupportedAppProtocol_002

> TC_SECC_CMN_VTB_SupportedAppProtocol_002.
>
> Test objective: GoodCase; Test System sends SupportedAppProtocolReq with an invalid ISO namespace and checks the SUT replies Failed_NoNegotiation and terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout (5 s).

### DUT (dut) PASS 6.39 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |
| ResponseCode = Failed_NoNegotiation for an invalid ISO namespace | got 'Failed_NoNegotiation' | ✓ |
| SUT closes the TCP connection after the Failed_NoNegotiation response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_SupportedAppProtocol_003

> TC_SECC_CMN_VTB_SupportedAppProtocol_003.
>
> Test objective: GoodCase; after SDP + TCP the Test System stays silent until V2G_Sequence_Timeout expires (sends no SupportedAppProtocolReq). Checks the SUT terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout after the timeout. LONG TEST (waits the ~60 s sequence timeout).

**DUT** (dut) **PASS** 23.83 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT closes the TCP connection after V2G_Sequence_Timeout (within 70s) | closed 18.1s after TCP connect | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 2.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| TCP connection close after failure/timeout | 18096.9 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_CMN_VTB_SupportedAppProtocol_004

TC_SECC_CMN_VTB_SupportedAppProtocol_004.

Test objective: GoodCase (eIM / no-TLS); Test System sends a valid SupportedAppProtocolReq including the ISO namespace urn:iso:15118:2:2013:MsgDef and all mandatory parameters, and checks the SUT replies SupportedAppProtocolRes with ResponseCode OK_SuccessfulNegotiation or OK_SuccessfulNegotiationWithMinorDeviation and a valid SchemaID.

**DUT** (dut) **PASS** 6.77 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |
| ResponseCode is an OK_* negotiation code | got 'OK_SuccessfulNegotiation' | ✓ |
| supportedAppProtocolRes carries the negotiated SchemaID | got 1 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| Negotiated SchemaID | 1 | - | info |

## TC_SECC_CMN_VTB_SupportedAppProtocol_005

> TC_SECC_CMN_VTB_SupportedAppProtocol_005.
>
> Test objective: GoodCase; Test System lets V2G_SECC_CommunicationSetup_Timeout (20 s) expire on the SUT side after the SDP response WITHOUT establishing the TCP connection, then attempts a TCP handshake and checks the SUT no longer accepts it / does not respond (and emits no further SDP response). LONG TEST.

**DUT (dut)** PASS 30.64 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT stops waiting after V2G_SECC_CommunicationSetup_Timeout (rejects the late TCP handshake / SupportedAppProtocol) | | ✓ |
| SUT emits no further SDP response after V2G_SECC_CommunicationSetup_Timeout | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 418.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |

## TC_SECC_CMN_VTB_SupportedAppProtocol_006

> TC_SECC_CMN_VTB_SupportedAppProtocol_006.
>
> Test objective: GoodCase; Test System sends valid AppProtocols where VersionNumberMinor does not match the SUT's, and checks the SUT replies exactly OK_SuccessfulNegotiationWithMinorDeviation.

### DUT (dut) PASS 5.75 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |
| SUT does not return plain OK_SuccessfulNegotiation for a mismatched VersionNumberMinor | | ✓ |
| ResponseCode = OK_SuccessfulNegotiationWithMinorDeviation | got 'OK_SuccessfulNegotiationWithMinorDeviation' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_CMN_VTB_SessionSetup_001

> TC_SECC_CMN_VTB_SessionSetup_001.
>
> Test objective: GoodCase; Test System sends SessionSetupReq with SessionID 0, a valid EvccID and all mandatory parameters, and checks the SUT replies SessionSetupRes with a NEW non-zero SessionID, ResponseCode OK_NewSessionEstablished and a valid EvseId.

**DUT (dut)** PASS 5.56 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a SessionSetupRes message | got 'SessionSetupRes' | ✓ |
| ResponseCode = OK_NewSessionEstablished | got 'OK_NewSessionEstablished' | ✓ |
| SessionSetupRes carries a new non-zero SessionID | got 9eadbc921882e5f9 | ✓ |
| SessionSetupRes carries a valid EVSEID | got 'DE*PNX*E12345*1' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_CMN_VTB_SessionSetup_002

> TC_SECC_CMN_VTB_SessionSetup_002.
>
> Test objective: GoodCase; Test System sends a valid ServiceDiscoveryReq instead of SessionSetupReq (sequence error) and checks the SUT replies ServiceDiscoveryRes with the current SessionID, ResponseCode FAILED_SequenceError and all mandatory parameters, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout (5 s).

**DUT (dut)** PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_SessionSetup_003

> TC_SECC_CMN_VTB_SessionSetup_003.
>
> Test objective: GoodCase; after SupportedAppProtocolRes the Test System stays silent until V2G_Sequence_Timeout expires (sends no SessionSetupReq). Checks the SUT terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout after the timeout. LONG TEST (waits the ~60 s sequence timeout).

**DUT (dut)** **PASS** 23.02 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT closes the TCP connection after V2G_Sequence_Timeout (within 70s) | closed 17.1s after SupportedAppProtocolRes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/timeout | 17102.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_CMN_VTB_SessionSetup_004

> TC_SECC_CMN_VTB_SessionSetup_004.
>
> Test objective: GoodCase; after SupportedAppProtocolRes the Test System lets V2G_SECC_CommunicationSetup_Timeout (20 s) expire (sends no SessionSetupReq). Checks the SUT terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout after the comm-setup timeout. Distinguishes the comm-setup timeout from the sequence timeout of _003. LONG TEST.

**DUT (dut)** PASS 23.01 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT closes the TCP connection within V2G_SECC_CommunicationSetup_Timeout+termination (26.0s) | closed after 17.1s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 2.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/timeout | 17097.5 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_DC_VTB_SessionSetup_001

> TC_SECC_DC_VTB_SessionSetup_001.
>
> Test objective: pause/resume; GoodCase → SessionStop(Pause) → sleep phase → wake (BCB toggle) → resume → SessionSetup with the previous SessionID, expecting OK_OldSessionJoined.

**DUT** (dut) **PASS** 20.19 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionSetupRes message | got 'SessionSetupRes' | ✓ |
| ResponseCode = OK_OldSessionJoined | got 'OK_OldSessionJoined' | ✓ |
| resumed SessionID = paused SessionID | resumed=00a10721ceae7c59 paused=00a10721ceae7c59 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| PowerDeliveryRes(Start) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 102.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1998.8 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_DC_VTB_SessionSetup_002

TC_SECC_DC_VTB_SessionSetup_002.

Test objective: pause/resume; as _001 but resume with a NEW SessionID, expecting OK_NewSessionEstablished (wake by BCB toggle).

**DUT** (dut) **PASS** 20.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionSetupRes message | got 'SessionSetupRes' | ✓ |
| ResponseCode = OK_NewSessionEstablished | got 'OK_NewSessionEstablished' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SessionSetupRes carries a new non-zero SessionID | got 806f3d13d2fa9d95 | ✓ |
| new session does not reuse the paused SessionID | new=806f3d13d2fa9d95 paused=aeea0e130c7afb72 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 103.8 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.4 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 102.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1998.7 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_DC_VTB_SessionSetup_003

TC_SECC_DC_VTB_SessionSetup_003.

Test objective: pause/resume; as _001 but resume with SessionID 0, expecting OK_NewSessionEstablished (wake by BCB toggle).

**DUT** (dut) **PASS** 20.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionSetupRes message | got 'SessionSetupRes' | ✓ |
| ResponseCode = OK_NewSessionEstablished | got 'OK_NewSessionEstablished' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SessionSetupRes carries a new non-zero SessionID | got ca7a3bb7a185f876 | ✓ |
| new session does not reuse the paused SessionID | new=ca7a3bb7a185f876 paused=a6afd3efb9489710 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 2.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.5 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.2 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1998.7 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_CMN_VTB_ServiceDiscovery_001

TC_SECC_CMN_VTB_ServiceDiscovery_001.

Test objective: GoodCase; Test System sends ServiceDiscoveryReq with the current SessionID and all mandatory parameters, and checks the SUT replies ServiceDiscoveryRes with the current SessionID, ResponseCode OK, a PaymentOptionList, a valid SupportedEnergyTransferMode and all mandatory parameters.

**DUT** (dut) **PASS** 6.09 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| PaymentOptionList offers at least one PaymentOption | got ['ExternalPayment'] | ✓ |
| ChargeService advertises a SupportedEnergyTransferMode | got ['DC_extended'] | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscovery PaymentOptionList entries | 1 | ≥ 1 | ✓ |

## TC_SECC_CMN_VTB_ServiceDiscovery_002

> TC_SECC_CMN_VTB_ServiceDiscovery_002.
>
> Test objective: GoodCase; Test System sends a valid AuthorizationReq instead of ServiceDiscoveryReq (sequence error) and checks the SUT replies AuthorizationRes with the current SessionID, ResponseCode FAILED_SequenceError and all mandatory parameters, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout (5 s).

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 101.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| AuthorizationRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_ServiceDiscovery_003

> TC_SECC_CMN_VTB_ServiceDiscovery_003.
>
> Test objective: GoodCase; Test System sends ServiceDiscoveryReq with an invalid SessionID and all mandatory parameters, and checks the SUT replies ServiceDiscoveryRes with a valid SessionID, ResponseCode FAILED_UnknownSession and all mandatory parameters, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout (5 s).

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT does not answer OK to an unknown SessionID | got 'FAILED_UnknownSession' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_ServiceDiscovery_004

> TC_SECC_CMN_VTB_ServiceDiscovery_004.
>
> Test objective: GoodCase; after SessionSetupRes (new SessionID, OK_NewSessionEstablished) the Test System stays silent until V2G_Sequence_Timeout expires (sends no ServiceDiscoveryReq). Checks the SUT terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout after the timeout. LONG TEST (waits the ~60 s sequence timeout).

**DUT (dut)** PASS 67.90 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT closes the TCP connection after V2G_Sequence_Timeout (within 70s) | closed 62.0s after SessionSetupRes | ✓ |
| SUT does not close before V2G_Sequence_Timeout expires (>= 55s) | closed after 62.0s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 2.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/timeout | 62022.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_DC_VTB_ServiceDiscovery_001

> TC_SECC_DC_VTB_ServiceDiscovery_001.
>
> Test objective: pause/resume; GoodCase → SessionStop(Pause) → sleep phase → wake (BCB toggle) → resume → ServiceDiscovery, expecting the previously paused session's PaymentOption/ChargeService.

**DUT** (dut) **PASS** 20.32 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| Resumed ServiceDiscoveryRes reuses the paused session's ChargeService | paused={'FreeService': True, 'ServiceCategory': 'EVCharging', 'ServiceID': 1, 'ServiceName': 'AC_DC_Charging', 'SupportedEnergyTransferMode': ['DC_extended']}, resumed={'FreeService': True, 'ServiceCategory': 'EVCharging', 'ServiceID': 1, 'ServiceName': 'AC_DC_Charging', 'SupportedEnergyTransferMode': ['DC_extended']} | ✓ |
| Resumed ServiceDiscoveryRes reuses the paused session's PaymentOptionList | paused=['ExternalPayment'], resumed=['ExternalPayment'] | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 420.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SET_KEY.CNF response time | 0.5 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.7 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.2 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1998.8 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentServiceSelection_001

TC_SECC_CMN_VTB_PaymentServiceSelection_001.

Test objective: GoodCase; Test System sends PaymentServiceSelectionReq with the current SessionID, a SelectedPaymentOption, a valid ServiceID and all mandatory parameters, and checks the SUT replies PaymentServiceSelectionRes with the current SessionID, ResponseCode OK and all mandatory parameters.

### DUT (dut) PASS 5.80 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PaymentServiceSelectionRes message | got 'PaymentServiceSelectionRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentServiceSelection_003

> TC_SECC_CMN_VTB_PaymentServiceSelection_003.
>
> Test objective: as _002 but SelectedPaymentOption Contract → OK; PnC.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PaymentServiceSelectionRes message | got 'PaymentServiceSelectionRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 2.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 419.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 104.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 103.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentServiceSelection_004

> TC_SECC_CMN_VTB_PaymentServiceSelection_004.
>
> Test objective: GoodCase; Test System sends a valid ServiceDiscoveryReq instead of PaymentServiceSelectionReq (sequence error) and checks the SUT replies ServiceDiscoveryRes with the current SessionID, FAILED_SequenceError, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT (dut)** PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 422.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.5 ms | - | info |
| supportedAppProtocolRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentServiceSelection_005

> TC_SECC_CMN_VTB_PaymentServiceSelection_005.
>
> Test objective: GoodCase; Test System sends PaymentServiceSelectionReq with an invalid SessionID and checks the SUT replies with a valid SessionID, FAILED_UnknownSession, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

### DUT (dut) PASS 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 418.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.5 ms | - | info |
| supportedAppProtocolRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 103.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentServiceSelection_006

TC_SECC_CMN_VTB_PaymentServiceSelection_006.

Test objective: GoodCase; Test System sends PaymentServiceSelectionReq with the current SessionID, a SelectedPaymentOption and an invalid (unoffered) ServiceID, and checks the SUT replies with the current SessionID, FAILED_ServiceSelectionInvalid, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT (dut)** **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_ServiceSelectionInvalid | got 'FAILED_ServiceSelectionInvalid' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 418.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.5 ms | - | info |
| supportedAppProtocolRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 103.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 103.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentServiceSelection_007

TC_SECC_CMN_VTB_PaymentServiceSelection_007.

Test objective: GoodCase; Test System sends PaymentServiceSelectionReq with the current SessionID and all mandatory parameters but WITHOUT selecting the charge service (no ServiceID equal to 1), and checks the SUT replies with the current SessionID, FAILED_NoChargeServiceSelected, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_NoChargeServiceSelected | got 'FAILED_NoChargeServiceSelected' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 418.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 103.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentServiceSelection_008

> TC_SECC_CMN_VTB_PaymentServiceSelection_008.
>
> Test objective: GoodCase; after a ServiceDetailRes (OK) the Test System stays silent until V2G_Sequence_Timeout expires → SUT terminates the session (TCP close). Requires the ServiceDetail (VAS) state.

**DUT** (dut) **PASS** 68.10 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT answers the ServiceDetail precondition with ServiceDetailRes(OK) | got name='ServiceDetailRes', code='OK' | ✓ |
| SUT closes the TCP connection within 70s of silence | closed after 62.0s | ✓ |
| SUT does not close before V2G_Sequence_Timeout expires (>= 55s) | closed after 62.0s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 418.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.5 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 103.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 103.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDetailRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/timeout | 62003.5 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_CMN_VTB_PaymentServiceSelection_010

> TC_SECC_CMN_VTB_PaymentServiceSelection_010.
>
> Test objective: mirror of _009; Test System selects externalPayment while the SUT is configured for contract/PnC. Requires the SUT configured for contract/PnC.

**DUT** (dut) **PASS** 5.90 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PaymentServiceSelectionRes message | got 'PaymentServiceSelectionRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 419.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentDetails_001

TC_SECC_CMN_VTB_PaymentDetails_001.

Test objective: GoodCase; Test System sends PaymentDetailsReq with the current SessionID, a valid EMAID and contract certificate chain, and checks the SUT replies PaymentDetailsRes with the current SessionID, OK and a valid GenChallenge. PnC.

### DUT (dut) PASS 5.98 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PaymentDetailsRes message | got 'PaymentDetailsRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| GenChallenge is 16 bytes | got 16 bytes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentDetails GenChallenge length | 16 bytes | = 16 bytes | ✓ |

## TC_SECC_CMN_VTB_PaymentDetails_002

> TC_SECC_CMN_VTB_PaymentDetails_002.
>
> Test objective: as _001 but after a certificate installation provisioning. PnC.

### DUT (dut) PASS 6.01 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PaymentDetailsRes message | got 'PaymentDetailsRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| GenChallenge is 16 bytes | got 16 bytes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateInstallationRes response time | 106.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentDetails GenChallenge length (post-install) | 16 bytes | = 16 bytes | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:28:30.266 | CS → CSMS | StatusNotification | {"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:28:30.265Z"} |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:28:30.266 | CSMS → CS | StatusNotification (response) | {} |
| 06:28:32.705 | CS → CSMS | Get15118EVCertificate | {"action":"Install","exiRequest":"<1044 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"} |
| 06:28:32.721 | CSMS → CS | Get15118EVCertificate (response) | {"exiResponse":"<5572 base64 chars>","status":"Accepted"} |

## TC_SECC_CMN_VTB_PaymentDetails_003

> TC_SECC_CMN_VTB_PaymentDetails_003.
>
> Test objective: as _001 but after a certificate update provisioning. PnC.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PaymentDetailsRes message | got 'PaymentDetailsRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| GenChallenge is 16 bytes | got 16 bytes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 106.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentDetails GenChallenge length (post-update) | 16 bytes | = 16 bytes | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:28:36.267 | CS → CSMS | StatusNotification | {"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:28:36.266Z"} |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:28:36.268 | CSMS → CS | StatusNotification (response) | {} |
| 06:28:38.710 | CS → CSMS | Get15118EVCertificate | {"action":"Update","exiRequest":"<2916 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"} |
| 06:28:38.718 | CSMS → CS | Get15118EVCertificate (response) | {"exiResponse":"<5572 base64 chars>","status":"Accepted"} |

## TC_SECC_CMN_VTB_PaymentDetails_004

> TC_SECC_CMN_VTB_PaymentDetails_004.
>
> Test objective: GoodCase; Test System sends a valid ServiceDiscoveryReq instead of PaymentDetailsReq → ServiceDiscoveryRes, FAILED_SequenceError, TCP close. PnC (reachable only via the contract PaymentServiceSelection path).

### DUT (dut) PASS 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentDetails_005

> TC_SECC_CMN_VTB_PaymentDetails_005.
>
> Test objective: GoodCase; PaymentDetailsReq with an invalid SessionID → FAILED_UnknownSession, TCP close. PnC.

### DUT (dut) PASS 6.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PaymentDetailsRes message | got 'PaymentDetailsRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentDetails_006

> TC_SECC_CMN_VTB_PaymentDetails_006.
>
> Test objective: GoodCase; wait until V2G_Sequence_Timeout expires after a PaymentServiceSelectionRes (OK) → SUT terminates the session. PnC. LONG TEST.

### DUT (dut) PASS 68.70 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection after V2G_Sequence_Timeout (within 70s) with no PaymentDetailsReq | closed after 62.0s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 62028.4 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_CMN_VTB_PaymentDetails_007

> TC_SECC_CMN_VTB_PaymentDetails_007.
>
> Test objective: GoodCase; PaymentDetailsReq with a valid EMAID but an invalid (Expired) contract certificate chain → FAILED_CertificateExpired, TCP close. PnC.

**DUT** (dut) **PASS** 6.11 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT rejects the expired contract leaf certificate (FAILED_CertificateExpired) | got 'FAILED_CertificateExpired' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 419.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 103.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentDetails_008

> TC_SECC_CMN_VTB_PaymentDetails_008.
>
> Test objective: GoodCase; invalid contract certificate chain (invalid content)
> → one of FAILED / FAILED_NoCertificateAvailable /
> FAILED_CertificateNotAllowedAtThisEVSE, TCP close. PnC.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT rejects the corrupt contract leaf content (one of ('FAILED', 'FAILED_NoCertificateAvailable', 'FAILED_CertificateNotAllowedAtThisEVSE')) | got 'FAILED' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 2.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 420.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 104.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 103.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 103.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentDetails_009

> TC_SECC_CMN_VTB_PaymentDetails_009.
>
> Test objective: GoodCase; invalid contract certificate chain (expired issuer Sub CA 1) → one of FAILED / FAILED_NoCertificateAvailable / FAILED_CertificateNotAllowedAtThisEVSE, TCP close. PnC.

**DUT** (dut) **PASS** 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT rejects the chain with an expired issuer (MO Sub-CA 1) (one of ('FAILED', 'FAILED_NoCertificateAvailable', 'FAILED_CertificateNotAllowedAtThisEVSE')) | got 'FAILED' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 103.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 103.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.5 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_PaymentDetails_010

> TC_SECC_CMN_VTB_PaymentDetails_010.
>
> Test objective: GoodCase; invalid contract certificate chain (expired issuer Sub CA 2) → one of FAILED / FAILED_NoCertificateAvailable / FAILED_CertificateNotAllowedAtThisEVSE, TCP close. PnC.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT rejects the chain with an expired issuer (MO Sub-CA 2) (one of ('FAILED', 'FAILED_NoCertificateAvailable', 'FAILED_CertificateNotAllowedAtThisEVSE')) | got 'FAILED' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 2.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 103.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 103.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_Authorization_001

> TC_SECC_CMN_VTB_Authorization_001.
>
> Test objective: GoodCase; Test System sends AuthorizationReq with the current SessionID and all mandatory parameters, and checks the SUT replies AuthorizationRes with the current SessionID, ResponseCode OK and EVSEProcessing Finished. If an earlier AuthorizationRes had EVSEProcessing Ongoing[_WaitingForCustomerInteraction] the Test System repeats the AuthorizationReq until authorization finishes or V2G_EVCC_Ongoing_Timer expires.

**DUT** (dut) **PASS** 7.17 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 2 poll(s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| Authorization polls to EVSEProcessing=Finished | 2 | - | info |

## TC_SECC_CMN_VTB_Authorization_002

> TC_SECC_CMN_VTB_Authorization_002.
>
> Test objective: as _001 but PnC (AuthorizationReq with a valid signature and GenChallenge) → OK, EVSEProcessing Finished.

**DUT** (dut) **PASS** 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 2 poll(s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PnC Authorization polls to EVSEProcessing=Finished | 2 | - | info |

## TC_SECC_CMN_VTB_Authorization_003

> TC_SECC_CMN_VTB_Authorization_003.
>
> Test objective: GoodCase; Test System sends a valid ServiceDiscoveryReq instead of AuthorizationReq (sequence error) and checks the SUT replies ServiceDiscoveryRes with the current SessionID, FAILED_SequenceError, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

### DUT (dut) PASS 6.40 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_Authorization_004

> TC_SECC_CMN_VTB_Authorization_004.
>
> Test objective: as _003 (ServiceDiscoveryReq instead → FAILED_SequenceError, TCP close) but reached via the PnC PaymentDetails precondition.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_Authorization_005

> TC_SECC_CMN_VTB_Authorization_005.
>
> Test objective: GoodCase; Test System sends AuthorizationReq with an invalid SessionID and checks the SUT replies with a valid SessionID, FAILED_UnknownSession, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| AuthorizationRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_Authorization_006

> TC_SECC_CMN_VTB_Authorization_006.
>
> Test objective: PnC; AuthorizationReq with a valid signature, invalid SessionID, valid GenChallenge → FAILED_UnknownSession, TCP close.

**DUT** (dut) **PASS** 6.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| AuthorizationRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_Authorization_007

> TC_SECC_CMN_VTB_Authorization_007.
>
> Test objective: PnC; AuthorizationReq with a valid signature, current SessionID, invalid GenChallenge → FAILED_ChallengeInvalid, TCP close.

**DUT** (dut) **PASS** 6.80 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_ChallengeInvalid | got 'FAILED_ChallengeInvalid' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| AuthorizationRes response time | 103.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_Authorization_008

> TC_SECC_CMN_VTB_Authorization_008.
>
> Test objective: GoodCase; after a PaymentServiceSelectionRes (OK) the Test System stays silent until V2G_Sequence_Timeout expires (sends no AuthorizationReq). Checks the SUT terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout after the timeout. LONG TEST.

**DUT (dut)** PASS 67.88 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection after V2G_Sequence_Timeout (>= 59s, within 70s) | closed 62.0s after PaymentServiceSelectionRes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 62020.6 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_CMN_VTB_Authorization_009

> TC_SECC_CMN_VTB_Authorization_009.
>
> Test objective: as _008 but waits V2G_Sequence_Timeout after a PaymentDetailsRes (OK, valid GenChallenge) → TCP close. PnC. LONG TEST.

**DUT (dut)** **PASS** 68.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection after V2G_Sequence_Timeout (>= 59s, within 70s) | closed 62.0s after PaymentDetailsRes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 62029.4 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_CMN_VTB_Authorization_010

> TC_SECC_CMN_VTB_Authorization_010.
>
> Test objective: PnC; AuthorizationReq with an invalid signature (wrong private key) → FAILED_SignatureError, TCP close.

**DUT** (dut) **PASS** 6.11 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_SignatureError | got 'FAILED_SignatureError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| AuthorizationRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_Authorization_011

> TC_SECC_CMN_VTB_Authorization_011.
>
> Test objective: PnC; AuthorizationReq with an invalid signature (signed body changed after signature calculation) → FAILED_SignatureError, TCP close.

**DUT (dut)** **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_SignatureError (body tampered after signing) | got 'FAILED_SignatureError' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| AuthorizationRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_Authorization_012

> TC_SECC_CMN_VTB_Authorization_012.
>
> Test objective: PnC happy path (valid signature, GenChallenge → OK, Finished) after a certificate installation provisioning.

### DUT (dut) PASS 6.81 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 2 poll(s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateInstallationRes response time | 104.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PnC Authorization polls to Finished (post-install) | 2 | - | info |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:12:46.266 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:12:46.265Z"}` |
| 06:12:46.266 | CSMS → CS | StatusNotification (response) | `{}` |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:12:48.699 | CS → CSMS | Get15118EVCertificate | {"action":"Install","exiRequest":"<1044 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"} |
| 06:12:48.723 | CSMS → CS | Get15118EVCertificate (response) | {"exiResponse":"<5572 base64 chars>","status":"Accepted"} |
| 06:12:48.932 | CS → CSMS | Authorize | {"idToken":{"idToken":"UKSWI123456791A","type":"eMAID"},"iso15118CertificateHashData":[{"hashAlgorithm":"SHA256","issuerKeyHash":"0480f6d9754088df63bb0d82a7263951afc9e9eb756fff66c688c69906343bd8","issuerNameHash":"efc5a55e6e1e59c7605b48d7f260f2321aba004f97f5872986c3388f8756dce5","responderURL":"https://www.example.com/","serialNumber":"3044"},{"hashAlgorithm":"SHA256","issuerKeyHash":"0cdd2ec66a1034a4de1b35c10927e4ca23d1ffd3c6396d2abc146cb3d176ddfc","issuerNameHash":"0fd741fe2020f72842fa212a4dc59f00d8d9e04a48d6eb01ea68e63978ac0c4a","responderURL":"https://www.example.com/","serialNumber":"3043 … |
| 06:12:48.933 | CSMS → CS | Authorize (response) | {"certificateStatus":"Accepted","idTokenInfo":{"status":"Accepted"}} |
| 06:12:48.973 | CS → CSMS | TransactionEvent | {"eventType":"Started","evse":{"connectorId":1,"id":1},"idToken":{"additionalInfo":[{"additionalIdToken":"B6:44:63:86:A2:CD","type":"EVCCID"}],"idToken":"UKSWI123456791A","type":"eMAID"},"meterValue":[{"sampledValue":[{"context":"Transaction.Begin","location":"Outlet","measurand":"Energy.Active.Import.Register","unitOfMeasure":{"unit":"Wh"},"value":515.8055419921875},{"context":"Transaction.Begin","location":"Outlet","measurand":"Current.Import","unitOfMeasure":{"unit":"A"},"value":0.0}],"timestamp":"2026-10-28T06:12:48.576Z"}],"offline":false,"seqNo":0,"timestamp":"2026-10-28T06:12:48.970Z"," … |
| 06:12:48.974 | CSMS → CS | TransactionEvent (response) | {"idTokenInfo":{"status":"Accepted"}} |

## TC_SECC_CMN_VTB_Authorization_013

> TC_SECC_CMN_VTB_Authorization_013.
>
> Test objective: as _012 but after a certificate update provisioning → OK, Finished. PnC.

### DUT (dut) PASS 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 105.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PnC Authorization polls to Finished (post-update) | 2 | - | info |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:12:53.262 | CS → CSMS | StatusNotification | {"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:12:53.262Z"} |
| 06:12:53.263 | CSMS → CS | StatusNotification (response) | {} |
| 06:12:55.699 | CS → CSMS | Get15118EVCertificate | {"action":"Update","exiRequest":"<2916 base64 chars>","iso15118SchemaVersion":"urn:iso: 15118:2:2013:MsgDef"} |
| 06:12:55.707 | CSMS → CS | Get15118EVCertificate (response) | {"exiResponse":"<5572 base64 chars>","status":"Accepted"} |
| 06:12:55.931 | CS → CSMS | Authorize | {"idToken": {"idToken":"UKSWI123456791A","type":"eMAID"},"iso1511 8CertificateHashData": [{"hashAlgorithm":"SHA256","issuerKeyHash":"0480f6d97 54088df63bb0d82a7263951afc9e9eb756fff66c688c69906343 bd8","issuerNameHash":"efc5a55e6e1e59c7605b48d7f260f2 321aba004f97f5872986c3388f8756dce5","responderURL":" https://www.example.com/","serialNumber":"3044"}, {"hashAlgorithm":"SHA256","issuerKeyHash":"0cdd2ec66 a1034a4de1b35c10927e4ca23d1ffd3c6396d2abc146cb3d176d dfc","issuerNameHash":"0fd741fe2020f72842fa212a4dc59 f00d8d9e04a48d6eb01ea68e63978ac0c4a","responderURL":" https://www.example.com/","serialNumber":"3043 … |
| 06:12:55.932 | CSMS → CS | Authorize (response) | {"certificateStatus":"Accepted","idTokenInfo": {"status":"Accepted"}} |
| 06:12:55.969 | CS → CSMS | TransactionEvent | {"eventType":"Started","evse":{"connectorId":1,"id": 1},"idToken":{"additionalInfo": [{"additionalIdToken":"B6:44:63:86:A2:CD","type":"EV CCID"}],"idToken":"UKSWI123456791A","type":"eMAID"}," meterValue":[{"sampledValue": [{"context":"Transaction.Begin","location":"Outlet"," measurand":"Energy.Active.Import.Register","unitOfMe asure":{"unit":"Wh"},"value":515.8055419921875}, {"context":"Transaction.Begin","location":"Outlet"," measurand":"Current.Import","unitOfMeasure": {"unit":"A"},"value": 0.0}],"timestamp":"2026-10-28T06:12:55.579Z"}],"offl ine":false,"seqNo": 0,"timestamp":"2026-10-28T06:12:55.966Z"," … |
| 06:12:55.970 | CSMS → CS | TransactionEvent (response) | {"idTokenInfo":{"status":"Accepted"}} |

## TC_SECC_CMN_VTB_ChargeParameterDiscovery_001

TC_SECC_CMN_VTB_ChargeParameterDiscovery_001.

Test objective: GoodCase; the Test System sends a valid ServiceDiscoveryReq instead of a ChargeParameterDiscoveryReq (sequence error) and checks the SUT replies ServiceDiscoveryRes with the current SessionID, FAILED_SequenceError, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

### DUT (dut) PASS 6.60 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_SequenceError | got FAILED_SequenceError (name=ServiceDiscoveryRes) | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_CMN_VTB_ChargeParameterDiscovery_002

TC_SECC_CMN_VTB_ChargeParameterDiscovery_002.

Test objective: GoodCase; after an AuthorizationRes (OK, EVSEProcessing Finished) the Test System stays silent until V2G_Sequence_Timeout expires (sends no ChargeParameterDiscoveryReq). Checks the SUT terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout after the timeout. LONG TEST.

**DUT** (dut) **PASS** 69.70 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection after V2G_Sequence_Timeout (>= 59s, within 70s) | closed 62.0s after AuthorizationRes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 62038.4 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_DC_VTB_ChargeParameterDiscovery_001

TC_SECC_DC_VTB_ChargeParameterDiscovery_001.

Test objective: GoodCase; the Test System sends a ChargeParameterDiscoveryReq with the current SessionID, a valid RequestedEnergyTransferMode (DC_extended) and a valid DC_EVChargeParameter, without MaxEntriesSAScheduleTuple and DepartureTime. Checks the SUT replies ChargeParameterDiscoveryRes with the current SessionID, ResponseCode OK, EVSEProcessing Finished and a valid SAScheduleList; if an earlier response had EVSEProcessing Ongoing the Test System repeats until Finished or V2G_EVCC_Ongoing_Timer. Primary happy-path ChargeParameterDiscovery TC (DC).

**DUT** (dut) **PASS** 7.30 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ChargeParameterDiscoveryRes message | got 'ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 1 poll(s) | ✓ |
| ChargeParameterDiscoveryRes(Finished) carries a SAScheduleList | SAScheduleList=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| ChargeParameterDiscovery polls to EVSEProcessing=Finished | 1 | - | info |
| ChargeParameterDiscoveryRes SAScheduleList entries | 1 | ≥ 1 | ✓ |

## TC_SECC_DC_VTB_ChargeParameterDiscovery_002

TC_SECC_DC_VTB_ChargeParameterDiscovery_002.

Test objective: GoodCase; the Test System sends a ChargeParameterDiscoveryReq with an invalid SessionID and a valid RequestedEnergyTransferMode. Checks the SUT replies ChargeParameterDiscoveryRes with a valid SessionID, FAILED_UnknownSession, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT** (dut) **PASS** 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_UnknownSession | got FAILED_UnknownSession (name=ChargeParameterDiscoveryRes) | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ChargeParameterDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_ChargeParameterDiscovery_003

TC_SECC_DC_VTB_ChargeParameterDiscovery_003.

Test objective: GoodCase; the Test System sends a ChargeParameterDiscoveryReq with an invalid / mismatched RequestedEnergyTransferMode (a mode the SUT does not offer). Checks the SUT replies ChargeParameterDiscoveryRes with FAILED_WrongEnergyTransferMode, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT** (dut) **PASS** 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_WrongEnergyTransferMode | got FAILED_WrongEnergyTransferMode (name=ChargeParameterDiscoveryRes) | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ChargeParameterDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_ChargeParameterDiscovery_004

TC_SECC_DC_VTB_ChargeParameterDiscovery_004.

Test objective: GoodCase; the Test System sends a ChargeParameterDiscoveryReq with a valid RequestedEnergyTransferMode but an invalid DC_EVChargeParameter (EVMaximumCurrentLimit -100). Checks the SUT replies ChargeParameterDiscoveryRes with FAILED_WrongChargeParameter, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT (dut)** **PASS** 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_WrongChargeParameter | got FAILED_WrongChargeParameter (name=ChargeParameterDiscoveryRes) | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ChargeParameterDiscoveryRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_ChargeParameterDiscovery_006

TC_SECC_DC_VTB_ChargeParameterDiscovery_006.

Test objective: GoodCase; ChargeParameterDiscoveryReq with a valid mode, without MaxEntriesSAScheduleTuple, WITH DepartureTime → ChargeParameterDiscoveryRes OK, EVSEProcessing Finished, valid SAScheduleList (Ongoing repeats allowed).

**DUT** (dut) **PASS** 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ChargeParameterDiscoveryRes message | got 'ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 1 poll(s) | ✓ |
| ChargeParameterDiscoveryRes(Finished) carries a SAScheduleList | SAScheduleList=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 7200, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ChargeParameterDiscovery polls to EVSEProcessing=Finished | 1 | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| ChargeParameterDiscoveryRes SAScheduleList entries | 1 | ≥ 1 | ✓ |

## TC_SECC_DC_VTB_ChargeParameterDiscovery_007

TC_SECC_DC_VTB_ChargeParameterDiscovery_007.

Test objective: pause/resume (BCB toggle); GoodCase → SessionStop(Pause) → sleep → wake → resume → ChargeParameterDiscoveryReq reusing the paused session's SAScheduleTuple.

**DUT** (dut) **PASS** 21.22 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Pre-pause ChargeParameterDiscovery selects a SAScheduleList | got [{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| SessionStop(Pause) → SessionStopRes(OK) | got {'Signature_isUsed': False, 'msg': {'ResponseCode': 'OK', 'ResponseCodeValue': 0}, 'name': 'SessionStopRes', 'ok': True, 'proto': 'iso2', 'session_id': 'c3a81e5f68c601fd'} | ✓ |
| Resume rejoins the paused session (OK_OldSessionJoined) | got {'Signature_isUsed': False, 'msg': {'EVSEID': 'DE*PNX*E12345*1', 'ResponseCode': 'OK_OldSessionJoined', 'ResponseCodeValue': 2}, 'name': 'SessionSetupRes', 'ok': True, 'proto': 'iso2', 'session_id': 'c3a81e5f68c601fd'} | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ChargeParameterDiscoveryRes message | got 'ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 1 poll(s) | ✓ |
| ChargeParameterDiscoveryRes(Finished) carries a SAScheduleList | SAScheduleList=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |
| Resumed session offers the SAScheduleTuple selected in the paused session (SAScheduleTupleID 1) | paused tuple 1, resumed tuples [1]; paused=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] resumed=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.8 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 101.9 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1999.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ChargeParameterDiscovery polls to EVSEProcessing=Finished | 1 | - | info |
| ChargeParameterDiscoveryRes SAScheduleList entries | 1 | ≥ 1 | ✓ |

## TC_SECC_DC_VTB_ChargeParameterDiscovery_012

TC_SECC_DC_VTB_ChargeParameterDiscovery_012.

Test objective: GoodCase; ChargeParameterDiscoveryReq valid mode, WITH MaxEntriesSAScheduleTuple, without DepartureTime → ChargeParameterDiscoveryRes OK, Finished, valid SAScheduleList (MaxEntries considered).

**DUT (dut)** PASS 6.77 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ChargeParameterDiscoveryRes message | got 'ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 1 poll(s) | ✓ |
| ChargeParameterDiscoveryRes(Finished) carries a SAScheduleList | SAScheduleList=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |
| SAScheduleList honours MaxEntriesSAScheduleTuple (<= 3) | got 1 tuple(s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ChargeParameterDiscovery polls to EVSEProcessing=Finished | 1 | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| ChargeParameterDiscoveryRes SAScheduleList entries | 1 | ≥ 1 | ✓ |

## TC_SECC_DC_VTB_ChargeParameterDiscovery_016

TC_SECC_DC_VTB_ChargeParameterDiscovery_016.

Test objective: GoodCase; ChargeParameterDiscoveryReq valid mode, WITH MaxEntries and DepartureTime → ChargeParameterDiscoveryRes OK, Finished, valid SAScheduleList (MaxEntries considered).

**DUT (dut)** **PASS** 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ChargeParameterDiscoveryRes message | got 'ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 1 poll(s) | ✓ |
| ChargeParameterDiscoveryRes(Finished) carries a SAScheduleList | SAScheduleList=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 7200, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |
| SAScheduleList honours MaxEntriesSAScheduleTuple (<= 3) | got 1 tuple(s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ChargeParameterDiscovery polls to EVSEProcessing=Finished | 1 | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| ChargeParameterDiscoveryRes SAScheduleList entries | 1 | ≥ 1 | ✓ |

## TC_SECC_DC_VTB_ChargeParameterDiscovery_017

TC_SECC_DC_VTB_ChargeParameterDiscovery_017.

Test objective: pause/resume (BCB toggle); ChargeParameterDiscoveryReq without MaxEntries/DepartureTime, SAScheduleTuple from the paused session.

**DUT** (dut) **PASS** 21.22 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Pre-pause ChargeParameterDiscovery selects a SAScheduleList | got [{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SessionStop(Pause) → SessionStopRes(OK) | got {'Signature_isUsed': False, 'msg': {'ResponseCode': 'OK', 'ResponseCodeValue': 0}, 'name': 'SessionStopRes', 'ok': True, 'proto': 'iso2', 'session_id': 'c0b9606c054f4bcf'} | ✓ |
| Resume rejoins the paused session (OK_OldSessionJoined) | got {'Signature_isUsed': False, 'msg': {'EVSEID': 'DE*PNX*E12345*1', 'ResponseCode': 'OK_OldSessionJoined', 'ResponseCodeValue': 2}, 'name': 'SessionSetupRes', 'ok': True, 'proto': 'iso2', 'session_id': 'c0b9606c054f4bcf'} | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ChargeParameterDiscoveryRes message | got 'ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 1 poll(s) | ✓ |
| ChargeParameterDiscoveryRes(Finished) carries a SAScheduleList | SAScheduleList=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |
| Resumed session offers the SAScheduleTuple selected in the paused session (SAScheduleTupleID 1) | paused tuple 1, resumed tuples [1]; paused=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] resumed=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.7 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 101.7 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1999.0 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ChargeParameterDiscovery polls to EVSEProcessing=Finished | 1 | - | info |
| ChargeParameterDiscoveryRes SAScheduleList entries | 1 | ≥ 1 | ✓ |

## TC_SECC_DC_VTB_ChargeParameterDiscovery_019

> TC_SECC_DC_VTB_ChargeParameterDiscovery_019.
>
> Test objective: pause/resume (BCB toggle); with MaxEntries.

**DUT** (dut) **PASS** 21.40 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Pre-pause ChargeParameterDiscovery selects a SAScheduleList | got [{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SessionStop(Pause) → SessionStopRes(OK) | got {'Signature_isUsed': False, 'msg': {'ResponseCode': 'OK', 'ResponseCodeValue': 0}, 'name': 'SessionStopRes', 'ok': True, 'proto': 'iso2', 'session_id': 'eaf56074dd9ab9af'} | ✓ |
| Resume rejoins the paused session (OK_OldSessionJoined) | got {'Signature_isUsed': False, 'msg': {'EVSEID': 'DE*PNX*E12345*1', 'ResponseCode': 'OK_OldSessionJoined', 'ResponseCodeValue': 2}, 'name': 'SessionSetupRes', 'ok': True, 'proto': 'iso2', 'session_id': 'eaf56074dd9ab9af'} | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ChargeParameterDiscoveryRes message | got 'ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 1 poll(s) | ✓ |
| ChargeParameterDiscoveryRes(Finished) carries a SAScheduleList | SAScheduleList=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |
| SAScheduleList honours MaxEntriesSAScheduleTuple (<= 3) | got 1 tuple(s) | ✓ |
| Resumed session offers the SAScheduleTuple selected in the paused session (SAScheduleTupleID 1) | paused tuple 1, resumed tuples [1]; paused=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] resumed=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.9 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.2 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1998.8 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ChargeParameterDiscovery polls to EVSEProcessing=Finished | 1 | - | info |
| ChargeParameterDiscoveryRes SAScheduleList entries | 1 | ≥ 1 | ✓ |

## TC_SECC_DC_VTB_ChargeParameterDiscovery_022

TC_SECC_DC_VTB_ChargeParameterDiscovery_022.

Test objective: pause/resume (BCB toggle); with MaxEntries and DepartureTime.
The ATS selects this TC with PIXIT_SECC_CMN_SalesTariff := unknown, so no
SalesTariff is required in the SAScheduleList (the salesTariff twins with
PIXIT_SECC_CMN_SalesTariff := salesTariff are _020 / _034 / _035).

**DUT (dut)** PASS 22.15 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Pre-pause ChargeParameterDiscovery selects a SAScheduleList | got [{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |
| Response echoes the current SessionID | | ✓ |
| SessionStop(Pause) → SessionStopRes(OK) | got {'Signature_isUsed': False, 'msg': {'ResponseCode': 'OK', 'ResponseCodeValue': 0}, 'name': 'SessionStopRes', 'ok': True, 'proto': 'iso2', 'session_id': '03a7bef45fb54836'} | ✓ |
| Resume rejoins the paused session (OK_OldSessionJoined) | got {'Signature_isUsed': False, 'msg': {'EVSEID': 'DE*PNX*E12345*1', 'ResponseCode': 'OK_OldSessionJoined', 'ResponseCodeValue': 2}, 'name': 'SessionSetupRes', 'ok': True, 'proto': 'iso2', 'session_id': '03a7bef45fb54836'} | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ChargeParameterDiscoveryRes message | got 'ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 1 poll(s) | ✓ |
| ChargeParameterDiscoveryRes(Finished) carries a SAScheduleList | SAScheduleList=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 7200, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |
| SAScheduleList honours MaxEntriesSAScheduleTuple (<= 3) | got 1 tuple(s) | ✓ |
| Resumed session offers the SAScheduleTuple selected in the paused session (SAScheduleTupleID 1) | paused tuple 1, resumed tuples [1]; paused=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 86400, 'start': 0}], 'SAScheduleTupleID': 1}] resumed=[{'PMaxScheduleEntry': [{'PMax': {'multiplier': 2, 'unit': 'W', 'value': 1500}, 'duration': 7200, 'start': 0}], 'SAScheduleTupleID': 1}] | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.7 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1999.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ChargeParameterDiscovery polls to EVSEProcessing=Finished | 1 | - | info |
| ChargeParameterDiscoveryRes SAScheduleList entries | 1 | ≥ 1 | ✓ |

## TC_SECC_DC_VTB_CableCheck_001

TC_SECC_DC_VTB_CableCheck_001.

Test objective: GoodCase; Test System signals CP State C and sends a CableCheckReq with the current SessionID and all mandatory parameters. Checks the SUT replies CableCheckRes with the current SessionID, ResponseCode OK, EVSEProcessing Finished and all mandatory parameters. If an earlier CableCheckRes had EVSEProcessing Ongoing the Test System repeats the CableCheckReq until the SUT's cable check has finished or V2G_EVCC_CableCheck_Timer expires. Primary happy-path CableCheck TC (CP C).

### DUT (dut) PASS 11.40 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' (after 12 poll(s), 4.0s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CableCheck polls to EVSEProcessing=Finished | 12 | - | info |
| CableCheck IMD self-test completion time | 3969.4 ms | - | info |

## TC_SECC_DC_VTB_CableCheck_003

TC_SECC_DC_VTB_CableCheck_003.

Test objective: GoodCase; the Test System signals CP State C or D and sends a valid ServiceDiscoveryReq instead of a CableCheckReq (sequence error). Checks the SUT replies ServiceDiscoveryRes with the current SessionID, FAILED_SequenceError, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT** (dut) **PASS** 7.99 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_SequenceError | got FAILED_SequenceError (name=ServiceDiscoveryRes) | ✓ |
| SUT closes the TCP connection within par_CMN_TCP_Connection_Termination_Timeout | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_CableCheck_004

TC_SECC_DC_VTB_CableCheck_004.

Test objective: GoodCase; CP State C or D; the Test System sends a CableCheckReq with an invalid SessionID and all mandatory parameters. Checks the SUT replies CableCheckRes with a valid SessionID, FAILED_UnknownSession, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

### DUT (dut) PASS 8.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_UnknownSession | got FAILED_UnknownSession (name=CableCheckRes) | ✓ |
| SUT closes the TCP connection within par_CMN_TCP_Connection_Termination_Timeout | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CableCheckRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_CableCheck_005

TC_SECC_DC_VTB_CableCheck_005.

Test objective: GoodCase; after a ChargeParameterDiscoveryRes (OK, EVSEProcessing Finished, valid scheduleList) the Test System stays silent until V2G_Sequence_Timeout expires (sends no CableCheckReq). Checks the SUT terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout after the timeout. LONG TEST.

**DUT (dut)** **PASS** 69.49 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection after V2G_Sequence_Timeout (within 70s) | closed 62.0s after ChargeParameterDiscoveryRes | ✓ |
| SUT does not close before V2G_Sequence_Timeout expires (>= 55s) | closed after 62.0s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 62028.0 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_DC_VTB_CableCheck_006

TC_SECC_DC_VTB_CableCheck_006.

Test objective: GoodCase; CableCheckReq with the current SessionID → CableCheckRes OK, EVSEProcessing Finished. The repetition of the CableCheck message sequence is checked (EVSEProcessing Ongoing) until the SUT's cable check has finished or V2G_EVCC_CableCheck_Timer expires. (PICS CableCheckOngoing=ongoing; exercises the repeated-Ongoing loop.)

**DUT** (dut) **PASS** 10.71 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| CableCheckRes = OK, EVSEProcessing Finished | got ResponseCode='OK', EVSEProcessing='Finished' after 10 poll(s) | ✓ |
| At least one CableCheckRes had EVSEProcessing=Ongoing (repeated-Ongoing loop exercised) | reached Finished after 10 poll(s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CableCheck polls to EVSEProcessing=Finished | 10 | - | info |
| CableCheck IMD self-test completion time | 3276.0 ms | - | info |

## TC_SECC_DC_VTB_CableCheck_008

TC_SECC_DC_VTB_CableCheck_008.

Test objective: GoodCase; the Test System signals NO CP State C or D and sends a CableCheckReq with the current SessionID and all mandatory parameters. Checks the SUT replies CableCheckRes with the current SessionID, ResponseCode FAILED, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout. (Requires the bench to hold the CP off both State C and State D -- the EV stays in State B here.)

[V2G2-916]..[V2G2-918]: on the first CableCheckReq the SECC waits V2G_SECC_CPState_Detection_Timeout for CP State C/D before responding; with the CP held in B it answers FAILED and closes. The SECC parks the request until the CP state is reported, which absorbs the CP-state/CableCheckReq cross-transport ordering so the outcome is deterministic.

### DUT (dut) PASS 8.80 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode is FAILED (CP not in State C/D) | got 'FAILED' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CableCheckRes response time | 1525.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_PreCharge_001

TC_SECC_DC_VTB_PreCharge_001.

Test objective: GoodCase; Test System sends a PreChargeReq with the current SessionID, target voltage and current and all mandatory parameters. Checks the SUT replies PreChargeRes with the current SessionID, ResponseCode OK, EVSEStatusCode EVSE_Ready and all mandatory parameters. If the Test System sends another valid PreChargeReq the SUT shall respond again; the sequence repeats until the required voltage has been reached or V2G_EVCC_PreCharge_Timer expires. "Reached" is asserted as EVSEPresentVoltage within +/-20 V of the EV target voltage (the tester's reached-tolerance).
Primary happy-path PreCharge TC.

### DUT (dut) PASS 11.80 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PreChargeRes message | got 'PreChargeRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEPresentVoltage reaches the 400 V target (+/-20 V) | got 400.0 V after 2 poll(s), 0.4s | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVSEStatusCode = EVSE_Ready | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| EVSEPresentVoltage at PreCharge target | 400.0 V | 400 ± 20.0 V | ✓ |
| PreCharge polls to within-tolerance voltage | 2 | - | info |

## TC_SECC_DC_VTB_PreCharge_002

TC_SECC_DC_VTB_PreCharge_002.

Test objective: GoodCase; the Test System sends a valid ServiceDiscoveryReq instead of a PreChargeReq (sequence error). Checks the SUT replies ServiceDiscoveryRes with the current SessionID, FAILED_SequenceError, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

### DUT (dut) PASS 11.59 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_SequenceError | got FAILED_SequenceError (name=ServiceDiscoveryRes) | ✓ |
| SUT closes the TCP connection within par_CMN_TCP_Connection_Termination_Timeout | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_PreCharge_003

TC_SECC_DC_VTB_PreCharge_003.

Test objective: GoodCase; the Test System sends a PreChargeReq with an invalid SessionID, target voltage and current and all mandatory parameters. Checks the SUT replies PreChargeRes with a valid SessionID, FAILED_UnknownSession, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT** (dut) **PASS** 11.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_UnknownSession | got FAILED_UnknownSession | ✓ |
| SUT closes the TCP connection within par_CMN_TCP_Connection_Termination_Timeout | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PreChargeRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_PreCharge_004

TC_SECC_DC_VTB_PreCharge_004.

Test objective: GoodCase; after a CableCheckRes (OK, EVSEProcessing Finished) the Test System stays silent until V2G_Sequence_Timeout expires (sends no PreChargeReq). Checks the SUT terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout after the timeout. LONG TEST. (The -4 table text names CableCheckRes as the preceding message; the PreCharge precondition is CableCheck done -- see doc/tc_catalog_iso4_wave4.md.)

### DUT (dut) PASS 72.90 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection after V2G_Sequence_Timeout (within 70s) | closed 62.0s after CableCheckRes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 62013.5 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_DC_VTB_PowerDelivery_001

TC_SECC_DC_VTB_PowerDelivery_001.

Test objective: GoodCase; Test System sends a PowerDeliveryReq with the current SessionID, ChargeProgress=Start, a valid ChargingProfile and a valid SAScheduleTupleID. Checks the SUT replies PowerDeliveryRes with the current SessionID, ResponseCode OK. Primary happy-path PowerDelivery TC (Start).

### DUT (dut) PASS 11.41 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = OK | got 'OK' (0.1s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 421.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 103.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 103.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 103.0 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_002

TC_SECC_DC_VTB_PowerDelivery_002.

Test objective: GoodCase; the Test System sends a PowerDeliveryReq with the current SessionID, ChargeProgress=Stop; after receiving the valid PowerDeliveryRes it signals CP State B. Checks the SUT replies PowerDeliveryRes with the current SessionID, ResponseCode OK.

**DUT** (dut) **PASS** 13.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = OK | got 'OK' (0.1s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 419.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.5 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.2 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_003

TC_SECC_DC_VTB_PowerDelivery_003.

Test objective: GoodCase; the Test System sends a valid ServiceDiscoveryReq instead of a PowerDeliveryReq (ChargeProgress Start context, sequence error). Checks the SUT replies ServiceDiscoveryRes with the current SessionID, FAILED_SequenceError, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT** (dut) **PASS** 12.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' (name='ServiceDiscoveryRes') | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 421.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 103.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_004

TC_SECC_DC_VTB_PowerDelivery_004.

Test objective: as _003 but in the Stop context -- GoodCase through the charge loop, then a valid ServiceDiscoveryReq instead of a PowerDeliveryReq (ChargeProgress Stop context). Checks ServiceDiscoveryRes with the current SessionID, FAILED_SequenceError, then TCP close.

**DUT (dut)** PASS 12.79 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' (name='ServiceDiscoveryRes') | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| ServiceDiscoveryRes response time | 102.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_005

TC_SECC_DC_VTB_PowerDelivery_005.

Test objective: GoodCase; the Test System sends a PowerDeliveryReq with an invalid SessionID, ChargeProgress=Start, a valid ChargingProfile and SAScheduleTupleID. Checks the SUT replies PowerDeliveryRes with a valid SessionID, FAILED_UnknownSession, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT** (dut) **PASS** 11.80 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT does not answer OK to an unknown SessionID | SUT did not validate the (unknown) SessionID and answered OK instead of FAILED_UnknownSession; the -4 ATS requires FAILED_UnknownSession and a TCP close. | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' (name='PowerDeliveryRes') | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.5 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_006

TC_SECC_DC_VTB_PowerDelivery_006.

Test objective: as _005 but ChargeProgress=Stop -- GoodCase through the charge loop, then a PowerDeliveryReq with an invalid SessionID, ChargeProgress=Stop. Checks PowerDeliveryRes with a valid SessionID, FAILED_UnknownSession, then TCP close.

**DUT (dut)** **PASS** 13.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT does not answer OK to an unknown SessionID | SUT did not validate the (unknown) SessionID and answered OK instead of FAILED_UnknownSession; the -4 ATS requires FAILED_UnknownSession and a TCP close. | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' (name='PowerDeliveryRes') | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 418.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.5 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 103.6 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| PowerDeliveryRes(Stop) response time | 103.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_007

TC_SECC_DC_VTB_PowerDelivery_007.

Test objective: GoodCase; the Test System sends a PowerDeliveryReq with the current SessionID, ChargeProgress=Start, an invalid ChargingProfile (its ChargingProfileEntryMaxPower exceeds EVSEMaximumPowerLimit) and a valid SAScheduleTupleID. Checks the SUT replies PowerDeliveryRes with the current SessionID, FAILED_ChargingProfileInvalid, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT** (dut) **PASS** 11.79 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_ChargingProfileInvalid | got 'FAILED_ChargingProfileInvalid' (name='PowerDeliveryRes') | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 104.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.2 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_008

TC_SECC_DC_VTB_PowerDelivery_008.

Test objective: GoodCase; the Test System sends a PowerDeliveryReq with the current SessionID, ChargeProgress=Start, a valid ChargingProfile and an invalid SAScheduleTupleID. Checks the SUT replies PowerDeliveryRes with the current SessionID, FAILED_TariffSelectionInvalid, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT** (dut) **PASS** 12.41 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_TariffSelectionInvalid | got 'FAILED_TariffSelectionInvalid' (name='PowerDeliveryRes') | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 418.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.2 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_009

TC_SECC_DC_VTB_PowerDelivery_009.

Test objective: GoodCase; after a PreChargeRes (current SessionID, ResponseCode OK) the Test System stays silent until V2G_Sequence_Timeout expires (sends no PowerDeliveryReq). Checks the SUT terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout. LONG TEST.

**DUT (dut)** PASS 73.89 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection after V2G_Sequence_Timeout (>= 59s, within 70s) | closed 62.0s after PreChargeRes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 62010.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_DC_VTB_PowerDelivery_010

TC_SECC_DC_VTB_PowerDelivery_010.

Test objective: GoodCase; after a valid CurrentDemandRes (or MeteringReceipt) the Test System stays silent until V2G_Sequence_Timeout expires (sends no PowerDeliveryReq). Checks the SUT terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout. LONG TEST.

**DUT (dut)** PASS 75.01 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection after V2G_Sequence_Timeout (>= 59s, within 70s) | closed 61.8s after CurrentDemandRes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 61798.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_DC_VTB_PowerDelivery_013

TC_SECC_DC_VTB_PowerDelivery_013.

Test objective: GoodCase; the Test System sends a PowerDeliveryReq with the current SessionID, ChargeProgress=Renegotiate but no PowerDeliveryReq with ChargeProgress=Start was sent before. Checks the SUT replies PowerDeliveryRes with the current SessionID, ResponseCode FAILED, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT** (dut) **PASS** 12.09 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT rejects ChargeProgress=Renegotiate without a prior Start (not OK) | got 'FAILED' | ✓ |
| ResponseCode = FAILED | got 'FAILED' | ✓ |
| SUT closes the TCP connection after the FAILED response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Renegotiate) response time | 102.5 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_014

> TC_SECC_DC_VTB_PowerDelivery_014.
>
> Test objective: GoodCase; PowerDeliveryReq ChargeProgress=Stop with DC_EVStatus.EVErrorCode=FAILED_RESSTemperatureInhibit. The EVErrorCode must not influence the charging process → PowerDeliveryRes OK.

**DUT (dut)** **PASS** 13.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVErrorCode FAILED_RESSTemperatureInhibit ignored on Stop → ResponseCode = OK | got 'OK' (0.1s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.5 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.6 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_015

> TC_SECC_DC_VTB_PowerDelivery_015.
>
> Test objective: GoodCase; PowerDeliveryReq ChargeProgress=Stop with DC_EVStatus.EVErrorCode=FAILED_EVShiftPosition. Must be ignored → PowerDeliveryRes OK.

**DUT** (dut) **PASS** 13.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVErrorCode FAILED_EVShiftPosition ignored on Stop → ResponseCode = OK | got 'OK' (0.1s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.8 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_016

> TC_SECC_DC_VTB_PowerDelivery_016.
>
> Test objective: GoodCase; PowerDeliveryReq ChargeProgress=Stop with DC_EVStatus.EVErrorCode=FAILED_ChargerConnectorLockFault. Must be ignored → PowerDeliveryRes OK.

**DUT** (dut) **PASS** 13.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVErrorCode FAILED_ChargerConnectorLockFault ignored on Stop → ResponseCode = OK | got 'OK' (0.1s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.0 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.9 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_017

> TC_SECC_DC_VTB_PowerDelivery_017.
>
> Test objective: GoodCase; PowerDeliveryReq ChargeProgress=Stop with DC_EVStatus.EVErrorCode=FAILED_EVRESSMalfunction. Must be ignored → PowerDeliveryRes OK.

**DUT** (dut) **PASS** 12.60 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVErrorCode FAILED_EVRESSMalfunction ignored on Stop → ResponseCode = OK | got 'OK' (0.1s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_018

> TC_SECC_DC_VTB_PowerDelivery_018.
>
> Test objective: GoodCase; PowerDeliveryReq ChargeProgress=Stop with DC_EVStatus.EVErrorCode=FAILED_ChargingCurrentdifferential. Must be ignored → PowerDeliveryRes OK.

**DUT** (dut) **PASS** 13.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVErrorCode FAILED_ChargingCurrentdifferential ignored on Stop → ResponseCode = OK | got 'OK' (0.1s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.4 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_019

> TC_SECC_DC_VTB_PowerDelivery_019.
>
> Test objective: GoodCase; PowerDeliveryReq ChargeProgress=Stop with DC_EVStatus.EVErrorCode=FAILED_ChargingVoltageOutOfRange. Must be ignored → PowerDeliveryRes OK.

**DUT** (dut) **PASS** 13.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVErrorCode FAILED_ChargingVoltageOutOfRange ignored on Stop → ResponseCode = OK | got 'OK' (0.1s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 103.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_020

> TC_SECC_DC_VTB_PowerDelivery_020.
>
> Test objective: GoodCase; PowerDeliveryReq ChargeProgress=Stop with DC_EVStatus.EVErrorCode=FAILED_ChargingSystemIncompatibility. Must be ignored → PowerDeliveryRes OK.

**DUT** (dut) **PASS** 13.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVErrorCode FAILED_ChargingSystemIncompatibility ignored on Stop → ResponseCode = OK | got 'OK' (0.1s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.0 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.6 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

## TC_SECC_DC_VTB_PowerDelivery_021

TC_SECC_DC_VTB_PowerDelivery_021.

Test objective: GoodCase; PowerDeliveryReq ChargeProgress=Stop with DC_EVStatus.EVErrorCode=NoData (no error). → PowerDeliveryRes OK.

**DUT** (dut) **PASS** 13.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| EVErrorCode NoData ignored on Stop → ResponseCode = OK | got 'OK' (0.1s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.9 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

## TC_SECC_DC_VTB_CurrentDemand_001

TC_SECC_DC_VTB_CurrentDemand_001.

Test objective: GoodCase; the Test System sends a CurrentDemandReq with the current SessionID, target voltage and current and all mandatory parameters. Checks the SUT replies CurrentDemandRes with the current SessionID, ResponseCode OK, ReceiptRequired false and all mandatory parameters. Primary happy-path CurrentDemand TC.

**DUT** (dut) **PASS** 12.24 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CurrentDemandRes message | got 'CurrentDemandRes' | ✓ |
| ResponseCode = OK | got 'OK' (0.1s) | ✓ |
| CurrentDemandRes carries mandatory EVSEPresentVoltage | present | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| CurrentDemandRes carries mandatory EVSEPresentCurrent | present | ✓ |
| CurrentDemandRes carries mandatory EVSEID | present | ✓ |
| CurrentDemandRes carries mandatory SAScheduleTupleID | present | ✓ |
| CurrentDemandRes carries mandatory DC_EVSEStatus | present | ✓ |
| EVSEPresentVoltage is sourcing (> 0 V) | got 400.0 V | ✓ |
| ReceiptRequired is false (eIM free service) | ReceiptRequired=None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.9 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.2 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| CurrentDemandRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CurrentDemand EVSEPresentVoltage | 400.0 V | ≥ 0 V | ✓ |
| CurrentDemand EVSEPresentCurrent | 10.0 A | - | info |

## TC_SECC_DC_VTB_CurrentDemand_002

TC_SECC_DC_VTB_CurrentDemand_002.

Test objective: as _001 but pnC with ReceiptRequired true → CurrentDemandRes OK, ReceiptRequired true. Boots config-test-conformance-pnc.yaml with ev_receipt_required=true and drives a real Contract DC session into the CurrentDemand loop; fails honestly if the SUT never sets ReceiptRequired.

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sets ReceiptRequired=true in a CurrentDemandRes(OK) | | ✓ |
| SUT sends a CurrentDemandRes message | got 'CurrentDemandRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.5 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

## TC_SECC_DC_VTB_CurrentDemand_003

TC_SECC_DC_VTB_CurrentDemand_003.

Test objective: GoodCase; the Test System sends a valid ServiceDiscoveryReq instead of a CurrentDemandReq (sequence error). Checks the SUT replies ServiceDiscoveryRes with the current SessionID, FAILED_SequenceError, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT (dut)** PASS 12.39 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_SequenceError | got FAILED_SequenceError (name=ServiceDiscoveryRes) | ✓ |
| SUT closes the TCP connection within par_CMN_TCP_Connection_Termination_Timeout | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| ServiceDiscoveryRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_CurrentDemand_004

TC_SECC_DC_VTB_CurrentDemand_004.

Test objective: GoodCase; the Test System sends a CurrentDemandReq with an invalid SessionID. Checks the SUT replies CurrentDemandRes with a valid SessionID, FAILED_UnknownSession, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT (dut)** **PASS** 11.60 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_UnknownSession | got FAILED_UnknownSession (name=CurrentDemandRes) | ✓ |
| SUT closes the TCP connection within par_CMN_TCP_Connection_Termination_Timeout | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 103.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.6 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| CurrentDemandRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_CurrentDemand_005

TC_SECC_DC_VTB_CurrentDemand_005.

Test objective: GoodCase; after a PowerDeliveryRes (current SessionID, ResponseCode OK) the Test System stays silent until V2G_Sequence_Timeout expires (sends no CurrentDemandReq). Checks the SUT terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout. LONG TEST.

**DUT** (dut) **PASS** 73.90 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection after V2G_Sequence_Timeout (within 70s) | closed 62.0s after PowerDeliveryRes | ✓ |
| SUT does not close before V2G_Sequence_Timeout expires (>= 55s) | closed after 62.0s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.9 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 62009.8 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_DC_VTB_CurrentDemand_006

TC_SECC_DC_VTB_CurrentDemand_006.

Test objective: GoodCase; CurrentDemandReq (current SessionID, target V and I, all mandatory). Checks CurrentDemandRes OK, ReceiptRequired false, including a MeterInfo. (MeterInfo is PIXIT-driven; this free-service SUT may omit it -- the reachable OK + present-voltage assertion is checked, MeterInfo presence is N/A.)

**DUT** (dut) **PASS** 12.30 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CurrentDemandRes message | got 'CurrentDemandRes' | ✓ |
| ResponseCode = OK | got 'OK' (0.1s) | ✓ |
| CurrentDemandRes carries mandatory EVSEPresentVoltage | present | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| CurrentDemandRes carries mandatory EVSEPresentCurrent | present | ✓ |
| CurrentDemandRes carries mandatory EVSEID | present | ✓ |
| CurrentDemandRes carries mandatory SAScheduleTupleID | present | ✓ |
| CurrentDemandRes carries mandatory DC_EVSEStatus | present | ✓ |
| EVSEPresentVoltage is sourcing (> 0 V) | got 400.0 V | ✓ |
| ReceiptRequired is false (eIM free service) | ReceiptRequired=None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.9 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| CurrentDemandRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CurrentDemand EVSEPresentVoltage | 400.0 V | ≥ 0 V | ✓ |
| CurrentDemand EVSEPresentCurrent | 10.0 A | - | info |

## TC_SECC_DC_VTB_CurrentDemand_007

TC_SECC_DC_VTB_CurrentDemand_007.

Test objective: as _006 (MeterInfo) but pnC with ReceiptRequired true. Boots the PnC config with ev_receipt_required=true and drives a real Contract DC session; fails honestly if the SUT never sets ReceiptRequired.

**DUT** (dut) **PASS** 11.99 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sets ReceiptRequired=true in a CurrentDemandRes(OK) | | ✓ |
| SUT sends a CurrentDemandRes message | got 'CurrentDemandRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.5 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| CurrentDemand MeterReading | 622 Wh | - | info |

## TC_SECC_DC_VTB_CurrentDemand_008

TC_SECC_DC_VTB_CurrentDemand_008.

Test objective: GoodCase; CurrentDemandReq (current SessionID, target V and I, all mandatory). Checks CurrentDemandRes OK, ReceiptRequired false, MeterInfo including MeterReading. (MeterInfo/MeterReading are PIXIT-driven; N/A on this free-service SUT -- the reachable OK + present-voltage assertion is checked.)

**DUT** (dut) **PASS** 12.40 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CurrentDemandRes message | got 'CurrentDemandRes' | ✓ |
| ResponseCode = OK | got 'OK' (0.1s) | ✓ |
| CurrentDemandRes carries mandatory EVSEPresentVoltage | present | ✓ |
| CurrentDemandRes carries mandatory EVSEPresentCurrent | present | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| CurrentDemandRes carries mandatory EVSEID | present | ✓ |
| CurrentDemandRes carries mandatory SAScheduleTupleID | present | ✓ |
| CurrentDemandRes carries mandatory DC_EVSEStatus | present | ✓ |
| EVSEPresentVoltage is sourcing (> 0 V) | got 400.0 V | ✓ |
| ReceiptRequired is false (eIM free service) | ReceiptRequired=None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.0 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| CurrentDemandRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CurrentDemand EVSEPresentVoltage | 400.0 V | ≥ 0 V | ✓ |
| CurrentDemand EVSEPresentCurrent | 10.0 A | - | info |

## TC_SECC_DC_VTB_CurrentDemand_009

TC_SECC_DC_VTB_CurrentDemand_009.

Test objective: as _008 (MeterInfo incl MeterReading) but pnC with ReceiptRequired true. Boots the PnC config with ev_receipt_required=true and drives a real Contract DC session; fails honestly if the SUT never sets ReceiptRequired.

**DUT** (dut) **PASS** 12.60 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sets ReceiptRequired=true in a CurrentDemandRes(OK) | | ✓ |
| SUT sends a CurrentDemandRes message | got 'CurrentDemandRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.6 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| CurrentDemand MeterReading | 628 Wh | - | info |

## TC_SECC_DC_VTB_MeteringReceipt_001

> TC_SECC_DC_VTB_MeteringReceipt_001.
>
> GoodCase; MeteringReceiptReq carrying a valid signature, the current SessionID, a valid SAScheduleTupleID and MeterInfo → MeteringReceiptRes OK.

**DUT** (dut) **PASS** 12.40 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a MeteringReceiptRes message | got 'MeteringReceiptRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.9 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

## TC_SECC_DC_VTB_MeteringReceipt_002

TC_SECC_DC_VTB_MeteringReceipt_002.

A valid ServiceDiscoveryReq instead of the expected MeteringReceiptReq → the SUT answers the received-type response with FAILED_SequenceError and closes the TCP connection [V2G2-539].

**DUT** (dut) **PASS** 12.59 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 101.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_MeteringReceipt_003

> TC_SECC_DC_VTB_MeteringReceipt_003.
>
> MeteringReceiptReq with a valid signature but an invalid SessionID → MeteringReceiptRes FAILED_UnknownSession + TCP close [V2G2-388].

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a MeteringReceiptRes message | got 'MeteringReceiptRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.0 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_MeteringReceipt_004

> TC_SECC_DC_VTB_MeteringReceipt_004.
>
> MeteringReceiptReq whose signed content no longer matches the transmitted MeterReading (the receipt body was altered after the signature was calculated) → MeteringReceiptRes FAILED_MeteringSignatureNotValid + TCP close.

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_MeteringSignatureNotValid | got 'FAILED_MeteringSignatureNotValid' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_MeteringReceipt_005

TC_SECC_DC_VTB_MeteringReceipt_005.

After a CurrentDemandRes(OK) that requested a receipt, the EV sends nothing further; once V2G_SECC_Sequence_Timeout expires the SUT closes the TCP connection.

**DUT** (dut) **PASS** 74.10 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection on V2G_SECC_Sequence_Timeout | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_ATTEN_CHAR.IND after sounding | 415.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.0 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 62040.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_DC_VTB_MeteringReceipt_006

> TC_SECC_DC_VTB_MeteringReceipt_006.
>
> MeteringReceiptReq signed with a private key that does not belong to the contract leaf → MeteringReceiptRes FAILED_MeteringSignatureNotValid + TCP close.

**DUT** (dut) **PASS** 12.09 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a MeteringReceiptRes message | got 'MeteringReceiptRes' | ✓ |
| ResponseCode = FAILED_MeteringSignatureNotValid | got 'FAILED_MeteringSignatureNotValid' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.8 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_MeteringReceipt_007

> TC_SECC_DC_VTB_MeteringReceipt_007.
>
> MeteringReceiptReq whose body was changed after the signature calculation (the SignedInfo Reference digest no longer matches the transmitted element) → MeteringReceiptRes FAILED_MeteringSignatureNotValid + TCP close.

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a MeteringReceiptRes message | got 'MeteringReceiptRes' | ✓ |
| ResponseCode = FAILED_MeteringSignatureNotValid | got 'FAILED_MeteringSignatureNotValid' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_WeldingDetection_001

TC_SECC_DC_VTB_WeldingDetection_001.

Test objective: GoodCase; after PowerDelivery(Stop) the Test System sends a WeldingDetectionReq with the current SessionID and all mandatory parameters (DC_EVStatus). Checks the SUT replies WeldingDetectionRes with the current SessionID, ResponseCode OK, DC_EVSEStatus and EVSEPresentVoltage. Primary happy-path WeldingDetection TC.

**DUT** (dut) **PASS** 13.40 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a WeldingDetectionRes message | got 'WeldingDetectionRes' | ✓ |
| ResponseCode = OK | got 'OK' (0.1s) | ✓ |
| WeldingDetectionRes carries DC_EVSEStatus and EVSEPresentVoltage | keys=['DC_EVSEStatus', 'EVSEPresentVoltage', 'ResponseCode', 'ResponseCodeValue'] | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.2 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| WeldingDetectionRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| WeldingDetection EVSEPresentVoltage | 0.0 V | - | info |

## TC_SECC_DC_VTB_WeldingDetection_002

TC_SECC_DC_VTB_WeldingDetection_002.

Test objective: GoodCase; the Test System sends a valid ServiceDiscoveryReq instead of a WeldingDetectionReq (sequence error). Checks the SUT replies ServiceDiscoveryRes with the current SessionID, FAILED_SequenceError, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT (dut)** **PASS** 12.99 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' (name='ServiceDiscoveryRes') | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.8 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.5 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| ServiceDiscoveryRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_WeldingDetection_003

TC_SECC_DC_VTB_WeldingDetection_003.

Test objective: GoodCase; the Test System sends a WeldingDetectionReq with an invalid SessionID. Checks the SUT replies WeldingDetectionRes with a valid SessionID, FAILED_UnknownSession, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

**DUT (dut)** **PASS** 13.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' (name='WeldingDetectionRes') | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.0 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| WeldingDetectionRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_WeldingDetection_004

TC_SECC_DC_VTB_WeldingDetection_004.

Test objective: GoodCase; after a PowerDeliveryRes (current SessionID, ResponseCode OK) the Test System stays silent until V2G_Sequence_Timeout expires (sends no WeldingDetectionReq). Checks the SUT terminates the session by closing TCP within par_CMN_TCP_Connection_Termination_Timeout. LONG TEST.

**DUT** (dut) **PASS** 75.51 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT closes the TCP connection within V2G_Sequence_Timeout+termination (70s) with no WeldingDetectionReq | closed 61.7s after PowerDeliveryRes(Stop) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.8 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| TCP connection close after failure/ timeout | 61727.9 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_DC_VTB_WeldingDetection_005

TC_SECC_DC_VTB_WeldingDetection_005.

Test objective: GoodCase; the Test System sends a WeldingDetectionReq with the current SessionID and all mandatory parameters WITHOUT signalling CP State B before. Checks the SUT replies WeldingDetectionRes with the current SessionID, ResponseCode FAILED, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout. (Requires the bench to hold the CP off State B after Stop -- the EV stays in State C here.)

[V2G2-920]..[V2G2-922]: after PowerDelivery(Stop) the SECC waits V2G_SECC_CPState_Detection_Timeout for CP State B before answering the WeldingDetectionReq; with no B it responds FAILED and closes.

**DUT** (dut) **PASS** 14.70 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED (no preceding CP State B) | got 'FAILED' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.2 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 101.9 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| WeldingDetectionRes response time | 1526.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_SessionStop_002

TC_SECC_DC_VTB_SessionStop_002.

Test objective: as _001 but after a WeldingDetection sequence (PICS_SECC_DC_WeldingDetection=true): GoodCase through WeldingDetection, then SessionStopReq ChargingSession=Terminate → SessionStopRes OK, data-link teardown within par_TP_match_leave.

**DUT** (dut) **PASS** 13.21 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' (0.1s) | ✓ |
| SUT terminates the data link within par_TP_match_leave (1.25s) | left after 0.10s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.9 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.6 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| WeldingDetectionRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionStopRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SUT left the logical network after SessionStop(Terminate) | 0.102 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_SessionStop_006

TC_SECC_DC_VTB_SessionStop_006.

Test objective: GoodCase; the Test System sends a SessionStopReq with the current SessionID and ChargingSession=Terminate WITHOUT signalling CP State B before. Checks the SUT replies SessionStopRes with the current SessionID, ResponseCode FAILED, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout. (Requires the bench to hold the CP off State B -- the EV stays in State C here.)

[V2G2-920]..[V2G2-922]: after PowerDelivery(Stop) the SECC waits V2G_SECC_CPState_Detection_Timeout for CP State B before answering the SessionStopReq; with no B it responds FAILED and closes.

**DUT** (dut) **PASS** 13.98 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED (no preceding CP State B) | got 'FAILED' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 420.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 1.2 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 105.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.4 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 1527.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.7 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_SessionStop_007

> TC_SECC_DC_VTB_SessionStop_007.
>
> Test objective: as _003 (ServiceDiscoveryReq instead of SessionStopReq, FAILED_SequenceError + TCP close) but after a WeldingDetection sequence (PICS_SECC_DC_WeldingDetection=true).

**DUT** (dut) **PASS** 13.29 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' (name='ServiceDiscoveryRes') | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.9 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.4 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| WeldingDetectionRes response time | 102.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_SessionStop_008

> TC_SECC_DC_VTB_SessionStop_008.
>
> Test objective: as _004 (invalid SessionID → FAILED_UnknownSession + TCP close) but after a WeldingDetection sequence (PICS_SECC_DC_WeldingDetection=true).

**DUT** (dut) **PASS** 13.90 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' (name='SessionStopRes') | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 418.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 103.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 103.4 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.8 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| WeldingDetectionRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionStopRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_SessionStop_009

TC_SECC_DC_VTB_SessionStop_009.

Test objective: as _005 (sequence timeout → TCP close) but after a WeldingDetection sequence (PICS_SECC_DC_WeldingDetection=true). LONG TEST.

**DUT** (dut) **PASS** 75.90 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT does not close before V2G_Sequence_Timeout expires (>= 55s) | closed after 62.0s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.5 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| WeldingDetectionRes response time | 102.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 62000.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC_SECC_CMN_VTB_V2GTPSessionSetup_001

> TC_SECC_CMN_VTB_V2GTPSessionSetup_001.
>
> Test objective: GoodCase; the Test System sends a SessionSetupReq with the V2GTP header protocolVersion=0×01, invProtocolVersion=0xFE and payloadType=0×8001 (valid header for V2G message content), and checks the SUT answers a SessionSetupRes framed with protocolVersion=0×01, invProtocolVersion=0xFE and payloadType=0×8001.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| response V2GTP protocolVersion = 0×01 | got 0×01 | ✓ |
| response V2GTP invProtocolVersion = 0xFE | got 0xfe | ✓ |
| response V2GTP payloadType = 0×8001 | got 0×8001 | ✓ |
| framed response is a SessionSetupRes | got 'SessionSetupRes' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_CMN_VTB_V2GTPSessionSetup_002

TC_SECC_CMN_VTB_V2GTPSessionSetup_002.

Test objective: the Test System sends a SessionSetupReq whose V2GTP header carries an invalid payloadLength=0×00000000, and checks the SUT ignores the message (detected by the Test System as a V2G_Msg_Timeout -- no SessionSetupRes).

**DUT (dut)** PASS 9.99 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT ignores the malformed-V2GTP-header SessionSetupReq (no SessionSetupRes) | ignored (got None) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_CMN_VTB_V2GTPSessionSetup_003

TC_SECC_CMN_VTB_V2GTPSessionSetup_003.

Test objective: the Test System sends a SessionSetupReq whose V2GTP header carries an invalid payloadType=0×7001, and checks the SUT ignores the message (V2G_Msg_Timeout).

**DUT (dut)** PASS 10.01 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT ignores the malformed-V2GTP-header SessionSetupReq (no SessionSetupRes) | ignored (got None) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_CMN_VTB_V2GTPSessionSetup_004

TC_SECC_CMN_VTB_V2GTPSessionSetup_004.

Test objective: the Test System sends a SessionSetupReq whose V2GTP header carries an invalid protocolVersion=0xFF, and checks the SUT ignores the message (V2G_Msg_Timeout).

**DUT (dut)** PASS 10.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT ignores the malformed-V2GTP-header SessionSetupReq (no SessionSetupRes) | ignored (got None) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_CMN_VTB_V2GTPSessionSetup_005

TC_SECC_CMN_VTB_V2GTPSessionSetup_005.

Test objective: the Test System sends a SessionSetupReq whose V2GTP header carries an invalid invProtocolVersion=0xFF, and checks the SUT ignores the message (V2G_Msg_Timeout).

**DUT (dut)** PASS 10.60 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT ignores the malformed-V2GTP-header SessionSetupReq (no SessionSetupRes) | ignored (got None) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_CMN_VTB_V2GTPSDP_001

> TC_SECC_CMN_VTB_V2GTPSDP_001.
>
> Test objective: GoodCase; the Test System sends a valid SDP request with the V2GTP header protocolVersion=0×01, invProtocolVersion=0xFE and payloadType=0×9000, and checks the SUT answers an SDP response with protocolVersion=0×01, invProtocolVersion=0xFE, payloadType=0×9001 and payloadLength=0×00000014 (20).

### DUT (dut) PASS 6.99 s

| Conformance check | Observed | Result |
|---|---|---|
| SDP response V2GTP protocolVersion = 0×01 | got 0×01 | ✓ |
| SDP response V2GTP invProtocolVersion = 0xFE | got 0xfe | ✓ |
| SDP response payloadType = 0×9001 | got 0×9001 | ✓ |
| SDP response payloadLength = 20 | got 20 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SDP response payloadLength | 20 byte | = 20 byte | ✓ |
| SDP advertised TCP port | 50000 | - | info |

## TC_SECC_CMN_VTB_V2GTPSDP_002

TC_SECC_CMN_VTB_V2GTPSDP_002.

Test objective: the Test System sends an SDP request whose V2GTP header carries an invalid payloadLength=0×00000000, and checks the SUT ignores it (no SDP response, detected as an SDP message timeout).

**DUT** (dut) **PASS** 8.20 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT ignores the SDP request with invalid payloadLength=0 (no SDP response) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_CMN_VTB_V2GTPSDP_003

TC_SECC_CMN_VTB_V2GTPSDP_003.

Test objective: the Test System sends an SDP request whose V2GTP header carries an invalid payloadType=0×7000, and checks the SUT ignores it (SDP message timeout).

**DUT** (dut) **PASS** 8.80 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT ignores the SDP request with invalid payloadType=0×7000 (no SDP response) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_CMN_VTB_V2GTPSDP_004

TC_SECC_CMN_VTB_V2GTPSDP_004.

Test objective: the Test System sends an SDP request whose V2GTP header carries an invalid protocolVersion=0xFF, and checks the SUT ignores it (SDP message timeout).

**DUT** (dut) **PASS** 8.21 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT ignores the SDP request with invalid protocolVersion=0xFF (no SDP response) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |

## TC_SECC_CMN_VTB_V2GTPSDP_005

TC_SECC_CMN_VTB_V2GTPSDP_005.

Test objective: the Test System sends an SDP request whose V2GTP header carries an invalid invProtocolVersion=0xFF, and checks the SUT ignores it (SDP message timeout).

**DUT** (dut) **PASS** 8.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT ignores the SDP request with invalid invProtocolVersion=0xFF (no SDP response) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |

## TC_SECC_CMN_VTB_SDP_001

> TC_SECC_CMN_VTB_SDP_001.
>
> Test objective: GoodCase (eIM / no-TLS); the Test System sends an SDP request with Security=0×10 (no transport security) and TransportProtocol=0×00 (TCP), and checks the SUT answers an SDP response with Security=0×10, TransportProtocol=0×00 and a valid port + IP address.

**DUT (dut)** PASS 5.80 s

| Conformance check | Observed | Result |
|---|---|---|
| SDP response Security = 0×10 (no-TLS) | got 0×10 | ✓ |
| SDP response TransportProtocol = 0×00 (TCP) | got 0×00 | ✓ |
| SDP response carries a valid port + IP | port=50000 addr=fe80::4c20:35ff:fed4:ed22 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| SDP advertised TCP port | 50000 | - | info |

## TC_SECC_CMN_VTB_SDP_002

TC_SECC_CMN_VTB_SDP_002.

Test objective: GoodCase; SDP request with Security=0×00 (TLS) and TransportProtocol=0×00, expecting an SDP response announcing Security=0×00 (TLS). Requires PIXIT_SECC_CMN_TLS := true.

Runs against the PnC/TLS SUT config: the SUT runs a TLS server, so a TLS-requesting SDP must be answered with Security=0×00 and the TLS port.

### DUT (dut) PASS 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| SDP response TransportProtocol = 0×00 (TCP) | got 0×00 | ✓ |
| SDP response Security = 0×00 (TLS) from the TLS-configured SUT | got 0×00 | ✓ |
| SDP response carries a valid port + IP | port=50000 addr=fe80::4c20:35ff:fed4:ed22 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| SDP advertised TLS port | 50000 | - | info |

## TC_SECC_CMN_VTB_SDP_003

> TC_SECC_CMN_VTB_SDP_003.
>
> Test objective: GoodCase; SDP request with Security=0×00 (TLS) and TransportProtocol=0×00 under Plug-and-Charge (PICS IdentificationMode := pnC), expecting an SDP response announcing Security=0×00 (TLS).
>
> Runs against the PnC/TLS SUT config (Contract + TLS): same TLS SDP advertisement as _002, under the Plug-and-Charge configuration.

**DUT (dut)** **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SDP response TransportProtocol = 0×00 (TCP) | got 0×00 | ✓ |
| SDP response Security = 0×00 (TLS) under PnC | got 0×00 | ✓ |
| SDP response carries a valid port + IP | port=50000 addr=fe80::4c20:35ff:fed4:ed22 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_CMN_VTB_SDP_004

TC_SECC_CMN_VTB_SDP_004.

Test objective: GoodCase; the Test System waits until V2G_SECC_CommunicationSetup_Timeout (20 s) has expired on the SUT side without opening a TCP connection, then sends a valid SDP request with Security=0×10 and TransportProtocol=0×00 and checks the SUT has stopped the SDP server and does not respond. LONG TEST.

**DUT** (dut) **PASS** 40.11 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT stops the SDP server after V2G_SECC_CommunicationSetup_Timeout (no SDP response) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_CMN_VTB_SDP_005

TC_SECC_CMN_VTB_SDP_005.

Test objective: as SDP_004 (comm-setup-timeout → SDP server stops) but for the TLS-configured run (Security=0×00, PIXIT_SECC_CMN_TLS := true). LONG TEST.

**DUT** (dut) **PASS** 40.03 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT stops the SDP server after V2G_SECC_CommunicationSetup_Timeout (TLS run, no SDP response) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_CMN_VTB_SDP_006

> TC_SECC_CMN_VTB_SDP_006.
>
> Test objective: as SDP_004 (comm-setup-timeout → SDP server stops) but under Plug-and-Charge (PICS IdentificationMode := pnC). LONG TEST.

**DUT (dut)** **PASS** 40.04 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT stops the SDP server after V2G_SECC_CommunicationSetup_Timeout (under PnC, no SDP response) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_DC_VTB_DinChargeParameterDiscovery_008

**DUT (dut)** **NOT RUN** not applicable (ATS selection) 0.00 s

Not-run reason:

```
('tests/test_secc_din_charge_parameter_discovery.py', 321, "Skipped: NOT APPLICABLE —
deselected by the ATS selection rules (PIXIT_SECC_CMN_ChargeParameterDiscoveryOngoing :=
unknown).\n\nThis TC verifies the SECC-side give-up path of ChargeParameterDiscovery: the SUT
repeatedly answers EVSEProcessing 'Ongoing' until its internal V2G_SECC_Ongoing_Timer expires,
then terminates the V2G session by closing the TCP connection within
par_CMN_TCP_Connection_Termination_Timeout (ISO 15118-4:2018, Table 241 / Table 116 test
objective).\n\nWhy this TC is not executed against this SUT:\n1. EVSEProcessing in
ChargeParameterDiscoveryRes is SUT-internal state (readiness of the charge parameters / SA
schedules). No content of a ChargeParameterDiscoveryReq can force the SECC into 'Ongoing' — the
precondition cannot be established from the EV side or from outside a black-box SUT.\n2. ISO
15118-4 acknowledges this: the TC carries the PIXIT selection
'PIXIT_SECC_CMN_ChargeParameterDiscoveryOngoing := ongoing' — a vendor DECLARATION that such an
operating condition exists (Table 10). The ATS control part executes the TC only inside 'if
(PIXIT_SECC_CMN_ChargeParameterDiscoveryOngoing == ongoing)'; with the declaration 'unknown'
the TC is deselected, not failed.\n3. The requirement-coverage annex states for the underlying
requirement [V2G2-688]: 'Requirement is only testable if a corresponding PIXIT is considered.
Internal timings cannot be tested explicitly in SECC black box configuration, so the
requirement can only be tested partially.'\n\nThis SUT computes its charge parameters
synchronously and always answers EVSEProcessing 'Finished', so the declaration is 'unknown' and
the TC is not applicable. (Note: even when executed, the ATS behavior function passes (DC) /
sets inconc (AC) if the first response is already 'Finished' — a FAIL verdict is not reachable
without the declared operating condition.)")
```

## TC_SECC_CMN_VTB_CertificateInstallation_001

TC_SECC_CMN_VTB_CertificateInstallation_001.

Test objective: GoodCase; Test System sends a CertificateInstallationReq with a valid signature, current SessionID, OEMProvisioning certificate and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateInstallationRes with a valid signature, current SessionID, ResponseCode 'OK', SAProvisioning certificate chain, contract certificate chain, ContractSignatureEncryptedPrivateKey, DHpublickey, EMAID and all additional mandatory parameters. PnC, PICS_CertificateHandling := certInstall.

### DUT (dut) PASS 6.80 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| CertificateInstallationRes within V2G_EVCC_Msg_Timeout (5 s) | 105 ms | ✓ |
| response carries the current SessionID | got 'da14a9556df9aa6f', expected 'da14a9556df9aa6f' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| response signature verifies (four References, SA provisioning leaf) | signature verifies with the SA provisioning leaf over all four elements | ✓ |
| SAProvisioningCertificateChain verifies against the V2G root (CPS) | SA provisioning chain verifies against the V2G root (leaf DC CPS) | ✓ |
| delivered contract certificate is valid | contract certificate valid until 2028-10-27 | ✓ |
| delivered EMAID has valid syntax and matches the contract certificate | eMAID UKSWI123456791A well-formed and equal to the contract certificate CN | ✓ |
| contract private key decrypts with the OEM provisioning key and matches the contract certificate | contract private key decrypts and matches the contract certificate | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| ServiceDiscoveryRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateInstallationRes response time | 105.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:14:57.264 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:14:57.263Z"}` |
| 06:14:57.265 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:14:59.702 | CS → CSMS | Get15118EVCertificate | `{"action":"Install","exiRequest":"<1136 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:14:59.709 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<5572 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateInstallation_002

TC_SECC_CMN_VTB_CertificateInstallation_002.

Test objective: GoodCase; Test System sends a valid ServiceDiscoveryReq instead of a CertificateInstallationReq. Test System then checks that the SUT sends a ServiceDiscoveryRes with the current SessionID, ResponseCode 'FAILED_SequenceError' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:15:03.264 | CS → CSMS | StatusNotification | {"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:15:03.264Z"} |
| 06:15:03.265 | CSMS → CS | StatusNotification (response) | {} |

## TC_SECC_CMN_VTB_CertificateInstallation_003

TC_SECC_CMN_VTB_CertificateInstallation_003.

Test objective: GoodCase; Test System sends a CertificateInstallationReq with a valid signature, invalid SessionID, OEMProvisioning certificate and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateInstallationRes with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and terminates the session (TCP close within 'par_CMN_TCP_Connection_Termination_Timeout').

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| CertificateInstallationRes within V2G_EVCC_Msg_Timeout (5 s) | 103 ms | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateInstallationRes response time | 102.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:15:09.262 | CS → CSMS | StatusNotification | {"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:15:09.262Z"} |
| 06:15:09.263 | CSMS → CS | StatusNotification (response) | {} |
| 06:15:10.879 | CS → CSMS | Heartbeat | {} |
| 06:15:10.880 | CSMS → CS | Heartbeat (response) | {"currentTime":"2026-10-28T06:15:10.000Z"} |

## TC_SECC_CMN_VTB_CertificateInstallation_004

> TC_SECC_CMN_VTB_CertificateInstallation_004.
>
> Test objective: GoodCase; Test System waits until the 'V2G_Sequence_Timeout' timer has expired after receiving PaymentServiceSelectionRes (OK). Test System then checks that the SUT terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST.

**DUT** (dut) **PASS** 67.89 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection after V2G_Sequence_Timeout (within 70s) with no CertificateInstallationReq | closed after 62.0s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 62030.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:15:15.262 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:15:15.262Z"}` |
| 06:15:15.263 | CSMS → CS | StatusNotification (response) | `{}` |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:15:40.880 | CS → CSMS | Heartbeat | `{}` |
| 06:15:40.881 | CSMS → CS | Heartbeat (response) | `{"currentTime":"2026-10-28T06:15:40.000Z"}` |
| 06:16:10.880 | CS → CSMS | Heartbeat | `{}` |
| 06:16:10.881 | CSMS → CS | Heartbeat (response) | `{"currentTime":"2026-10-28T06:16:10.000Z"}` |

## TC_SECC_CMN_VTB_CertificateInstallation_005

> TC_SECC_CMN_VTB_CertificateInstallation_005.
>
> Test objective: GoodCase; Test System sends a CertificateInstallationReq with an invalid signature (wrong private key used -- the ATS signs with the V2G root's key), current SessionID, OEMProvisioning certificate and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateInstallationRes with the current SessionID, ResponseCode 'FAILED_SignatureError' and terminates the session (TCP close).

### DUT (dut) PASS 6.11 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| CertificateInstallationRes within V2G_EVCC_Msg_Timeout (5 s) | 106 ms | ✓ |
| response carries the current SessionID | got '15e3536a8e84f303', expected '15e3536a8e84f303' | ✓ |
| ResponseCode = FAILED_SignatureError | got 'FAILED_SignatureError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateInstallationRes response time | 105.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:16:23.267 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:16:23.266Z"}` |
| 06:16:23.268 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:16:25.709 | CS → CSMS | Get15118EVCertificate | `{"action":"Install","exiRequest":"<1136 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:16:25.714 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateInstallation_006

TC_SECC_CMN_VTB_CertificateInstallation_006.

Test objective: GoodCase; Test System sends a CertificateInstallationReq with an invalid signature (signed body changed after signature calculation: the element Id is rewritten to "contentChanged"), current SessionID, OEMProvisioning certificate and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateInstallationRes with the current SessionID, ResponseCode 'FAILED_SignatureError' and terminates the session (TCP close).

### DUT (dut) PASS 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| CertificateInstallationRes within V2G_EVCC_Msg_Timeout (5 s) | 105 ms | ✓ |
| response carries the current SessionID | got 'f6907684c0749551', expected 'f6907684c0749551' | ✓ |
| ResponseCode = FAILED_SignatureError | got 'FAILED_SignatureError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CertificateInstallationRes response time | 104.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:16:29.264 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:16:29.263Z"}` |
| 06:16:29.265 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:16:31.699 | CS → CSMS | Get15118EVCertificate | `{"action":"Install","exiRequest":"<1120 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:16:31.704 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateInstallation_007

TC_SECC_CMN_VTB_CertificateInstallation_007.

Test objective: GoodCase; Test System sends a CertificateInstallationReq with a valid signature, current SessionID, invalid OEMProvisioning certificate (Expired) and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateInstallationRes with the current SessionID, ResponseCode 'FAILED_CertificateExpired' ([V2G2-468]) and terminates the session (TCP close).

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| CertificateInstallationRes within V2G_EVCC_Msg_Timeout (5 s) | 105 ms | ✓ |
| response carries the current SessionID | got '562c9b610b0c474e', expected '562c9b610b0c474e' | ✓ |
| ResponseCode = FAILED_CertificateExpired | got 'FAILED_CertificateExpired' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateInstallationRes response time | 105.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:16:35.263 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:16:35.263Z"}` |
| 06:16:35.264 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:16:37.701 | CS → CSMS | Get15118EVCertificate | `{"action":"Install","exiRequest":"<1136 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:16:37.706 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateInstallation_008

TC_SECC_CMN_VTB_CertificateInstallation_008.

Test objective: GoodCase; Test System sends a CertificateInstallationReq with a valid signature, current SessionID, invalid OEMProvisioning certificate (Invalid content → Domain component changed) and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateInstallationRes with the current SessionID, ResponseCode 'FAILED' or 'FAILED_NoCertificateAvailable' ([V2G2-894]: the CPS requires DC "OEM") and terminates the session (TCP close).

### DUT (dut) PASS 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| CertificateInstallationRes within V2G_EVCC_Msg_Timeout (5 s) | 104 ms | ✓ |
| response carries the current SessionID | got 'b88e37f412dab24a', expected 'b88e37f412dab24a' | ✓ |
| ResponseCode = FAILED or FAILED_NoCertificateAvailable | got 'FAILED_NoCertificateAvailable' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CertificateInstallationRes response time | 104.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:16:41.263 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:16:41.262Z"}` |
| 06:16:41.264 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:16:43.698 | CS → CSMS | Get15118EVCertificate | `{"action":"Install","exiRequest":"<1160 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:16:43.704 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_001

TC_SECC_CMN_VTB_CertificateUpdate_001.

Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, current SessionID, contract certificate chain, EMAID and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with a valid signature, current SessionID, ResponseCode 'OK', SAProvisioning certificate chain, contract certificate chain, ContractSignatureEncryptedPrivateKey, DHpublickey, EMAID and all additional mandatory parameters. PnC, PICS_CertificateHandling := certUpdate.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 107 ms | ✓ |
| response carries the current SessionID | got 'ae77991eefe8e922', expected 'ae77991eefe8e922' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| response signature verifies (four References, SA provisioning leaf) | signature verifies with the SA provisioning leaf over all four elements | ✓ |
| SAProvisioningCertificateChain verifies against the V2G root (CPS) | SA provisioning chain verifies against the V2G root (leaf DC CPS) | ✓ |
| renewed contract certificate is valid | contract certificate valid until 2028-10-27 | ✓ |
| EMAID has valid syntax and matches the renewed contract certificate | eMAID UKSWI123456791A well-formed and equal to the contract certificate CN | ✓ |
| EMAID equals the requested contract's EMAID | got 'UKSWI123456791A' | ✓ |
| renewed contract private key decrypts with the old contract key and matches the renewed contract certificate | contract private key decrypts and matches the contract certificate | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 107.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:16:47.265 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:16:47.264Z"}` |
| 06:16:47.266 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:16:49.708 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<2988 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:16:49.718 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<5572 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_002

TC_SECC_CMN_VTB_CertificateUpdate_002.

Test objective: GoodCase; Test System sends a valid ServiceDiscoveryReq instead of a CertificateUpdateReq. Test System then checks that the SUT sends a ServiceDiscoveryRes with the current SessionID, ResponseCode 'FAILED_SequenceError' and terminates the session (TCP close within 'par_CMN_TCP_Connection_Termination_Timeout').

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:16:53.263 | CS → CSMS | StatusNotification | {"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:16:53.262Z"} |
| 06:16:53.264 | CSMS → CS | StatusNotification (response) | {} |

## TC_SECC_CMN_VTB_CertificateUpdate_003

TC_SECC_CMN_VTB_CertificateUpdate_003.

Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, invalid SessionID, contract certificate chain, valid EMAID and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and terminates the session (TCP close).

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 104 ms | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 104.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:16:59.264 | CS → CSMS | StatusNotification | {"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:16:59.263Z"} |
| 06:16:59.265 | CSMS → CS | StatusNotification (response) | {} |

## TC_SECC_CMN_VTB_CertificateUpdate_004

> TC_SECC_CMN_VTB_CertificateUpdate_004.
>
> Test objective: GoodCase; Test System waits until the 'V2G_Sequence_Timeout' timer has expired after receiving PaymentServiceSelectionRes (OK). Test System then checks that the SUT terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST.

**DUT (dut)** **PASS** 68.30 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection after V2G_Sequence_Timeout (within 70s) with no CertificateUpdateReq | closed after 62.0s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 62038.4 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:17:05.265 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:17:05.264Z"}` |
| 06:17:05.266 | CSMS → CS | StatusNotification (response) | `{}` |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:17:10.880 | CS → CSMS | Heartbeat | `{}` |
| 06:17:10.881 | CSMS → CS | Heartbeat (response) | `{"currentTime":"2026-10-28T06:17:10.000Z"}` |
| 06:17:40.880 | CS → CSMS | Heartbeat | `{}` |
| 06:17:40.881 | CSMS → CS | Heartbeat (response) | `{"currentTime":"2026-10-28T06:17:40.000Z"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_005

> TC_SECC_CMN_VTB_CertificateUpdate_005.
>
> Test objective: GoodCase; Test System sends a CertificateUpdateReq with an invalid signature (wrong private key used -- the ATS signs with the V2G root's key), current SessionID, contract certificate chain, valid EMAID and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED_SignatureError' and terminates the session (TCP close).

### DUT (dut) PASS 6.70 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 106 ms | ✓ |
| response carries the current SessionID | got 'f480a52a7da5ede5', expected 'f480a52a7da5ede5' | ✓ |
| ResponseCode = FAILED_SignatureError | got 'FAILED_SignatureError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 105.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:18:14.264 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:18:14.264Z"}` |
| 06:18:14.265 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:18:16.702 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<2988 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:18:16.708 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_006

TC_SECC_CMN_VTB_CertificateUpdate_006.

Test objective: GoodCase; Test System sends a CertificateUpdateReq with an invalid signature (signed body changed after signature calculation: the element Id is rewritten), current SessionID, contract certificate chain, valid EMAID and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED_SignatureError' and terminates the session (TCP close).

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 107 ms | ✓ |
| response carries the current SessionID | got '8958adc14250a4ab', expected '8958adc14250a4ab' | ✓ |
| ResponseCode = FAILED_SignatureError | got 'FAILED_SignatureError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 107.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:18:20.265 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:18:20.265Z"}` |
| 06:18:20.267 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:18:22.704 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<2980 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:18:22.710 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_007

> TC_SECC_CMN_VTB_CertificateUpdate_007.
>
> Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, current SessionID, invalid contract certificate chain (order of Sub CAs is inverted), valid EMAID and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED_CertChainError' ([V2G2-470]) and terminates the session (TCP close).

**DUT (dut)** PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 107 ms | ✓ |
| response carries the current SessionID | got '9784bd8cbe32515b', expected '9784bd8cbe32515b' | ✓ |
| ResponseCode = FAILED_CertChainError | got 'FAILED_CertChainError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 106.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:18:26.268 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:18:26.267Z"}` |
| 06:18:26.270 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:18:28.711 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<2988 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:18:28.717 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_008

> TC_SECC_CMN_VTB_CertificateUpdate_008.
>
> Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, current SessionID, invalid contract certificate chain (leaf certificate from wrong PKI is selected), valid EMAID and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED_CertChainError' and terminates the session (TCP close).

**DUT (dut)** PASS 6.03 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 122 ms | ✓ |
| response carries the current SessionID | got '0b299748809636e6', expected '0b299748809636e6' | ✓ |
| ResponseCode = FAILED_CertChainError | got 'FAILED_CertChainError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 4.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 428.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 105.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 105.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 105.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 122.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:18:32.309 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:18:32.307Z"}` |
| 06:18:32.311 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:18:34.812 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<2968 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:18:34.834 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_009

> TC_SECC_CMN_VTB_CertificateUpdate_009.
>
> Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, current SessionID, invalid contract certificate chain (Sub2 certificate from wrong PKI is selected), valid EMAID and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED_CertChainError' and terminates the session (TCP close).

### DUT (dut) PASS 5.97 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 106 ms | ✓ |
| response carries the current SessionID | got 'f2f0345ef770c78d', expected 'f2f0345ef770c78d' | ✓ |
| ResponseCode = FAILED_CertChainError | got 'FAILED_CertChainError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 106.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:18:38.266 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:18:38.265Z"}` |
| 06:18:38.267 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:18:40.709 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<2968 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:18:40.717 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_010

> TC_SECC_CMN_VTB_CertificateUpdate_010.
>
> Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, current SessionID, invalid contract certificate chain (Sub1 certificate from wrong PKI is selected), valid EMAID and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED_CertChainError' and terminates the session (TCP close).

**DUT (dut)** PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 107 ms | ✓ |
| response carries the current SessionID | got 'c29f733814c20710', expected 'c29f733814c20710' | ✓ |
| ResponseCode = FAILED_CertChainError | got 'FAILED_CertChainError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 106.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:18:44.264 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:18:44.263Z"}` |
| 06:18:44.264 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:18:46.709 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<2968 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:18:46.716 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_011

> TC_SECC_CMN_VTB_CertificateUpdate_011.
>
> Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, current SessionID, invalid contract certificate chain (certificate chain from wrong PKI is selected), valid EMAID and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED_CertChainError' and terminates the session (TCP close).

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 108 ms | ✓ |
| response carries the current SessionID | got '8c86725b3ffd9746', expected '8c86725b3ffd9746' | ✓ |
| ResponseCode = FAILED_CertChainError | got 'FAILED_CertChainError' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 107.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:18:50.268 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:18:50.267Z"}` |
| 06:18:50.269 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:18:52.713 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<2928 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:18:52.721 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_012

TC_SECC_CMN_VTB_CertificateUpdate_012.

Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, current SessionID, contract certificate chain, wrong EMAID (BodyType CertificateUpdateReq) and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED', 'FAILED_NoCertificateAvailable' or 'FAILED_ContractCanceled' and terminates the session (TCP close).

### DUT (dut) PASS 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 106 ms | ✓ |
| response carries the current SessionID | got '2eb78ed75fe0a61d', expected '2eb78ed75fe0a61d' | ✓ |
| ResponseCode = FAILED or FAILED_ContractCanceled or FAILED_NoCertificateAvailable | got 'FAILED_ContractCanceled' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 106.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:18:56.264 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:18:56.263Z"}` |
| 06:18:56.265 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:18:58.700 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<2988 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:18:58.707 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_013

TC_SECC_CMN_VTB_CertificateUpdate_013.

Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, current SessionID, contract certificate chain, invalid EMAID (BodyType CertificateUpdateReq) and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED', 'FAILED_NoCertificateAvailable' or 'FAILED_ContractCanceled' and terminates the session (TCP close).

### DUT (dut) PASS 6.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 107 ms | ✓ |
| response carries the current SessionID | got '53eee9a72a756380', expected '53eee9a72a756380' | ✓ |
| ResponseCode = FAILED or FAILED_ContractCanceled or FAILED_NoCertificateAvailable | got 'FAILED_ContractCanceled' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 106.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:19:02.263 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:19:02.262Z"}` |
| 06:19:02.264 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:19:04.708 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<2988 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:19:04.717 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_014

TC_SECC_CMN_VTB_CertificateUpdate_014.

Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, current SessionID, contract certificate chain, wrong EMAID (contract certificate) and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED', 'FAILED_NoCertificateAvailable' or 'FAILED_ContractCanceled' and terminates the session (TCP close).

### DUT (dut) PASS 6.80 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 107 ms | ✓ |
| response carries the current SessionID | got '72aed47b760a2a6e', expected '72aed47b760a2a6e' | ✓ |
| ResponseCode = FAILED or FAILED_ContractCanceled or FAILED_NoCertificateAvailable | got 'FAILED_ContractCanceled' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 107.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:19:09.265 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:19:09.264Z"}` |
| 06:19:09.266 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:19:10.880 | CS → CSMS | Heartbeat | `{}` |
| 06:19:10.881 | CSMS → CS | Heartbeat (response) | `{"currentTime":"2026-10-28T06:19:10.000Z"}` |
| 06:19:11.706 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<3012 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:19:11.714 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_015

TC_SECC_CMN_VTB_CertificateUpdate_015.

Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, current SessionID, contract certificate chain, invalid EMAID (contract certificate) and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED', 'FAILED_NoCertificateAvailable' or 'FAILED_ContractCanceled' and terminates the session (TCP close).

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 107 ms | ✓ |
| response carries the current SessionID | got 'aaa139070fa36b6b', expected 'aaa139070fa36b6b' | ✓ |
| ResponseCode = FAILED or FAILED_ContractCanceled or FAILED_NoCertificateAvailable | got 'FAILED_ContractCanceled' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 101.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 107.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:19:15.266 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:19:15.265Z"}` |
| 06:19:15.267 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:19:17.716 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<3008 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:19:17.726 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_016

> TC_SECC_CMN_VTB_CertificateUpdate_016.
>
> Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, current SessionID, invalid contract certificate chain (Expired), valid EMAID and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED_CertificateExpired' ([V2G2-473]) and terminates the session (TCP close).

### DUT (dut) PASS 6.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 107 ms | ✓ |
| response carries the current SessionID | got 'b1be56e990f9196f', expected 'b1be56e990f9196f' | ✓ |
| ResponseCode = FAILED_CertificateExpired | got 'FAILED_CertificateExpired' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CertificateUpdateRes response time | 106.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/ timeout | 0.1 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:19:21.273 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:19:21.272Z"}` |
| 06:19:21.274 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:19:23.720 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<2988 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:19:23.727 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

## TC_SECC_CMN_VTB_CertificateUpdate_017

TC_SECC_CMN_VTB_CertificateUpdate_017.

Test objective: GoodCase; Test System sends a CertificateUpdateReq with a valid signature, current SessionID, invalid contract certificate chain (Invalid content → Country name was set), valid EMAID and all additional mandatory parameters. Test System then checks that the SUT sends a CertificateUpdateRes with the current SessionID, ResponseCode 'FAILED' or 'FAILED_NoCertificateAvailable' (Table F.4: the contract leaf subject carries no Country) and terminates the session (TCP close).

### DUT (dut) PASS 6.79 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateUpdateRes message | got 'CertificateUpdateRes' | ✓ |
| CertificateUpdateRes within V2G_EVCC_Msg_Timeout (5 s) | 107 ms | ✓ |
| response carries the current SessionID | got '30260f50e27a1862', expected '30260f50e27a1862' | ✓ |
| ResponseCode = FAILED or FAILED_NoCertificateAvailable | got 'FAILED_NoCertificateAvailable' | ✓ |
| SUT closes the TCP connection after the error response | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 101.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CertificateUpdateRes response time | 107.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 0.9 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:19:28.265 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:19:28.264Z"}` |
| 06:19:28.265 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:19:30.713 | CS → CSMS | Get15118EVCertificate | `{"action":"Update","exiRequest":"<3032 base64 chars>","iso15118SchemaVersion":"urn:iso:15118:2:2013:MsgDef"}` |
| 06:19:30.721 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<4704 base64 chars>","status":"Accepted"}` |

# Test case details — DIN SPEC 70121 DC charging (DIN 70122)

## TC_SECC_DC_VTB_DinV2gtpSessionSetup_001

> TC_SECC_DC_VTB_DinV2gtpSessionSetup_001.
>
> Test objective: Test System executes GoodCase procedure and sends a SessionSetupReq message with the V2GTP header information 'protocolVersion' equals to '0×01'H, 'invProtocolVersion' equals to 'FE'H and 'payloadType' equals to '0×8001'H. (V2GTP Header is matched for V2G message content) . Test System then checks that the SUT sends a SessionSetupRes message with the V2GTP header information 'protocolVersion' equals to '0×01'H, 'invProtocolVersion' equals to 'FE'H and 'payloadType' equals to '0×8001'H. (Native DIN id: TC_SECC_VTB_V2GTPSessionSetup_001.)

**DUT** (dut) **PASS** 6.20 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a SessionSetupRes message | got 'SessionSetupRes' | ✓ |
| ResponseCode = OK_NewSessionEstablished | got 'OK_NewSessionEstablished' | ✓ |
| SessionSetupRes V2GTP protocolVersion = 0×01 | got 0×01 | ✓ |
| SessionSetupRes V2GTP invProtocolVersion = 0xFE | got 0xfe | ✓ |
| SessionSetupRes V2GTP payloadType = 0×8001 | got 0×8001 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_DC_VTB_DinV2gtpSessionSetup_002

TC_SECC_DC_VTB_DinV2gtpSessionSetup_002.

Test objective: Test System executes GoodCase procedure and sends an invalid 'protocolVersion' equals to 'FF'H in the V2GTP header of the SessionSetupReq message. Test System then checks that the SUT closes the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_V2GTPSessionSetup_002.)

**DUT (dut)** **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT closes the TCP connection after the invalid V2GTP header | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_DinV2gtpSessionSetup_003

> TC_SECC_DC_VTB_DinV2gtpSessionSetup_003.
>
> Test objective: Test System executes GoodCase procedure and sends an invalid 'invProtocolVersion' equals to 'FF'H in the V2GTP header of the SessionSetupReq message. Test System then checks that the SUT closes the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
> (Native DIN id: TC_SECC_VTB_V2GTPSessionSetup_003.)

**DUT** (dut) **PASS** 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT closes the TCP connection after the invalid V2GTP header | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_DinV2gtpSessionSetup_004

TC_SECC_DC_VTB_DinV2gtpSessionSetup_004.

Test objective: Test System executes GoodCase procedure and sends an invalid 'payloadLength' equals to '00000000'H in the V2GTP header of the SessionSetupReq message. Test System then checks that the SUT closes the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_V2GTPSessionSetup_004.)

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT closes the TCP connection after the invalid V2GTP header | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| TCP connection close after failure/ timeout | 0.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_DinV2gtpSdp_001

TC_SECC_DC_VTB_DinV2gtpSdp_001.

Test objective: Test System executes GoodCase procedure, establishes a new AVLN and sends a valid SDP request message with the V2GTP header information 'protocolVersion' equals to '0×01'H, 'invProtocolVersion' equals to 'FE'H and 'payloadType' equals to '0×9000'H. (V2GTP Header is matched for SDP message content). Test System then checks that the SUT sends a SDP response message with the V2GTP header information 'protocolVersion' equals to '0×01'H, 'invProtocolVersion' equals to 'FE'H, payloadType' equals to '0×9001'H and 'payloadLength' equals to '00000014'H.
(Native DIN id: TC_SECC_VTB_V2GTPSDP_001.)

**DUT** (dut) **PASS** 5.80 s

| Conformance check | Observed | Result |
|---|---|---|
| SDP response V2GTP protocolVersion = 0×01 | got 0×01 | ✓ |
| SDP response V2GTP invProtocolVersion = 0xFE | got 0xfe | ✓ |
| SDP response V2GTP payloadLength = 0×14 (20) | got 20 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SDP response payloadLength | 20 byte | = 20 byte | ✓ |
| SDP advertised TCP port | 50000 | - | info |

## TC_SECC_DC_VTB_DinSdp_001

> TC_SECC_DC_VTB_DinSdp_001.
>
> Test objective: Test System executes GoodCase procedure, establishes a new AVLN and sends an SDP request message with 'Security' equals to '0×10'H and 'TransportProtocol' equals to '0×00'H. Test System then checks that the SUT sends an SDP response message with 'Security' equals to '0×10'H, 'TransportProtocol' equals to '0×00'H and a valid port and IP address.
> (Native DIN id: TC_SECC_VTB_SDP_001.)

### DUT (dut) PASS 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SDP response Security = 0×10 (no transport security) | got 0×10 | ✓ |
| SDP response TransportProtocol = 0×00 (TCP) | got 0×00 | ✓ |
| SDP response carries a valid port + IP address | got port=50000 addr='fe80::4c20:35ff:fed4:ed22' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SDP advertised TCP port | 50000 | - | info |

## TC_SECC_DC_VTB_DinSdp_002

> TC_SECC_DC_VTB_DinSdp_002.
>
> Test objective: Test System executes GoodCase procedure, establishes a new AVLN and sends an SDP request message with 'Security' equals to '0×10'H and 'TransportProtocol' equals to '0×00'H. Test System then checks that the SUT sends an SDP response message with 'Security' equals to '0×10'H, 'TransportProtocol' equals to '0×00'H and a valid port and IP address.
> (Native DIN id: TC_SECC_VTB_SDP_002.)

### DUT (dut) PASS 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SDP response Security = 0×10 (no transport security) | got 0×10 | ✓ |
| SDP response TransportProtocol = 0×00 (TCP) | got 0×00 | ✓ |
| SDP response carries a valid port + IP address | got port=50000 addr='fe80::4c20:35ff:fed4:ed22' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| SDP advertised TCP port | 50000 | - | info |

## TC_SECC_DC_VTB_DinSupportedAppProtocol_001

TC_SECC_DC_VTB_DinSupportedAppProtocol_001.

Test objective: Test System executes GoodCase procedure and sends a SupportedAppProtocolReq message with a list of valid AppProtocols including DIN namespace and all additional mandatory parameters. Test System then checks that the SUT sends a SupportedAppProtocolRes message with ResponseCode 'OK_SuccessfulNegotiation' or 'OK_SuccessfulNegotiationWithMinorDeviation' and all additional mandatory parameters. (Native DIN id: TC_SECC_VTB_SupportedAppProtocol_001.)

### DUT (dut) PASS 6.42 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |
| ResponseCode is an OK_* negotiation code | got 'OK_SuccessfulNegotiation' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |

## TC_SECC_DC_VTB_DinSupportedAppProtocol_002

TC_SECC_DC_VTB_DinSupportedAppProtocol_002.

Test objective: Test System executes GoodCase procedure and sends a SupportedAppProtocolReq message with an invalid DIN namespace and all additional mandatory parameters. Test System then checks that the SUT sends a SupportedAppProtocolRes message with ResponseCode 'FAILED_NoNegotiation' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_SupportedAppProtocol_002.)

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |
| ResponseCode = Failed_NoNegotiation | got 'Failed_NoNegotiation' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 2.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinSupportedAppProtocol_003

TC_SECC_DC_VTB_DinSupportedAppProtocol_003.

Test objective: Test System executes GoodCase procedure and waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a SDP response message with 'security' equals to '0×10'H, 'transportProtocol' equals to '0×00'H and a valid port and IP address. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST (waits the ~60 s sequence timeout). (Native DIN id: TC_SECC_VTB_SupportedAppProtocol_003.)

### DUT (dut) PASS 23.82 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 18.1s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.00s after teardown | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinSupportedAppProtocol_004

TC_SECC_DC_VTB_DinSupportedAppProtocol_004.

Test objective: Test System executes GoodCase procedure, waits for 'par_CMN_remainingTime2Timeout' seconds before V2G_EVCC_CommunicationSetup_Timeout and sends a SupportedAppProtocolReq message with a list of valid AppProtocols including DIN namespace and all additional mandatory parameters. Test System then checks that the SUT sends a SupportedAppProtocolRes message with ResponseCode 'OK_SuccessfulNegotiation' or 'OK_SuccessfulNegotiationWithMinorDeviation' and all additional mandatory parameters. Furthermore Test system checks if the SUT waits for TCP connection until the SECC enters CP State A. (Native DIN id: TC_SECC_VTB_SupportedAppProtocol_004.)

Timing caveat: this harness/rig does not expose independent control over 'par_CMN_remainingTime2Timeout' relative to the SUT's V2G_EVCC_CommunicationSetup_Timeout, so the "send just before the deadline" precondition is not driven precisely. This TC implements and asserts the reachable subset of the objective -- a valid SupportedAppProtocolReq sent on the already-connected session gets negotiated OK -- without waiting a specific pre-timeout margin.

**DUT** (dut) **PASS** 6.17 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |
| ResponseCode is an OK_* negotiation code | got 'OK_SuccessfulNegotiation' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinSupportedAppProtocol_005

TC_SECC_DC_VTB_DinSupportedAppProtocol_005.

Test objective: Test System executes GoodCase procedure and sends a SupportedAppProtocolReq message with a list of valid AppProtocols (VersionNumberMinor does not match VersionNumberMinor of the SUT) including DIN namespace and all additional mandatory parameters. Test System then checks that the SUT sends a SupportedAppProtocolRes message with ResponseCode 'OK_SuccessfulNegotiationWithMinorDeviation' and all additional mandatory parameters.
(Native DIN id: TC_SECC_VTB_SupportedAppProtocol_005.)

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |
| ResponseCode = OK_SuccessfulNegotiationWithMinorDeviation | got 'OK_SuccessfulNegotiationWithMinorDeviation' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinSessionSetup_001

> TC_SECC_DC_VTB_DinSessionSetup_001.
>
> Test objective: GoodCase; Test System sends a SessionSetupReq message with SessionID '0', valid EvccID and all additional mandatory parameters. Test System then checks that the SUT sends a SessionSetupRes message with a new SessionID, ResponseCode 'OK_NewSessionEstablished', valid EvseId and all additional mandatory parameters. (Native DIN id: TC_SECC_VTB_SessionSetup_001.)

**DUT** (dut) **PASS** 6.37 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a SessionSetupRes message | got 'SessionSetupRes' | ✓ |
| ResponseCode = OK_NewSessionEstablished | got 'OK_NewSessionEstablished' | ✓ |
| SessionSetupRes carries a new non-zero SessionID | got c4b7b822a1a59c90 | ✓ |
| SessionSetupRes carries a valid EVSEID | got '49a80737a45678' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_DC_VTB_DinSessionSetup_002

TC_SECC_DC_VTB_DinSessionSetup_002.

Test objective: GoodCase; Test System sends a valid ServiceDiscoveryReq message instead of a SessionSetupReq message. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_SessionSetup_002.)

**DUT** (dut) **PASS** 6.40 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 408.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinSessionSetup_003

TC_SECC_DC_VTB_DinSessionSetup_003.

Test objective: GoodCase; Test System waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a supportedAppProtocolRes message with ResponseCode 'OK_SuccessfulNegotiation', a valid schema and all additional mandatory parameters (sends no SessionSetupReq). Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST (waits the ~60 s sequence timeout). (Native DIN id: TC_SECC_VTB_SessionSetup_003.)

**DUT** (dut) **PASS** 23.02 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 17.1s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.00s after teardown | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinSessionSetup_004

TC_SECC_DC_VTB_DinSessionSetup_004.

Test objective: GoodCase; Test System sets CP State A after receipt of a valid supportedAppProtocolRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A. The leave is observed at the Test System's PLC node via CM_NW_STATS (f_SECC_getPLCLinkTermination), the same mechanism as the SLAC-family PLCLinkStatus network-leave checks; the acceptance bound is par_TP_match_leave + par_CMN_Transmission_Delay = 1.25 s. (Native DIN id: TC_SECC_VTB_SessionSetup_004.)

**DUT** (dut) **PASS** 21.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT leaves the logical network within TP_match_leave (+transmission delay, 1.25s) after CP State A | left after 0.101s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| SUT left the logical network after CP State A | 0.101 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_DinServiceDiscovery_001

> TC_SECC_DC_VTB_DinServiceDiscovery_001.
>
> Test objective: GoodCase; Test System sends a ServiceDiscoveryReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'Ok', PaymentOption 'ExternalPayment', SupportedEnergyTransferMode 'dC_extended' and all additional mandatory parameters. (Native DIN id: TC_SECC_VTB_ServiceDiscovery_001.)

**DUT (dut)** PASS 6.60 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| PaymentOption 'ExternalPayment' offered | got ['ExternalPayment'] | ✓ |
| SupportedEnergyTransferMode = DC_extended | got 'DC_extended' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinServiceDiscovery_002

> TC_SECC_DC_VTB_DinServiceDiscovery_002.
>
> Test objective: GoodCase; Test System sends a valid ContractAuthenticationReq message instead of a ServiceDiscoveryReq message. Test System then checks that the SUT sends a ContractAuthenticationRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
> (Native DIN id: TC_SECC_VTB_ServiceDiscovery_002.)

**DUT (dut)** PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| SUT sends a ContractAuthenticationRes message | got 'ContractAuthenticationRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinServiceDiscovery_003

TC_SECC_DC_VTB_DinServiceDiscovery_003.

Test objective: GoodCase; Test System sends a ServiceDiscoveryReq message with an invalid SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a ServiceDiscoveryRes message with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_ServiceDiscovery_003.)

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinServiceDiscovery_004

TC_SECC_DC_VTB_DinServiceDiscovery_004.

Test objective: GoodCase; Test System waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a SessionSetupRes message with a new SessionID, ResponseCode 'OK_NewSessionEstablished', valid EvseId and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST (waits the ~60 s sequence timeout).
(Native DIN id: TC_SECC_VTB_ServiceDiscovery_004.)

**DUT (dut)** **FAIL** 68.29 s

**FAILURE REASON**

harness/test_ev.py:2635

```
AssertionError: sequence-timeout teardown incomplete: tcp_closed=True,
oscillator_off=False
```

| Conformance check | Observed | Result |
|---|---|---|
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 62.0s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.50s after teardown | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

Failure detail (excerpt):

```
E   AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

## TC_SECC_DC_VTB_DinServiceDiscovery_005

TC_SECC_DC_VTB_DinServiceDiscovery_005.

Test objective: GoodCase; Test System sends a SessionStopReq message with the current SessionID and all additional mandatory parameters terminating the V2G communication session for a non-critical reason. Test System then checks that the SUT sends a SessionStopRes message with the current SessionID, ResponseCode 'OK' and all additional mandatory parameters. Furthermore Test System checks that the SUT turns off the oscillator after 'V2G_SECC_CPOscillator_Retain_Timeout'.
(Native DIN id: TC_SECC_VTB_ServiceDiscovery_005.)

**DUT** (dut) **PASS** 8.11 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| SUT retains the CP oscillator for V2G_SECC_CPOscillator_Retain_Timeout (1.5s) | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (2.2s) | off (X1) 0.40s after teardown | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinServiceDiscovery_006

TC_SECC_DC_VTB_DinServiceDiscovery_006.

Test objective: GoodCase; Test System signals CP State A and sends a ServiceDiscoveryReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_ServiceDiscovery_006.)

**DUT** (dut) **PASS** 22.02 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after CP State A | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.5 ms | - | info |

## TC_SECC_DC_VTB_DinServiceDiscovery_008

> TC_SECC_DC_VTB_DinServiceDiscovery_008.
>
> Test objective: GoodCase; Test System sets CP State A after receipt of a valid SessionSetupRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A.
> (Native DIN id: TC_SECC_VTB_ServiceDiscovery_008.)

**DUT** (dut) **PASS** 21.50 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.10s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| SUT left the logical network after CP State A | 0.102 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_DinServicePaymentSelection_001

> TC_SECC_DC_VTB_DinServicePaymentSelection_001.
>
> Test objective: Test System executes GoodCase procedure and sends a ServicePaymentSelectionReq message with the current SessionID, SelectedPaymentOption 'ExternalPayment' and all additional mandatory parameters. Test System then checks that the SUT sends a ServicePaymentSelectionRes message with the current SessionID, ResponseCode 'OK' and all additional mandatory parameters. (Native DIN id: TC_SECC_VTB_ServicePaymentSelection_001.)

**DUT** (dut) **PASS** 6.07 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServicePaymentSelectionRes message | got 'ServicePaymentSelectionRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinServicePaymentSelection_002

TC_SECC_DC_VTB_DinServicePaymentSelection_002.

Test objective: Test System executes GoodCase procedure and sends a valid ServiceDiscoveryReq message instead of a ServicePaymentSelectionReq message. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_ServicePaymentSelection_002.)

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinServicePaymentSelection_003

TC_SECC_DC_VTB_DinServicePaymentSelection_003.

Test objective: Test System executes GoodCase procedure and sends a ServicePaymentSelectionReq message with an invalid SessionID, SelectedPaymentOption 'ExternalPayment' and all additional mandatory parameters. Test System then checks that the SUT sends a ServicePaymentSelectionRes message with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_ServicePaymentSelection_003.)

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServicePaymentSelectionRes message | got 'ServicePaymentSelectionRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinServicePaymentSelection_004

TC_SECC_DC_VTB_DinServicePaymentSelection_004.

Test objective: Test System executes GoodCase procedure and sends a ServicePaymentSelectionReq message with the current SessionID, SelectedPaymentOption 'ExternalPayment', invalid serviceID and all additional mandatory parameters. Test System then checks that the SUT sends a ServicePaymentSelectionRes message with the current SessionID, ResponseCode 'FAILED_ServiceSelectionInvalid' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_ServicePaymentSelection_004.)

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServicePaymentSelectionRes message | got 'ServicePaymentSelectionRes' | ✓ |
| ResponseCode = FAILED_ServiceSelectionInvalid | got 'FAILED_ServiceSelectionInvalid' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinServicePaymentSelection_005

TC_SECC_DC_VTB_DinServicePaymentSelection_005.

Test objective: Test System executes GoodCase procedure and waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'Ok', PaymentOption 'ExternalPayment', valid SupportedEnergyTransferMode and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST (waits the ~60 s sequence timeout). (Native DIN id: TC_SECC_VTB_ServicePaymentSelection_005.)

**DUT** (dut) **FAIL** 68.48 s

**FAILURE REASON**

harness/test_ev.py:2635

```
AssertionError: sequence-timeout teardown incomplete: tcp_closed=True,
oscillator_off=False
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 62.0s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.50s after teardown | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

Failure detail (excerpt):

```
E   AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

## TC_SECC_DC_VTB_DinServicePaymentSelection_006

TC_SECC_DC_VTB_DinServicePaymentSelection_006.

Test objective: Test System executes GoodCase procedure, signals CP State A and sends a ServicePaymentSelectionReq message with the current SessionID, SelectedPaymentOption 'ExternalPayment' and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_ServicePaymentSelection_006.)

**DUT** (dut) **PASS** 21.34 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after CP State A | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_DC_VTB_DinServicePaymentSelection_007

> TC_SECC_DC_VTB_DinServicePaymentSelection_007.
>
> Test objective: Test System executes GoodCase procedure and sends a ServicePaymentSelectionReq message with the current SessionID, SelectedPaymentOption 'Contract', valid serviceID and all additional mandatory parameters. Test System then checks that the SUT sends a ServicePaymentSelectionRes message with the current SessionID, ResponseCode 'FAILED_PaymentSelectionInvalid' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_ServicePaymentSelection_007.)

**DUT** (dut) **PASS** 6.17 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServicePaymentSelectionRes message | got 'ServicePaymentSelectionRes' | ✓ |
| ResponseCode = FAILED_PaymentSelectionInvalid | got 'FAILED_PaymentSelectionInvalid' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinServicePaymentSelection_008

TC_SECC_DC_VTB_DinServicePaymentSelection_008.

Test objective: Test System executes GoodCase procedure, sets CP State A after receipt of a valid ServiceDiscoveryRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A. (Native DIN id: TC_SECC_VTB_ServicePaymentSelection_008.)

**DUT** (dut) **PASS** 21.03 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.103s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| SUT left the logical network after CP State A | 0.103 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_DinContractAuthentication_001

TC_SECC_DC_VTB_DinContractAuthentication_001.

Test objective: Test System executes GoodCase procedure, initiates an EIM authorization and sends a ContractAuthenticationReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a ContractAuthenticationRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished' and all additional mandatory parameters. If a valid ContractAuthenticationRes message with EVSEProcessing 'Ongoing' was received before, the Test System sends another ContractAuthenticationReq message. This message sequence will be repeated until the authorization has finished or 'V2G_EVCC_ReadyToCharge_Timer' has expired. (Native DIN id: TC_SECC_VTB_ContractAuthentication_001.)

### DUT (dut) PASS 6.96 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ContractAuthenticationRes message | got 'ContractAuthenticationRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 2 poll(s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| ContractAuthentication polls to EVSEProcessing=Finished | 2 | - | info |

## TC_SECC_DC_VTB_DinContractAuthentication_002

TC_SECC_DC_VTB_DinContractAuthentication_002.

Test objective: Test System executes GoodCase procedure and sends a valid ServiceDiscoveryReq message instead of a ContractAuthenticationReq message. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_ContractAuthentication_002.)

**DUT (dut)** PASS 6.39 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinContractAuthentication_003

TC_SECC_DC_VTB_DinContractAuthentication_003.

Test objective: Test System executes GoodCase procedure and sends a ContractAuthenticationReq message with an invalid SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a ContractAuthenticationRes message with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_ContractAuthentication_003.)

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ContractAuthenticationRes message | got 'ContractAuthenticationRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinContractAuthentication_004

TC_SECC_DC_VTB_DinContractAuthentication_004.

Test objective: Test System executes GoodCase procedure and waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a ServicePaymentSelectionRes message with the current SessionID, ResponseCode 'OK' and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST (waits the ~60 s sequence timeout). (Native DIN id: TC_SECC_VTB_ContractAuthentication_004.)

**DUT** (dut) **FAIL** 69.29 s

**FAILURE REASON**

harness/test_ev.py:2635

```
AssertionError: sequence-timeout teardown incomplete: tcp_closed=True,
oscillator_off=False
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 62.0s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.50s after teardown | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

Failure detail (excerpt):

```
E   AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

## TC_SECC_DC_VTB_DinContractAuthentication_005

TC_SECC_DC_VTB_DinContractAuthentication_005.

Test objective: Test System executes GoodCase procedure, signals CP State A and sends a ContractAuthenticationReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_ContractAuthentication_005.)

### DUT (dut) PASS 21.93 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after CP State A | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinContractAuthentication_006

TC_SECC_DC_VTB_DinContractAuthentication_006.

Test objective: Test System executes GoodCase procedure and sends a ContractAuthenticationReq message with the current SessionID and all additional mandatory parameters without initiating an EIM authorization before. Test System then checks that the SUT sends a ContractAuthenticationRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Ongoing' and all additional mandatory parameters. Furthermore Test System checks an additional repetition of this message sequence. (Native DIN id: TC_SECC_VTB_ContractAuthentication_006.)

**DUT** (dut) **PASS** 6.97 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ContractAuthenticationRes message | got 'ContractAuthenticationRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| first ContractAuthenticationRes reports EVSEProcessing = Ongoing | got 'Ongoing' | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ContractAuthenticationRes message on repetition | got 'ContractAuthenticationRes' | ✓ |
| ResponseCode = OK on repetition | got 'OK' | ✓ |
| EVSEProcessing = Ongoing on repetition | got 'Ongoing' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinContractAuthentication_007

TC_SECC_DC_VTB_DinContractAuthentication_007.

Test objective: Test System executes GoodCase procedure, sets CP State A after receipt of a valid ServicePaymentSelectionRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A.
(Native DIN id: TC_SECC_VTB_ContractAuthentication_007.)

**DUT** (dut) **PASS** 21.73 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.102s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SUT left the logical network after CP State A | 0.102 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_DinChargeParameterDiscovery_001

TC_SECC_DC_VTB_DinChargeParameterDiscovery_001.

Test objective: Test System executes GoodCase procedure and sends a ChargeParameterDiscoveryReq message with the current SessionID, a valid requestedEnergyTransferType and all additional mandatory parameters. Test System then checks that the SUT sends a ChargeParameterDiscoveryRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', a valid scheduleList and all additional mandatory parameters. If a valid ChargeParameterDiscoveryRes message with EVSEProcessing 'Ongoing' was received before, the Test System sends another ChargeParameterDiscoveryReq message. This message sequence will be repeated until the processing of the charging information has finished or 'V2G_EVCC_ReadyToCharge_Timer' has expired. (Native DIN id: TC_SECC_VTB_ChargeParameterDiscovery_001.)

**DUT** (dut) **PASS** 6.86 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ChargeParameterDiscoveryRes message | got 'ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 1 poll(s) | ✓ |
| ChargeParameterDiscoveryRes(Finished) carries a scheduleList | got 1 entries | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| DIN ChargeParameterDiscovery polls to EVSEProcessing=Finished | 1 | - | info |
| DIN ChargeParameterDiscoveryRes SAScheduleList entries | 1 | ≥ 1 | ✓ |

## TC_SECC_DC_VTB_DinChargeParameterDiscovery_002

TC_SECC_DC_VTB_DinChargeParameterDiscovery_002.

Test objective: Test System executes GoodCase procedure and sends a valid ServiceDiscoveryReq message instead of a ChargeParameterDiscoveryReq message. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_ChargeParameterDiscovery_002.)

### DUT (dut) PASS 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_DC_VTB_DinChargeParameterDiscovery_003

TC_SECC_DC_VTB_DinChargeParameterDiscovery_003.

Test objective: Test System executes GoodCase procedure and sends a ChargeParameterDiscoveryReq message with an invalid SessionID, a valid requestedEnergyTransferType and all additional mandatory parameters. Test System then checks that the SUT sends a ChargeParameterDiscoveryRes message with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_ChargeParameterDiscovery_003.)

**DUT** (dut) **PASS** 7.60 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ChargeParameterDiscoveryRes message | got 'ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinChargeParameterDiscovery_004

TC_SECC_DC_VTB_DinChargeParameterDiscovery_004.

Test objective: Test System executes GoodCase procedure and sends a ChargeParameterDiscoveryReq message with the current SessionID, an invalid requestedEnergyTransferType and all additional mandatory parameters. Test System then checks that the SUT sends a ChargeParameterDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_WrongEnergyTransferMode' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_ChargeParameterDiscovery_004.)

Note: the objective above is the DIN ATS text, which says 'FAILED_WrongEnergyTransferMode' because the DIN ATS reuses ISO 15118-2's ATS descriptions verbatim. The DIN 70121 schema names ResponseCode 22 'FAILED_WrongEnergyTransferType' (no '...Mode' spelling exists in DIN), so the assertion checks 'Type' -- the actual DIN wire enum a conformant SECC emits.

### DUT (dut) PASS 7.40 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ChargeParameterDiscoveryRes message | got 'ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = FAILED_WrongEnergyTransferType | got 'FAILED_WrongEnergyTransferType' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinChargeParameterDiscovery_005

TC_SECC_DC_VTB_DinChargeParameterDiscovery_005.

Test objective: Test System executes GoodCase procedure and sends a ChargeParameterDiscoveryReq message with the current SessionID, a valid requestedEnergyTransferType, EVMaximumCurrentLimit '-100' and all additional mandatory parameters. Test System then checks that the SUT sends a ChargeParameterDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_WrongChargeParameter' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_ChargeParameterDiscovery_005.)

**DUT (dut)** PASS 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ChargeParameterDiscoveryRes message | got 'ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = FAILED_WrongChargeParameter | got 'FAILED_WrongChargeParameter' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinChargeParameterDiscovery_006

TC_SECC_DC_VTB_DinChargeParameterDiscovery_006.

Test objective: Test System executes GoodCase procedure and waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a ContractAuthenticationRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', and all additional mandatory parameters (sends no ChargeParameterDiscoveryReq). Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST (waits the ~60 s sequence timeout).
(Native DIN id: TC_SECC_VTB_ChargeParameterDiscovery_006.)

### DUT (dut) FAIL 69.69 s

**FAILURE REASON**

harness/test_ev.py:2635

```
AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 62.0s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.50s after teardown | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

Failure detail (excerpt):

```
E   AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

## TC_SECC_DC_VTB_DinChargeParameterDiscovery_007

TC_SECC_DC_VTB_DinChargeParameterDiscovery_007.

Test objective: Test System executes GoodCase procedure, signals CP State A and sends a ChargeParameterDiscoveryReq message with the current SessionID, a valid requestedEnergyTransferType and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_ChargeParameterDiscovery_007.)

**DUT** (dut) **PASS** 22.93 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after CP State A | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_DC_VTB_DinChargeParameterDiscovery_009

TC_SECC_DC_VTB_DinChargeParameterDiscovery_009.

Test objective: Test System executes GoodCase procedure, sets CP State A after receipt of a valid ContractAuthenticationRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A.
(Native DIN id: TC_SECC_VTB_ChargeParameterDiscovery_009.)

**DUT** (dut) **PASS** 22.41 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.10s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| SUT left the logical network after CP State A | 0.101 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_DinCableCheck_001

TC_SECC_DC_VTB_DinCableCheck_001.

Test objective: Test System executes GoodCase procedure, signals CP State C and sends a CableCheckReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. If a CableCheckRes message with EVSEProcessing 'Ongoing', EVSEStatusCode 'EVSE_IsolationMonitoringActive' was received before, the Test System sends another valid CableCheckReq message. This message sequence will be repeated until the cable check of the SUT has finished or 'V2G_EVCC_CableCheck_Timer' has expired.
(Native DIN id: TC_SECC_VTB_CableCheck_001.)

**DUT** (dut) **PASS** 10.99 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_003

TC_SECC_DC_VTB_DinCableCheck_003.

Test objective: Test System executes GoodCase procedure, signals CP State C or D and sends a valid ServiceDiscoveryReq message instead of a CableCheckReq message. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_CableCheck_003.)

**DUT (dut)** PASS 7.39 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_004

TC_SECC_DC_VTB_DinCableCheck_004.

Test objective: Test System executes GoodCase procedure, signals CP State C or D and sends a CableCheckReq message with an invalid SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a CableCheckRes message with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_CableCheck_004.)

### DUT (dut) PASS 8.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_005

TC_SECC_DC_VTB_DinCableCheck_005.

Test objective: Test System executes GoodCase procedure and waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a ChargeParameterDiscoveryRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', a valid scheduleList and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_CableCheck_005.)

**DUT** (dut) **FAIL** 69.88 s

**FAILURE REASON**

harness/test_ev.py:2635

```
AssertionError: sequence-timeout teardown incomplete: tcp_closed=True,
oscillator_off=False
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 62.0s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.50s after teardown | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

Failure detail (excerpt):

```
E   AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

## TC_SECC_DC_VTB_DinCableCheck_006

TC_SECC_DC_VTB_DinCableCheck_006.

Test objective: Test System executes GoodCase procedure, signals no CP State C or D and sends a CableCheckReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a CableCheckRes message with the current SessionID, ResponseCode 'FAILED' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_CableCheck_006.)

**DUT** (dut) **PASS** 8.11 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = FAILED | got 'FAILED' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_007

TC_SECC_DC_VTB_DinCableCheck_007.

Test objective: Test System executes GoodCase procedure, signals CP State C or D and sends a CableCheckReq message with the current SessionID, EVErrorCode 'FAILED_RESSTemperatureInhibit' and all additional mandatory parameters. Test System then checks that the EVErrorCode not influences the EVSE charging process by receiving a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters.
(Native DIN id: TC_SECC_VTB_CableCheck_007.)

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVErrorCode must not influence) | got 'OK' | ✓ |
| EVSEProcessing = Finished (EVErrorCode must not influence) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVErrorCode must not influence) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_008

TC_SECC_DC_VTB_DinCableCheck_008.

Test objective: Test System executes GoodCase procedure, signals CP State C or D and sends a CableCheckReq message with the current SessionID, EVErrorCode 'FAILED_EVShiftPosition' and all additional mandatory parameters. Test System then checks that the EVErrorCode not influences the EVSE charging process by receiving a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters.
(Native DIN id: TC_SECC_VTB_CableCheck_008.)

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVErrorCode must not influence) | got 'OK' | ✓ |
| EVSEProcessing = Finished (EVErrorCode must not influence) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVErrorCode must not influence) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_009

TC_SECC_DC_VTB_DinCableCheck_009.

Test objective: Test System executes GoodCase procedure, signals CP State C or D and sends a CableCheckReq message with the current SessionID, EVErrorCode 'FAILED_ChargerConnectorLockFault' and all additional mandatory parameters. Test System then checks that the EVErrorCode not influences the EVSE charging process by receiving a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters.
(Native DIN id: TC_SECC_VTB_CableCheck_009.)

**DUT** (dut) **PASS** 11.80 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVErrorCode must not influence) | got 'OK' | ✓ |
| EVSEProcessing = Finished (EVErrorCode must not influence) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVErrorCode must not influence) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_010

TC_SECC_DC_VTB_DinCableCheck_010.

Test objective: Test System executes GoodCase procedure, signals CP State C or D and sends a CableCheckReq message with the current SessionID, EVErrorCode 'FAILED_EVRESSMalfunction' and all additional mandatory parameters. Test System then checks that the EVErrorCode not influences the EVSE charging process by receiving a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters.
(Native DIN id: TC_SECC_VTB_CableCheck_010.)

**DUT** (dut) **PASS** 11.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVErrorCode must not influence) | got 'OK' | ✓ |
| EVSEProcessing = Finished (EVErrorCode must not influence) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVErrorCode must not influence) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_011

TC_SECC_DC_VTB_DinCableCheck_011.

Test objective: Test System executes GoodCase procedure, signals CP State C or D and sends a CableCheckReq message with the current SessionID, EVErrorCode 'FAILED_ChargingCurrentdifferential' and all additional mandatory parameters.
Test System then checks that the EVErrorCode not influences the EVSE charging process by receiving a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters.
(Native DIN id: TC_SECC_VTB_CableCheck_011.)

**DUT** (dut) **PASS** 11.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVErrorCode must not influence) | got 'OK' | ✓ |
| EVSEProcessing = Finished (EVErrorCode must not influence) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVErrorCode must not influence) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_012

TC_SECC_DC_VTB_DinCableCheck_012.

Test objective: Test System executes GoodCase procedure, signals CP State C or D and sends a CableCheckReq message with the current SessionID, EVErrorCode 'FAILED_ChargingVoltageOutOfRange' and all additional mandatory parameters. Test System then checks that the EVErrorCode not influences the EVSE charging process by receiving a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters.
(Native DIN id: TC_SECC_VTB_CableCheck_012.)

**DUT** (dut) **PASS** 11.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVErrorCode must not influence) | got 'OK' | ✓ |
| EVSEProcessing = Finished (EVErrorCode must not influence) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVErrorCode must not influence) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_013

TC_SECC_DC_VTB_DinCableCheck_013.

Test objective: Test System executes GoodCase procedure, signals CP State C or D and sends a CableCheckReq message with the current SessionID, EVErrorCode 'FAILED_ChargingSystemIncompatibility' and all additional mandatory parameters. Test System then checks that the EVErrorCode not influences the EVSE charging process by receiving a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters.
(Native DIN id: TC_SECC_VTB_CableCheck_013.)

**DUT** (dut) **PASS** 11.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVErrorCode must not influence) | got 'OK' | ✓ |
| EVSEProcessing = Finished (EVErrorCode must not influence) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVErrorCode must not influence) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_014

TC_SECC_DC_VTB_DinCableCheck_014.

Test objective: Test System executes GoodCase procedure, signals CP State C or D and sends a CableCheckReq message with the current SessionID, EVErrorCode 'NoData' and all additional mandatory parameters. Test System then checks that the EVErrorCode not influences the EVSE charging process by receiving a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters.
(Native DIN id: TC_SECC_VTB_CableCheck_014.)

**DUT** (dut) **PASS** 11.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVErrorCode must not influence) | got 'OK' | ✓ |
| EVSEProcessing = Finished (EVErrorCode must not influence) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVErrorCode must not influence) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_015

TC_SECC_DC_VTB_DinCableCheck_015.

Test objective: Test System executes GoodCase procedure, signals CP State A and sends a CableCheckReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_CableCheck_015.)

### DUT (dut) PASS 21.61 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after CP State A | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_016

TC_SECC_DC_VTB_DinCableCheck_016.

Test objective: Test System executes GoodCase procedure, signals CP State C and sends a CableCheckReq message with the current SessionID, valid EVStatus with EVReady 'false', no EVCabinConditioning, no EVRESSConiditioning and all additional mandatory parameters. Test System then checks that the SUT sends a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. If a CableCheckRes message with EVSEProcessing 'Ongoing', EVSEStatusCode 'EVSE_IsolationMonitoringActive' was received before, the Test System sends another valid CableCheckReq message. This message sequence will be repeated until the cable check of the SUT has finished or 'V2G_EVCC_CableCheck_Timer' has expired. Furthermore Test system checks that the elements EVReady, EVCabinConditioning, and EVRESSConditioning have not affected the charging session.
(Native DIN id: TC_SECC_VTB_CableCheck_016.)

**DUT** (dut) **PASS** 11.38 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVReady=false must not affect) | got 'OK' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVSEProcessing = Finished (EVReady=false must not affect) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVReady=false must not affect) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_017

TC_SECC_DC_VTB_DinCableCheck_017.

Test objective: Test System executes GoodCase procedure, signals CP State C and sends a CableCheckReq message with the current SessionID, valid EVStatus with EVReady 'true', EVCabinConditioning 'false', no EVRESSConiditioning and all additional mandatory parameters. Test System then checks that the SUT sends a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. If a CableCheckRes message with EVSEProcessing 'Ongoing', EVSEStatusCode 'EVSE_IsolationMonitoringActive' was received before, the Test System sends another valid CableCheckReq message. This message sequence will be repeated until the cable check of the SUT has finished or 'V2G_EVCC_CableCheck_Timer' has expired. Furthermore Test system checks that the elements EVReady, EVCabinConditioning, and EVRESSConditioning have not affected the charging session.
(Native DIN id: TC_SECC_VTB_CableCheck_017.)

**DUT** (dut) **PASS** 11.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVCabinConditioning=false must not affect) | got 'OK' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVSEProcessing = Finished (EVCabinConditioning=false must not affect) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVCabinConditioning=false must not affect) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_018

TC_SECC_DC_VTB_DinCableCheck_018.

Test objective: Test System executes GoodCase procedure, signals CP State C and sends a CableCheckReq message with the current SessionID, valid EVStatus with EVReady 'true', EVCabinConditioning 'true', no EVRESSConiditioning and all additional mandatory parameters. Test System then checks that the SUT sends a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. If a CableCheckRes message with EVSEProcessing 'Ongoing', EVSEStatusCode 'EVSE_IsolationMonitoringActive' was received before, the Test System sends another valid CableCheckReq message. This message sequence will be repeated until the cable check of the SUT has finished or 'V2G_EVCC_CableCheck_Timer' has expired. Furthermore Test system checks that the elements EVReady, EVCabinConditioning, and EVRESSConditioning have not affected the charging session.
(Native DIN id: TC_SECC_VTB_CableCheck_018.)

**DUT** (dut) **PASS** 11.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVCabinConditioning=true must not affect) | got 'OK' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVSEProcessing = Finished (EVCabinConditioning=true must not affect) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVCabinConditioning=true must not affect) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_019

TC_SECC_DC_VTB_DinCableCheck_019.

Test objective: Test System executes GoodCase procedure, signals CP State C and sends a CableCheckReq message with the current SessionID, valid EVStatus with EVReady 'true', no EVCabinConditioning, EVRESSConiditioning 'false' and all additional mandatory parameters. Test System then checks that the SUT sends a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. If a CableCheckRes message with EVSEProcessing 'Ongoing', EVSEStatusCode 'EVSE_IsolationMonitoringActive' was received before, the Test System sends another valid CableCheckReq message. This message sequence will be repeated until the cable check of the SUT has finished or 'V2G_EVCC_CableCheck_Timer' has expired. Furthermore Test system checks that the elements EVReady, EVCabinConditioning, and EVRESSConditioning have not affected the charging session.
(Native DIN id: TC_SECC_VTB_CableCheck_019.)

**DUT** (dut) **PASS** 11.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVRESSConditioning=false must not affect) | got 'OK' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVSEProcessing = Finished (EVRESSConditioning=false must not affect) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVRESSConditioning=false must not affect) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_020

TC_SECC_DC_VTB_DinCableCheck_020.

Test objective: Test System executes GoodCase procedure, signals CP State C and sends a CableCheckReq message with the current SessionID, valid EVStatus with EVReady 'true', no EVCabinConditioning, EVRESSConiditioning 'true' and all additional mandatory parameters. Test System then checks that the SUT sends a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. If a CableCheckRes message with EVSEProcessing 'Ongoing', EVSEStatusCode 'EVSE_IsolationMonitoringActive' was received before, the Test System sends another valid CableCheckReq message. This message sequence will be repeated until the cable check of the SUT has finished or 'V2G_EVCC_CableCheck_Timer' has expired. Furthermore Test system checks that the elements EVReady, EVCabinConditioning, and EVRESSConditioning have not affected the charging session.
(Native DIN id: TC_SECC_VTB_CableCheck_020.)

**DUT** (dut) **PASS** 11.80 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK (EVRESSConditioning=true must not affect) | got 'OK' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| EVSEProcessing = Finished (EVRESSConditioning=true must not affect) | got 'Finished' | ✓ |
| EVSEStatusCode = EVSE_Ready (EVRESSConditioning=true must not affect) | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_021

TC_SECC_DC_VTB_DinCableCheck_021.

Test objective: Test System executes GoodCase procedure, signals CP State C and sends a CableCheckReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Ongoing', EVSEStatusCode 'EVSE_IsolationMonitoringActive' and all additional mandatory parameters. Furthermore Test System checks an additional repetition of this message sequence.
(Native DIN id: TC_SECC_VTB_CableCheck_021.)

**DUT** (dut) **PASS** 7.39 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message | got 'CableCheckRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| SUT reports an intermediate EVSEProcessing = Ongoing | got 'Ongoing' (IMD self-test may have completed before Ongoing could be observed on this SIL rig) | ✓ |
| EVSEStatusCode = EVSE_IsolationMonitoringActive while Ongoing | got 'EVSE_IsolationMonitoringActive' | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CableCheckRes message on repetition | got 'CableCheckRes' | ✓ |
| ResponseCode = OK on repetition | got 'OK' | ✓ |
| SUT reports EVSEProcessing = Ongoing on repetition | got 'Ongoing' | ✓ |
| EVSEStatusCode = EVSE_IsolationMonitoringActive on repetition | got 'EVSE_IsolationMonitoringActive' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCableCheck_023

TC_SECC_DC_VTB_DinCableCheck_023.

Test objective: Test System executes GoodCase procedure, sets CP State A after receipt of a valid ChargeParameterDiscoveryRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A.
(Native DIN id: TC_SECC_VTB_CableCheck_023.)

**DUT** (dut) **PASS** 22.34 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.10s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SUT left the logical network after CP State A | 0.101 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_DinPreCharge_001

TC_SECC_DC_VTB_DinPreCharge_001.

Test objective: GoodCase; Test System sends a PreChargeReq message with the current SessionID, target voltage and current and all additional mandatory parameters. Test System then checks that the SUT sends a PreChargeRes message with the current SessionID, ResponseCode 'OK', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. If the Test System sends another valid PreChargeReq message, the SUT shall respond to this request message again. This message sequence will be repeated until the required voltage has been reached or 'V2G_EVCC_PreCharge_Timer' has expired. (Native DIN id: TC_SECC_VTB_PreCharge_001.)

**DUT** (dut) **PASS** 11.74 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PreChargeRes message | got 'PreChargeRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEStatusCode = EVSE_Ready | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| PreCharge polls to within-tolerance voltage | 2 | - | info |
| PreCharge loop total latency | 402.8 ms | V2G_EVCC_PreCharge_Timer (informational) | ✓ |

## TC_SECC_DC_VTB_DinPreCharge_002

TC_SECC_DC_VTB_DinPreCharge_002.

Test objective: GoodCase; Test System sends a valid ServiceDiscoveryReq message instead of a PreChargeReq message. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_PreCharge_002.)

**DUT (dut)** PASS 11.60 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinPreCharge_003

TC_SECC_DC_VTB_DinPreCharge_003.

Test objective: GoodCase; Test System sends a PreChargeReq message with an invalid SessionID, target voltage and current and all additional mandatory parameters. Test System then checks that the SUT sends a PreChargeRes message with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_PreCharge_003.)

**DUT** (dut) **PASS** 11.40 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PreChargeRes message | got 'PreChargeRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinPreCharge_004

TC_SECC_DC_VTB_DinPreCharge_004.

Test objective: GoodCase; Test System waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST (waits the ~60 s sequence timeout). (Native DIN id: TC_SECC_VTB_PreCharge_004.)

**DUT (dut)** **FAIL** 74.09 s

**FAILURE REASON**

harness/test_ev.py:2635

```
AssertionError: sequence-timeout teardown incomplete: tcp_closed=True,
oscillator_off=False
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 62.0s after the last message | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.50s after teardown | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

Failure detail (excerpt):

```
E   AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

## TC_SECC_DC_VTB_DinPreCharge_005

TC_SECC_DC_VTB_DinPreCharge_005.

Test objective: GoodCase; Test System signals CP State B after receiving the CableCheckRes message with the current SessionID, ResponseCode 'OK', EVSEProcessing 'Finished', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. Test System then checks that the SUT terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_PreCharge_005.)

**DUT** (dut) **PASS** 11.51 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TCP connection after CP State B (par_CMN_TCP_Connection_Termination_Timeout) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| TCP connection close after failure/ timeout | 48.5 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |

## TC_SECC_DC_VTB_DinPreCharge_006

TC_SECC_DC_VTB_DinPreCharge_006.

Test objective: GoodCase; Test System sends a SessionStopReq message with the current SessionID and all additional mandatory parameters terminating V2G communication session for non-critical reason. Test System then checks that the SUT sends a SessionStopRes message with the current SessionID, ResponseCode 'OK' and all additional mandatory parameters. Furthermore Test System checks that the SUT turns off the oscillator after 'V2G_SECC_CPOscillator_Retain_Timeout'. (Native DIN id: TC_SECC_VTB_PreCharge_006.)

**DUT** (dut) **PASS** 12.60 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| SUT retains the CP oscillator for V2G_SECC_CPOscillator_Retain_Timeout (1.5s) | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (2.2s) | off (X1) 0.40s after teardown | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinPreCharge_007

TC_SECC_DC_VTB_DinPreCharge_007.

Test objective: GoodCase; Test System signals CP State A and sends a PreChargeReq message with the current SessionID, target voltage and current and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_PreCharge_007.)

**DUT** (dut) **PASS** 26.22 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after CP State A | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinPreCharge_008

TC_SECC_DC_VTB_DinPreCharge_008.

Test objective: GoodCase; Test System sets CP State A after receipt of a valid CableCheckRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A. (Native DIN id: TC_SECC_VTB_PreCharge_008.)

**DUT** (dut) **PASS** 26.57 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.102s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| SUT left the logical network after CP State A | 0.102 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_DinPowerDelivery_001

> TC_SECC_DC_VTB_DinPowerDelivery_001.
>
> Test objective: Test System executes GoodCase procedure and sends a PowerDeliveryReq message with the current sessionID, ReadyToChargeState 'true' and all additional mandatory parameters. Test System then checks that the SUT sends an PowerDeliveryRes message with the current SessionID, ResponseCode 'OK', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. (Native DIN id: TC_SECC_VTB_PowerDelivery_001.)

**DUT** (dut) **PASS** 11.85 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEStatusCode = EVSE_Ready | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinPowerDelivery_002

TC_SECC_DC_VTB_DinPowerDelivery_002.

Test objective: Test System executes GoodCase procedure, sends a PowerDeliveryReq message with the current sessionID, ReadyToChargeState 'false' and all additional mandatory parameters. Furthermore Test System signals CP State B after received valid PowerDeliveryRes message. Test System then checks that the SUT sends an PowerDeliveryRes message with the current SessionID, ResponseCode 'OK', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. (Native DIN id: TC_SECC_VTB_PowerDelivery_002.)

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEStatusCode = EVSE_Ready | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinPowerDelivery_003

TC_SECC_DC_VTB_DinPowerDelivery_003.

Test objective: Test System executes GoodCase procedure and sends a valid ServiceDiscoveryReq message instead of a PowerDeliveryReq message with ReadyToChargeState 'true'. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_PowerDelivery_003.)

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_DC_VTB_DinPowerDelivery_004

TC_SECC_DC_VTB_DinPowerDelivery_004.

Test objective: Test System executes GoodCase procedure and sends a valid ServiceDiscoveryReq message instead of a PowerDeliveryReq message with ReadyToChargeState 'false'. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_PowerDelivery_004.)

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinPowerDelivery_005

TC_SECC_DC_VTB_DinPowerDelivery_005.

Test objective: Test System executes GoodCase procedure and sends a PowerDeliveryReq message with invalid sessionID, ReadyToChargeState 'true' and all additional mandatory parameters. Test System then checks that the SUT sends a PowerDeliveryRes message with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_PowerDelivery_005.)

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinPowerDelivery_006

TC_SECC_DC_VTB_DinPowerDelivery_006.

Test objective: Test System executes GoodCase procedure and sends a PowerDeliveryReq message with invalid sessionID, ReadyToChargeState 'false' and all additional mandatory parameters. Test System then checks that the SUT sends a PowerDeliveryRes message with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_PowerDelivery_006.)

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinPowerDelivery_007

TC_SECC_DC_VTB_DinPowerDelivery_007.

Test objective: Test System executes GoodCase procedure and sends a PowerDeliveryReq message the current sessionID, ReadyToChargeState 'true', invalid charging profile and all additional mandatory parameters. Test System then checks that the SUT sends a PowerDeliveryRes message with the current SessionID, ResponseCode 'FAILED_ChargingProfileInvalid' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_PowerDelivery_007.)

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = FAILED_ChargingProfileInvalid | got 'FAILED_ChargingProfileInvalid' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinPowerDelivery_008

TC_SECC_DC_VTB_DinPowerDelivery_008.

Test objective: Test System executes GoodCase procedure and sends a PowerDeliveryReq message with the current sessionID, invalid tupleID, ReadyToChargeState 'true' and all additional mandatory parameters. Test System then checks that the SUT sends a PowerDeliveryRes message with the current SessionID, ResponseCode 'FAILED_TariffSelectionInvalid' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_PowerDelivery_008.)

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = FAILED_TariffSelectionInvalid | got 'FAILED_TariffSelectionInvalid' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_DC_VTB_DinPowerDelivery_009

TC_SECC_DC_VTB_DinPowerDelivery_009.

Test objective: Test System executes GoodCase procedure and waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a PreChargeRes message with the current SessionID, ResponseCode 'OK', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST (waits the ~60 s sequence timeout). (Native DIN id: TC_SECC_VTB_PowerDelivery_009.)

**DUT (dut)** **FAIL** 34.03 s

**FAILURE REASON**

harness/test_ev.py:2635

```
AssertionError: sequence-timeout teardown incomplete: tcp_closed=True,
oscillator_off=False
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 21.6s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.50s after teardown | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

Failure detail (excerpt):

```
E   AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

## TC_SECC_DC_VTB_DinPowerDelivery_010

TC_SECC_DC_VTB_DinPowerDelivery_010.

Test objective: Test System executes GoodCase procedure and waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a CurrentDemandRes message with the current SessionID, ResponseCode 'OK', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST (waits the ~60 s sequence timeout). (Native DIN id: TC_SECC_VTB_PowerDelivery_010.)

**DUT** (dut) **FAIL** 22.58 s

**FAILURE REASON**

harness/test_ev.py:2635

```
AssertionError: sequence-timeout teardown incomplete: tcp_closed=True,
oscillator_off=False
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 7.0s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.50s after teardown | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

Failure detail (excerpt):

```
E   AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

## TC_SECC_DC_VTB_DinPowerDelivery_011

TC_SECC_DC_VTB_DinPowerDelivery_011.

Test objective: Test System executes GoodCase procedure, signals CP State A and sends a PowerDeliveryReq message with the current sessionID, ReadyToChargeState 'true' and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_PowerDelivery_011.)

**DUT** (dut) **PASS** 26.21 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after CP State A | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_DC_VTB_DinPowerDelivery_012

TC_SECC_DC_VTB_DinPowerDelivery_012.

Test objective: Test System executes GoodCase procedure, signals CP State A and sends a PowerDeliveryReq message with the current sessionID, ReadyToChargeState 'false' and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_PowerDelivery_012.)

**DUT** (dut) **PASS** 26.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after CP State A | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinPowerDelivery_013

TC_SECC_DC_VTB_DinPowerDelivery_013.

Test objective: Test System executes GoodCase procedure, sets CP State A after receipt of a valid PreChargeRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A.
(Native DIN id: TC_SECC_VTB_PowerDelivery_013.)

**DUT** (dut) **PASS** 27.03 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.101s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| SUT left the logical network after CP State A | 0.101 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_DinPowerDelivery_014

TC_SECC_DC_VTB_DinPowerDelivery_014.

Test objective: Test System executes GoodCase procedure, sets CP State A after receipt of a valid CurrentDemandRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A.
(Native DIN id: TC_SECC_VTB_PowerDelivery_014.)

**DUT** (dut) **PASS** 27.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.102s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SUT left the logical network after CP State A | 0.102 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_DinCurrentDemand_001

> TC_SECC_DC_VTB_DinCurrentDemand_001.
>
> Test objective: Test System executes GoodCase procedure and sends a CurrentDemandReq message with the current SessionID, target voltage and current and all additional mandatory parameters. Test System then checks that the SUT sends a CurrentDemandRes message with the current SessionID, ResponseCode 'OK', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. (Native DIN id: TC_SECC_VTB_CurrentDemand_001.)

**DUT** (dut) **PASS** 11.47 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CurrentDemandRes message | got 'CurrentDemandRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEStatusCode = EVSE_Ready | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCurrentDemand_002

> TC_SECC_DC_VTB_DinCurrentDemand_002.
>
> Test objective: Test System executes GoodCase procedure and sends a valid ServiceDiscoveryReq message instead of a CurrentDemandReq message. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
> (Native DIN id: TC_SECC_VTB_CurrentDemand_002.)

**DUT** (dut) **PASS** 12.40 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCurrentDemand_003

TC_SECC_DC_VTB_DinCurrentDemand_003.

Test objective: Test System executes GoodCase procedure and sends a CurrentDemandReq message with an invalid SessionID, target voltage and current and all additional mandatory parameters. Test System then checks that the SUT sends a CurrentDemandRes message with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_CurrentDemand_003.)

**DUT** (dut) **PASS** 11.60 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CurrentDemandRes message | got 'CurrentDemandRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCurrentDemand_004

TC_SECC_DC_VTB_DinCurrentDemand_004.

Test objective: Test System executes GoodCase procedure and waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a PowerDeliveryRes message with the current SessionID, ResponseCode 'OK', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST (waits the ~60 s sequence timeout). (Native DIN id: TC_SECC_VTB_CurrentDemand_004.)

**DUT** (dut) **FAIL** 74.28 s

**FAILURE REASON**

harness/test_ev.py:2635

```
AssertionError: sequence-timeout teardown incomplete: tcp_closed=True,
oscillator_off=False
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 62.0s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.50s after teardown | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

Failure detail (excerpt):

```
E   AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

## TC_SECC_DC_VTB_DinCurrentDemand_005

TC_SECC_DC_VTB_DinCurrentDemand_005.

Test objective: Test System executes GoodCase procedure, signals CP State A and sends a CurrentDemandReq message with the current SessionID, target voltage and current and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_CurrentDemand_005.)

**DUT** (dut) **PASS** 26.53 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after CP State A | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinCurrentDemand_006

TC_SECC_DC_VTB_DinCurrentDemand_006.

Test objective: Test System executes GoodCase procedure, sets CP State A after receipt of a valid PowerDeliveryRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A.
(Native DIN id: TC_SECC_VTB_CurrentDemand_006.)

**DUT** (dut) **PASS** 27.13 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.10s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SUT left the logical network after CP State A | 0.101 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_DinWeldingDetection_001

TC_SECC_DC_VTB_DinWeldingDetection_001.

Test objective: Test System executes GoodCase procedure and sends a WeldingDetectionReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a WeldingDetectionRes message with the current SessionID, ResponseCode 'OK', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. (Native DIN id: TC_SECC_VTB_WeldingDetection_001.)

**DUT** (dut) **PASS** 13.41 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a WeldingDetectionRes message | got 'WeldingDetectionRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEStatusCode = EVSE_Ready | got 'EVSE_Ready' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinWeldingDetection_002

TC_SECC_DC_VTB_DinWeldingDetection_002.

Test objective: Test System executes GoodCase procedure and sends a valid ServiceDiscoveryReq message instead of a WeldingDetectionReq message. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_WeldingDetection_002.)

**DUT** (dut) **PASS** 13.30 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinWeldingDetection_003

TC_SECC_DC_VTB_DinWeldingDetection_003.

Test objective: Test System executes GoodCase procedure and sends a WeldingDetectionReq message with an invalid SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a WeldingDetectionRes message with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_WeldingDetection_003.)

**DUT** (dut) **PASS** 13.70 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a WeldingDetectionRes message | got 'WeldingDetectionRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinWeldingDetection_004

TC_SECC_DC_VTB_DinWeldingDetection_004.

Test objective: Test System executes GoodCase procedure and waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a PowerDeliveryRes message with the current SessionID, ResponseCode 'OK', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. LONG TEST (waits the ~60 s sequence timeout). (Native DIN id: TC_SECC_VTB_WeldingDetection_004.)

**DUT** (dut) **FAIL** 75.48 s

**FAILURE REASON**

harness/test_ev.py:2635

```
AssertionError: sequence-timeout teardown incomplete: tcp_closed=True,
oscillator_off=False
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 62.0s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.50s after teardown | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

Failure detail (excerpt):

```
E   AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

## TC_SECC_DC_VTB_DinWeldingDetection_005

TC_SECC_DC_VTB_DinWeldingDetection_005.

Test objective: Test System executes GoodCase procedure, signals no CP State B after receipt of a valid PowerDeliveryReq message and sends a WeldingDetectionReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a WeldingDetectionRes message with the current SessionID, ResponseCode 'FAILED' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_WeldingDetection_005.)

**DUT** (dut) **PASS** 15.31 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a WeldingDetectionRes message | got 'WeldingDetectionRes' | ✓ |
| ResponseCode = FAILED | got 'FAILED' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.1 ms | - | info |

## TC_SECC_DC_VTB_DinWeldingDetection_006

TC_SECC_DC_VTB_DinWeldingDetection_006.

Test objective: Test System executes GoodCase procedure, signals CP State A and sends a WeldingDetectionReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'.
(Native DIN id: TC_SECC_VTB_WeldingDetection_006.)

**DUT** (dut) **PASS** 28.42 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after CP State A | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinWeldingDetection_007

TC_SECC_DC_VTB_DinWeldingDetection_007.

Test objective: Test System executes GoodCase procedure, sets CP State A after receipt of a valid PowerDeliveryRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A.
(Native DIN id: TC_SECC_VTB_WeldingDetection_007.)

**DUT** (dut) **PASS** 29.15 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.10s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SUT left the logical network after CP State A | 0.101 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_DinSessionStop_001

TC_SECC_DC_VTB_DinSessionStop_001.

Test objective: Test System executes GoodCase procedure without welding detection process and sends a SessionStopReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a SessionStopRes message with the current SessionID, ResponseCode 'OK' and all additional mandatory parameters. Furthermore Test System checks that the SUT turns off the oscillator after 'V2G_SECC_CPOscillator_Retain_Timeout'. (Native DIN id: TC_SECC_VTB_SessionStop_001.)

**DUT** (dut) **PASS** 14.99 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| SUT retains the CP oscillator for V2G_SECC_CPOscillator_Retain_Timeout (1.5s) | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (2.2s) | off (X1) 0.40s after teardown | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.5 ms | - | info |

## TC_SECC_DC_VTB_DinSessionStop_002

TC_SECC_DC_VTB_DinSessionStop_002.

Test objective: Test System executes GoodCase procedure with welding detection process and sends a SessionStopReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a SessionStopRes message with the current SessionID, ResponseCode 'OK' and all additional mandatory parameters. Furthermore Test System checks that the SUT turns off the oscillator after 'V2G_SECC_CPOscillator_Retain_Timeout'. (Native DIN id: TC_SECC_VTB_SessionStop_002.)

**DUT** (dut) **PASS** 15.19 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| SUT retains the CP oscillator for V2G_SECC_CPOscillator_Retain_Timeout (1.5s) | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (2.2s) | off (X1) 0.40s after teardown | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinSessionStop_003

TC_SECC_DC_VTB_DinSessionStop_003.

Test objective: Test System executes GoodCase procedure without welding detection process and sends a valid ServiceDiscoveryReq message instead of a SessionStopReq message. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_SessionStop_003.)

**DUT** (dut) **PASS** 13.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinSessionStop_004

TC_SECC_DC_VTB_DinSessionStop_004.

Test objective: Test System executes GoodCase procedure without welding detection process and sends a SessionStopReq message with an invalid SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a SessionStopRes message with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_SessionStop_004.)

**DUT** (dut) **PASS** 13.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinSessionStop_005

TC_SECC_DC_VTB_DinSessionStop_005.

Test objective: Test System executes GoodCase procedure without welding detection process and waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a PowerDeliveryRes message with the current SessionID, ResponseCode 'OK', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_SessionStop_005.)

**DUT** (dut) **FAIL** 75.49 s

**FAILURE REASON**

harness/test_ev.py:2635

```
AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 62.0s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.50s after teardown | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

Failure detail (excerpt):

```
E   AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

## TC_SECC_DC_VTB_DinSessionStop_006

TC_SECC_DC_VTB_DinSessionStop_006.

Test objective: Test System executes GoodCase procedure with welding detection process and sends a valid ServiceDiscoveryReq message instead of a SessionStopReq message. Test System then checks that the SUT sends a ServiceDiscoveryRes message with the current SessionID, ResponseCode 'FAILED_SequenceError' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_SessionStop_006.)

**DUT** (dut) **PASS** 13.71 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinSessionStop_007

TC_SECC_DC_VTB_DinSessionStop_007.

Test objective: Test System executes GoodCase procedure with welding detection process and sends a SessionStopReq message with an invalid SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a SessionStopRes message with a valid SessionID, ResponseCode 'FAILED_UnknownSession' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_SessionStop_007.)

**DUT** (dut) **PASS** 14.00 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC_SECC_DC_VTB_DinSessionStop_008

TC_SECC_DC_VTB_DinSessionStop_008.

Test objective: Test System executes GoodCase procedure with welding detection process and waits until the 'V2G_Sequence_Timeout' timer has expired after receiving a WeldingDetectionRes message with the current SessionID, ResponseCode 'OK', EVSEStatusCode 'EVSE_Ready' and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_SessionStop_008.)

**DUT** (dut) **FAIL** 36.23 s

**FAILURE REASON**

harness/test_ev.py:2635

```
AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the V2G session by closing the TCP connection after V2G_Sequence_Timeout (within 66s) | closed 21.9s after the last message | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.50s after teardown | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

Failure detail (excerpt):

```
E   AssertionError: sequence-timeout teardown incomplete: tcp_closed=True, oscillator_off=False
```

## TC_SECC_DC_VTB_DinSessionStop_009

TC_SECC_DC_VTB_DinSessionStop_009.

Test objective: Test System executes GoodCase procedure, signals no CP State B after receipt of a valid PowerDeliveryReq message and sends a SessionStopReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT sends a SessionStopRes message with the current SessionID, ResponseCode 'FAILED' and all additional mandatory parameters, turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_SessionStop_009.)

**DUT** (dut) **PASS** 14.97 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = FAILED | got 'FAILED' | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after the error response | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinSessionStop_010

TC_SECC_DC_VTB_DinSessionStop_010.

Test objective: Test System executes GoodCase procedure without welding detection process, signals CP State A and sends a SessionStopReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_SessionStop_010.)

**DUT** (dut) **PASS** 28.42 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after CP State A | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.1 ms | - | info |

## TC_SECC_DC_VTB_DinSessionStop_011

TC_SECC_DC_VTB_DinSessionStop_011.

Test objective: Test System executes GoodCase procedure with welding detection process, signals CP State A and sends a SessionStopReq message with the current SessionID and all additional mandatory parameters. Test System then checks that the SUT turns off the CP oscillator within 'par_SECC_CPOscillator_Shutdown_Timeout' and terminates the V2G communication session by closing the TCP connection within 'par_CMN_TCP_Connection_Termination_Timeout'. (Native DIN id: TC_SECC_VTB_SessionStop_011.)

**DUT** (dut) **PASS** 28.20 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT turns off the CP oscillator within par_SECC_CPOscillator_Shutdown_Timeout (0.2s) | off (X1) 0.02s after teardown | ✓ |
| SUT terminates the V2G session by closing the TCP connection within par_CMN_TCP_Connection_Termination_Timeout (6.0s) | closed 0.00s after CP State A | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC_SECC_DC_VTB_DinSessionStop_012

TC_SECC_DC_VTB_DinSessionStop_012.

Test objective: Test System executes GoodCase procedure, sets CP State A after receipt of a valid PowerDeliveryRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A. (Native DIN id: TC_SECC_VTB_SessionStop_012.)

**DUT** (dut) **PASS** 28.05 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.10s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SUT left the logical network after CP State A | 0.102 s | ≤ 1.25 s | ✓ |

## TC_SECC_DC_VTB_DinSessionStop_013

TC_SECC_DC_VTB_DinSessionStop_013.

Test objective: Test System executes GoodCase procedure, sets CP State A after receipt of a valid WeldingDetectionRes message and observes the data link connection. Test System then checks that the SUT leaves the logical network within 'TP_match_leave' after detecting CP State A. (Native DIN id: TC_SECC_VTB_SessionStop_013.)

**DUT** (dut) **PASS** 28.10 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT leaves the logical network within TP_match_leave (1.25s) after CP State A | left after 0.10s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SUT left the logical network after CP State A | 0.101 s | ≤ 1.25 s | ✓ |

# Test case details — ISO 15118-20 (ISO 15118-21 ATS)

## TC21_SECC_SDP_001

> TC21_SECC_SDP_001.
>
> Test objective: once the data link is established, send an SDP request built from the default parameter set (Security=TLS, TransportProtocol=TCP) up to 50 times, waiting up to the SDP_Msg_Timeout (0.25 s) between attempts and stopping at the first response. Passes if an SDP response is received before the retries are exhausted, with Security=TLS, TransportProtocol=TCP and a usable SECC IP address + port.

**DUT** (dut) **PASS** 5.60 s

| Conformance check | Observed | Result |
|---|---|---|
| SDP response Security = 0×00 (TLS) | got 0×00 | ✓ |
| SDP response TransportProtocol = 0×00 (TCP) | got 0×00 | ✓ |
| SDP response carries a valid SECC IP address + port | got port=50000 addr='fe80::4c20:35ff:fed4:ed22' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| SDP advertised TLS port | 50000 | - | info |

## TC21_SECC_SDP_002

TC21_SECC_SDP_002.

Test objective: once the data link is established, send a default-parameter-set SDP request from a client UDP port drawn from the dynamic range (49152-65535) to the well-known SDP destination port, retrying up to 50 times at the SDP_Msg_Timeout (0.25 s) cadence and stopping at the first response. Passes if the response datagram's own UDP source port is the fixed SDP server port (15118) -- distinct from the SECC TCP endpoint port carried inside the response payload, which _001 already checks.

### DUT (dut) PASS 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| SDP response received within the SDP_Msg_Timeout retry window (<= 50 requests) | SdpResponse(addr=fe80::4c20:35ff:fed4:ed22, port=50000, security=0×00, transport=0×00, header=0×01/0xfe/0×9001, len=20) | ✓ |
| SDP response UDP source port = V2G_UDP_SDP_SERVER (15118) | got src port=15118 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| TS SDP request client-side (V2G_UDP_SDP_CLIENT) port | 52170 | - | info |

## TC21_SECC_SDP_003

TC21_SECC_SDP_003.

Test objective: once the data link is established, send the default-parameter-set SDP request 50 times from the SAME fixed client UDP port (drawn from the dynamic range 49152-65535), ignoring every response and always waiting the full SDP_Msg_Timeout (0.25 s) before the next retransmission -- unlike _001/_002, the sequence never stops early on the first response. Passes if more than one SDP response datagram is observed on that fixed client port over the whole sequence (fewer than 50, even though 50 requests are sent, is expected and acceptable given UDP's unreliable delivery).

### DUT (dut) PASS 18.82 s

| Conformance check | Observed | Result |
|---|---|---|
| more than one SDP response received on the re-used V2G_UDP_SDP_CLIENT port across the full retry sequence | count=50 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SDP responses received on the re-used client port | 50 | - | info |

## TC21_SECC_SDP_004

TC21_SECC_SDP_004.

Test objective: once the data link is established, send the default-parameter-set SDP request 50 times, each retransmission from a FRESH, distinct client UDP port drawn from the dynamic range (49152-65535), ignoring every response and always waiting the full SDP_Msg_Timeout (0.25 s) before switching to the next port -- the sequence never stops early. Passes if more than one SDP response is observed; each one is inherently addressed back to the exact client port its corresponding request was sent from (a UDP socket only ever receives datagrams sent to the port it is bound to), so counting responses across all per-request sockets already proves the per-request port match. Fewer than 50 responses, even though 50 requests are sent, is expected and acceptable given UDP's unreliable delivery.

### DUT (dut) PASS 18.80 s

| Conformance check | Observed | Result |
|---|---|---|
| more than one SDP response received, each on the distinct V2G_UDP_SDP_CLIENT port its request was sent from, across the full retry sequence | count=50 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SDP responses received across the 50 distinct client ports | 50 | - | info |

## TC21_SECC_V2GTPSDP_001

TC21_SECC_V2GTPSDP_001.

Test objective: with the data link already established, the Test System sends a fully default SDP request and checks that the SUT answers, within SDP_Msg_Timeout, with an SDP response carrying the standard V2GTP framing: ProtocolVersion 0×01 and InvProtocolVersion 0xFE (PayloadType 0×9001 is implicit -- see below). The request is retransmitted on every timeout, up to the retry ceiling, until a response arrives.

### DUT (dut) PASS 5.97 s

| Conformance check | Observed | Result |
|---|---|---|
| SDP response V2GTP ProtocolVersion = 0×01 | got 0×01 | ✓ |
| SDP response V2GTP InvProtocolVersion = 0xFE | got 0xfe | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SDP response V2GTP PayloadLength | 20 byte | - | info |

## TC21_SECC_V2GTPSDP_002

TC21_SECC_V2GTPSDP_002.

Test objective: with the data link already established, the Test System sends an otherwise-default SDP request but sets the V2GTP header's ProtocolVersion field to the invalid value 0xFF, repeating the send on every SDP_Msg_Timeout up to the retry ceiling. The SUT must never answer an SDP response to this malformed request.

**DUT** (dut) **PASS** 18.62 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT does not respond to an SDP request with an invalid V2GTP ProtocolVersion (0xFF) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPSDP_003

TC21_SECC_V2GTPSDP_003.

Test objective: with the data link already established, the Test System sends an otherwise-default SDP request but sets the V2GTP header's InvProtocolVersion field to the invalid value 0xFF, repeating the send on every SDP_Msg_Timeout up to the retry ceiling. The SUT must never answer an SDP response to this malformed request.

**DUT** (dut) **PASS** 17.99 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT does not respond to an SDP request with an invalid V2GTP InvProtocolVersion (0xFF) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPSDP_004

TC21_SECC_V2GTPSDP_004.

Test objective: with the data link already established, the Test System sends an otherwise-default SDP request but sets the V2GTP header's PayloadType field to the invalid value 0×7001 (neither the SDP request nor any known application payload type), repeating the send on every SDP_Msg_Timeout up to the retry ceiling. The SUT must never answer an SDP response to this malformed request.

**DUT (dut)** PASS 18.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT does not respond to an SDP request with an invalid V2GTP PayloadType (0×7001) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 408.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPSDP_005

TC21_SECC_V2GTPSDP_005.

Test objective: with the data link already established, the Test System sends an otherwise-default SDP request but declares the V2GTP header's PayloadLength field as 0 (the actual SDP request payload is 2 bytes), repeating the send on every SDP_Msg_Timeout up to the retry ceiling. The SUT must never answer an SDP response to this malformed request.

**DUT (dut)** PASS 18.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT does not respond to an SDP request with V2GTP PayloadLength = 0×00000000 | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPSDP_006

> TC21_SECC_V2GTPSDP_006.
>
> Test objective: with the data link already established, the Test System sends an otherwise-default SDP request but declares the V2GTP header's PayloadLength field as one byte smaller than the actual SDP request payload (2 bytes, so 1 is declared), repeating the send on every SDP_Msg_Timeout up to the retry ceiling. The SUT must never answer an SDP response to this malformed request.

### DUT (dut) PASS 18.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT does not respond to an SDP request whose V2GTP PayloadLength is one byte smaller than the actual payload | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 408.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPSDP_007

> TC21_SECC_V2GTPSDP_007.
>
> Test objective: with the data link already established, the Test System sends an otherwise-default SDP request but declares the V2GTP header's PayloadLength field as one byte larger than the actual SDP request payload (2 bytes, so 3 is declared), repeating the send on every SDP_Msg_Timeout up to the retry ceiling. The SUT must never answer an SDP response to this malformed request.

**DUT (dut)** PASS 18.00 s

| Conformance check | Observed | Result |
|---|---|---|
| SUT does not respond to an SDP request whose V2GTP PayloadLength is one byte larger than the actual payload | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPSupportedAppProtocol_001

> TC21_SECC_V2GTPSupportedAppProtocol_001.
>
> Test objective: with the data link and TLS 1.3 connection already established, the Test System sends a default-parameter-set SupportedAppProtocolReq and checks that a SupportedAppProtocolRes arrives within the message response window, framed with the standard V2GTP header: ProtocolVersion 0×01, InvProtocolVersion 0xFE and PayloadType 0×8001.

### DUT (dut) PASS 6.18 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| response V2GTP ProtocolVersion = 0×01 | got 0×01 | ✓ |
| response V2GTP InvProtocolVersion = 0xFE | got 0xfe | ✓ |
| response V2GTP PayloadType = 0×8001 | got 0×8001 | ✓ |
| framed response is a SupportedAppProtocolRes | got 'supportedAppProtocolRes' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SupportedAppProtocolRes response time | 103.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_V2GTPSupportedAppProtocol_002

> TC21_SECC_V2GTPSupportedAppProtocol_002.
>
> Test objective: with the data link and TLS 1.3 connection already established, the Test System sends a default-parameter-set SupportedAppProtocolReq and checks that the SECC's SDP-advertised V2G data port -- the TCP destination port the whole V2G connection (and hence the SupportedAppProtocolRes) rides on -- lies in the Dynamic Ports range (49152-65535), rather than a well-known/reserved port.

### DUT (dut) PASS 6.40 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |
| SECC-advertised V2G data TCP destination port is in the Dynamic Ports range (49152-65535) | got 50000 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SECC-advertised (V2G_DST_TCP_DATA) TCP port | 50000 | - | info |

## TC21_SECC_V2GTPSupportedAppProtocol_003

> TC21_SECC_V2GTPSupportedAppProtocol_003.
>
> Test objective: with the TLS 1.3 connection already established, the Test System sends a SupportedAppProtocolReq whose V2GTP header carries an invalid ProtocolVersion (0xFF). Checks that the SUT neither answers within the message response window nor tears down the TLS connection within the CommunicationSetup performance budget -- the malformed frame is dropped, the session is not.

**DUT (dut)** PASS 23.62 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT does not respond to a SupportedAppProtocolReq with an invalid V2GTP ProtocolVersion (0xFF) within V2G_EVCC_Msg_Timeout | | ✓ |
| SUT does not terminate the TLS connection within V2G_SECC_CommunicationSetup_Performance_Time | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC21_SECC_V2GTPSupportedAppProtocol_004

TC21_SECC_V2GTPSupportedAppProtocol_004.

Test objective: with the TLS 1.3 connection already established, the Test System sends a SupportedAppProtocolReq whose V2GTP header carries an invalid InvProtocolVersion (0xFF). Checks that the SUT neither answers within the message response window nor tears down the TLS connection within the CommunicationSetup performance budget.

**DUT** (dut) **PASS** 23.20 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT does not respond to a SupportedAppProtocolReq with an invalid V2GTP InvProtocolVersion (0xFF) within V2G_EVCC_Msg_Timeout | | ✓ |
| SUT does not terminate the TLS connection within V2G_SECC_CommunicationSetup_Performance_Time | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |

## TC21_SECC_V2GTPSupportedAppProtocol_005

TC21_SECC_V2GTPSupportedAppProtocol_005.

Test objective: with the TLS 1.3 connection already established, the Test System sends a SupportedAppProtocolReq whose V2GTP header carries an invalid PayloadType (0×7001, neither the SAP payload type nor any known application payload type). Checks that the SUT neither answers within the message response window nor tears down the TLS connection within the CommunicationSetup performance budget.

**DUT (dut)** PASS 23.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT does not respond to a SupportedAppProtocolReq with an invalid V2GTP PayloadType (0×7001) within V2G_EVCC_Msg_Timeout | | ✓ |
| SUT does not terminate the TLS connection within V2G_SECC_CommunicationSetup_Performance_Time | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPSupportedAppProtocol_006

> TC21_SECC_V2GTPSupportedAppProtocol_006.
>
> Test objective: with the TLS 1.3 connection already established, the Test System sends a SupportedAppProtocolReq whose V2GTP header declares a PayloadLength of 0, although the actual EXI payload is longer. Checks that the SUT neither answers within the message response window nor tears down the TLS connection within the CommunicationSetup performance budget.

**DUT (dut)** PASS 23.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT does not respond to a SupportedAppProtocolReq with V2GTP PayloadLength = 0×00000000 within V2G_EVCC_Msg_Timeout | | ✓ |
| SUT does not terminate the TLS connection within V2G_SECC_CommunicationSetup_Performance_Time | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC21_SECC_V2GTPSupportedAppProtocol_007

> TC21_SECC_V2GTPSupportedAppProtocol_007.
>
> Test objective: with the TLS 1.3 connection already established, the Test System sends a SupportedAppProtocolReq whose V2GTP header declares a PayloadLength one byte smaller than the actual EXI payload (Annex D function setSmallerV2GTPPayloadLength). Checks that the SUT neither answers within the message response window nor tears down the TLS connection within the CommunicationSetup performance budget.

**DUT (dut)** PASS 23.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT does not respond to a SupportedAppProtocolReq whose V2GTP PayloadLength is one byte smaller than the actual payload within V2G_EVCC_Msg_Timeout | | ✓ |
| SUT does not terminate the TLS connection within V2G_SECC_CommunicationSetup_Performance_Time | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC21_SECC_V2GTPSupportedAppProtocol_008

TC21_SECC_V2GTPSupportedAppProtocol_008.

Test objective: with the TLS 1.3 connection already established, the Test System sends a SupportedAppProtocolReq whose V2GTP header declares a PayloadLength one byte larger than the actual EXI payload (Annex D function setLargerV2GTPPayloadLength). Checks that the SUT neither answers within the message response window nor tears down the TLS connection within the CommunicationSetup performance budget.

**DUT (dut)** PASS 23.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT does not respond to a SupportedAppProtocolReq whose V2GTP PayloadLength is one byte larger than the actual payload within V2G_EVCC_Msg_Timeout | | ✓ |
| SUT does not terminate the TLS connection within V2G_SECC_CommunicationSetup_Performance_Time | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPCommonMessages_001

> TC21_SECC_V2GTPCommonMessages_001.
>
> Test objective: GoodCase up to the SupportedAppProtocol handshake; the Test System sends a fully default SessionSetupReq and checks that the SUT answers, within the message observation window, with a SessionSetupRes carrying the standard V2GTP framing for a mainstream CommonMessages payload: ProtocolVersion 0×01, InvProtocolVersion 0xFE and PayloadType 0×8002.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT sends a SessionSetupRes message | got 'SessionSetupRes' | ✓ |
| SessionSetupRes V2GTP ProtocolVersion = 0×01 | got 0×01 | ✓ |
| SessionSetupRes V2GTP InvProtocolVersion = 0xFE | got 0xfe | ✓ |
| SessionSetupRes V2GTP PayloadType = 0×8002 | got 0×8002 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPCommonMessages_002

> TC21_SECC_V2GTPCommonMessages_002.
>
> Test objective: GoodCase up to the SupportedAppProtocol handshake; the Test System sends a fully default SessionSetupReq and checks that a SessionSetupRes arrives, then checks that the TCP destination port used for the whole V2G data connection (established earlier from the SDP response and reused for every CommonMessages exchange) falls within the dynamic port range (49152-65535).

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT sends a SessionSetupRes message | got 'SessionSetupRes' | ✓ |
| V2G data connection TCP destination port is in the dynamic port range (49152-65535) | port=50000 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SessionSetupRes response time | 99.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_V2GTPCommonMessages_003

TC21_SECC_V2GTPCommonMessages_003.

Test objective: GoodCase up to the SupportedAppProtocol handshake; the Test System sends an otherwise-default SessionSetupReq but sets the V2GTP header's ProtocolVersion field to the invalid value 0xFF. Checks the SUT neither answers nor tears down the TLS connection for the whole CommunicationSetup budget.

**DUT** (dut) **PASS** 22.82 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT does not respond to a SessionSetupReq with an invalid V2GTP ProtocolVersion (0xFF) within V2G_EVCC_Msg_Timeout | | ✓ |
| SUT does not terminate the TLS connection within V2G_SECC_CommunicationSetup_Performance_Time after a SessionSetupReq with an invalid V2GTP ProtocolVersion (0xFF) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPCommonMessages_004

TC21_SECC_V2GTPCommonMessages_004.

Test objective: GoodCase up to the SupportedAppProtocol handshake; the Test System sends an otherwise-default SessionSetupReq but sets the V2GTP header's InvProtocolVersion field to the invalid value 0xFF. Checks the SUT neither answers nor tears down the TLS connection for the whole CommunicationSetup budget.

**DUT (dut)** **PASS** 23.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT does not respond to a SessionSetupReq with an invalid V2GTP InvProtocolVersion (0xFF) within V2G_EVCC_Msg_Timeout | | ✓ |
| SUT does not terminate the TLS connection within V2G_SECC_CommunicationSetup_Performance_Time after a SessionSetupReq with an invalid V2GTP InvProtocolVersion (0xFF) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPCommonMessages_005

TC21_SECC_V2GTPCommonMessages_005.

Test objective: GoodCase up to the SupportedAppProtocol handshake; the Test System sends an otherwise-default SessionSetupReq but sets the V2GTP header's PayloadType field to the invalid value 0×7001 (neither the mainstream CommonMessages id nor any other known application payload type). Checks the SUT neither answers nor tears down the TLS connection for the whole CommunicationSetup budget.

### DUT (dut) PASS 23.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT does not respond to a SessionSetupReq with an invalid V2GTP PayloadType (0×7001) within V2G_EVCC_Msg_Timeout | | ✓ |
| SUT does not terminate the TLS connection within V2G_SECC_CommunicationSetup_Performance_Time after a SessionSetupReq with an invalid V2GTP PayloadType (0×7001) | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPCommonMessages_006

TC21_SECC_V2GTPCommonMessages_006.

Test objective: GoodCase up to the SupportedAppProtocol handshake; the Test System sends an otherwise-default SessionSetupReq but declares the V2GTP header's PayloadLength field as 0 (the actual EXI payload is longer). Checks the SUT neither answers nor tears down the TLS connection for the whole CommunicationSetup budget.

**DUT (dut)** **PASS** 23.80 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT does not respond to a SessionSetupReq with V2GTP PayloadLength = 0×00000000 within V2G_EVCC_Msg_Timeout | | ✓ |
| SUT does not terminate the TLS connection within V2G_SECC_CommunicationSetup_Performance_Time after a SessionSetupReq with V2GTP PayloadLength = 0×00000000 | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPCommonMessages_007

TC21_SECC_V2GTPCommonMessages_007.

Test objective: GoodCase up to the SupportedAppProtocol handshake; the Test System sends an otherwise-default SessionSetupReq but declares the V2GTP header's PayloadLength field as one byte smaller than the actual EXI payload. Checks the SUT neither answers nor tears down the TLS connection for the whole CommunicationSetup budget.

**DUT** (dut) **PASS** 23.19 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT does not respond to a SessionSetupReq with a V2GTP PayloadLength one byte smaller than the actual payload within V2G_EVCC_Msg_Timeout | | ✓ |
| SUT does not terminate the TLS connection within V2G_SECC_CommunicationSetup_Performance_Time after a SessionSetupReq with a V2GTP PayloadLength one byte smaller than the actual payload | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_V2GTPCommonMessages_008

TC21_SECC_V2GTPCommonMessages_008.

Test objective: GoodCase up to the SupportedAppProtocol handshake; the Test System sends an otherwise-default SessionSetupReq but declares the V2GTP header's PayloadLength field as one byte larger than the actual EXI payload. Checks the SUT neither answers nor tears down the TLS connection for the whole CommunicationSetup budget.

**DUT** (dut) **PASS** 23.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT does not respond to a SessionSetupReq with a V2GTP PayloadLength one byte larger than the actual payload within V2G_EVCC_Msg_Timeout | | ✓ |
| SUT does not terminate the TLS connection within V2G_SECC_CommunicationSetup_Performance_Time after a SessionSetupReq with a V2GTP PayloadLength one byte larger than the actual payload | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_SupportedAppProtocol_001

TC21_SECC_SupportedAppProtocol_001.

Test objective: happy-path SupportedAppProtocol handshake. With the TLS 1.3 connection already established, the Test System offers the default single-entry AppProtocol list (the -20 namespace matching the configured energy-transfer type, at its default version/schema/priority values) and checks that a SupportedAppProtocolRes arrives before the message observation window elapses. Neither the ResponseCode nor the SchemaID value is constrained by this TC (the ATS format leaves both as wildcards here) -- only the message type and its timeliness are graded.

### DUT (dut) PASS 6.19 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SupportedAppProtocolRes response time | 103.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ResponseCode | OK_SuccessfulNegotiation | - | info |
| SchemaID | 1 | - | info |

## TC21_SECC_SupportedAppProtocol_002

TC21_SECC_SupportedAppProtocol_002.

Test objective: negative SupportedAppProtocol handshake. The Test System offers a single AppProtocol whose namespace names the -20 CommonMessages schema directly -- not a valid energy-transfer-service namespace -- and checks that the SUT rejects the offer with ResponseCode FAILED_NoNegotiation, in time, and omits SchemaID from the response.

**DUT (dut)** **PASS** 6.40 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |
| ResponseCode = FAILED_NoNegotiation for a CommonMessages namespace offer | got 'Failed_NoNegotiation' | ✓ |
| SupportedAppProtocolRes omits SchemaID | got None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SupportedAppProtocolRes response time | 103.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_SupportedAppProtocol_003

TC21_SECC_SupportedAppProtocol_003.

Test objective: SupportedAppProtocol priority selection. The Test System offers two AppProtocol entries in one request -- a legacy namespace at the lower priority (Priority 2, listed first) and the -20 energy-transfer namespace at the highest priority (Priority 1, listed second) -- and checks that the SUT negotiates the highest-priority entry (the -20 namespace) rather than simply the first list entry, replying OK_SuccessfulNegotiation and echoing that entry's SchemaID. Note that a lower Priority number means a higher priority: Priority 1 is the numerically lowest value and thus the highest priority.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |
| ResponseCode = OK_SuccessfulNegotiation | got 'OK_SuccessfulNegotiation' | ✓ |
| SUT selects the highest-priority protocol's SchemaID (2), not the first list entry's | got 2 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_SupportedAppProtocol_004

> TC21_SECC_SupportedAppProtocol_004.
>
> Test objective: SupportedAppProtocol minor-version tolerance. The Test System offers the default single-entry AppProtocol list but with a VersionNumberMinor that deliberately differs from the SUT's, and checks that the SUT still negotiates successfully -- replying OK_SuccessfulNegotiation and echoing back the SchemaID of the offered (and only) entry.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT sends a supportedAppProtocolRes message | got 'supportedAppProtocolRes' | ✓ |
| ResponseCode = OK_SuccessfulNegotiation | got 'OK_SuccessfulNegotiation' | ✓ |
| SupportedAppProtocolRes echoes the offered entry's SchemaID (1) | got 1 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_SessionSetup_002

> TC21_SECC_SessionSetup_002.
>
> Test objective: GoodCase up to the SupportedAppProtocol handshake (TLS connected, no session yet); the Test System then sends a valid ServiceDiscoveryReq (SessionID all-zero) in place of the expected SessionSetupReq. Checks the SUT answers with a ServiceDiscoveryRes within the message response window, carrying ResponseCode FAILED_SequenceError, no Header signature and no VASList.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| response Header carries no Signature | None | ✓ |
| ServiceDiscoveryRes omits the VASList | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| ServiceDiscoveryRes response time | 100.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_SessionSetup_003

> TC21_SECC_SessionSetup_003.
>
> Test objective: GoodCase up to the SupportedAppProtocol handshake; the Test System then withholds the SessionSetupReq entirely. Checks the SUT terminates the TLS connection between the CommunicationSetup performance time and the following TLS connection termination timeout -- distinguishing an SUT that enforces the tighter CommunicationSetup budget from one that only reacts to the much longer general sequence timeout.

**DUT (dut)** PASS 23.83 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT terminates the TLS connection between V2G_SECC_CommunicationSetup_Performance_Time and TLS_Connection_Termination_Timeout (17-24s) | closed 18.1s after TLS establishment | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| TCP connection close after failure/timeout | 17981.0 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |
| TLS termination after V2G_SECC_CommunicationSetup_Performance_Time | 18.1 s | - | info |

## TC21_SECC_SessionSetup_004

TC21_SECC_SessionSetup_004.

Test objective: GoodCase up to the SupportedAppProtocol handshake; the Test System sends a SessionSetupReq built from the default parameter set. Checks a SessionSetupRes is received within the message response window and that it carries a syntactically valid (non-empty) EVSEID; the ResponseCode value itself is not constrained by this TC.

**DUT** (dut) **PASS** 6.17 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT sends a SessionSetupRes message | got 'SessionSetupRes' | ✓ |
| SessionSetupRes carries a syntactically valid EVSEID | got 'DE*PNX*E12345*1' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SessionSetupRes response time | 100.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_SessionSetup_005

> TC21_SECC_SessionSetup_005.
>
> Test objective: GoodCase up to the SupportedAppProtocol handshake; the Test System sends a SessionSetupReq built from the default parameter set. Checks a SessionSetupRes is received in time carrying ResponseCode OK_NewSessionEstablished and a newly assigned, non-zero SessionID.

**DUT (dut)** PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT sends a SessionSetupRes message | got 'SessionSetupRes' | ✓ |
| ResponseCode = OK_NewSessionEstablished | got 'OK_NewSessionEstablished' | ✓ |
| SessionSetupRes carries a new non-zero SessionID | got c1d981f9e35c6133 | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SessionSetupRes response time | 99.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_AuthorizationSetup_001

TC21_SECC_AuthorizationSetup_001.

Test objective: GoodCase up to a received SessionSetupRes; the Test System then sends an AuthorizationSetupReq built from the default parameter set. Checks an AuthorizationSetupRes is received within the message response window, carrying ResponseCode OK, the list of advertised authorization services, the certificate-installation-service flag, and -- because the SUT offers PnC -- a PnC authorization-mode block whose GenChallenge is a present, 16-byte value.

**DUT (dut)** **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| AuthorizationSetupRes received within V2G_EVCC_Msg_Timeout (2 s) | 100 ms | ✓ |
| SUT sends an AuthorizationSetupRes message | got 'AuthorizationSetupRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| AuthorizationSetupRes advertises the authorization services | got ['EIM', 'PnC'] | ✓ |
| AuthorizationSetupRes carries the CertificateInstallationService flag | got True | ✓ |
| PnC offered: PnC_ASResAuthorizationMode carries a GenChallenge | got 'ef5355e5f5da0407292deb5ab7cb4fc4' | ✓ |
| GenChallenge is 16 bytes long [V2G20-2110] | got 16 byte(s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| AuthorizationSetupRes response time | 99.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| advertised authorization services | ['EIM', 'PnC'] | - | info |

## TC21_SECC_AuthorizationSetup_002

TC21_SECC_AuthorizationSetup_002.

Test objective: GoodCase up to a received SessionSetupRes; the Test System then sends an AuthorizationSetupReq whose SessionID is a freshly generated nonce not matching the established session. Checks an AuthorizationSetupRes is received within the message response window, carrying ResponseCode FAILED_UnknownSession and no Header signature.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| AuthorizationSetupRes received within V2G_EVCC_Msg_Timeout (2 s) | 100 ms | ✓ |
| SUT sends an AuthorizationSetupRes message | got 'AuthorizationSetupRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| AuthorizationSetupRes response time | 99.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_AuthorizationSetup_003

TC21_SECC_AuthorizationSetup_003.

Test objective: GoodCase up to a received SessionSetupRes; the Test System then sends a valid ServiceDiscoveryReq in place of the expected AuthorizationSetupReq. Checks the SUT answers with a ServiceDiscoveryRes within the message response window, carrying ResponseCode FAILED_SequenceError, no Header signature and no VASList.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| ServiceDiscoveryRes received within V2G_EVCC_Msg_Timeout (2 s) | 100 ms | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| response Header carries no Signature | None | ✓ |
| ServiceDiscoveryRes omits the VASList | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| ServiceDiscoveryRes response time | 100.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_AuthorizationSetup_004

> TC21_SECC_AuthorizationSetup_004.
>
> Test objective: GoodCase up to a received SessionSetupRes; the Test System then withholds the AuthorizationSetupReq entirely. Checks the SUT reacts to the V2G sequence timeout by terminating the TLS connection between the sequence timeout and the following TLS connection termination timeout (i.e. the connection comes down no earlier than the sequence timeout and within the termination budget on top of it).

### DUT (dut) PASS 67.88 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| SUT terminates the TLS connection between V2G_SECC_Sequence_Timeout and TLS_Connection_Termination_Timeout (59-66s) | closed 62.0s after SessionSetupRes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| TCP connection close after failure/ timeout | 62019.6 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |
| TLS termination after V2G_SECC_Sequence_Timeout | 62.0 s | - | info |

## TC21_SECC_AuthorizationSetup_005

> TC21_SECC_AuthorizationSetup_005.
>
> Test objective: GoodCase up to a received SessionSetupRes; the Test System then sends a SessionStopReq built from the default parameter set, modelling an EV-initiated shutdown for a non-critical reason (Terminate). Checks a SessionStopRes is received within the message response window carrying ResponseCode OK.

**DUT** (dut) **PASS** 6.11 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| SessionStopRes received within V2G_EVCC_Msg_Timeout (2 s) | 100 ms | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SessionStopRes response time | 99.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_001

TC21_SECC_CertificateInstallation_001.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq built from the default parameter set but with MaximumContractCertificateChains raised to 2, with more than one contract available at the backend for this vehicle's provisioning identity. Checks the SUT answers a CertificateInstallationRes in time, ultimately with EVSEProcessing Finished, ResponseCode OK, a verifiable SignedInstallationData signature, and RemainingContractCertificateChains == 1 (two available, one delivered). Requires more than one contract at the backend (PIXIT_CMN_MultipleContracts).

### DUT (dut) PASS 6.61 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 97 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| RemainingContractCertificateChains == 1 | got 1 | ✓ |
| SignedInstallationData signature verifies against the CPS leaf | SignedInstallationData signature verified against CPS leaf | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_ATTEN_CHAR.IND after sounding | 411.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 96.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 05:40:07.269 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T05:40:07.269Z"}` |
| 05:40:07.271 | CSMS → CS | StatusNotification (response) | `{}` |
| 05:40:09.612 | CS → CSMS | Get15118EVCertificate | `{"action":"Install","exiRequest":"<3084 base64 chars>","iso15118SchemaVersion":"urn:iso:std:iso:15118:-20:CommonMessages"}` |
| 05:40:09.643 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<6364 base64 chars>","remainingContracts": 1,"status":"Accepted"}` |

## TC21_SECC_CertificateInstallation_002

TC21_SECC_CertificateInstallation_002.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq from the default parameter set with exactly one contract available at the backend for this provisioning identity. Checks the SUT answers a CertificateInstallationRes in time, ultimately with EVSEProcessing Finished, ResponseCode OK, a verifiable SignedInstallationData signature, and RemainingContractCertificateChains == 0 (the single available contract delivered). Requires a single contract at the backend (PIXIT_CMN_MultipleContracts = False).

### DUT (dut) PASS 6.99 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 97 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| RemainingContractCertificateChains == 0 | got 0 | ✓ |
| SignedInstallationData signature verifies against the CPS leaf | SignedInstallationData signature verified against CPS leaf | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 05:40:14.266 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T05:40:14.265Z"}` |
| 05:40:14.267 | CSMS → CS | StatusNotification (response) | `{}` |
| 05:40:16.607 | CS → CSMS | Get15118EVCertificate | `{"action":"Install","exiRequest":"<3084 base64 chars>","iso15118SchemaVersion":"urn:iso:std:iso:15118:-20:CommonMessages"}` |
| 05:40:16.617 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<6364 base64 chars>","remainingContracts": 0,"status":"Accepted"}` |

## TC21_SECC_CertificateInstallation_003

TC21_SECC_CertificateInstallation_003.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose PrioritizedEMAIDs list names an EMAID whose eMSP is not registered at the backend. Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_eMSPUnknown. Backend condition PIXIT_SECC_UnknownEMAID.

**DUT** (dut) **PASS** 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 98 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_eMSPUnknown | got 'WARNING_eMSPUnknown' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 05:40:21.266 | CS → CSMS | StatusNotification | {"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T05:40:21.266Z"} |
| 05:40:21.267 | CSMS → CS | StatusNotification (response) | {} |
| 05:40:23.610 | CS → CSMS | Get15118EVCertificate | {"action":"Install","exiRequest":"<3104 base64 chars>","iso15118SchemaVersion":"urn:iso:std:iso:15118:-20:CommonMessages"} |
| 05:40:23.614 | CSMS → CS | Get15118EVCertificate (response) | {"exiResponse":"<224 base64 chars>","remainingContracts": 0,"status":"Accepted"} |

## TC21_SECC_CertificateInstallation_004

TC21_SECC_CertificateInstallation_004.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq for a provisioning identity (PCID) for which the backend has no registered contract. Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_NoContractMatchingPCIDFound. The "no contract for this PCID" condition is realised by the backend / SA configuration (PIXIT_SECC_UnknownPCID); the wire request uses the default OEM provisioning chain.

**DUT** (dut) **PASS** 6.99 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 98 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_NoContractMatchingPCIDFound | got 'WARNING_NoContractMatchingPCIDFound' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| CertificateInstallationRes response time | 97.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 05:40:28.262 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T05:40:28.262Z"}` |
| 05:40:28.263 | CSMS → CS | StatusNotification (response) | `{}` |
| 05:40:30.607 | CS → CSMS | Get15118EVCertificate | `{"action":"Install","exiRequest":"<3084 base64 chars>","iso15118SchemaVersion":"urn:iso:std:iso:15118:-20:CommonMessages"}` |
| 05:40:30.611 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<224 base64 chars>","remainingContracts": 0,"status":"Accepted"}` |

## TC21_SECC_CertificateInstallation_005

TC21_SECC_CertificateInstallation_005.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq while the SUT cannot retrieve a new contract certificate from the backend within the message performance time. Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_NoCertificateAvailable. The unreachable-backend condition is realised by the SA configuration (PIXIT_SECC_CommunicationToSA = False); the wire request uses the default OEM provisioning chain.

### DUT (dut) PASS 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 98 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_NoCertificateAvailable | got 'WARNING_NoCertificateAvailable' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 05:40:35.263 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T05:40:35.262Z"}` |
| 05:40:35.264 | CSMS → CS | StatusNotification (response) | `{}` |
| 05:40:37.607 | CS → CSMS | Get15118EVCertificate | `{"action":"Install","exiRequest":"<3084 base64 chars>","iso15118SchemaVersion":"urn:iso:std:iso:15118:-20:CommonMessages"}` |
| 05:40:37.608 | CSMS → CS | Get15118EVCertificate (response) | `{"exiResponse":"<0 base64 chars>","status":"Failed","statusInfo": {"reasonCode":"BackendUnreachable"}}` |

## TC21_SECC_CertificateInstallation_006

TC21_SECC_CertificateInstallation_006.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose OEM provisioning chain has an expired leaf certificate (sub-CAs valid). Checks the SUT answers a CertificateInstallation Res in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateExpired.

**DUT** (dut) **PASS** 6.40 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 97 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_007

TC21_SECC_CertificateInstallation_007.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose OEM provisioning chain has an expired SubCA2 certificate (leaf and SubCA1 valid). Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateExpired.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 97 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_008

TC21_SECC_CertificateInstallation_008.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose OEM provisioning chain has an expired SubCA1 certificate (leaf and SubCA2 valid). Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateExpired.

**DUT (dut)** **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 98 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_009

TC21_SECC_CertificateInstallation_009.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose OEM provisioning chain has an expired leaf and expired SubCA2 (SubCA1 valid). Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateExpired.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 98 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 98.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_010

TC21_SECC_CertificateInstallation_010.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose OEM provisioning chain has an expired leaf and expired SubCA1 (SubCA2 valid). Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateExpired.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 98 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_011

TC21_SECC_CertificateInstallation_011.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose OEM provisioning chain has an expired SubCA2 and expired SubCA1 (leaf valid). Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateExpired.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 97 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_012

TC21_SECC_CertificateInstallation_012.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose entire OEM provisioning chain (leaf, SubCA2 and SubCA1) is expired. Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateExpired.

**DUT** (dut) **PASS** 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 98 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_013

TC21_SECC_CertificateInstallation_013.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose OEM provisioning chain has a not-yet-valid leaf certificate (sub-CAs valid). Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateNotYetValid.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 97 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_014

TC21_SECC_CertificateInstallation_014.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose OEM provisioning chain has a not-yet-valid SubCA2 certificate (leaf and SubCA1 valid). Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateNotYetValid.

**DUT (dut)** PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 97 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_015

TC21_SECC_CertificateInstallation_015.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose OEM provisioning chain has a not-yet-valid SubCA1 certificate (leaf and SubCA2 valid). Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateNotYetValid.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 97 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| CertificateInstallationRes response time | 97.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_016

TC21_SECC_CertificateInstallation_016.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose OEM provisioning chain has a not-yet-valid leaf and not-yet-valid SubCA2 (SubCA1 valid). Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateNotYetValid.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 97 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_017

TC21_SECC_CertificateInstallation_017.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose OEM provisioning chain has a not-yet-valid leaf and not-yet-valid SubCA1 (SubCA2 valid). Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateNotYetValid.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 98 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_018

TC21_SECC_CertificateInstallation_018.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose OEM provisioning chain has a not-yet-valid SubCA2 and not-yet-valid SubCA1 (leaf valid). Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateNotYetValid.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 98 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_019

TC21_SECC_CertificateInstallation_019.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq whose entire OEM provisioning chain (leaf, SubCA2 and SubCA1) is not yet valid. Checks the SUT answers a CertificateInstallationRes in time with EVSEProcessing Finished and ResponseCode WARNING_CertificateNotYetValid.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 98 ms | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| CertificateInstallationRes response time | 98.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_020

TC21_SECC_CertificateInstallation_020.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq from the default parameter set but with an unknown SessionID. Checks the SUT answers a CertificateInstallationRes in time with ResponseCode FAILED_UnknownSession.

### DUT (dut) PASS 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 97 ms | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallation_021

TC21_SECC_CertificateInstallation_021.

Test objective: GoodCase up to AuthorizationSetupRes; the Test System sends a CertificateInstallationReq from the default parameter set whose Header signature is computed with a private key other than the OEM provisioning leaf's -- the ATS's f_setSignature(toBeSigned_value, contractPrivateKey_value): a well-formed signature over the correct content, made with the MO contract leaf key instead. Checks the SUT answers a CertificateInstallationRes in time with ResponseCode FAILED_SignatureError.

### DUT (dut) PASS 6.40 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a CertificateInstallationRes message | got 'CertificateInstallationRes' | ✓ |
| every CertificateInstallationRes received within V2G_EVCC_Msg_Timeout (5 s) | slowest 97 ms | ✓ |
| ResponseCode = FAILED_SignatureError | got 'FAILED_SignatureError' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| CertificateInstallationRes response time | 97.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_Authorization_001

TC21_SECC_Authorization_001.

Test objective: GoodCase up to AuthorizationSetup; the Test System sends the AuthorizationReq from the ATS Annex C default parameter set for the authorization service the SUT offers: when the SUT offered PnC, the signed PnC request (Id "ram_id", the SUT's GenChallenge, the valid contract chain, signed with the contract private key); otherwise the bare EIM request after an external-identification token was presented at the charger. Checks every AuthorizationRes arrives within the message response window and that, following the Ongoing → Finished repetition, the final one carries ResponseCode OK with EVSEProcessing Finished.

### DUT (dut) PASS 7.02 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 93 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 2 poll(s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| authorization service used (Annex C default set) | PnC | - | info |
| PnC Authorization polls to EVSEProcessing=Finished | 2 | - | info |

## TC21_SECC_Authorization_003

TC21_SECC_Authorization_003.

Test objective: GoodCase up to AuthorizationSetup; the Test System sends the default signed PnC AuthorizationReq (valid contract chain). The eMAID of that contract leaf is one the SUT's secondary actor does not know -- an SUT-side precondition (ATS PIXIT_SECC_UnknownEMAID = True), not a wire-level fault: the chain and signature verify. The TC therefore needs the tester's CSMS as the DUT's backend (PIXIT_SECC_TesterCSMS, marker requires_csms, SUT config config-test-conformance-iso20-pnc-ocpp.yaml); there the condition is produced by the backend itself: the OCPP Authorize for the eMAID is answered idTokenInfo.status Unknown ("identifier is unknown") with the certificate accepted, which is the OCPP rendering of [V2G20-2211]. Checks the AuthorizationRes carries ResponseCode WARNING_eMSPUnknown with EVSEProcessing Finished.

### DUT (dut) PASS 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 97 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_eMSPUnknown | got 'WARNING_eMSPUnknown' after 2 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 05:36:37.263 | CS → CSMS | StatusNotification | {"connectorId":1,"connectorStatus":"Occupied","evseId":1,"timestamp":"2026-10-28T05:36:37.262Z"} |
| 05:36:37.264 | CSMS → CS | StatusNotification (response) | {} |
| 05:36:39.640 | CS → CSMS | Authorize | {"idToken":{"idToken":"UKSWI123456791A","type":"eMAID"},"iso15118CertificateHashData":[{"hashAlgorithm":"SHA256","issuerKeyHash":"598e41a2ffeaca261de38bf138bb2ab2ae278417454a1d9b88d2978063386793","issuerNameHash":"efc5a55e6e1e59c7605b48d7f260f2321aba004f97f5872986c3388f8756dce5","responderURL":"https://www.example.com/","serialNumber":"3044"},{"hashAlgorithm":"SHA256","issuerKeyHash":"fa68992bc964afd096760aa8dc0147773cca7a58a87b524dd8c0e58cb1e4dfad","issuerNameHash":"0fd741fe2020f72842fa212a4dc59f00d8d9e04a48d6eb01ea68e63978ac0c4a","responderURL":"https://www.example.com/","serialNumber":"3043 … |
| 05:36:39.642 | CSMS → CS | Authorize (response) | {"certificateStatus":"Accepted","idTokenInfo":{"status":"Unknown"}} |

## TC21_SECC_Authorization_004

> TC21_SECC_Authorization_004.
>
> Test objective: signed PnC AuthorizationReq whose contract leaf certificate has a validity window entirely in the past (Sub-CA2/Sub-CA1 valid). Checks the AuthorizationRes reports WARNING_CertificateExpired, EVSEProcessing Finished.

**DUT (dut)** PASS 6.40 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 84 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_Authorization_005

> TC21_SECC_Authorization_005.
>
> Test objective: signed PnC AuthorizationReq whose eMSP Sub-CA2 certificate is expired (leaf/Sub-CA1 valid) → WARNING_CertificateExpired, Finished.

### DUT (dut) PASS 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 96 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC21_SECC_Authorization_006

> TC21_SECC_Authorization_006.
>
> Test objective: signed PnC AuthorizationReq whose eMSP Sub-CA1 certificate is expired (leaf/Sub-CA2 valid) → WARNING_CertificateExpired, Finished.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 96 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_Authorization_007

> TC21_SECC_Authorization_007.
>
> Test objective: signed PnC AuthorizationReq with both the contract leaf and eMSP Sub-CA2 expired (Sub-CA1 valid) → WARNING_CertificateExpired, Finished.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 95 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_Authorization_008

> TC21_SECC_Authorization_008.
>
> Test objective: signed PnC AuthorizationReq with both the contract leaf and eMSP Sub-CA1 expired (Sub-CA2 valid) → WARNING_CertificateExpired, Finished.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 95 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_Authorization_009

> TC21_SECC_Authorization_009.
>
> Test objective: signed PnC AuthorizationReq with both eMSP Sub-CA2 and Sub-CA1 expired (leaf valid) → WARNING_CertificateExpired, Finished.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 95 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC21_SECC_Authorization_010

> TC21_SECC_Authorization_010.
>
> Test objective: signed PnC AuthorizationReq with the contract leaf, eMSP Sub-CA2 and Sub-CA1 all expired → WARNING_CertificateExpired, Finished.

### DUT (dut) PASS 6.40 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 94 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateExpired | got 'WARNING_CertificateExpired' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC21_SECC_Authorization_011

> TC21_SECC_Authorization_011.
>
> Test objective: signed PnC AuthorizationReq whose contract leaf certificate has a validity window entirely in the future (Sub-CA2/Sub-CA1 valid) → WARNING_CertificateNotYetValid, EVSEProcessing Finished.

**DUT** (dut) **PASS** 6.60 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 96 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_Authorization_012

> TC21_SECC_Authorization_012.
>
> Test objective: signed PnC AuthorizationReq whose eMSP Sub-CA2 is not yet valid (leaf/Sub-CA1 valid) → WARNING_CertificateNotYetValid, Finished.

### DUT (dut) PASS 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 96 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_Authorization_013

> TC21_SECC_Authorization_013.
>
> Test objective: signed PnC AuthorizationReq whose eMSP Sub-CA1 is not yet valid (leaf/Sub-CA2 valid) → WARNING_CertificateNotYetValid, Finished.

**DUT (dut)** **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 96 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_Authorization_014

> TC21_SECC_Authorization_014.
>
> Test objective: signed PnC AuthorizationReq with both the contract leaf and eMSP Sub-CA2 not yet valid (Sub-CA1 valid) → WARNING_CertificateNotYetValid, Finished.

**DUT (dut)** **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 95 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_Authorization_015

TC21_SECC_Authorization_015.

Test objective: signed PnC AuthorizationReq with both the contract leaf and eMSP Sub-CA1 not yet valid (Sub-CA2 valid) → WARNING_CertificateNotYetValid, Finished.

**DUT (dut)** PASS 5.99 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 95 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_Authorization_016

> TC21_SECC_Authorization_016.
>
> Test objective: signed PnC AuthorizationReq with both eMSP Sub-CA2 and Sub-CA1 not yet valid (leaf valid) → WARNING_CertificateNotYetValid, Finished.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 95 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_Authorization_017

> TC21_SECC_Authorization_017.
>
> Test objective: signed PnC AuthorizationReq with the contract leaf, eMSP Sub-CA2 and Sub-CA1 all not yet valid → WARNING_CertificateNotYetValid, Finished.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 94 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_CertificateNotYetValid | got 'WARNING_CertificateNotYetValid' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_Authorization_018

TC21_SECC_Authorization_018.

Test objective: GoodCase up to AuthorizationSetup; the Test System sends a signed PnC AuthorizationReq whose GenChallenge was NOT the one delivered in the preceding AuthorizationSetupRes (a fresh random value the signature is computed over). Checks the AuthorizationRes reports WARNING_ChallengeInvalid with EVSEProcessing Finished.

**DUT** (dut) **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 97 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = WARNING_ChallengeInvalid | got 'WARNING_ChallengeInvalid' after 1 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC21_SECC_Authorization_021

TC21_SECC_Authorization_021.

Test objective: GoodCase up to AuthorizationSetup; the Test System sends a PnC AuthorizationReq whose Header signature is computed with a private key other than the contract leaf's (the OEM provisioning key), so the SUT's signature verification over PnC_AReqAuthorizationMode fails. Checks the AuthorizationRes reports ResponseCode FAILED_SignatureError.

**DUT (dut)** **PASS** 6.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | 97 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = FAILED_SignatureError | got 'FAILED_SignatureError' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| AuthorizationRes response time | 97.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_Authorization_023

TC21_SECC_Authorization_023.

Test objective: GoodCase up to AuthorizationSetup; the Test System sends a signed PnC AuthorizationReq whose contract leaf is still valid but expires in 14 days or fewer. Checks the AuthorizationRes carries ResponseCode OK_CertificateExpiresSoon with EVSEProcessing Finished.

**DUT (dut)** PASS 7.40 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every AuthorizationRes received within V2G_EVCC_Msg_Timeout (2 s) | slowest 96 ms | ✓ |
| SUT sends an AuthorizationRes message | got 'AuthorizationRes' | ✓ |
| ResponseCode = OK_CertificateExpiresSoon | got 'OK_CertificateExpiresSoon' after 2 poll(s) | ✓ |
| EVSEProcessing = Finished | got 'Finished' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_Authorization_024

TC21_SECC_Authorization_024.

Test objective: GoodCase up to AuthorizationSetup; the Test System sends a SessionStopReq from the default parameter set (an EV-initiated shutdown for a non-critical reason). Checks a SessionStopRes is received within the message response window carrying ResponseCode OK.

**DUT (dut)** PASS 6.60 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SessionStopRes received within V2G_EVCC_Msg_Timeout (2 s) | 100 ms | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| SessionStopRes response time | 99.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallationOrAuthorization_001

TC21_SECC_CertificateInstallationOrAuthorization_001.

Test objective: GoodCase up to AuthorizationSetup (the SUT now expects either an AuthorizationReq or a CertificateInstallationReq); the Test System instead sends a valid ServiceDiscoveryReq from the default parameter set. Checks the SUT answers a ServiceDiscoveryRes within the message response window carrying ResponseCode FAILED_SequenceError and no Header signature.

**DUT (dut)** **PASS** 6.59 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| response Header carries no Signature | None | ✓ |
| VASList omitted (ATS: VASList := omit) | got None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| ServiceDiscoveryRes response time | 100.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_CertificateInstallationOrAuthorization_002

TC21_SECC_CertificateInstallationOrAuthorization_002.

Test objective: GoodCase up to AuthorizationSetup; the Test System then sends neither an AuthorizationReq nor a CertificateInstallationReq and stays silent past the SECC sequence timeout. Checks the SUT reacts to the sequence timeout by terminating the TLS connection within the following TLS termination timeout. LONG TEST.

**DUT (dut)** **PASS** 67.89 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the TLS connection between V2G_SECC_Sequence_Timeout and TLS_Connection_Termination_Timeout (59..66s) | closed 62.0s after AuthorizationSetupRes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| TCP connection close after failure/timeout | 62030.5 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC21_SECC_ServiceDiscovery_001

TC21_SECC_ServiceDiscovery_001.

Test objective: after AuthorizationRes(Finished), send a ServiceDiscoveryReq built from the default parameter set and check the SUT replies ServiceDiscoveryRes with ResponseCode OK, an EnergyTransferServiceList and a stated ServiceRenegotiationSupported flag; any value-added-service list the SUT chooses to advertise must be well-formed.

**DUT** (dut) **PASS** 6.79 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| ServiceDiscoveryRes states ServiceRenegotiationSupported | got False | ✓ |
| ServiceDiscoveryRes offers at least one energy-transfer service | got {'Service': [{'FreeService': False, 'ServiceID': 2}, {'FreeService': False, 'ServiceID': 6}]} | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| ServiceDiscoveryRes response time | 100.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_ServiceDiscovery_002

TC21_SECC_ServiceDiscovery_002.

Test objective: after AuthorizationRes(Finished), send a ServiceDiscoveryReq carrying a SessionID the SECC never assigned and check the SUT replies ServiceDiscoveryRes with ResponseCode FAILED_UnknownSession and no VASList.

### DUT (dut) PASS 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| ServiceDiscoveryRes carries no VASList on the failure path | got None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_ServiceDiscovery_003

> TC21_SECC_ServiceDiscovery_003.
>
> Test objective: after AuthorizationRes(Finished), send a valid AuthorizationSetupReq instead of the expected ServiceDiscoveryReq (sequence error) and check the SUT replies AuthorizationSetupRes with ResponseCode FAILED_SequenceError.

**DUT (dut)** **PASS** 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends an AuthorizationSetupRes message | got 'AuthorizationSetupRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC21_SECC_ServiceDiscovery_004

> TC21_SECC_ServiceDiscovery_004.
>
> Test objective: after AuthorizationRes(Finished) the Test System sends no further request and waits out V2G_SECC_Sequence_Timeout instead of sending ServiceDiscoveryReq; checks the SUT terminates the session by closing the TLS connection between V2G_SECC_Sequence_Timeout and TLS_Connection_Termination_Timeout. LONG TEST (waits the ~60 s sequence timeout).

### DUT (dut) PASS 68.89 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TLS connection after V2G_SECC_Sequence_Timeout (within 66s) | closed 62.0s after AuthorizationRes(Finished) | ✓ |
| SUT does not close before V2G_SECC_Sequence_Timeout expires (>= 59s) | closed after 62.0s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/timeout | 62028.4 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC21_SECC_ServiceDiscovery_005

> TC21_SECC_ServiceDiscovery_005.
>
> Test objective: after AuthorizationRes(Finished), send a SessionStopReq built from the default parameter set (an EV-initiated shutdown for a non-critical reason) and check the SUT replies SessionStopRes with ResponseCode OK.

### DUT (dut) PASS 7.11 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.6 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SessionStopRes response time | 99.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_ServiceDetail_001

TC21_SECC_ServiceDetail_001.

Test objective: after ServiceDiscoveryRes, send a ServiceDetailReq for the energy-transfer service the SUT offered (matching the configured AC/DC transfer mode) and check the SUT replies ServiceDetailRes with ResponseCode OK, the echoed ServiceID and a ServiceParameterList.

**DUT** (dut) **PASS** 6.58 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B. 4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDetailRes message | got 'ServiceDetailRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |
| ServiceDetailRes echoes the requested ServiceID 2 | got 2 | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| ServiceDetailRes carries a ServiceParameterList | got {'ParameterSet': [{'Parameter': [{'Name': 'Connector', 'intValue': 2}, {'Name': 'ControlMode', 'intValue': 2}, {'Name': 'MobilityNeedsMode', 'intValue': 1}, {'Name': 'Pricing', 'intValue': 0}], 'ParameterSetID': 0}, {'Parameter': [{'Name': 'Connector', 'intValue': 2}, {'Name': 'ControlMode', 'intValue': 1}, {'Name': 'MobilityNeedsMode', 'intValue': 1}, {'Name': 'Pricing', 'intValue': 0}], 'ParameterSetID': 1}]} | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| ServiceDetailRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_ServiceDetail_002

> TC21_SECC_ServiceDetail_002.
>
> Test objective: after ServiceDiscoveryRes, send a ServiceDetailReq for the offered energy-transfer service carrying a SessionID the SECC never assigned and check the SUT replies ServiceDetailRes with ResponseCode FAILED_UnknownSession.

**DUT (dut)** **PASS** 7.60 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDetailRes message | got 'ServiceDetailRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_ServiceDetail_003

> TC21_SECC_ServiceDetail_003.
>
> Test objective: after ServiceDiscoveryRes, send a second, valid ServiceDiscoveryReq instead of the expected ServiceDetailReq (sequence error) and check the SUT replies ServiceDiscoveryRes with ResponseCode FAILED_SequenceError and no VASList.

**DUT (dut)** **PASS** 7.40 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| ServiceDiscoveryRes carries no VASList on the failure path | got None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC21_SECC_ServiceDetail_004

TC21_SECC_ServiceDetail_004.

Test objective: after ServiceDiscoveryRes, send a ServiceDetailReq naming a valid ServiceID the SUT did NOT offer in ServiceDiscoveryRes and check the SUT replies ServiceDetailRes with ResponseCode FAILED_ServiceIDInvalid.

### DUT (dut) PASS 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDetailRes message | got 'ServiceDetailRes' | ✓ |
| ResponseCode = FAILED_ServiceIDInvalid | got 'FAILED_ServiceIDInvalid' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_ServiceDetail_005

> TC21_SECC_ServiceDetail_005.
>
> Test objective: after ServiceDiscoveryRes the Test System sends no further request and waits out V2G_SECC_Sequence_Timeout instead of sending ServiceDetailReq; checks the SUT terminates the session by closing the TLS connection between V2G_SECC_Sequence_Timeout and TLS_Connection_Termination_Timeout. LONG TEST (waits the ~60 s sequence timeout).

### DUT (dut) PASS 68.89 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TLS connection after V2G_SECC_Sequence_Timeout (within 66s) | closed 62.0s after ServiceDiscoveryRes | ✓ |
| SUT does not close before V2G_SECC_Sequence_Timeout expires (>= 59s) | closed after 62.0s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/timeout | 62027.2 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC21_SECC_ServiceDetail_006

> TC21_SECC_ServiceDetail_006.
>
> Test objective: after ServiceDiscoveryRes, send a SessionStopReq built from the default parameter set (an EV-initiated shutdown for a non-critical reason) and check the SUT replies SessionStopRes with ResponseCode OK.

**DUT (dut)** PASS 7.11 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SessionStopRes response time | 99.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_ServiceSelection_001

> TC21_SECC_ServiceSelection_001.
>
> Test objective: GoodCase; after ServiceDetailRes the Test System sends a ServiceSelectionReq selecting the energy-transfer service and parameter set the SUT offered, with all other elements at their default parameter-set values, and checks that the SUT replies with a ServiceSelectionRes carrying ResponseCode OK within the message-response timeout.

**DUT** (dut) **PASS** 7.20 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceSelectionRes message | got 'ServiceSelectionRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_ServiceSelection_002

TC21_SECC_ServiceSelection_002.

Test objective: GoodCase; after ServiceDetailRes the Test System sends a ServiceSelectionReq that otherwise matches the default parameter set but carries a SessionID the SUT never assigned, and checks that the SUT replies with a ServiceSelectionRes carrying ResponseCode FAILED_UnknownSession within the message-response timeout.

**DUT (dut)** **PASS** 7.60 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceSelectionRes message | got 'ServiceSelectionRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

## TC21_SECC_ServiceSelection_003

TC21_SECC_ServiceSelection_003.

Test objective: GoodCase; after ServiceDetailRes the Test System sends a valid ServiceDiscoveryReq (default parameter set) rather than the expected ServiceSelectionReq, and checks that the SUT replies with a ServiceDiscoveryRes carrying ResponseCode FAILED_SequenceError and no VASList within the message-response timeout.

**DUT** (dut) **PASS** 7.40 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| ServiceDiscoveryRes carries no VASList on the failure path | got None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_ServiceSelection_004

> TC21_SECC_ServiceSelection_004.
>
> Test objective: GoodCase; after ServiceDetailRes the Test System sends a ServiceSelectionReq naming a ServiceID that was never offered in ServiceDiscoveryRes together with a ParameterSetID that was never offered in ServiceDetailRes, and checks that the SUT replies with a ServiceSelectionRes carrying ResponseCode FAILED_ServiceSelectionInvalid within the message-response timeout.

**DUT (dut)** PASS 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceSelectionRes message | got 'ServiceSelectionRes' | ✓ |
| ResponseCode = FAILED_ServiceSelectionInvalid | got 'FAILED_ServiceSelectionInvalid' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_ServiceSelection_006

> TC21_SECC_ServiceSelection_006.
>
> Test objective: GoodCase up to ServiceDetailRes, then the Test System sends nothing further (no ServiceSelectionReq) and waits. The pass criterion is that the SUT closes the TLS connection no earlier than the SECC sequence timeout and no later than that timeout plus the TLS connection termination bound.

**DUT** (dut) **PASS** 68.89 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT closes the TLS connection within 66s of silence after ServiceDetailRes | closed after 62.0s | ✓ |
| SUT does not close before V2G_SECC_Sequence_Timeout expires (>= 59s) | closed after 62.0s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/timeout | 62029.9 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC21_SECC_ServiceSelection_007

TC21_SECC_ServiceSelection_007.

Test objective: GoodCase up to ServiceDetailRes, then the Test System sends a SessionStopReq for a non-critical, EV-initiated shutdown (default parameter set), and checks that the SUT replies with a SessionStopRes carrying ResponseCode OK within the message-response timeout.

**DUT (dut)** PASS 7.11 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_ChargeParameterDiscovery_001

TC21_SECC_ChargeParameterDiscovery_001.

Test objective: happy-path ChargeParameterDiscovery. GoodCase through the CommonMessages negotiation sequence up to ServiceSelection, then the Test System sends the ChargeParameterDiscoveryReq for the negotiated energy-transfer type (DC or AC), built from the default parameter set. Checks that the matching schema-specific ChargeParameterDiscoveryRes arrives within the message response window carrying ResponseCode OK. The presence of the energy-transfer-mode sub-element is recorded informationally (the ATS format leaves it a wildcard).

### DUT (dut) PASS 7.30 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a DC_ChargeParameterDiscoveryRes message | got 'DC_ChargeParameterDiscoveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_ATTEN_CHAR.IND after sounding | 409.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| ChargeParameterDiscoveryRes response time | 100.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| energy-transfer-mode element | DC_CPDResEnergyTransferMode | - | info |

## TC21_SECC_ChargeParameterDiscovery_002

TC21_SECC_ChargeParameterDiscovery_002.

Test objective: out-of-sequence check. GoodCase up to ServiceSelection; the Test System then sends a valid ServiceDiscoveryReq in place of the expected ChargeParameterDiscoveryReq. Checks the SUT answers with a ServiceDiscoveryRes within the message response window, carrying ResponseCode FAILED_SequenceError and no Header signature.

**DUT** (dut) **PASS** 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| ServiceDiscoveryRes response time | 100.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_ChargeParameterDiscovery_003

TC21_SECC_ChargeParameterDiscovery_003.

Test objective: sequence-timeout observation. GoodCase up to ServiceSelection; the Test System then stays silent, sending no ChargeParameterDiscoveryReq. Checks the SUT terminates the TLS connection after the sequence timeout elapses and within the following TLS connection termination timeout. LONG TEST.

**DUT** (dut) **PASS** 69.49 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the TLS connection between V2G_SECC_Sequence_Timeout and TLS_Connection_Termination_Timeout (59..66s) | closed 62.0s after ServiceSelection | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/timeout | 62030.3 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC21_SECC_ChargeParameterDiscovery_004

TC21_SECC_ChargeParameterDiscovery_004.

Test objective: EV-initiated shutdown for a non-critical reason. GoodCase up to ServiceSelection; the Test System then sends a SessionStopReq (ChargingSession Terminate) built from the default parameter set. Checks a SessionStopRes is received within the message response window carrying ResponseCode OK.

**DUT** (dut) **PASS** 7.51 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| SessionStopRes response time | 99.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_ScheduleExchange_002

TC21_SECC_ScheduleExchange_002.

Test objective: ScheduleExchange completion. GoodCase up to ChargeParameterDiscoveryRes; the Test System sends a ScheduleExchangeReq and, while the SUT answers EVSEProcessing Ongoing, repeats the ScheduleExchangeReq/Res exchange (bounded by the EVCC Ongoing timeout) until the SUT reports EVSEProcessing Finished. Checks the final ScheduleExchangeRes carries a ResponseCode starting with OK and EVSEProcessing Finished.

### DUT (dut) PASS 7.39 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ScheduleExchangeRes message | got 'ScheduleExchangeRes' | ✓ |
| ResponseCode starts with OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 1 poll(s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| ScheduleExchange polls to EVSEProcessing=Finished | 1 | - | info |

## TC21_SECC_ScheduleExchange_003

TC21_SECC_ScheduleExchange_003.

Test objective: unknown-SessionID negative. GoodCase up to ChargeParameterDiscoveryRes; the Test System then sends a single ScheduleExchangeReq (default parameter set, negotiated control mode) but stamps its header with a freshly generated SessionID the SUT never assigned. Checks the SUT answers a ScheduleExchangeRes within the message response window carrying ResponseCode FAILED_UnknownSession, with the response Header omitting the Signature and the response omitting GoToPause.

### DUT (dut) PASS 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ScheduleExchangeRes message | got 'ScheduleExchangeRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| response Header carries no Signature | None | ✓ |
| response omits GoToPause | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| ScheduleExchangeRes response time | 99.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_ScheduleExchange_004

TC21_SECC_ScheduleExchange_004.

Test objective: out-of-sequence check. GoodCase up to ChargeParameterDiscoveryRes; the Test System then sends a valid ServiceDiscoveryReq in place of the expected ScheduleExchangeReq. Checks the SUT answers a ServiceDiscoveryRes within the message response window, carrying ResponseCode FAILED_SequenceError and no Header signature.

**DUT** (dut) **PASS** 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| ServiceDiscoveryRes response time | 99.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_ScheduleExchange_005

TC21_SECC_ScheduleExchange_005.

Test objective: sequence-timeout observation. GoodCase up to ChargeParameterDiscoveryRes; the Test System then stays silent, sending no ScheduleExchangeReq. Checks the SUT terminates the TLS connection after the sequence timeout elapses and within the following TLS connection termination timeout. LONG TEST.

**DUT** (dut) **PASS** 68.89 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the TLS connection between V2G_SECC_Sequence_Timeout and TLS_Connection_Termination_Timeout (59..66s) | closed 62.0s after ChargeParameterDiscoveryRes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| TCP connection close after failure/timeout | 62028.5 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC21_SECC_ScheduleExchange_006

TC21_SECC_ScheduleExchange_006.

Test objective: Scheduled-control-mode schedule tuple. GoodCase up to ChargeParameterDiscoveryRes, selecting a Scheduled parameter set in ServiceSelection (the precondition fails if the SUT offers none); the Test System sends a ScheduleExchangeReq in Scheduled control mode and drives the Ongoing→Finished processing loop. Checks the final ScheduleExchangeRes carries ResponseCode starting with OK, EVSEProcessing Finished, and a Scheduled_SEResControlMode whose first ScheduleTuple has a ScheduleTupleID in the range 1..255 and a present ChargingSchedule.

**DUT** (dut) **PASS** 7.11 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| SUT sends a ScheduleExchangeRes message | got 'ScheduleExchangeRes' | ✓ |
| ResponseCode starts with OK | got 'OK' | ✓ |
| EVSEProcessing = Finished | got 'Finished' after 1 poll(s) | ✓ |
| Scheduled_SEResControlMode carries at least one ScheduleTuple | got {'ScheduleTuple': [{'ChargingSchedule': {'PowerSchedule': {'PowerScheduleEntries': {'PowerScheduleEntry': [{'Duration': 86400, 'Power': {'Exponent': 1, 'Value': 1104}}]}, 'TimeAnchor': 1793166769604}}, 'ScheduleTupleID': 1}]} | ✓ |
| ScheduleTupleID is in the range 1..255 | got 1 | ✓ |
| ScheduleTuple carries a ChargingSchedule | got {'PowerSchedule': {'PowerScheduleEntries': {'PowerScheduleEntry': [{'Duration': 86400, 'Power': {'Exponent': 1, 'Value': 1104}}]}, 'TimeAnchor': 1793166769604}} | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| ScheduleExchange polls to EVSEProcessing=Finished | 1 | - | info |

## TC21_SECC_ScheduleExchange_008

TC21_SECC_ScheduleExchange_008.

Test objective: EV-initiated shutdown for a non-critical reason. GoodCase up to ChargeParameterDiscoveryRes; the Test System then sends a SessionStopReq (ChargingSession Terminate) built from the default parameter set. Checks a SessionStopRes is received within the message response window carrying ResponseCode OK.

**DUT (dut)** **PASS** 7.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| SessionStopRes response time | 99.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_PowerDelivery_001

TC21_SECC_PowerDelivery_001.

Test objective: GoodCase to the PowerDelivery precondition (DC_PreChargeRes on DC, ScheduleExchangeRes(Finished) on AC); the Test System sends a PowerDeliveryReq with ChargeProgress=Start from the default parameter set, and on AC signals CP State C right after sending it ([V2G20-847]). Checks a PowerDeliveryRes carrying ResponseCode OK is received within the message response window.

**DUT** (dut) **PASS** 12.41 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| PowerDeliveryRes response time | 99.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_PowerDelivery_002

TC21_SECC_PowerDelivery_002.

Test objective: GoodCase through PowerDelivery(Start) and several ChargeLoop exchanges; the Test System then sends a PowerDeliveryReq with ChargeProgress=Stop from the default parameter set, and on AC signals CP State B right after sending it ([V2G20-848]). Checks a PowerDeliveryRes carrying ResponseCode OK is received within the message response window.

**DUT (dut)** **PASS** 13.40 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| PowerDeliveryRes response time | 50.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_PowerDelivery_003

TC21_SECC_PowerDelivery_003.

Test objective: GoodCase to the PowerDelivery precondition; the Test System sends a PowerDeliveryReq with ChargeProgress=Start but a freshly generated, unknown SessionID (and on AC signals CP State C right after sending it). Checks the SUT answers a PowerDeliveryRes carrying ResponseCode FAILED_UnknownSession within the message response window, with no EVSEStatus.

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| PowerDeliveryRes carries no EVSEStatus | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| PowerDeliveryRes response time | 99.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_PowerDelivery_004

TC21_SECC_PowerDelivery_004.

Test objective: as _003 but in the Stop context -- GoodCase through the charge loop, then a PowerDeliveryReq with ChargeProgress=Stop and an unknown SessionID (on AC, CP State B right after sending it). Checks the SUT answers a PowerDeliveryRes carrying ResponseCode FAILED_UnknownSession within the message response window, with no EVSEStatus.

**DUT** (dut) **PASS** 13.60 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| PowerDeliveryRes carries no EVSEStatus | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| PowerDeliveryRes response time | 1.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_PowerDelivery_005

TC21_SECC_PowerDelivery_005.

Test objective: GoodCase to the PowerDelivery precondition; the Test System sends a valid ServiceDiscoveryReq in place of the expected PowerDeliveryReq(Start). Checks the SUT answers a ServiceDiscoveryRes carrying ResponseCode FAILED_SequenceError (and no Header signature) within the message response window.

**DUT** (dut) **PASS** 12.60 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 2.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| ServiceDiscoveryRes response time | 99.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_PowerDelivery_006

TC21_SECC_PowerDelivery_006.

Test objective: as _005 but in the Stop context -- GoodCase through the charge loop, then a valid ServiceDiscoveryReq in place of the expected PowerDeliveryReq(Stop). Checks the SUT answers a ServiceDiscoveryRes carrying ResponseCode FAILED_SequenceError (and no Header signature) within the message response window.

**DUT** (dut) **PASS** 13.40 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| ServiceDiscoveryRes response time | 1.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_PowerDelivery_007

TC21_SECC_PowerDelivery_007.

Test objective: GoodCase to the PowerDelivery precondition; the Test System sends a PowerDeliveryReq with ChargeProgress=Start whose EVPowerProfile requests a power beyond the limit the SUT offered in ScheduleExchangeRes. Checks the SUT answers a PowerDeliveryRes carrying ResponseCode FAILED_EVPowerProfileInvalid. The EVPowerProfile is encoded on the wire. Only a Scheduled ScheduleExchangeRes carries a power limit ([V2G20-478], [V2G20-1559]; in Dynamic mode [V2G20-1070] makes the SECC accept any profile), so the GoodCase selects a Scheduled parameter set in ServiceSelection (the precondition fails if the SUT offers none).

**DUT** (dut) **FAIL** 12.60 s

**FAILURE REASON**

harness/test_ev.py:434

```
AssertionError: ResponseCode = FAILED_EVPowerProfileInvalid — got 'OK'
```

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = FAILED_EVPowerProfileInvalid | got 'OK' | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| PowerDeliveryRes response time | 99.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

Failure detail (excerpt):

```
E   AssertionError: ResponseCode = FAILED_EVPowerProfileInvalid — got 'OK'
```

## TC21_SECC_PowerDelivery_008

TC21_SECC_PowerDelivery_008.

Test objective: GoodCase to the PowerDelivery precondition (Scheduled control mode); the Test System sends a PowerDeliveryReq with ChargeProgress=Start and PowerToleranceAcceptance=PowerToleranceNotConfirmed. Checks the SUT answers either PowerDeliveryRes(FAILED_PowerToleranceNotConfirmed, EVSENotification Terminate) or PowerDeliveryRes(WARNING_PowerToleranceNotConfirmed, EVSENotification Renegotiation) within the message response window; the EVSEStatus's NotificationMaxDelay is recorded. The EVPowerProfile carrying the tolerance flag is encoded on the wire and stays within the offered PowerSchedule, so only the tolerance flag deviates from the default. The GoodCase selects a Scheduled parameter set in ServiceSelection (the precondition fails if the SUT offers none).

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode/EVSENotification = FAILED_PowerToleranceNotConfirmed/ Terminate or WARNING_PowerToleranceNotConfirmed/ Renegotiation | got 'FAILED_PowerToleranceNotConfirmed'/'Terminate' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| PowerDeliveryRes response time | 99.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| EVSEStatus NotificationMaxDelay | 0.0 ms | V2G_SECC_NotificationMaxDelay_Timeout (informational) | ✓ |

## TC21_SECC_PowerDelivery_009

TC21_SECC_PowerDelivery_009.

Test objective: GoodCase to the PowerDelivery precondition (DC_PreChargeRes on DC, ScheduleExchangeRes(Finished) on AC); the Test System then withholds the PowerDeliveryReq entirely. Checks the SUT terminates the TLS connection between V2G_SECC_Sequence_Timeout and the following TLS_Connection_Termination_Timeout. LONG TEST.

**DUT** (dut) **PASS** 73.49 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the TLS connection between V2G_SECC_Sequence_Timeout and TLS_Connection_Termination_Timeout | closed 62.0s after the PowerDelivery precondition | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| TCP connection close after failure/timeout | 62023.9 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC21_SECC_PowerDelivery_010

TC21_SECC_PowerDelivery_010.

Test objective: GoodCase through PowerDelivery(Start) and several ChargeLoop exchanges; the Test System then withholds the next PowerDeliveryReq entirely. Checks the SUT terminates the TLS connection between V2G_SECC_Sequence_Timeout and the following TLS_Connection_Termination_Timeout. LONG TEST.

**DUT (dut)** PASS 74.60 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the TLS connection between V2G_SECC_Sequence_Timeout and TLS_Connection_Termination_Timeout | closed 61.8s after the last ChargeLoopRes | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| TCP connection close after failure/timeout | 61779.8 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC21_SECC_PowerDelivery_011

TC21_SECC_PowerDelivery_011.

Test objective: GoodCase to the PowerDelivery precondition (Scheduled control mode); the Test System sends a PowerDeliveryReq(Start) whose EVPowerProfile Scheduled control mode names a SelectedScheduleTupleID that was not offered in ScheduleExchangeRes. Checks the SUT answers a PowerDeliveryRes carrying the tuple-selection failure ResponseCode. The EVPowerProfile is encoded on the wire. The GoodCase selects a Scheduled parameter set in ServiceSelection (the precondition fails if the SUT offers none).

**DUT** (dut) **PASS** 11.51 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = FAILED_ScheduleSelectionInvalid (ATS: FAILED_TariffSelectionInvalid, not schema-encodable) | got 'FAILED_ScheduleSelectionInvalid' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| PowerDeliveryRes response time | 99.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_PowerDelivery_012

TC21_SECC_PowerDelivery_012.

Test objective: as _011 -- GoodCase to the PowerDelivery precondition (Scheduled control mode), then a PowerDeliveryReq(Start) selecting a schedule tuple id that ScheduleExchangeRes never contained. Checks a PowerDeliveryRes carrying the tuple-selection failure ResponseCode (no EVSEStatus). The EVPowerProfile is encoded on the wire. The GoodCase selects a Scheduled parameter set in ServiceSelection (the precondition fails if the SUT offers none).

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a PowerDeliveryRes message | got 'PowerDeliveryRes' | ✓ |
| ResponseCode = FAILED_ScheduleSelectionInvalid (ATS: FAILED_TariffSelectionInvalid, not schema-encodable) | got 'FAILED_ScheduleSelectionInvalid' | ✓ |
| PowerDeliveryRes carries no EVSEStatus | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 409.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| PowerDeliveryRes response time | 99.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_PowerDelivery_013

TC21_SECC_PowerDelivery_013.

Test objective: GoodCase to the PowerDelivery precondition; instead of the PowerDeliveryReq the Test System sends a SessionStopReq from the default parameter set (an EV-initiated shutdown for a non-critical reason). Checks the SUT answers a SessionStopRes carrying ResponseCode OK within the message response window.

**DUT** (dut) **PASS** 12.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SessionStopRes response time | 99.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_ChargeLoop_001

TC21_SECC_ChargeLoop_001.

Test objective: GoodCase up to PowerDeliveryRes(Start), i.e. inside the charge loop; the Test System then sends a valid ServiceDiscoveryReq in place of the expected (DC|AC)_ChargeLoopReq. Checks the SUT answers a ServiceDiscoveryRes carrying ResponseCode FAILED_SequenceError (and no Header signature) within the message response window.

**DUT** (dut) **PASS** 11.92 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| ServiceDiscoveryRes response time | 99.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_ChargeLoop_002

TC21_SECC_ChargeLoop_002.

Test objective: GoodCase up to PowerDeliveryRes(Start), i.e. inside the charge loop; the Test System then withholds the (DC|AC)_ChargeLoopReq entirely. Checks the SUT terminates the TLS connection between V2G_SECC_Sequence_Timeout and the following TLS_Connection_Termination_Timeout. LONG TEST.

**DUT** (dut) **PASS** 73.89 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the TLS connection between V2G_SECC_Sequence_Timeout and TLS_Connection_Termination_Timeout | closed 62.0s after PowerDeliveryRes(Start) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| TCP connection close after failure/timeout | 62022.7 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC21_SECC_SessionStop_001

TC21_SECC_SessionStop_001.

Test objective: GoodCase up to the point a SessionStop is legal (DC: DC_WeldingDetectionRes Finished). The Test System sends a SessionStopReq built from the default parameter set (ChargingSession=Terminate). Checks a SessionStopRes is received within the message response window carrying ResponseCode OK.

**DUT (dut)** **PASS** 13.21 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = OK | got 'OK' | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 408.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| SessionStopRes response time | 100.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_SessionStop_002

TC21_SECC_SessionStop_002.

Test objective: unknown-SessionID negative. GoodCase up to the point a SessionStop is legal (DC: DC_WeldingDetectionRes Finished); the Test System then sends a SessionStopReq (default parameter set, ChargingSession=Terminate) but stamps its header with a freshly generated SessionID the SUT never assigned. Checks the SUT answers a SessionStopRes within the message response window carrying ResponseCode FAILED_UnknownSession, with the response Header omitting the Signature.

**DUT (dut)** PASS 13.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = FAILED_UnknownSession | got 'FAILED_UnknownSession' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SessionStopRes response time | 100.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_SessionStop_003

TC21_SECC_SessionStop_003.

Test objective: GoodCase up to the point a SessionStop is legal; the Test System then sends a valid ServiceDiscoveryReq in place of the expected SessionStopReq. Checks the SUT answers a ServiceDiscoveryRes within the message response window, carrying ResponseCode FAILED_SequenceError and no Header signature.

**DUT** (dut) **PASS** 13.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a ServiceDiscoveryRes message | got 'ServiceDiscoveryRes' | ✓ |
| ResponseCode = FAILED_SequenceError | got 'FAILED_SequenceError' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| ServiceDiscoveryRes response time | 99.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_SessionStop_004

TC21_SECC_SessionStop_004.

Test objective: GoodCase up to the point a SessionStop is legal; the Test System then withholds the SessionStopReq entirely and stays silent. Checks the SUT terminates the TLS connection between V2G_SECC_Sequence_Timeout and the following TLS connection termination timeout (i.e. it does not tear the connection down early, and does so within the sequence timeout plus the TLS termination budget). LONG TEST.

**DUT (dut)** PASS 75.09 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT terminates the TLS connection within V2G_SECC_Sequence_Timeout + TLS_Connection_Termination_Timeout (66s) with no SessionStopReq | closed 62.0s after the final response | ✓ |
| SUT does not terminate before V2G_SECC_Sequence_Timeout expires (>= 55s) | closed after 62.0s | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| TCP connection close after failure/timeout | 62027.4 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✗ |

## TC21_SECC_SessionStop_005

TC21_SECC_SessionStop_005.

Test objective: GoodCase driving the full message flow for the selected charging mode through to SessionStop, with every request built from the default parameter set. Checks that the SessionID in the header of each received response equals the value the SUT assigned in the SessionSetupRes, for the entire session.

**DUT** (dut) **PASS** 13.11 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| every response echoes the SessionID assigned in SessionSetupRes | assigned '809991f8a9179a83'; all responses matched | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |

## TC21_SECC_SessionStop_006

TC21_SECC_SessionStop_006.

Test objective: GoodCase up to the point a SessionStop is legal, on an SUT that does not support service renegotiation. The Test System sends a SessionStopReq with ChargingSession=ServiceRenegotiation. Checks the SUT answers a SessionStopRes within the message response window carrying ResponseCode FAILED_NoServiceRenegotiationSupported and no Header signature.

**DUT** (dut) **PASS** 13.00 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = FAILED_NoServiceRenegotiationSupported | got 'FAILED_NoServiceRenegotiationSupported' | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SessionStopRes response time | 100.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC21_SECC_SessionStop_007

TC21_SECC_SessionStop_007.

Test objective: GoodCase up to the point a SessionStop is legal, under a dynamic control mode, with the SUT never having signalled EVSENotification= Pause. The Test System sends a SessionStopReq with ChargingSession=Pause. Checks the SUT answers a SessionStopRes within the message response window carrying ResponseCode FAILED_PauseNotAllowed and no Header signature.

**DUT** (dut) **PASS** 12.99 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SUT sends a SessionStopRes message | got 'SessionStopRes' | ✓ |
| ResponseCode = FAILED_PauseNotAllowed | got 'FAILED_PauseNotAllowed' (control mode 'Dynamic') | ✓ |
| response Header carries no Signature | None | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| SessionStopRes response time | 100.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

# Test case details — PIONIX interop testing (vendor-defined)

## TC_SECC_CMN_VTB_PionixInterop_010

TC_SECC_CMN_VTB_PionixInterop_010.

Test objective: Interop; after CM_SLAC_PARM the Test System runs the sounding phase (3x CM_START_ATTEN_CHAR.IND + 10x CM_MNBC_SOUND.IND) and measures the delay from the END of sounding to the SUT's CM_ATTEN_CHAR.IND result. Reference timestamp: the last CM_ATTEN_PROFILE.IND observed on the EV-side wire where the rig exposes it; on the tandem SIL the virtual modem injects the profile INDs on the EVSE tap only, so the transmission of the last CM_MNBC_SOUND.IND (which the modem answers with the profile IND within its forwarding latency) is the observable proxy. Pass band (PIONIX interop requirement, strict inequalities): more than 40 ms and less than 100 ms. The IND is also checked against the Annex F template and answered with CM_ATTEN_CHAR.RSP.

**DUT (dut)** PASS 4.38 s

| Conformance check | Observed | Result |
|---|---|---|
| CM_ATTEN_CHAR.IND delay after end of sounding in 40..100 ms (PIONIX interop requirement, strict) | arrived 46.1 ms after the end of sounding (last CM_MNBC_SOUND.IND (tx)) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.3 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_ATTEN_CHAR.IND after end of sounding | 46.1 ms | [40.0, 100.0] ms | ✓ |

## TC_SECC_DC_VTB_PionixInterop_001

TC_SECC_DC_VTB_PionixInterop_001.

Test objective: Interop; the Test System runs a complete DC ISO 15118-2 session (eIM, TLS off) up to and including PowerDelivery(Start) and measures the EVSE response time of EVERY exchange before the CurrentDemand loop: supportedAppProtocol, SessionSetup, ServiceDiscovery, PaymentServiceSelection, Authorization, ChargeParameterDiscovery, CableCheck, PreCharge and PowerDelivery -- polled messages (EVSEProcessing Ongoing) contribute one sample per poll. The measurement starts with the supportedAppProtocol handshake: the preceding SLAC matching is a precondition only and expressly NOT measured. Requests are timestamped at V2GTP frame send, responses at V2GTP frame arrival. Checks per message: the minimum shall be >= 90 ms (the same PIONIX interop minimum as the CurrentDemand-loop TC) and the maximum <= the standard's per-message limit V2G_SECC_Msg_Performance_Time (ISO 15118-2 Table 109: 1.5 s, resp. 4.5 s for PowerDeliveryRes). Per-message min/max are recorded; on failure every violating exchange is named exactly (message + sample number) with whether it came too quickly or took too long.

**DUT** (dut) **PASS** 12.41 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Every DC session-setup response time within bounds (min 90 ms / Table 109 max per message) | all exchanges within bounds | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.2 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| V2G exchanges measured up to PowerDelivery(Start) | 21 | - | info |
| AuthorizationRes response time (min of 2) | 100.6 ms | ≥ 90.0 ms | ✓ |
| AuthorizationRes response time (max of 2) | 100.7 ms | ≤ 1500.0 ms | ✓ |
| CableCheckRes response time (min of 11) | 100.5 ms | ≥ 90.0 ms | ✓ |
| CableCheckRes response time (max of 11) | 100.8 ms | ≤ 1500.0 ms | ✓ |
| ChargeParameterDiscoveryRes response time (min of 1) | 100.4 ms | ≥ 90.0 ms | ✓ |
| ChargeParameterDiscoveryRes response time (max of 1) | 100.4 ms | ≤ 1500.0 ms | ✓ |
| PaymentServiceSelectionRes response time (min of 1) | 100.6 ms | ≥ 90.0 ms | ✓ |
| PaymentServiceSelectionRes response time (max of 1) | 100.6 ms | ≤ 1500.0 ms | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| PowerDeliveryRes response time (min of 1) | 100.7 ms | ≥ 90.0 ms | ✓ |
| PowerDeliveryRes response time (max of 1) | 100.7 ms | ≤ 4500.0 ms | ✓ |
| PreChargeRes response time (min of 2) | 100.5 ms | ≥ 90.0 ms | ✓ |
| PreChargeRes response time (max of 2) | 100.9 ms | ≤ 1500.0 ms | ✓ |
| ServiceDiscoveryRes response time (min of 1) | 100.4 ms | ≥ 90.0 ms | ✓ |
| ServiceDiscoveryRes response time (max of 1) | 100.4 ms | ≤ 1500.0 ms | ✓ |
| SessionSetupRes response time (min of 1) | 100.6 ms | ≥ 90.0 ms | ✓ |
| SessionSetupRes response time (max of 1) | 100.6 ms | ≤ 1500.0 ms | ✓ |
| supportedAppProtocolRes response time (min of 1) | 100.6 ms | ≥ 90.0 ms | ✓ |
| supportedAppProtocolRes response time (max of 1) | 100.6 ms | ≤ 1500.0 ms | ✓ |
| PowerDeliveryRes(Stop) response time | 102.1 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_DC_VTB_PionixInterop_003

TC_SECC_DC_VTB_PionixInterop_003.

Test objective: Interop; the EVSE's SLAC must be ready to answer the moment it signals HLC availability. Per cycle the Test System plugs in (CP A→B), waits for the 5 % PWM (X2) and IMMEDIATELY sends a single CM_SLAC_PARM.REQ: the cycle passes if a template-valid CM_SLAC_PARM.CNF arrives within the per-attempt window (TT_match_response + transmission delay), and fails if the REQ goes unanswered (got lost) -- deliberately NO C_EV_match_retry retransmissions, stricter than ISO 15118-3's retry allowance, because a swallowed first REQ is exactly the interop flaw probed here. The rest of the SLAC sequence is not performed; the Test System then unplugs, waits for the 5 % duty to disappear (so the next cycle cannot read a stale 5 %), and dwells the operator's global inter-TC sleep (--dut-settle-s, the web UI setting) before the next plug-in. 10 cycles probe for flakiness; the TC passes only if every cycle got its CNF, and the failing cycle numbers are named otherwise. Per-cycle CNF response times are recorded for the report.

**DUT** (dut) **PASS** 52.57 s

| Conformance check | Observed | Result |
|---|---|---|
| Every cycle's immediate first CM_SLAC_PARM.REQ answered (10 cycles) | all 10 cycles answered | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF response time | 1.4 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| SLAC-ready cycles (immediate first CM_SLAC_PARM.REQ answered) | 10 | = 10 | ✓ |

## TC_SECC_DC_VTB_PionixInterop_004

TC_SECC_DC_VTB_PionixInterop_004.

Test objective: Interop; measure how long after plug-in (CP A→B) the EVSE turns on the 5 % PWM (X2, HLC offer). PIONIX interop requirement: < 1 s; the normative ISO 15118-3 bound (par_T_conn_max_comm) is a far looser 8 s. Timing is taken from the Test System's CP duty observation (plug-in command to the first ~5 % duty report), which includes the rig's own reporting latency -- strict-side: a PASS is trustworthy, the measured value can only overstate the true one. Precondition guards that no stale 5 % duty from an earlier session is being read.

### DUT (dut) PASS 4.62 s

| Conformance check | Observed | Result |
|---|---|---|
| 5% PWM within 1000 ms of plug-in (PIONIX interop requirement) | 5% PWM showed 510 ms after plug-in (ISO 15118-3 par_T_conn_max_comm would allow 8 s) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| 5% PWM after plug-in | 510.3 ms | ≤ 1000.0 ms | ✓ |

## TC_SECC_DC_VTB_PionixInterop_007

TC_SECC_DC_VTB_PionixInterop_007.

Test objective: Interop; DIN SPEC 70121 twin of the ISO-2 session-setup timing TC. The Test System negotiates urn:din:70121:2012:MsgDef in the supportedAppProtocol handshake and runs a complete DIN DC session (EIM) up to and including PowerDelivery(ReadyToChargeState=true), measuring the EVSE response time of EVERY exchange: supportedAppProtocol, SessionSetup, ServiceDiscovery, ServicePaymentSelection, ContractAuthentication, ChargeParameterDiscovery, CableCheck, PreCharge and PowerDelivery -- polled messages contribute one sample per poll. The measurement starts with the supportedAppProtocol handshake: SLAC is a precondition only and expressly NOT measured. Requests are timestamped at V2GTP frame send, responses at V2GTP frame arrival. Checks per message: the minimum shall be >= 90 ms (PIONIX interop requirement) and the maximum <= the DIN per-message V2G_SECC_Msg_Performance_Time (DIN TS 70121 Ed.3 Table 75: 1.5 s for every response, incl. PowerDeliveryRes -- stricter than ISO 15118-2's 4.5 s there). On failure every violating exchange is named exactly (message + sample number) with whether it came too quickly or took too long.

**DUT** (dut) **PASS** 11.60 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Every DIN 70121 session-setup response time within bounds (min 90 ms / DIN timing-table max per message) | all exchanges within bounds | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| V2G exchanges measured up to PowerDelivery(Start) | 21 | - | info |
| CableCheckRes response time (min of 11) | 100.6 ms | ≥ 90.0 ms | ✓ |
| CableCheckRes response time (max of 11) | 100.8 ms | ≤ 1500.0 ms | ✓ |
| ChargeParameterDiscoveryRes response time (min of 1) | 100.5 ms | ≥ 90.0 ms | ✓ |
| ChargeParameterDiscoveryRes response time (max of 1) | 100.5 ms | ≤ 1500.0 ms | ✓ |
| ContractAuthenticationRes response time (min of 2) | 100.6 ms | ≥ 90.0 ms | ✓ |
| ContractAuthenticationRes response time (max of 2) | 100.7 ms | ≤ 1500.0 ms | ✓ |
| PowerDeliveryRes response time (min of 1) | 100.6 ms | ≥ 90.0 ms | ✓ |
| PowerDeliveryRes response time (max of 1) | 100.6 ms | ≤ 1500.0 ms | ✓ |
| PreChargeRes response time (min of 2) | 100.6 ms | ≥ 90.0 ms | ✓ |
| PreChargeRes response time (max of 2) | 100.7 ms | ≤ 1500.0 ms | ✓ |
| ServiceDiscoveryRes response time (min of 1) | 100.4 ms | ≥ 90.0 ms | ✓ |
| ServiceDiscoveryRes response time (max of 1) | 100.4 ms | ≤ 1500.0 ms | ✓ |
| ServicePaymentSelectionRes response time (min of 1) | 100.5 ms | ≥ 90.0 ms | ✓ |
| ServicePaymentSelectionRes response time (max of 1) | 100.5 ms | ≤ 1500.0 ms | ✓ |
| SessionSetupRes response time (min of 1) | 100.5 ms | ≥ 90.0 ms | ✓ |
| SessionSetupRes response time (max of 1) | 100.5 ms | ≤ 1500.0 ms | ✓ |
| supportedAppProtocolRes response time (min of 1) | 100.6 ms | ≥ 90.0 ms | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| supportedAppProtocolRes response time (max of 1) | 100.6 ms | ≤ 1500.0 ms | ✓ |

## TC_SECC_DC_VTB_PionixInterop_014

TC_SECC_DC_VTB_PionixInterop_014.

Test objective: Interop; a real EV needs tens of seconds for its TLS handshake (certificate-chain checks and key agreement on a slow controller). The Test System runs the good case through SLAC and SDP (Security = TLS), connects to the SECC's TLS port and sends the ClientHello at once, but releases its own handshake flight (ClientKeyExchange / ChangeCipherSpec / Finished) only 15 s after the SECC's ServerHello flight arrived; nothing is read in between, so the SECC sits mid-handshake. Checks: (1) the SECC keeps the half-open TLS handshake for the whole 15 s (no TCP close); (2) the TLS 1.2 handshake then completes; (3) the ISO 15118-2 DC eIM session runs through supportedAppProtocol, SessionSetup, ServiceDiscovery, PaymentServiceSelection(ExternalPayment), Authorization(Finished), ChargeParameterDiscovery(Finished), CableCheck(Finished), PreCharge, PowerDelivery(Start) with OK, the CurrentDemand loop answers OK, and the session ends cleanly (PowerDelivery(Stop), WeldingDetection, SessionStop(Terminate) OK). The handshake timing (server flight after the ClientHello, hold, completion) is recorded.

Note: ISO 15118-2 Table 111 gives the SECC 18 s from D-LINK ready to SessionSetupReq (V2G_SECC_CommunicationSetup_Performance_Time, [V2G2-714]..[V2G2-716]); only on its expiry may it terminate the connection. The 15 s hold plus the SLAC-match → SDP → connect and supportedAppProtocol → SessionSetupReq exchanges around it stay inside that budget, so a SECC that closes the half-open handshake earlier applies an own TLS-handshake timeout the standard does not grant.

**DUT** (dut) **PASS** 28.43 s

| Conformance check | Observed | Result |
|---|---|---|
| SECC keeps the half-open TLS 1.2 handshake for the EV's 15 s hold (within V2G_SECC_CommunicationSetup_Performance_Time 18 s) | held 15 s, no close | ✓ |
| TLS 1.2 handshake completes after the hold | TLSv1.2 ('ECDHE-ECDSA-AES128-SHA256', 'TLSv1.2', 128) after 15.0 s | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| SessionSetupRes OK after the slow handshake | OK_NewSessionEstablished | ✓ |
| AuthorizationRes OK / EVSEProcessing = Finished | OK / Finished | ✓ |
| ChargeParameterDiscoveryRes OK / Finished | OK / Finished | ✓ |
| CableCheckRes OK / Finished | OK / Finished | ✓ |
| PreChargeRes OK | OK | ✓ |
| PowerDeliveryRes(Start) OK: the charging session starts | OK | ✓ |
| CurrentDemandRes OK in the charge loop | OK, 3 loop(s) | ✓ |
| Clean end: PowerDeliveryRes(Stop), WeldingDetectionRes, SessionStopRes all OK | OK / OK / OK | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.5 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 417.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| TLS ServerHello flight after the ClientHello | 0.9 ms | - | info |
| TLS handshake duration (EV holds its flight 15 s) | 15.0 s | - | info |
| supportedAppProtocolRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| WeldingDetectionRes response time | 103.2 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionStopRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_DC_VTB_PionixInterop_017

TC_SECC_DC_VTB_PionixInterop_017.

Test objective: Interop; a real EV takes a long time after plug-in before it starts the matching. The Test System plugs in (CP State B), waits for the 5 % PWM and then does nothing for 30 s. It then sends CM_SLAC_PARM.REQ (the standard's C_EV_match_retry retries apply) and runs the good case through sounding, CM_ATTEN_CHAR, CM_SLAC_MATCH, the link, SDP, TCP, supportedAppProtocol, SessionSetup, ServiceDiscovery, PaymentServiceSelection(ExternalPayment), Authorization(Finished), ChargeParameterDiscovery(Finished), CableCheck(Finished), PreCharge, PowerDelivery(Start) and the CurrentDemand loop. Checks: the 5 % duty was held for the whole 30 s (no X1 or E/F period), the CM_SLAC_PARM.REQ 30 s after plug-in is answered with a valid CNF ([V2G3-A09-11]), and the DC ISO 15118-2 eIM session reaches a CurrentDemandRes(OK). Ends the session cleanly. ISO 15118-3 Table A.1 allows TT_EVSE_SLAC_init anywhere in [20, 50] s and [V2G3-A09-13] lets the EVSE stop expecting SLAC at its expiry, so an EVSE with a timer below 30 s fails this TC while being conformant: the 30 s is the PIONIX interop requirement.

**DUT** (dut) **PASS** 42.80 s

| Conformance check | Observed | Result |
|---|---|---|
| 5 % duty held for the whole 30 s before the EV's first CM_SLAC_PARM.REQ (TT_EVSE_SLAC_init >= 30 s, PIONIX interop; -3 Table A.1 allows >= 20 s) | 5 % throughout | ✓ |
| CM_SLAC_PARM.REQ 30 s after plug-in answered with a valid CM_SLAC_PARM.CNF [V2G3-A09-11] | CNF after 2.2 ms, REQ 30.0 s after plug-in | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| DC ISO 15118-2 session after the late SLAC reaches CurrentDemandRes OK | OK, 3 loop(s) | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 2.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.0 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 101.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 103.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.4 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.0 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 102.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_DC_VTB_PionixInterop_019

TC_SECC_DC_VTB_PionixInterop_019.

Test objective: Interop; poor CP contact during a slow physical plug-in, the CP oscillating between States A and B a couple of times before the plug seats. From CP State A the Test System applies State B for 50 ms and returns to A for 100 ms, then repeats with B dwells of 100 ms, 200 ms, 500 ms and 1 s (100 ms of A after each). It then applies State B for good and runs a normal DC ISO 15118-2 eIM session: SLAC (C_EV_match_retry retries on the first CM_SLAC_PARM.REQ, as the good case), link, SDP, TCP, supportedAppProtocol … PowerDelivery(Start) and the CurrentDemand loop. Checks: the 5 % PWM is applied within par_T_conn_max_comm (8 s) of the final plug-in (the EVSE has sorted out the bounces, no pending E/F or reinit swallows the real plug-in), and the session reaches a CurrentDemandRes(OK). What the CP did during the bounces (duty periods) is recorded. Ends the session cleanly.

**DUT** (dut) **PASS** 15.03 s

| Conformance check | Observed | Result |
|---|---|---|
| 5 % PWM within par_T_conn_max_comm (8 s) of the plug-in that stays | 0.20 s | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| DC ISO 15118-2 session after the plug bounces reaches CurrentDemandRes OK | OK, 3 loop(s) | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CP duty periods during the plug bounces | 100 % at +0.00 s; 5 % at +0.51 s; 100 % at +0.52 s; 5 % at +1.02 s; 100 % at +1.03 s; 5 % at +1.53 s; 100 % at +1.54 s; 5 % at +2.04 s; 100 % at +2.05 s | - | info |
| 5 % PWM after the plug-in that stays | 201.7 ms | - | info |
| CM_SLAC_PARM.CNF response time | 1.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 416.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 103.3 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| SessionStopRes response time | 103.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## Test case details — PIONIX use cases (vendor-defined end-to-end scenarios)

## TC_SECC_DC_VTB_PionixUseCase_001

TC_SECC_DC_VTB_PionixUseCase_001.

Test objective: Use case; a DC EV pauses and resumes its session. The Test System runs the DC ISO 15118-2 eIM session into the CurrentDemand loop and ends the energy transfer with PowerDelivery(Stop) and CP State B, then SessionStop(ChargingSession = Pause) → OK (precondition). Then every step of the resume is a criterion: (1) the EVSE switches its PWM off (X1) for the sleep phase within 10 s; (2) after a 3 s sleep the EV wakes it with a BCB toggle (3 x B-C-B) and the PWM is back within par_T_conn_max_comm (8 s) of the toggle, as the 5 % HLC signal; (3) the PLC link stayed matched, so the SUT sends no SLAC frame across the resume and the EV runs no SLAC; (4) SDP, TCP, supportedAppProtocol(OK) and SessionSetupReq with the paused SessionID → OK_OldSessionJoined (the ISO 15118-2 pause/resume); (5) ServiceDiscovery, PaymentServiceSelection(ExternalPayment), Authorization(Finished), ChargeParameterDiscovery(Finished), CP State C, CableCheck(Finished), PreCharge, PowerDelivery(Start) OK and the CurrentDemand loop answers OK again; (6) a clean end: PowerDelivery(Stop), CP State B, WeldingDetection, SessionStop(Terminate), all OK.

**DUT** (dut) **PASS** 28.32 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| EVSE switches the PWM off (X1) within 10 s after the pause (SessionStopRes + TCP close) | 0.00 s | ✓ |
| PWM back within par_T_conn_max_comm (8 s) of the BCB toggle | 0.01 s, duty 5.0 % | ✓ |
| duty after the BCB toggle is the 5 % HLC signal (DC) | 5.0 % | ✓ |
| supportedAppProtocolRes OK on the resumed connection | OK_SuccessfulNegotiation | ✓ |
| SessionSetupReq with the paused SessionID → OK_OldSessionJoined | SessionSetupRes OK_OldSessionJoined | ✓ |
| no SLAC frame from the SUT across the pause/resume (PLC link stays matched; the EV goes straight to SDP) | silent | ✓ |
| Response echoes the current SessionID | | ✓ |
| ServiceDiscoveryRes OK (resumed session) | OK | ✓ |
| Response echoes the current SessionID | | ✓ |
| PaymentServiceSelectionRes OK (resumed session) | OK | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| AuthorizationRes OK / Finished (resumed session) | OK/Finished after 2 poll(s) | ✓ |
| Response echoes the current SessionID | | ✓ |
| ChargeParameterDiscoveryRes OK / Finished (resumed session) | OK/Finished after 1 poll(s) | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| CableCheckRes OK / Finished (resumed session) | OK/Finished after 12 poll(s) | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| PreChargeRes OK (resumed session) | OK | ✓ |
| Response echoes the current SessionID | | ✓ |
| PowerDeliveryRes(Start) OK (resumed session) | OK | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| CurrentDemandRes OK: the charge loop is reached again (resumed session) | OK, 3 loop(s) | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Clean end: PowerDeliveryRes(Stop), WeldingDetectionRes, SessionStopRes(Terminate) all OK | OK / OK / OK | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 418.7 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.4 ms | - | info |
| supportedAppProtocolRes response time | 100.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 102.1 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.7 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.0 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| SessionStopRes response time | 104.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| TCP connection close after failure/ timeout | 1996.7 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| PWM off (X1) after the pause (SessionStopRes + TCP close) | 0.0 ms | - | info |
| PWM back after the BCB toggle train | 9.3 ms | - | info |
| CP duty after the BCB toggle | 5.0 % | - | info |
| supportedAppProtocolRes response time | 100.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.3 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.7 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 101.8 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| WeldingDetectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionStopRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_DC_VTB_PionixUseCase_003

TC_SECC_DC_VTB_PionixUseCase_003.

Test objective: Use case; a DC EV ends its session and starts a new one on the still-plugged connector. The Test System runs the DC ISO 15118-2 eIM session into the CurrentDemand loop and ends it: PowerDelivery (Stop), CP State B, WeldingDetection, SessionStop(ChargingSession = Terminate) → OK, and closes TCP (precondition). Then every step of the restart is a criterion: (1) the EVSE switches its PWM off (X1) within 10 s of the termination; (2) after a 3 s dwell the EV performs a BCB toggle (3 x B-C-B) and the PWM is back within par_T_conn_max_comm (8 s) of the toggle, as the 5 % HLC signal -- the EVSE offers HLC again; (3) the previous link was terminated (D-LINK_TERMINATE), so the EV runs a complete NEW SLAC matching with a fresh RunID: CM_SLAC_PARM, sounding, CM_ATTEN_CHAR, CM_SLAC_MATCH.CNF, the link comes up; (4) SDP, TCP, supportedAppProtocol and SessionSetupReq with a zero SessionID → OK_NewSessionEstablished; (5) ServiceDiscovery, PaymentServiceSelection (ExternalPayment), Authorization(Finished), ChargeParameterDiscovery (Finished), CP State C, CableCheck(Finished), PreCharge, PowerDelivery (Start) OK and the CurrentDemand loop answers OK again; (6) a clean end: PowerDelivery(Stop), CP State B, WeldingDetection, SessionStop (Terminate), all OK.

**DUT** (dut) **PASS** 29.47 s

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| EVSE switches the PWM off (X1) within 10 s after SessionStop(Terminate) + TCP close | 1.61 s | ✓ |
| PWM back within par_T_conn_max_comm (8 s) of the BCB toggle | 0.01 s, duty 5.0 % | ✓ |
| duty after the BCB toggle is the 5 % HLC signal (DC) | 5.0 % | ✓ |
| duty after the BCB toggle is the 5 % HLC signal (a new matching is required) | 5.0 % | ✓ |
| new SLAC matching after the BCB-toggle restart: CM_SLAC_MATCH.CNF received | matched | ✓ |
| SessionSetupReq with a zero SessionID on the restarted session → OK_NewSessionEstablished | SessionSetupRes OK_NewSessionEstablished | ✓ |
| Response echoes the current SessionID | | ✓ |
| ServiceDiscoveryRes OK (restarted session) | OK | ✓ |
| Response echoes the current SessionID | | ✓ |
| PaymentServiceSelectionRes OK (restarted session) | OK | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| AuthorizationRes OK / Finished (restarted session) | OK/Finished after 2 poll(s) | ✓ |
| Response echoes the current SessionID | | ✓ |
| ChargeParameterDiscoveryRes OK / Finished (restarted session) | OK/Finished after 1 poll(s) | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| CableCheckRes OK / Finished (restarted session) | OK/Finished after 11 poll(s) | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| PreChargeRes OK (restarted session) | OK | ✓ |
| Response echoes the current SessionID | | ✓ |
| PowerDeliveryRes(Start) OK (restarted session) | OK | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| CurrentDemandRes OK: the charge loop is reached again (restarted session) | OK, 3 loop(s) | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Clean end: PowerDeliveryRes(Stop), WeldingDetectionRes, SessionStopRes(Terminate) all OK | OK / OK / OK | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 0.9 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 412.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| SessionSetupRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.4 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.7 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 101.9 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| WeldingDetectionRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionStopRes response time | 101.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PWM off (X1) after SessionStop(Terminate) + TCP close | 1606.3 ms | - | info |
| PWM back after the BCB toggle train | 10.5 ms | - | info |
| CP duty after the BCB toggle | 5.0 % | - | info |
| CM_SLAC_PARM.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 410.1 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| supportedAppProtocolRes response time | 100.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionSetupRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| CM_SLAC_MATCH.CNF of the restarted matching after the 5 % PWM returned | 1622.1 ms | - | info |
| ServiceDiscoveryRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.6 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 102.0 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| PowerDeliveryRes(Stop) response time | 102.4 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| WeldingDetectionRes response time | 102.0 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| SessionStopRes response time | 101.8 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

## TC_SECC_DC_VTB_PionixUseCase_005

TC_SECC_DC_VTB_PionixUseCase_005.

Test objective: Use case; the ISO 15118-20 version of _001. The Test System runs the DC ISO 15118-20 session (EIM, TLS 1.3 mTLS, Scheduled control mode) into the DC charge loop, PowerDelivery(Stop), CP State B, then SessionStop(ChargingSession = Pause) → OK and the SECC's connection close (precondition; Scheduled because ISO 15118-20 allows an EV pause there, 8.5.6.2, while a pause the SECC did not initiate is FAILED_PauseNotAllowed in Dynamic mode, [V2G20-1850]). Then every step of the resume is a criterion: (1) the EVSE switches its PWM off (X1) within 10 s; (2) after a 3 s sleep the EV wakes it with a BCB toggle and the 5 % PWM is back within par_T_conn_max_comm (8 s); (3) the PLC link stayed matched: no SLAC frame from the SUT, the EV runs no SLAC; (4) SDP(TLS), a new TLS 1.3 session ([V2G20-2073]), supportedAppProtocol and SessionSetupReq with the paused SessionID → OK_OldSessionJoined; (5) no AuthorizationSetup / Authorization and no service selection -- the paused session's stay valid ([V2G20-1844]) -- but straight to DC_ChargeParameterDiscovery ([V2G20-2098]), ScheduleExchange(Finished), CP State C, DC_CableCheck(Finished), DC_PreCharge, PowerDelivery(Start) and the DC_ChargeLoop answers OK again; (6) a clean end: PowerDelivery(Stop), CP State B, DC_WeldingDetection, SessionStop(Terminate), all OK.

**DUT** (dut) **PASS** 26.17 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| EVSE switches the PWM off (X1) within 10 s after the pause (SessionStopRes + connection close) | 0.00 s | ✓ |
| PWM back within par_T_conn_max_comm (8 s) of the BCB toggle | 0.01 s, duty 5.0 % | ✓ |
| duty after the BCB toggle is the 5 % HLC signal (DC) | 5.0 % | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| no SLAC frame from the SUT across the pause/ resume (PLC link stays matched; the EV goes straight to SDP) | silent | ✓ |
| SessionSetupReq with the paused SessionID → OK_OldSessionJoined (new TLS session, [V2G20-2073]) | SessionSetupRes OK_OldSessionJoined | ✓ |
| DC_ChargeLoopRes OK: the charge loop is reached again (resumed session) | DC_ChargeLoopRes OK | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Clean end: PowerDeliveryRes(Stop), DC_WeldingDetectionRes, SessionStopRes(Terminate) all OK | OK / OK / OK | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.3 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| TCP connection close after failure/ timeout | 1999.6 ms | ≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout) | ✓ |
| PWM off (X1) after the pause (SessionStopRes + connection close) | 0.0 ms | - | info |
| PWM back after the BCB toggle train | 9.9 ms | - | info |
| CP duty after the BCB toggle | 5.0 % | - | info |

## TC_SECC_DC_VTB_PionixUseCase_007

TC_SECC_DC_VTB_PionixUseCase_007.

Test objective: Use case; the ISO 15118-20 version of _003. The Test System runs the DC ISO 15118-20 EIM session (TLS 1.3 mTLS) into the DC charge loop and ends it: PowerDelivery(Stop), CP State B, DC_WeldingDetection, SessionStop(ChargingSession = Terminate) → OK, and closes the connection (precondition). Then every step of the restart is a criterion: (1) the EVSE switches its PWM off (X1) within 10 s; (2) after a 3 s dwell the EV performs a BCB toggle and the 5 % PWM is back within par_T_conn_max_comm (8 s) -- the EVSE offers HLC again; (3) the link was terminated, so the EV runs a complete NEW SLAC matching with a fresh RunID through CM_SLAC_MATCH.CNF and the link comes up; (4) SDP(TLS), TLS 1.3, supportedAppProtocol and SessionSetupReq with a zero SessionID → OK_NewSessionEstablished; (5) the -20 sequence (AuthorizationSetup … PowerDelivery(Start)) and the DC_ChargeLoop answers OK again; (6) a clean end: PowerDelivery(Stop), CP State B, DC_WeldingDetection, SessionStop(Terminate), all OK.

**DUT** (dut) **PASS** 27.50 s

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| EVSE switches the PWM off (X1) within 10 s after SessionStop(Terminate) + connection close | 1.61 s | ✓ |
| PWM back within par_T_conn_max_comm (8 s) of the BCB toggle | 0.01 s, duty 5.0 % | ✓ |
| duty after the BCB toggle is the 5 % HLC signal (DC) | 5.0 % | ✓ |
| duty after the BCB toggle is the 5 % HLC signal (a new matching is required) | 5.0 % | ✓ |
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| new SLAC matching after the BCB-toggle restart: CM_SLAC_MATCH.CNF received | matched | ✓ |
| SessionSetupReq with a zero SessionID on the restarted session → OK_NewSessionEstablished | SessionSetupRes OK_NewSessionEstablished | ✓ |
| DC_ChargeLoopRes OK: the charge loop is reached again (restarted session) | DC_ChargeLoopRes OK | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Clean end: PowerDeliveryRes(Stop), DC_WeldingDetectionRes, SessionStopRes(Terminate) all OK | OK / OK / OK | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.0 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 413.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| PWM off (X1) after SessionStop(Terminate) + connection close | 1606.0 ms | - | info |
| PWM back after the BCB toggle train | 8.9 ms | - | info |
| CP duty after the BCB toggle | 5.0 % | - | info |
| CM_SLAC_PARM.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.8 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| CM_SLAC_MATCH.CNF of the restarted matching after the 5 % PWM returned | 7688.4 ms | - | info |

## TC_SECC_DC_VTB_PionixUseCase_009

TC_SECC_DC_VTB_PionixUseCase_009.

Test objective: Use case; a DIN SPEC 70121 DC session first delivers the power the EVSE advertises, then follows two limits from the CSMS. Precondition: the DIN eIM session (TCP; authorized by the tester CSMS, OCPP RequestStartTransaction) through PowerDelivery (ReadyToChargeState = true). The Test System then runs the CurrentDemand loop (one request every 250 ms) at EVTargetVoltage 400 V and EVTargetCurrent = the EVSEMaximumCurrentLimit of ChargeParameterDiscoveryRes -- more than the EVSE can deliver once its power limit applies. Criteria: (1) within 10 s the delivered power (EVSEPresentVoltage x EVSEPresentCurrent of CurrentDemandRes) is within 3 % of the advertised EVSEMaximumPowerLimit of the same response, and EVSEPresentVoltage within 3 % of the 400 V target; (2) for a further 5 s the delivered power holds within 3 % of EVSEMaximumPowerLimit and the present current at or below EVSEMaximumCurrentLimit (+3 %) -- that advertised power is P_max; (3) with exactly one open OCPP transaction on the EVSE (precondition), for L = 75 % and then 50 % of P_max, with the CurrentDemand loop running throughout: the CSMS sends SetChargingProfile(evseId 1, TxProfile of the open transaction, Absolute, chargingRateUnit W, one period with limit L; the same profile id, so the second replaces the first) and the station answers Accepted; within 10 s of the request the CurrentDemandRes advertise EVSEMaximumPowerLimit <= L (+3 %) and the delivered power is within 3 % of L; it holds there for 3 s; (4) every CurrentDemandRes is OK with EVSEStatusCode = EVSE_Ready; (5) a clean end: PowerDelivery (ReadyToChargeState = false), CP State B, WeldingDetection, SessionStop, all OK. The charging profile is cleared afterwards; the levels and EVSEPowerLimitAchieved are recorded.

**DUT** (dut) **FAIL** 28.10 s

**FAILURE REASON**

harness/test_ev.py:434

```
AssertionError: within 10 s: EVSEMaximumPowerLimit <= 75% of P_max = 8.28 kW and the
delivered power within 3% of it — last: P 2.76 kW (U 400.00 V, I 6.90 A),
EVSEMaximumPowerLimit 2.76 kW, EVSEMaximumCurrentLimit 6.90 A
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| within 10 s of the charge loop start the delivered power (EVSEPresentVoltage x EVSEPresentCurrent) is within 3% of the advertised EVSEMaximumPowerLimit and EVSEPresentVoltage within 3% of the 400 V target | after 0.4 s: P 11.04 kW (U 400.00 V, I 27.60 A), EVSEMaximumPowerLimit 11.04 kW, EVSEMaximumCurrentLimit 200.00 A | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| for 5 s the delivered power holds within 3% of EVSEMaximumPowerLimit and EVSEPresentCurrent <= EVSEMaximumCurrentLimit | 21 responses, last P 11.04 kW (U 400.00 V, I 27.60 A), EVSEMaximumPowerLimit 11.04 kW, EVSEMaximumCurrentLimit 27.60 A | ✓ |
| Response echoes the current SessionID | | ✓ |
| SetChargingProfile(TxProfile, 75% of P_max = 8.28 kW) -> Accepted | {'status': 'Accepted'} | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| within 10 s: EVSEMaximumPowerLimit <= 75% of P_max = 8.28 kW and the delivered power within 3% of it | last: P 2.76 kW (U 400.00 V, I 6.90 A), EVSEMaximumPowerLimit 2.76 kW, EVSEMaximumCurrentLimit 6.90 A | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 415.5 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_MATCH.CNF response time | 0.7 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.2 ms | - | info |
| EV target current (EVSEMaximumCurrentLimit of ChargeParameterDiscoveryRes) | 200.0 A | - | info |
| delivered power at the advertised maximum after PowerDelivery(Start) | 353.4 ms | - | info |
| Delivered power (maximum power) | 11.04 kW | - | info |
| EVSEMaximumPowerLimit (maximum power) | 11.04 kW | - | info |
| EVSEPresentVoltage (maximum power) | 400.0 V | - | info |
| EVSEPresentCurrent (maximum power) | 27.6 A | - | info |
| EVSEPowerLimitAchieved (maximum power) | True | - | info |
| P_max (advertised EVSEMaximumPowerLimit) | 11.04 kW | - | info |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:58:00.270 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:58:00.269Z"}` |
| 06:58:00.271 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:58:02.701 | CSMS → CS | RequestStartTransaction | `{"evseId":1,"idToken": {"idToken":"DEADBEEF","type":"ISO14443"},"remoteStartId":1170682700}` |
| 06:58:02.702 | CS → CSMS | RequestStartTransaction (response) | `{"status":"Accepted"}` |
| 06:58:02.778 | CS → CSMS | TransactionEvent | `{"eventType":"Started","evse":{"connectorId": 1,"id":1},"idToken":{"additionalInfo": [{"additionalIdToken":"B6:44:63:86:A2:CD","type":"EVCCID"}],"idToken":"DEADBEEF","type":"ISO14443"}, "meterValue":[{"sampledValue": [{"context":"Transaction.Begin","location":"Outlet","measurand":"Energy.Active.Import.Register","unitOfMeasure":{"unit":"Wh"},"value": 1055.050048828125}, {"context":"Transaction.Begin","location":"Outlet","measurand":"Current.Import","unitOfMeasure": {"unit":"A"},"value": 0.0}],"timestamp":"2026-10-28T06:58:02.534Z"}],"offline":false,"seqNo": 0,"timestamp":"2026-10-28T06:58:02.774Z","tran …` |
| 06:58:02.780 | CSMS → CS | TransactionEvent (response) | `{"idTokenInfo":{"status":"Accepted"}}` |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:58:03.410 | CS → CSMS | NotifyEVChargingNeeds | {"chargingNeeds":{"dcChargingParameters": {"evMaxCurrent":300.0,"evMaxVoltage": 900.0,"stateOfCharge": 50},"requestedEnergyTransfer":"DC"},"evseId":1} |
| 06:58:03.411 | CSMS → CS | NotifyEVChargingNeeds (response) | {"status":"NoChargingProfile"} |
| 06:58:08.532 | CS → CSMS | TransactionEvent | {"eventType":"Updated","offline":false,"seqNo": 1,"timestamp":"2026-10-28T06:58:08.530Z","transac tionInfo": {"chargingState":"Charging","remoteStartId": 1170682700,"transactionId":"629fe7a4- ebf0-4ed6-92fe- f3fdcb3f049f"},"triggerReason":"ChargingStateChan ged"} |
| 06:58:08.533 | CSMS → CS | TransactionEvent (response) | {} |
| 06:58:10.889 | CS → CSMS | Heartbeat | {} |
| 06:58:10.890 | CSMS → CS | Heartbeat (response) | {"currentTime":"2026-10-28T06:58:10.000Z"} |
| 06:58:14.031 | CSMS → CS | SetChargingProfile | {"chargingProfile": {"chargingProfileKind":"Absolute","chargingProfil ePurpose":"TxProfile","chargingSchedule": [{"chargingRateUnit":"W","chargingSchedulePeriod": [{"limit":8280.0,"startPeriod":0}],"id": 1,"startSchedule":"2026-10-28T06:58:09Z"}],"id": 1001,"stackLevel":0,"transactionId":"629fe7a4- ebf0-4ed6-92fe-f3fdcb3f049f"},"evseId":1} |
| 06:58:14.034 | CS → CSMS | SetChargingProfile (response) | {"status":"Accepted"} |
| 06:58:24.272 | CSMS → CS | ClearChargingProfile | {"chargingProfileId":1001} |
| 06:58:24.275 | CS → CSMS | ClearChargingProfile (response) | {"status":"Accepted"} |

Failure detail (excerpt):

```
E   AssertionError: within 10 s: EVSEMaximumPowerLimit <= 75% of P_max = 8.28 kW and the delivered power within 3% of it — last: P 2.76 kW (U 400.00 V, I 6.90 A), EVSEMaximumPowerLimit 2.76 kW, EVSEMaximumCurrentLimit 6.90 A
```

## TC_SECC_DC_VTB_PionixUseCase_010

TC_SECC_DC_VTB_PionixUseCase_010.

Test objective: Use case; the ISO 15118-2 twin of _009. Precondition: the DC ISO 15118-2 eIM session (TCP; authorized by the tester CSMS, OCPP RequestStartTransaction) through PowerDelivery(Start). The Test System then runs the CurrentDemand loop (one request every 250 ms) at EVTargetVoltage 400 V and EVTargetCurrent = the EVSEMaximumCurrentLimit of ChargeParameterDiscoveryRes -- more than the EVSE can deliver once its power limit applies. Criteria: (1) within 10 s the delivered power (EVSEPresentVoltage x EVSEPresentCurrent of CurrentDemandRes) is within 3 % of the advertised EVSEMaximumPowerLimit of the same response, and EVSEPresentVoltage within 3 % of the 400 V target; (2) for a further 5 s the delivered power holds within 3 % of EVSEMaximumPowerLimit and the present current at or below EVSEMaximumCurrentLimit (+3 %) -- that advertised power is P_max; (3) with exactly one open OCPP transaction on the EVSE (precondition), for L = 75 % and then 50 % of P_max, with the CurrentDemand loop running throughout: the CSMS sends SetChargingProfile(evseId 1, TxProfile of the open transaction, Absolute, chargingRateUnit W, one period with limit L; the same profile id, so the second replaces the first) and the station answers Accepted; within 10 s of the request the CurrentDemandRes advertise EVSEMaximumPowerLimit <= L (+3 %) and the delivered power is within 3 % of L; it holds there for 3 s; (4) every CurrentDemandRes is OK with EVSEStatusCode = EVSE_Ready; (5) a clean end: PowerDelivery(Stop), CP State B, WeldingDetection, SessionStop(Terminate), all OK. The charging profile is cleared afterwards; the levels and EVSEPowerLimitAchieved are recorded.

**DUT** (dut) **FAIL** 27.99 s

**FAILURE REASON**

harness/test_ev.py:434

```
AssertionError: within 10 s: EVSEMaximumPowerLimit <= 75% of P_max = 8.28 kW and the
delivered power within 3% of it — last: P 2.76 kW (U 400.00 V, I 6.90 A),
EVSEMaximumPowerLimit 2.76 kW, EVSEMaximumCurrentLimit 6.90 A
```

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| within 10 s of the charge loop start the delivered power (EVSEPresentVoltage x EVSEPresentCurrent) is within 3% of the advertised EVSEMaximumPowerLimit and EVSEPresentVoltage within 3% of the 400 V target | after 0.4 s: P 11.04 kW (U 400.00 V, I 27.60 A), EVSEMaximumPowerLimit 11.04 kW, EVSEMaximumCurrentLimit 200.00 A | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| for 5 s the delivered power holds within 3% of EVSEMaximumPowerLimit and EVSEPresentCurrent <= EVSEMaximumCurrentLimit | 21 responses, last P 11.04 kW (U 400.00 V, I 27.60 A), EVSEMaximumPowerLimit 11.04 kW, EVSEMaximumCurrentLimit 27.60 A | ✓ |
| Response echoes the current SessionID | | ✓ |
| SetChargingProfile(TxProfile, 75% of P_max = 8.28 kW) -> Accepted | {'status': 'Accepted'} | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| within 10 s: EVSEMaximumPowerLimit <= 75% of P_max = 8.28 kW and the delivered power within 3% of it | last: P 2.76 kW (U 400.00 V, I 6.90 A), EVSEMaximumPowerLimit 2.76 kW, EVSEMaximumCurrentLimit 6.90 A | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 411.9 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.6 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| supportedAppProtocolRes response time | 100.5 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| SessionSetupRes response time | 101.7 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| ServiceDiscoveryRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PaymentServiceSelectionRes response time | 101.9 ms | ≤ 2000 ms (V2G_EVCC_Msg_Timeout) | ✓ |
| PowerDeliveryRes(Start) response time | 101.4 ms | V2G_SECC_Sequence_Performance_Time (informational) | ✓ |
| EV target current (EVSEMaximumCurrentLimit of ChargeParameterDiscoveryRes) | 200.0 A | - | info |
| delivered power at the advertised maximum after PowerDelivery(Start) | 352.8 ms | - | info |
| Delivered power (maximum power) | 11.04 kW | - | info |
| EVSEMaximumPowerLimit (maximum power) | 11.04 kW | - | info |
| EVSEPresentVoltage (maximum power) | 400.0 V | - | info |
| EVSEPresentCurrent (maximum power) | 27.6 A | - | info |
| EVSEPowerLimitAchieved (maximum power) | True | - | info |
| P_max (advertised EVSEMaximumPowerLimit) | 11.04 kW | - | info |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:58:28.267 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:58:28.266Z"}` |
| 06:58:28.268 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:58:30.692 | CSMS → CS | RequestStartTransaction | `{"evseId":1,"idToken": {"idToken":"DEADBEEF","type":"ISO14443"},"remoteStartId":1170710690}` |
| 06:58:30.693 | CS → CSMS | RequestStartTransaction (response) | `{"status":"Accepted"}` |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:58:30.774 | CS → CSMS | TransactionEvent | {"eventType":"Started","evse":{"connectorId": 1,"id":1},"idToken":{"additionalInfo": [{"additionalIdToken":"B6:44:63:86:A2:CD","type":" EVCCID"}],"idToken":"DEADBEEF","type":"ISO14443"}, "meterValue":[{"sampledValue": [{"context":"Transaction.Begin","location":"Outlet ","measurand":"Energy.Active.Import.Register","un itOfMeasure":{"unit":"Wh"},"value": 1097.927734375}, {"context":"Transaction.Begin","location":"Outlet" ,"measurand":"Current.Import","unitOfMeasure": {"unit":"A"},"value": 0.0}],"timestamp":"2026-10-28T06:58:30.552Z"}],"o ffline":false,"seqNo": 0,"timestamp":"2026-10-28T06:58:30.771Z","transac … |
| 06:58:30.776 | CSMS → CS | TransactionEvent (response) | {"idTokenInfo":{"status":"Accepted"}} |
| 06:58:31.402 | CS → CSMS | NotifyEVChargingNeeds | {"chargingNeeds":{"dcChargingParameters": {"evMaxCurrent":300.0,"evMaxVoltage": 900.0,"stateOfCharge": 50},"requestedEnergyTransfer":"DC"},"evseId":1} |
| 06:58:31.403 | CSMS → CS | NotifyEVChargingNeeds (response) | {"status":"NoChargingProfile"} |
| 06:58:36.518 | CS → CSMS | TransactionEvent | {"eventType":"Updated","offline":false,"seqNo": 1,"timestamp":"2026-10-28T06:58:36.517Z","transac tionInfo": {"chargingState":"Charging","remoteStartId": 1170710690,"transactionId":"f355e586-9170-4fb9- b858-5fed546bb47b"},"triggerReason":"ChargingStat eChanged"} |
| 06:58:36.520 | CSMS → CS | TransactionEvent (response) | {} |
| 06:58:40.889 | CS → CSMS | Heartbeat | {} |
| 06:58:40.890 | CSMS → CS | Heartbeat (response) | {"currentTime":"2026-10-28T06:58:40.000Z"} |
| 06:58:42.054 | CSMS → CS | SetChargingProfile | {"chargingProfile": {"chargingProfileKind":"Absolute","chargingProfil ePurpose":"TxProfile","chargingSchedule": [{"chargingRateUnit":"W","chargingSchedulePeriod": [{"limit":8280.0,"startPeriod":0}],"id": 1,"startSchedule":"2026-10-28T06:58:37Z"}],"id": 1001,"stackLevel": 0,"transactionId":"f355e586-9170-4fb9- b858-5fed546bb47b"},"evseId":1} |
| 06:58:42.059 | CS → CSMS | SetChargingProfile (response) | {"status":"Accepted"} |
| 06:58:52.298 | CSMS → CS | ClearChargingProfile | {"chargingProfileId":1001} |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:58:52.301 | CS → CSMS | ClearChargingProfile (response) | `{"status":"Accepted"}` |

Failure detail (excerpt):

```
E   AssertionError: within 10 s: EVSEMaximumPowerLimit <= 75% of P_max = 8.28 kW and the
delivered power within 3% of it — last: P 2.76 kW (U 400.00 V, I 6.90 A), EVSEMaximumPowerLimit
2.76 kW, EVSEMaximumCurrentLimit 6.90 A
```

## TC_SECC_DC_VTB_PionixUseCase_011

TC_SECC_DC_VTB_PionixUseCase_011.

Test objective: Use case; the ISO 15118-20 Scheduled-mode twin of _009 (unidirectional DC). Precondition: the DC ISO 15118-20 EIM session (TLS 1.3 mTLS, Scheduled control mode; authorized by the tester CSMS, OCPP RequestStartTransaction) through PowerDelivery(Start). The Test System then runs the DC charge loop (one DC_ChargeLoopReq every 250 ms) at EVTargetVoltage 400 V and EVTargetCurrent = the power of the agreed PowerSchedule (first entry of the first ScheduleTuple of ScheduleExchangeRes) / 400 V, at most the EVSEMaximumChargeCurrent of DC_ChargeParameterDiscoveryRes; the EV keeps that request throughout -- an EVSEMaximumChargePower below the PowerSchedule overrules it (ISO 15118-20 [V2G20-1011] NOTE 2) and a schedule renegotiation is the SECC's option ([V2G20-1845]), not required. Criteria: (1) within 10 s the delivered power (EVSEPresentVoltage x EVSEPresentCurrent of DC_ChargeLoopRes) is within 3 % of the advertised EVSEMaximumChargePower of the same response's Scheduled_DC_CLResControlMode, and EVSEPresentVoltage within 3 % of the 400 V target; (2) for a further 5 s the delivered power holds within 3 % of EVSEMaximumChargePower and the present current at or below EVSEMaximumChargeCurrent (+3 %) -- that advertised power is P_max; (3) with exactly one open OCPP transaction on the EVSE (precondition), for L = 75 % and then 50 % of P_max, with the charge loop running throughout: the CSMS sends SetChargingProfile(evseId 1, TxProfile of the open transaction, Absolute, chargingRateUnit W, one period with limit L; the same profile id, so the second replaces the first) and the station answers Accepted; within 10 s of the request the DC_ChargeLoopRes advertise EVSEMaximumChargePower <= L (+3 %) and the delivered power is within 3 % of L; it holds there for 3 s; (4) every DC_ChargeLoopRes is OK and carries Scheduled_DC_CLResControlMode; (5) a clean end: PowerDelivery(Stop), CP State B, DC_WeldingDetection, SessionStop(Terminate), all OK. The charging profile is cleared afterwards; the levels and EVSEPowerLimitAchieved are recorded.

**DUT** (dut) **FAIL** 28.60 s

**FAILURE REASON**

harness/test_ev.py:434

```
AssertionError: within 10 s: EVSEMaximumChargePower <= 75% of P_max = 8.28 kW and the delivered power within 3% of it — last: P 2.76 kW (U 400.00 V, I 6.90 A), EVSEMaximumChargePower 2.76 kW, EVSEMaximumChargeCurrent 6.90 A
```

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| within 10 s of the charge loop start the delivered power (EVSEPresentVoltage x EVSEPresentCurrent) is within 3% of the advertised EVSEMaximumChargePower and EVSEPresentVoltage within 3% of the 400 V target | after 1.3 s: P 11.04 kW (U 400.00 V, I 27.60 A), EVSEMaximumChargePower 11.04 kW, EVSEMaximumChargeCurrent 27.60 A | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| for 5 s the delivered power holds within 3% of EVSEMaximumChargePower and EVSEPresentCurrent <= EVSEMaximumChargeCurrent | 21 responses, last P 11.04 kW (U 400.00 V, I 27.60 A), EVSEMaximumChargePower 11.04 kW, EVSEMaximumChargeCurrent 27.60 A | ✓ |
| Response echoes the current SessionID | | ✓ |
| SetChargingProfile(TxProfile, 75% of P_max = 8.28 kW) → Accepted | {'status': 'Accepted'} | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| within 10 s: EVSEMaximumChargePower <= 75% of P_max = 8.28 kW and the delivered power within 3% of it | last: P 2.76 kW (U 400.00 V, I 6.90 A), EVSEMaximumChargePower 2.76 kW, EVSEMaximumChargeCurrent 6.90 A | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.2 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.4 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| EV target current (the agreed PowerSchedule / EVTargetVoltage, at most EVSEMaximumChargeCurrent) | 27.6 A | - | info |
| delivered power at the advertised maximum after PowerDelivery(Start) | 1256.5 ms | - | info |
| Delivered power (maximum power) | 11.04 kW | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| EVSEMaximumChargePower (maximum power) | 11.04 kW | - | info |
| EVSEPresentVoltage (maximum power) | 400.0 V | - | info |
| EVSEPresentCurrent (maximum power) | 27.6 A | - | info |
| EVSEPowerLimitAchieved (maximum power) | False | - | info |
| P_max (advertised EVSEMaximumChargePower) | 11.04 kW | - | info |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:58:56.266 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:58:56.265Z"}` |
| 06:58:56.266 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:58:58.612 | CSMS → CS | RequestStartTransaction | `{"evseId":1,"idToken": {"idToken":"DEADBEEF","type":"ISO14443"},"remoteStartId":1170738610}` |
| 06:58:58.613 | CS → CSMS | RequestStartTransaction (response) | `{"status":"Accepted"}` |
| 06:58:58.672 | CS → CSMS | TransactionEvent | `{"eventType":"Started","evse":{"connectorId": 1,"id":1},"idToken":{"additionalInfo": [{"additionalIdToken":"b6446386a2cd","type":"EVCCID"}],"idToken":"DEADBEEF","type":"ISO14443"},"meterValue":[{"sampledValue": [{"context":"Transaction.Begin","location":"Outlet","measurand":"Energy.Active.Import.Register","unitOfMeasure":{"unit":"Wh"},"value":1141.25}, {"context":"Transaction.Begin","location":"Outlet","measurand":"Current.Import","unitOfMeasure": {"unit":"A"},"value": 0.0}],"timestamp":"2026-10-28T06:58:58.569Z"}],"offline":false,"seqNo": 0,"timestamp":"2026-10-28T06:58:58.669Z","transactionInfo":{" …` |
| 06:58:58.675 | CSMS → CS | TransactionEvent (response) | `{"idTokenInfo":{"status":"Accepted"}}` |
| 06:58:59.613 | CS → CSMS | NotifyEVChargingNeeds | `{"chargingNeeds": {"controlMode":"ScheduledControl","mobilityNeedsMode":"EVCC","requestedEnergyTransfer":"DC","v2xChargingParameters":{"evMaxEnergyRequest": 20000.0,"evMinEnergyRequest": 5000.0,"evTargetEnergyRequest": 10000.0,"maxChargeCurrent":200.0,"maxChargePower": 11040.0,"maxVoltage":900.0,"minChargeCurrent": 1.0,"minChargePower":1000.0,"minVoltage": 200.0}},"evseId":1}` |
| 06:58:59.614 | CSMS → CS | NotifyEVChargingNeeds (response) | `{"status":"NoChargingProfile"}` |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:59:04.526 | CS → CSMS | TransactionEvent | {"eventType":"Updated","offline":false,"seqNo":1,"timestamp":"2026-10-28T06:59:04.524Z","transactionInfo":{"chargingState":"Charging","remoteStartId":1170738610,"transactionId":"4a75819f-5679-4ece-9af3-14785885e061"},"triggerReason":"ChargingStateChanged"} |
| 06:59:04.527 | CSMS → CS | TransactionEvent (response) | {} |
| 06:59:10.771 | CSMS → CS | SetChargingProfile | {"chargingProfile":{"chargingProfileKind":"Absolute","chargingProfilePurpose":"TxProfile","chargingSchedule":[{"chargingRateUnit":"W","chargingSchedulePeriod":[{"limit":8280.0,"startPeriod":0}],"id":1,"startSchedule":"2026-10-28T06:59:05Z"}],"id":1001,"stackLevel":0,"transactionId":"4a75819f-5679-4ece-9af3-14785885e061"},"evseId":1} |
| 06:59:10.775 | CS → CSMS | SetChargingProfile (response) | {"status":"Accepted"} |
| 06:59:10.889 | CS → CSMS | Heartbeat | {} |
| 06:59:10.890 | CSMS → CS | Heartbeat (response) | {"currentTime":"2026-10-28T06:59:10.000Z"} |
| 06:59:20.857 | CSMS → CS | ClearChargingProfile | {"chargingProfileId":1001} |
| 06:59:20.861 | CS → CSMS | ClearChargingProfile (response) | {"status":"Accepted"} |

Failure detail (excerpt):

```
E   AssertionError: within 10 s: EVSEMaximumChargePower <= 75% of P_max = 8.28 kW and the delivered power within 3% of it — last: P 2.76 kW (U 400.00 V, I 6.90 A), EVSEMaximumChargePower 2.76 kW, EVSEMaximumChargeCurrent 6.90 A
```

## TC_SECC_DC_VTB_PionixUseCase_012

TC_SECC_DC_VTB_PionixUseCase_012.

Test objective: Use case; the ISO 15118-20 Dynamic-mode twin of _009 (unidirectional DC). Precondition: the DC ISO 15118-20 EIM session (TLS 1.3 mTLS, Dynamic control mode; authorized by the tester CSMS, OCPP RequestStartTransaction) through PowerDelivery(Start). The Test System then runs the DC charge loop (one DC_ChargeLoopReq every 250 ms, EVPresentVoltage 400 V). In the Dynamic mode the EVSE controls the power within the EV's limits; the EV states EVMaximumChargePower 150 kW, EVMaximumChargeCurrent 300 A, EVMaximumVoltage 900 V and EVTargetEnergyRequest = EVMaximumEnergyRequest = 100 kWh -- more than the EVSE can deliver by the DepartureTime (7200 s) of the ScheduleExchange, so charging at its maximum is expected. Criteria: (1) within 10 s the delivered power (EVSEPresentVoltage x EVSEPresentCurrent of DC_ChargeLoopRes) is within 3 % of the advertised EVSEMaximumChargePower of the same response's Dynamic_DC_CLResControlMode (no voltage criterion: the Dynamic request carries no target voltage); (2) for a further 5 s the delivered power holds within 3 % of EVSEMaximumChargePower and the present current at or below EVSEMaximumChargeCurrent (+3 %) -- that advertised power is P_max; (3) with exactly one open OCPP transaction on the EVSE (precondition), for L = 75 % and then 50 % of P_max, with the charge loop running throughout: the CSMS sends SetChargingProfile(evseId 1, TxProfile of the open transaction, Absolute, chargingRateUnit W, one period with limit L; the same profile id, so the second replaces the first) and the station answers Accepted; within 10 s of the request the DC_ChargeLoopRes advertise EVSEMaximumChargePower <= L (+3 %) and the delivered power is within 3 % of L; it holds there for 3 s; (4) every DC_ChargeLoopRes is OK and carries Dynamic_DC_CLResControlMode; (5) a clean end: PowerDelivery(Stop), CP State B, DC_WeldingDetection, SessionStop(Terminate), all OK. The charging profile is cleared afterwards; the levels and EVSEPowerLimitAchieved are recorded.

**DUT** (dut) **FAIL** 26.90 s

**FAILURE REASON**

harness/test_ev.py:434

```
AssertionError: within 10 s: EVSEMaximumChargePower <= 75% of P_max = 8.28 kW and the
delivered power within 3% of it — last: P 2.76 kW (U 900.00 V, I 3.07 A),
EVSEMaximumChargePower 2.76 kW, EVSEMaximumChargeCurrent 3.07 A
```

| Conformance check | Observed | Result |
|---|---|---|
| TLS protocol version = TLSv1.3 [V2G20-2365] | negotiated TLSv1.3 | ✓ |
| TLS ciphersuite in ISO 15118-20 Table 6 [V2G20-2458] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |
| TLS ciphersuite = TLS_AES_256_GCM_SHA384 (most preferred offered suite) [V2G20-1857] | negotiated TLS_AES_256_GCM_SHA384 | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| TLS key share named group = secp521r1 (Table 7) [V2G20-1634] | client key_share pinned; handshake completed | ✓ |
| SECC certificate key on secp521r1 (ISO 15118-20 Annex B.4, [V2G20-2674] curve) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: key curve secp521r1 | ✓ |
| SECC certificate signed with ECDSA-SHA512 (ISO 15118-20 Annex B.4) | DC=CPO,C=DE,O=EVerest,CN=SECCCert: signature digest sha512 | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| within 10 s of the charge loop start the delivered power (EVSEPresentVoltage x EVSEPresentCurrent) is within 3% of the advertised EVSEMaximumChargePower | after 0.3 s: P 11.03 kW (U 900.00 V, I 12.26 A), EVSEMaximumChargePower 11.04 kW, EVSEMaximumChargeCurrent 12.26 A | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| for 5 s the delivered power holds within 3% of EVSEMaximumChargePower and EVSEPresentCurrent <= EVSEMaximumChargeCurrent | 21 responses, last P 11.03 kW (U 900.00 V, I 12.26 A), EVSEMaximumChargePower 11.04 kW, EVSEMaximumChargeCurrent 12.26 A | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| SetChargingProfile(TxProfile, 75% of P_max = 8.28 kW) → Accepted | {'status': 'Accepted'} | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |

| Conformance check | Observed | Result |
|---|---|---|
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
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| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| Response echoes the current SessionID | | ✓ |
| within 10 s: EVSEMaximumChargePower <= 75% of P_max = 8.28 kW and the delivered power within 3% of it | last: P 2.76 kW (U 900.00 V, I 3.07 A), EVSEMaximumChargePower 2.76 kW, EVSEMaximumChargeCurrent 3.07 A | ✗ |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| CM_SLAC_PARM.CNF response time | 1.1 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_ATTEN_CHAR.IND after sounding | 414.2 ms | ≤ 1200 ms (TT_EV_atten_results) | ✓ |
| CM_SLAC_MATCH.CNF response time | 0.8 ms | ≤ 200 ms (TT_match_response) | ✓ |
| CM_SET_KEY.CNF response time | 0.3 ms | - | info |
| EV request | Dynamic mode: the EVSE controls the power (EVMaximumChargePower 150 kW, energy request 100 kWh) | - | info |
| delivered power at the advertised maximum after PowerDelivery(Start) | 253.2 ms | - | info |
| Delivered power (maximum power) | 11.03 kW | - | info |
| EVSEMaximumChargePower (maximum power) | 11.04 kW | - | info |
| EVSEPresentVoltage (maximum power) | 900.0 V | - | info |

| Measurement | Measured | Target / range | Result |
|---|---|---|---|
| EVSEPresentCurrent (maximum power) | 12.3 A | - | info |
| EVSEPowerLimitAchieved (maximum power) | False | - | info |
| P_max (advertised EVSEMaximumChargePower) | 11.04 kW | - | info |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:59:24.397 | CS → CSMS | StatusNotification | `{"connectorId": 1,"connectorStatus":"Occupied","evseId": 1,"timestamp":"2026-10-28T06:59:24.396Z"}` |
| 06:59:24.398 | CSMS → CS | StatusNotification (response) | `{}` |
| 06:59:26.718 | CSMS → CS | RequestStartTransaction | `{"evseId":1,"idToken": {"idToken":"DEADBEEF","type":"ISO14443"},"remoteS tartId":1170766716}` |
| 06:59:26.719 | CS → CSMS | RequestStartTransaction (response) | `{"status":"Accepted"}` |
| 06:59:26.805 | CS → CSMS | TransactionEvent | `{"eventType":"Started","evse":{"connectorId": 1,"id":1},"idToken":{"additionalInfo": [{"additionalIdToken":"b6446386a2cd","type":"EVCC ID"}],"idToken":"DEADBEEF","type":"ISO14443"},"me terValue":[{"sampledValue": [{"context":"Transaction.Begin","location":"Outlet ","measurand":"Energy.Active.Import.Register","un itOfMeasure":{"unit":"Wh"},"value": 1172.62548828125}, {"context":"Transaction.Begin","location":"Outlet" ,"measurand":"Current.Import","unitOfMeasure": {"unit":"A"},"value": 0.0}],"timestamp":"2026-10-28T06:59:26.586Z"}],"o ffline":false,"seqNo": 0,"timestamp":"2026-10-28T06:59:26.801Z","transac tio …` |
| 06:59:26.807 | CSMS → CS | TransactionEvent (response) | `{"idTokenInfo":{"status":"Accepted"}}` |
| 06:59:27.721 | CS → CSMS | NotifyEVChargingNeeds | `{"chargingNeeds": {"controlMode":"DynamicControl","mobilityNeedsMode ":"EVCC","requestedEnergyTransfer":"DC","v2xCharg ingParameters":{"evMaxEnergyRequest": 20000.0,"evMinEnergyRequest": 5000.0,"evTargetEnergyRequest": 10000.0,"maxChargeCurrent":200.0,"maxChargePower": 11040.0,"maxVoltage":900.0,"minChargeCurrent": 1.0,"minChargePower":1000.0,"minVoltage": 200.0}},"evseId":1}` |
| 06:59:27.723 | CSMS → CS | NotifyEVChargingNeeds (response) | `{"status":"NoChargingProfile"}` |

| OCPP (DUT ↔ tester CSMS) | Direction | Message | Payload |
|---|---|---|---|
| 06:59:32.433 | CS → CSMS | TransactionEvent | {"eventType":"Updated","offline":false,"seqNo":1,"timestamp":"2026-10-28T06:59:32.432Z","transactionInfo":{"chargingState":"Charging","remoteStartId":1170766716,"transactionId":"d1fff99a-f3c8-41b0-9559-ac18b82c18b7"},"triggerReason":"ChargingStateChanged"} |
| 06:59:32.435 | CSMS → CS | TransactionEvent (response) | {} |
| 06:59:37.670 | CSMS → CS | SetChargingProfile | {"chargingProfile":{"chargingProfileKind":"Absolute","chargingProfilePurpose":"TxProfile","chargingSchedule":[{"chargingRateUnit":"W","chargingSchedulePeriod":[{"limit":8280.0,"startPeriod":0}],"id":1,"startSchedule":"2026-10-28T06:59:32Z"}],"id":1001,"stackLevel":0,"transactionId":"d1fff99a-f3c8-41b0-9559-ac18b82c18b7"},"evseId":1} |
| 06:59:37.673 | CS → CSMS | SetChargingProfile (response) | {"status":"Accepted"} |
| 06:59:40.889 | CS → CSMS | Heartbeat | {} |
| 06:59:40.890 | CSMS → CS | Heartbeat (response) | {"currentTime":"2026-10-28T06:59:40.000Z"} |
| 06:59:47.758 | CSMS → CS | ClearChargingProfile | {"chargingProfileId":1001} |
| 06:59:47.761 | CS → CSMS | ClearChargingProfile (response) | {"status":"Accepted"} |

Failure detail (excerpt):

```
E   AssertionError: within 10 s: EVSEMaximumChargePower <= 75% of P_max = 8.28 kW and the delivered power within 3% of it — last: P 2.76 kW (U 900.00 V, I 3.07 A), EVSEMaximumChargePower 2.76 kW, EVSEMaximumChargeCurrent 3.07 A
```

# Appendix

## Codec self-tests

No codec self-tests were included in this run.

## Toolchain / report metadata

| | |
|---|---|
| pytest version | unknown |
| Report created | 2026-10-28 07:02:41 |
| pytest exit code | 1 |
| Report generator | Pionix conformance tester report generator |
| Generation timestamp | 2026-10-28 08:13:11 |